

E70967_80

The equation $x^4 + xy + y^3 - y - 4 = 0$ has no integer solutions

Size $h = 34$, length $l = 12$

Theorem 1. *The equation*

$$x^4 + xy + y^3 - y - 4 = 0 \tag{1}$$

has no integer solutions.

The proof combines quadratic Hilbert symbols over $\mathbb{Q}$ and cubic Hilbert symbols over $K = \mathbb{Q}(\omega)$, where $\omega^2 + \omega + 1 = 0$. To a hypothetical integer solution we attach the values of auxiliary polynomials H_i with rational coefficients, linked by 2 graphs, and the values of auxiliary polynomials J_i with coefficients in $\mathbb{Z}[\omega]$, linked by 3 weighted graphs. By the product formula and the reciprocity law, the products of the quadratic symbols over all places are 1, and the sums of the cubic symbols over all places are 0. We show that the local contributions vanish at the primes outside a finite set S , determine the sign contributions at the real place, compute the local contributions jointly at the primes of S , and obtain a contradiction. Apart from polynomial identities, which can be checked by expanding both sides, finite computations with residues, and basic properties of the Legendre symbol and of the quadratic Hilbert symbol, the proof uses two standard facts from class field theory: the explicit formula for the cubic Hilbert symbol at the places not above 3, and the Hilbert reciprocity law.

1 Legendre symbols and Hilbert symbols

For a prime p and a non-zero integer N , we write $v_p(N)$ for the exponent of p in the prime factorization of N . For an integer a and an odd prime p , the Legendre symbol (a/p) is 0 if $p \mid a$, 1 if a is a non-zero square modulo p , and -1 otherwise. It is multiplicative in a , and it depends only on a modulo p [1, Chapter 5].

For a prime p , let $\mathbb{Q}_p$ be the field of p -adic numbers, and put $\mathbb{Q}_\infty = \mathbb{R}$. For v a prime or $v = \infty$, and non-zero $a, b \in \mathbb{Q}$, the Hilbert symbol $(a, b)_v$ is 1 if the equation $z^2 = ar^2 + bs^2$ has a solution $(r, s, z) \neq (0, 0, 0)$ in $\mathbb{Q}_v$, and -1 otherwise. In particular, $(a, b)_v = 1$ for every v if this equation has a non-zero solution in $\mathbb{Q}$. For an odd integer t , we put $\varepsilon(t) = (t - 1)/2$ and $\omega(t) = (t^2 - 1)/8$. We use the following facts [2, Chapter III, Proposition 2 and Theorems 1 and 3].

Theorem 2. *Let a, a' and b be non-zero integers, and let v be a prime or $v = \infty$.*

- (i) $(a, b)_\infty = -1$ if and only if $a < 0$ and $b < 0$.
- (ii) Let p be an odd prime, and write $a = p^\alpha u$ and $b = p^\beta w$ with $p \nmid uw$. Then $(a, b)_p = (-1)^{\alpha\beta\varepsilon(p)}(u/p)^\beta(w/p)^\alpha$.
- (iii) Write $a = 2^\alpha u$ and $b = 2^\beta w$ with u and w odd. Then $(a, b)_2 = (-1)^{\varepsilon(u)\varepsilon(w) + \alpha\omega(w) + \beta\omega(u)}$.
- (iv) $(a, b)_v = 1$ for all but finitely many v , and $\prod_v (a, b)_v = 1$, the product over all primes v and $v = \infty$.
- (v) $(a, b)_v = (b, a)_v$ and $(aa', b)_v = (a, b)_v(a', b)_v$.

Lemma 3. Let a and b be non-zero integers. If p is an odd prime with $p \nmid a$ and $(a/p) = 1$, then $(a, b)_p = 1$. If $a \equiv 1 \pmod{8}$, then $(a, b)_2 = 1$.

Proof. The first claim is Theorem 2(ii) with $\alpha = 0$. For the second, $\varepsilon(a)$ and $\omega(a)$ are even, and Theorem 2(iii) with $\alpha = 0$ gives $(a, b)_2 = 1$. $\square$

Lemma 4. Let p be a prime. For a non-zero integer X , write $X = p^j u$ with $p \nmid u$. If p is odd, put $c_p(X) = (j, e) \in \mathbb{F}_2^2$, where $(u/p) = (-1)^e$; if $p = 2$, put $c_2(X) = (j, e_1, e_2) \in \mathbb{F}_2^3$, where $u \equiv (-1)^{e_1} 5^{e_2} \pmod{8}$; all entries are taken modulo 2. Then $c_p(XX') = c_p(X) + c_p(X')$ and $(X, X')_p = (-1)^{\beta_p(c_p(X), c_p(X'))}$ for non-zero integers X, X' , where β_p is the symmetric bilinear form

$$\begin{aligned} \beta_p((j, e), (j', e')) &= jj' \varepsilon(p) + je' + j'e && \text{if } p \text{ is odd,} \\ \beta_2((j, e_1, e_2), (j', e'_1, e'_2)) &= e_1 e'_1 + je'_2 + j'e_2. \end{aligned}$$

Proof. For odd p , this is Theorem 2(ii), as the Legendre symbol is multiplicative. For $p = 2$, the odd residues 1, 3, 5, 7 modulo 8 are $(-1)^{e_1} 5^{e_2}$ with $(e_1, e_2) = (0, 0), (1, 1), (0, 1), (1, 0)$, respectively; in each case $\varepsilon(u) \equiv e_1$ and $\omega(u) \equiv e_2 \pmod{2}$, so Theorem 2(iii) gives the formula, and c_2 is additive because $(e_1, e_2) \mapsto (-1)^{e_1} 5^{e_2}$ is an isomorphism from $\mathbb{F}_2^2$ to the group of odd residues modulo 8. $\square$

Lemma 5. Let p be a prime, $k \geq 1$, and let X be a non-zero integer with $X \equiv r \pmod{p^k}$, where $0 \leq r < p^k$. Then $c_p(X) \in b + D$ for the vector b and the subspace D given as follows. If $r = 0$, then $b = 0$ and D is the whole space. Otherwise write $r = p^j r'$ with $p \nmid r'$, so that $j < k$. If p is odd, or if $p = 2$ and $k - j \geq 3$, then $b = c_p(r)$ and $D = 0$. If $p = 2$ and $k - j = 2$, then $b = (j, e_1, 0)$ with $r' \equiv (-1)^{e_1} \pmod{4}$, and D is spanned by $(0, 0, 1)$. If $p = 2$ and $k - j = 1$, then $b = (j, 0, 0)$, and D is spanned by $(0, 1, 0)$ and $(0, 0, 1)$.

Proof. If $r \neq 0$, then $v_p(X) = j$ and $X/p^j \equiv r' \pmod{p^{k-j}}$. For odd p , the Legendre symbol of X/p^j is therefore that of r' . For $p = 2$, the residue of $X/2^j$ modulo 8 is that of r' if $k - j \geq 3$, and its residue modulo 4, which determines e_1 , is that of r' if $k - j = 2$. $\square$

Lemma 6. Let p be a prime, and let E be a set of edges on a finite set of vertices. For every vertex i , let b_i be a vector and D_i a subspace of $\mathbb{F}_2^2$ (of $\mathbb{F}_2^3$ if $p = 2$), and put $\sigma_i = \sum_{j \sim i} b_j$, the sum over the edges $\{i, j\} \in E$. Suppose that $\beta_p(d, \sigma_i) = 0$ for every vertex i and every $d \in D_i$, and that $\beta_p(d, d') = 0$ for every edge $\{i, j\} \in E$ and all $d \in D_i$ and $d' \in D_j$. Then

$$\sum_{\{i, j\} \in E} \beta_p(c_i, c_j) = \sum_{\{i, j\} \in E} \beta_p(b_i, b_j) \quad \text{whenever } c_i \in b_i + D_i \text{ for every } i.$$

By bilinearity, it suffices to check the hypotheses for d and d' in spanning sets of D_i and D_j .

Proof. Write $c_i = b_i + d_i$ with $d_i \in D_i$. As β_p is bilinear and symmetric, the left side equals the right side plus $\sum_i \beta_p(d_i, \sigma_i) + \sum_{\{i, j\} \in E} \beta_p(d_i, d_j)$, and both sums vanish. $\square$

Lemma 7. Let $f(z) = az^3 + bz^2 + cz + d$ be a real polynomial with $a \neq 0$ and discriminant $\Delta < 0$. Then f has exactly one real root η , and $f(z)$ has the sign of $a(z - \eta)$ for every real $z \neq \eta$.

Proof. The discriminant is $\Delta = a^4 \prod_{i < j} (\rho_i - \rho_j)^2$, where ρ_1, ρ_2, ρ_3 are the complex roots of f . If all roots were real, then $\Delta \geq 0$. Hence f has a non-real root ρ ; then $\bar{\rho} \neq \rho$ is also a root, and the third root η is real. Thus $f(z) = a(z - \eta)(z - \rho)(z - \bar{\rho}) = a(z - \eta)|z - \rho|^2$ for real z , and $|z - \rho|^2 > 0$. $\square$

2 Cubic Hilbert symbols over $\mathbb{Q}(\omega)$

The field. Let $K = \mathbb{Q}(\omega)$, where $\omega^2 + \omega + 1 = 0$. Its ring of integers is $\mathbb{Z}[\omega]$, and every element of K can be written uniquely as $A + B\omega$ with $A, B \in \mathbb{Q}$. We write

$$\langle A, B \rangle = A + B\omega,$$

also when A and B are polynomials in x and y . Then $\langle A, B \rangle \langle C, D \rangle = \langle AC - BD, AD + BC - BD \rangle$, and the norm is $N(\langle A, B \rangle) = A^2 - AB + B^2$. In particular, a value $\langle A, B \rangle$ with $A, B \in \mathbb{Z}$ is zero if and only if $A = B = 0$.

Places. The finite places of K correspond to the non-zero prime ideals of $\mathbb{Z}[\omega]$. A prime $p \equiv 2 \pmod{3}$ is inert: there is one place above p , with uniformizer p and residue field $\mathbb{Z}[\omega]/p\mathbb{Z}[\omega]$ with p^2 elements. If $p \equiv 1 \pmod{3}$, the polynomial $t^2 + t + 1$ has two roots r modulo p , and for each of them there is a place v_r above p , at which ω is the root of $t^2 + t + 1$ in $\mathbb{Z}_p$ congruent to r modulo p ; the completion is $\mathbb{Q}_p$, the uniformizer is p , the residue field is $\mathbb{F}_p$, and ω reduces to r . The prime 3 is ramified: $\pi = 1 - \omega$ generates the unique prime ideal above 3, $\pi^2 = -3\omega$, and $\mathbb{Z}[\omega]/\pi\mathbb{Z}[\omega] = \mathbb{F}_3$, where $\langle A, B \rangle$ reduces to $A + B$. The only infinite place of K is complex. For a finite place v we write K_v for the completion and $v(a)$ for the normalized valuation, and we call a a *unit* at v if $v(a) = 0$.

Hilbert symbols. For a place v of K and $a, b \in K_v^\times$, the cubic Hilbert symbol $(a, b)_v$ is a cube root of unity, see [3, Chapter V, §3]. We write $(a, b)_v = \omega^{\theta_v(a, b)}$ with $\theta_v(a, b) \in \mathbb{Z}/3\mathbb{Z}$, and we use the following three facts [3, Chapter V, §3 and Chapter VI, §8].

- (H1) $\theta_v(a, bc) = \theta_v(a, b) + \theta_v(a, c)$, $\theta_v(a, b) = -\theta_v(b, a)$, and $\theta_v(a, c^3) = 0$. In particular, $\theta_v(a, b)$ depends only on the classes of a and b in $K_v^\times / K_v^{\times 3}$.
- (H2) Let v be a finite place not above 3, let q be the number of elements of its residue field, and let ϖ be a uniformizer at v . Write $a = \varpi^e u$ and $b = \varpi^f w$ with units u and w , and define $\chi(u) \in \mathbb{Z}/3\mathbb{Z}$ by $\bar{u}^{(q-1)/3} = \bar{\omega}^{\chi(u)}$, where the bar denotes the reduction to the residue field. Then $\theta_v(a, b) = e\chi(w) - f\chi(u)$. In particular, $\theta_v(a, b) = 0$ if a and b are both units.
- (H3) (Hilbert reciprocity.) For $a, b \in K^\times$, $\theta_v(a, b) = 0$ for all but finitely many places v , and $\sum_v \theta_v(a, b) = 0$, where the sum is over all places of K . At the complex place, $\theta_v(a, b) = 0$.

The general formula in (H2) contains an additional term $ef\chi(-1)$, which vanishes because $-1 = (-1)^3$ is a cube. Formula (H2) fixes the normalization of the symbols; with the opposite sign convention all values θ_v change sign, and the proof below is the same.

Classes at the places not above 3. Let $p \neq 3$ be a prime and v a place above p . For $a = p^e u \in K_v^\times$ with a unit u , put $\text{cl}_v(a) = (e \bmod 3, \chi(u)) \in \mathbb{F}_3^2$. By (H2) and (H1), cl_v is additive, $\text{cl}_v(a)$ determines the class of a modulo cubes, and $\theta_v(a, b) = e\chi(w) - f\chi(u)$ for $\text{cl}_v(a) = (e, \chi(u))$ and $\text{cl}_v(b) = (f, \chi(w))$. We put $\text{cl}_p(a) = \text{cl}_v(a) \in \mathbb{F}_3^2$ if $p \equiv 2 \pmod{3}$, and $\text{cl}_p(a) = (\text{cl}_{v_r}(a), \text{cl}_{v_{r'}}(a)) \in \mathbb{F}_3^4$ if $p \equiv 1 \pmod{3}$, where $r < r'$ are the roots of $t^2 + t + 1$ in $\{1, \dots, p-1\}$; accordingly, θ_p is the sum of the forms θ_v over the places v above p . For $p \equiv 2 \pmod{3}$ and a rational integer u prime to p , we have $\bar{u} \in \mathbb{F}_p$, so that $\bar{u}^{(p^2-1)/3} = (\bar{u}^{p-1})^{(p+1)/3} = 1$ and $\chi(u) = 0$.

The place above 3.

Lemma 8. *Let v be the place above 3. Every non-zero element of K_v is equal to $\pi^a \omega^b 2^c (1 + 3\omega)^d$ times a cube, for unique $a, b, c, d \in \mathbb{Z}/3\mathbb{Z}$, and every unit congruent to 1 modulo 9 is a cube. Put $\text{cl}_3(A) = (a, b, c, d) \in \mathbb{F}_3^4$. Then cl_3 is additive, and $\theta_v(A, B) = \text{cl}_3(A)^\top M_3 \text{cl}_3(B)$, where*

$$M_3 = \begin{pmatrix} 0 & 0 & 2 & 0 \\ 0 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \pmod{3}.$$

Proof. For $\alpha \in \mathbb{Z}[\omega]$, the polynomial $g(s) = s + 3s^2 + 3s^3 - \alpha$ satisfies $g'(s) \equiv 1 \pmod{\pi}$, so by Hensel's lemma it has a root s in the completion of $\mathbb{Z}[\omega]$, and then $(1 + 3s)^3 = 1 + 9(s + 3s^2 + 3s^3) = 1 + 9\alpha$. Hence every unit congruent to 1 modulo 9 is a cube in K_v , and the class of a unit modulo cubes depends only on its residue modulo 9. There are 54 units in $\mathbb{Z}[\omega]/(9)$, and a direct enumeration shows that the 27 products $\omega^b 2^c (1 + 3\omega)^d$ with $b, c, d \in \{0, 1, 2\}$ are pairwise distinct modulo 9, also after multiplication by $-1 = (-1)^3$. Hence they represent all 27 classes of units modulo cubes. Since π^3 is a cube times a unit, the exponent of π is the valuation modulo 3, and the first statement follows.

By (H1), it remains to compute θ_v on the 16 ordered pairs of the elements $\pi, \omega, 2$ and $1 + 3\omega$. Their norms are 3, 1, 4 and 7, hence, for each such pair, all the symbols at the places not above 2, 3 and 7 vanish by (H2), and, by (H3), θ_v at the place above 3 is minus the sum of the values at the place above 2 and at the two places above 7. These places correspond to $\bar{\omega} = 2$ and $\bar{\omega} = 4$ in $\mathbb{F}_7$. The classes (e, χ) of the four elements at these places are

	2	$\bar{\omega} = 2$	$\bar{\omega} = 4$
π	(0, 2)	(0, 0)	(0, 2)
ω	(0, 1)	(0, 2)	(0, 2)
2	(1, 0)	(0, 2)	(0, 1)
$1 + 3\omega$	(0, 2)	(1, 0)	(0, 0)

where, at the places above 7, $\chi(u)$ is defined by $\bar{u}^2 = \bar{\omega}^{\chi(u)}$ in $\mathbb{F}_7$, and 7 is a uniformizer. Formula (H2) and the sum over these three places give M_3 . $\square$

We write θ_3 for the form of Lemma 8. Since $3 = -\omega^2 \pi^2$ and $4 = 2^2$, we have $\text{cl}_3(3) = (2, 2, 0, 0)$ and $\text{cl}_3(4) = (0, 0, 2, 0)$.

Residue classes. The local computations use the following description of the classes of the values of a polynomial on a residue class.

Lemma 9. *Let p be a prime, $k \geq 1$, and let $z \in \mathbb{Z}[\omega]$ be non-zero with $z \equiv \zeta \pmod{p^k}$, where $\zeta = \langle A, B \rangle$ with $0 \leq A, B < p^k$. Then $\text{cl}_p(z) \in b + D$ for the vector b and the subspace D given as follows.*

- (i) $p \equiv 2 \pmod{3}$. If $\zeta \neq 0$, then $b = \text{cl}_p(\zeta)$ and $D = 0$. If $\zeta = 0$, then $b = 0$, and D is $\mathbb{F}_3^2$, or the span of $(1, 0)$ if $z \in \mathbb{Z}$.
- (ii) $p \equiv 1 \pmod{3}$. For each root r , let ρ be the root of $t^2 + t + 1$ modulo p^k congruent to r . If $A + B\rho \not\equiv 0 \pmod{p^k}$, the two coordinates of $\text{cl}_{v_r}(z)$ are those of $\text{cl}_{v_r}(\zeta)$; otherwise these two coordinates are arbitrary.
- (iii) $p = 3$, $z \notin \mathbb{Z}$. If $\zeta \neq 0$, let $a = v(\zeta) < 2k$ be its valuation at π and $m = 2k - a$. Then $b = \text{cl}_3(\zeta)$, and $D = 0$ if $m \geq 4$, while $D = D_m$ if $m \leq 3$, where D_1 is spanned by $(0, 1, 0, 0)$, $(0, 0, 1, 0)$ and $(0, 0, 0, 1)$, D_2 by $(0, 0, 1, 0)$ and $(0, 0, 0, 1)$, and D_3 by $(0, 0, 1, 1)$. If $\zeta = 0$, then $b = 0$ and $D = \mathbb{F}_3^4$.

(iv) $p = 3$, $z \in \mathbb{Z}$, so that $B = 0$. If $3^k \nmid A$, let $j = v_3(A)$. Then $b = \text{cl}_3(A)$ and $D = 0$ if $k - j \geq 2$, while $b = j \text{cl}_3(3)$ and D is spanned by $\text{cl}_3(4)$ if $k - j = 1$. If $3^k \mid A$, then $b = 0$ and D is spanned by $\text{cl}_3(3)$ and $\text{cl}_3(4)$.

Proof. (i) If $\zeta \neq 0$, then $e = v(z) = v(\zeta) < k$ and $z/p^e \equiv \zeta/p^e \pmod{p}$, so that χ is determined. If $z \in \mathbb{Z}$, its unit part is a rational integer, and $\chi = 0$. (ii) At v_r , z and ζ are the p -adic numbers $A' + B'\omega$ and $A + B\omega$ with $\omega \equiv \rho \pmod{p^k}$, and the argument of (i) applies. (iii) If $\zeta \neq 0$, then $v(z) = a$, and $z/\pi^a \equiv \zeta/\pi^a \pmod{\pi^m}$. By Lemma 8, the class of a unit depends only on its residue modulo $9 = \omega\pi^4$; the units congruent to 1 modulo π^m form a subgroup, and a direct enumeration of their residues modulo 9 shows that their classes form D_m . (iv) If $3^k \nmid A$, then $z = 3^j u$ with a rational unit $u \equiv A/3^j \pmod{3^{k-j}}$. The class of a rational unit depends only on its residue modulo 9, the units ± 1 are cubes, and the rational units congruent to 1 modulo 3 are 4^s times a unit congruent to 1 modulo 9; hence the classes are as stated, and if $3^k \mid A$ the valuation j is also unknown. $\square$

Lemma 10. *Let p be a prime, and consider vertices i with vectors b_i , subspaces D_i and vectors γ_i , and a set of edges (i, j) with $i < j$ and weights $w_{ij} \in \{1, 2\}$. Put $\rho_{ij} = w_{ij}$ and $\rho_{ji} = -w_{ij}$ for the edges, and $\sigma_i = \gamma_i + \sum_j \rho_{ij} b_j$. Suppose that $\theta_p(d, \sigma_i) = 0$ for every i and every $d \in D_i$, and that $\theta_p(d, d') = 0$ for every edge (i, j) and all $d \in D_i$, $d' \in D_j$. Then*

$$\sum_{(i,j)} w_{ij} \theta_p(c_i, c_j) + \sum_i \theta_p(c_i, \gamma_i) = \sum_{(i,j)} w_{ij} \theta_p(b_i, b_j) + \sum_i \theta_p(b_i, \gamma_i)$$

whenever $c_i \in b_i + D_i$ for every i . By bilinearity, it suffices to check the hypotheses for d and d' in spanning sets of D_i and D_j .

Proof. Write $c_i = b_i + d_i$ with $d_i \in D_i$. As θ_p is bilinear and $\theta_p(u, v) = -\theta_p(v, u)$, the left side equals the right side plus $\sum_i \theta_p(d_i, \sigma_i) + \sum_{(i,j)} w_{ij} \theta_p(d_i, d_j)$, and both sums vanish. $\square$

3 The quadratic certificate

Put $F(x, y) = x^4 + xy + y^3 - y - 4$, so that (1) is the equation $F(x, y) = 0$. The vertices are the polynomials

$$\begin{aligned} H_0 &= -y - 1, \\ H_1 &= x - 1, \\ H_2 &= 2x + 2y - 2, \\ H_3 &= x - 2, \\ H_4 &= 2x + 1, \\ H_5 &= x - y + 2, \\ H_6 &= 4x - y + 3, \\ H_7 &= 2x - y + 7, \\ H_8 &= 2x + y - 2, \\ H_9 &= 2x^2 - 2x + y - 1, \\ H_{10} &= x^2 + x + y + 1, \\ H_{11} &= x^2 - x + y - 1, \\ H_{12} &= x^2 - 3x + 2y - 1, \\ H_{13} &= x + y^2 - 1, \\ H_{14} &= 1, \end{aligned}$$

$$\begin{aligned}
H_{15} &= -3, \\
H_{16} &= 19, \\
H_{17} &= 38, \\
H_{18} &= -1, \\
H_{19} &= 2, \\
H_{20} &= 2x^2 - 2y - 3.
\end{aligned}$$

There are 2 graphs on these vertices. The edges of the first graph are $\{0, 2\}, \{0, 3\}, \{0, 7\}, \{1, 5\}, \{1, 8\}, \{1, 9\}, \{1, 12\}, \{1, 13\}, \{2, 8\}, \{2, 11\}, \{2, 13\}, \{3, 5\}, \{3, 9\}, \{4, 7\}, \{4, 8\}, \{4, 11\}, \{5, 10\}, \{5, 11\}, \{6, 9\}, \{6, 11\}, \{8, 11\}, \{8, 12\}, \{10, 13\}, \{1, 15\}, \{4, 15\}, \{6, 16\}, \{7, 17\}, \{8, 18\}, \{9, 18\}, \{11, 19\}, \{12, 18\}$. The edges of the second graph are $\{20, 19\}$. For a vertex i , we write $j \sim_g i$ if $\{i, j\}$ is an edge of the g -th graph. For every non-constant vertex i and every graph g in which i has a neighbour, Table 1 gives a positive integer $m_{g,i}$ and polynomials $W_{g,i}$, $A_{g,i}$ and $B_{g,i}$ with integer coefficients such that

$$m_{g,i}^2 \prod_{j \sim_g i} H_j - W_{g,i}^2 = A_{g,i}F + B_{g,i}H_i. \quad (2)$$

For every edge $\{i, j\}$ between two non-constant vertices, Table 2 gives a positive integer k_{ij} and polynomials α_{ij} , β_{ij} and γ_{ij} with integer coefficients such that

$$\alpha_{ij}F + \beta_{ij}H_i + \gamma_{ij}H_j = k_{ij}. \quad (3)$$

These identities can be checked by expanding both sides. For the signs of the other vertices, we regard F as a polynomial in y ; it has degree 3 and leading coefficient 1, and its discriminant is

$$\Delta(x) = -27x^8 + 216x^4 - 4x^3 + 12x^2 - 12x - 428.$$

For $i \in I = \{0, 2, 5, 6, 7, 8, 9, 10, 11, 12, 20\}$, the vertex H_i has the form $k_i y + r_i(x)$ with a non-zero integer k_i , and Table 3 lists the polynomial $\varphi_i(x) = k_i^3 F(x, -r_i(x)/k_i)$, which has integer coefficients.

4 The cubic certificates

There are 3 certificates. The g -th certificate consists of polynomials $J_i^{(g)}$ with coefficients in $\mathbb{Z}[\omega]$ and constants $c_i^{(g)} \in \mathbb{Z}[\omega]$, listed in Table 4 and Table 5 and Table 6, and of edges (i, j) with $i < j$ and weights $w_{ij}^{(g)} \in \{1, 2\}$. For an edge we put $\rho_{ij}^{(g)} = w_{ij}^{(g)}$ and $\rho_{ji}^{(g)} = -w_{ij}^{(g)} \bmod 3$, and $\rho_{ij}^{(g)} = 0$ otherwise. In the rest of the proof we usually suppress the index g . The edges $(i, j; w_{ij})$ of the first certificate are $(0, 6; 2), (0, 9; 2), (0, 16; 1), (0, 17; 2), (0, 32; 1), (0, 33; 1), (0, 34; 2), (0, 44; 1), (0, 58; 1), (0, 64; 1), (0, 65; 1), (0, 68; 1), (0, 82; 2), (0, 86; 2), (0, 94; 2), (0, 97; 2), (1, 9; 1), (1, 11; 2), (1, 14; 2), (1, 23; 2), (1, 29; 2), (1, 30; 2), (1, 31; 1), (1, 39; 2), (1, 40; 1), (1, 43; 1), (1, 44; 1), (1, 48; 2), (1, 57; 2), (1, 59; 2), (1, 63; 1), (1, 64; 1), (1, 68; 2), (1, 72; 1), (1, 77; 1), (1, 81; 1), (1, 85; 2), (1, 86; 2), (1, 90; 1), (1, 93; 1), (1, 95; 1), (1, 97; 2), (2, 23; 2), (2, 26; 1), (2, 27; 2), (2, 28; 2), (2, 33; 2), (2, 34; 1), (2, 38; 1), (2, 47; 1), (2, 54; 1), (2, 59; 2), (2, 60; 1), (2, 61; 2), (2, 62; 1), (2, 77; 1), (2, 80; 1), (2, 84; 2), (2, 89; 2), (2, 92; 2), (2, 95; 1), (2, 96; 2), (3, 6; 1), (3, 8; 1), (3, 18; 1), (3, 19; 2), (3, 20; 2), (3, 22; 1), (3, 29; 1), (3, 36; 2), (3, 37; 1), (3, 41; 1), (3, 42; 2), (3, 43; 1), (3, 52; 2), (3, 53; 1), (3, 54; 2), (3, 57; 1), (3, 60; 2), (3, 61; 1), (3, 62; 1), (3, 65; 2), (3, 67; 2), (3, 74; 1), (3, 76; 2), (3, 81; 2), (3, 85; 2), (3, 92; 1), (3, 93; 2), (3, 96; 2), (4, 14; 2), (4, 15; 1), (4, 16; 2), (4, 17; 1), (4, 22; 1), (4, 30; 2), (4, 31; 1), (4, 41; 1), (4, 42; 2), (4, 49; 1), (4, 50; 2), (4, 51; 1), (4, 64; 1), (4, 72; 1), (4, 73; 2), (4, 76; 2), (4, 91; 2), (4, 97; 2), (5, 9; 2), (5, 10; 1), (5, 11; 1), (5, 23; 1), (5, 26; 1), (5, 27; 2), (5, 43; 1), (5, 52; 2), (5, 53; 1), (5, 54; 2), (5, 63; 2),$

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 (0, 40; 1), (0, 41; 1), (0, 42; 2), (0, 44; 1), (0, 60; 2), (0, 61; 1), (0, 62; 1), (0, 65; 2), (0, 70; 2),
 (0, 72; 1), (0, 73; 2), (0, 78; 2), (0, 86; 2), (0, 96; 2), (1, 6; 1), (1, 8; 1), (1, 9; 2), (1, 14; 2), (1, 23; 2),
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 (20, 63; 2), (21, 36; 2), (21, 39; 1), (21, 41; 2), (21, 42; 1), (21, 43; 1), (21, 45; 2), (21, 59; 1),
 (21, 60; 1), (21, 61; 2), (22, 24; 1), (22, 31; 2), (22, 32; 2), (22, 39; 1), (22, 41; 1), (22, 57; 2),
 (22, 59; 1), (22, 61; 2), (22, 62; 2), (23, 25; 2), (23, 27; 2), (23, 31; 2), (23, 35; 2), (23, 40; 2),
 (23, 43; 2), (23, 47; 2), (23, 60; 1), (24, 37; 1), (24, 44; 1), (24, 56; 2), (24, 62; 2), (25, 39; 1),
 (25, 40; 1), (25, 41; 2), (25, 55; 1), (25, 57; 2), (25, 60; 2), (25, 61; 2), (25, 62; 2), (26, 29; 1),
 (26, 30; 1), (26, 32; 1), (26, 33; 2), (26, 38; 1), (26, 40; 1), (26, 45; 2), (26, 55; 2), (26, 64; 2),
 (26, 68; 1), (26, 72; 1), (26, 73; 1), (26, 77; 2), (26, 85; 1), (26, 87; 2), (27, 31; 2), (27, 34; 1),
 (27, 39; 2), (27, 43; 2), (27, 45; 1), (27, 55; 1), (27, 63; 1), (27, 67; 1), (27, 81; 2), (27, 82; 2),
 (27, 84; 2), (27, 87; 1), (27, 97; 1), (29, 37; 1), (29, 38; 2), (29, 56; 2), (30, 32; 1), (30, 39; 1),
 (30, 43; 2), (30, 61; 1), (30, 76; 2), (30, 77; 2), (30, 85; 1), (30, 87; 1), (30, 89; 2), (31, 40; 2),
 (31, 43; 2), (31, 45; 2), (31, 47; 1), (31, 60; 2), (31, 69; 1), (31, 74; 1), (31, 82; 2), (31, 85; 1),
 (32, 35; 1), (32, 55; 2), (32, 60; 1), (32, 61; 2), (33, 38; 2), (33, 41; 2), (33, 55; 2), (33, 59; 2),
 (33, 61; 1), (33, 69; 2), (33, 71; 1), (33, 73; 1), (33, 93; 1), (34, 42; 1), (34, 55; 1), (34, 57; 2),
 (34, 60; 2), (34, 69; 2), (34, 70; 2), (34, 73; 2), (34, 84; 1), (34, 95; 1), (36, 47; 2), (36, 68; 1),
 (36, 69; 1), (36, 70; 2), (36, 73; 2), (36, 78; 1), (36, 81; 1), (36, 85; 1), (36, 86; 1), (37, 43; 2),
 (37, 44; 2), (37, 67; 2), (37, 68; 1), (37, 70; 2), (37, 75; 2), (37, 89; 1), (38, 40; 2), (38, 61; 1),
 (39, 40; 2), (39, 41; 1), (39, 45; 2), (39, 47; 2), (39, 57; 2), (39, 61; 1), (39, 68; 2), (39, 70; 1),
 (39, 71; 2), (39, 77; 2), (39, 79; 1), (39, 84; 2), (39, 95; 1), (40, 42; 2), (40, 59; 2), (40, 60; 2),
 (40, 63; 2), (40, 75; 1), (40, 76; 1), (40, 79; 1), (40, 87; 1), (40, 89; 1), (40, 93; 1), (41, 42; 1),
 (41, 45; 1), (41, 55; 1), (41, 64; 2), (41, 67; 1), (41, 70; 1), (41, 73; 1), (41, 75; 1), (42, 55; 2),
 (42, 69; 2), (42, 73; 1), (42, 75; 2), (42, 76; 1), (42, 79; 2), (42, 87; 2), (42, 97; 1), (43, 45; 2),
 (45, 47; 2), (45, 52; 2), (47, 61; 1), (47, 62; 1), (47, 64; 1), (52, 71; 2), (53, 87; 2), (55, 57; 1),
 (55, 65; 1), (55, 68; 2), (55, 71; 1), (55, 72; 2), (55, 73; 2), (55, 74; 2), (55, 79; 1), (55, 82; 2),
 (55, 93; 2), (56, 62; 1), (56, 71; 2), (56, 78; 2), (56, 81; 2), (56, 96; 2), (57, 59; 1), (59, 61; 2),
 (59, 62; 1), (59, 63; 1), (60, 62; 1), (60, 63; 2), (60, 68; 1), (60, 69; 1), (60, 72; 1), (60, 73; 1),
 (60, 75; 2), (60, 76; 2), (60, 79; 2), (60, 82; 2), (60, 84; 1), (60, 89; 1), (60, 95; 2), (60, 96; 1),
 (60, 97; 2), (61, 62; 2), (61, 63; 1), (61, 64; 1), (61, 65; 2), (61, 67; 2), (61, 68; 2), (61, 71; 1),
 (61, 72; 1), (61, 74; 2), (61, 75; 1), (61, 77; 1), (61, 79; 2), (61, 82; 1), (61, 96; 2), (63, 64; 1),
 (63, 71; 2), (63, 73; 2), (63, 75; 1), (63, 76; 1), (63, 79; 2), (63, 82; 1), (63, 87; 2), (63, 95; 2),
 (63, 96; 1), (65, 67; 2), (65, 74; 2), (65, 82; 1), (65, 85; 2), (65, 86; 2), (65, 89; 1), (67, 68; 2),
 (67, 71; 2), (67, 79; 1), (67, 81; 2), (67, 84; 1), (68, 69; 1), (68, 71; 1), (68, 72; 2), (68, 78; 1),
 (68, 82; 2), (68, 87; 1), (68, 95; 1), (69, 71; 2), (69, 74; 1), (69, 79; 2), (69, 87; 1), (69, 95; 1),
 (69, 97; 1), (70, 71; 1), (70, 74; 2), (70, 78; 1), (71, 76; 2), (71, 89; 1), (71, 95; 1), (71, 97; 2),
 (72, 75; 2), (72, 82; 2), (72, 95; 2), (73, 76; 1), (73, 87; 1), (73, 95; 2), (74, 76; 2), (74, 82; 1),
 (75, 85; 2), (75, 87; 1), (75, 95; 2), (76, 78; 2), (76, 82; 1), (76, 93; 1), (76, 95; 2), (76, 96; 1),

$(77, 79; 1), (77, 83; 1), (77, 85; 1), (77, 89; 1), (78, 86; 2), (78, 96; 1), (79, 93; 1), (79, 96; 1),$
 $(81, 84; 1), (82, 83; 2), (85, 87; 1), (87, 89; 1), (89, 96; 2), (89, 97; 2), (93, 95; 2), (95, 96; 2).$ The
edges $(i, j; w_{ij})$ of the third certificate are $(0, 6; 1), (0, 11; 2), (0, 16; 1), (0, 17; 2), (0, 23; 1),$
 $(0, 32; 1), (0, 59; 1), (0, 60; 2), (0, 61; 1), (0, 63; 1), (0, 65; 2), (0, 77; 2), (0, 82; 2), (0, 95; 2), (1, 8; 2),$
 $(1, 11; 1), (1, 22; 1), (1, 39; 1), (1, 40; 2), (1, 47; 2), (1, 57; 1), (1, 59; 1), (1, 63; 2), (1, 64; 2),$
 $(1, 67; 1), (1, 76; 2), (1, 89; 1), (1, 93; 2), (1, 95; 2), (1, 97; 1), (2, 12; 1), (2, 57; 2), (2, 59; 2),$
 $(2, 60; 2), (2, 61; 1), (2, 70; 2), (2, 93; 1), (2, 95; 1), (3, 8; 2), (3, 18; 1), (3, 19; 2), (3, 20; 2),$
 $(3, 23; 2), (3, 29; 2), (3, 36; 1), (3, 37; 2), (3, 62; 1), (3, 67; 1), (3, 74; 1), (3, 77; 1), (3, 81; 1),$
 $(3, 96; 2), (4, 16; 1), (4, 17; 2), (4, 26; 1), (4, 27; 2), (4, 32; 2), (4, 39; 1), (4, 40; 2), (4, 47; 2),$
 $(4, 50; 1), (4, 51; 2), (4, 59; 1), (4, 64; 2), (4, 82; 1), (4, 89; 1), (4, 95; 2), (4, 97; 1), (5, 8; 1), (5, 10; 1),$
 $(5, 47; 1), (5, 64; 2), (5, 67; 2), (5, 69; 2), (5, 89; 2), (5, 97; 1), (6, 8; 2), (6, 13; 2), (6, 16; 2), (6, 17; 2),$
 $(6, 21; 2), (6, 26; 2), (6, 32; 1), (6, 33; 2), (6, 36; 2), (6, 40; 2), (6, 55; 1), (6, 57; 2), (6, 60; 2),$
 $(6, 62; 1), (6, 63; 2), (8, 10; 1), (8, 16; 1), (8, 21; 2), (8, 26; 2), (8, 29; 1), (8, 32; 1), (8, 36; 1),$
 $(8, 38; 1), (8, 60; 1), (9, 16; 2), (9, 18; 1), (9, 19; 1), (9, 24; 1), (9, 27; 1), (9, 34; 2), (9, 36; 2),$
 $(9, 37; 2), (9, 40; 1), (9, 47; 1), (9, 55; 2), (9, 60; 2), (9, 61; 2), (10, 13; 1), (10, 23; 1), (10, 61; 1),$
 $(10, 64; 2), (11, 19; 1), (11, 22; 1), (11, 32; 1), (11, 37; 1), (11, 55; 1), (11, 60; 2), (11, 65; 2),$
 $(11, 72; 1), (11, 77; 2), (12, 13; 2), (12, 19; 2), (12, 20; 1), (12, 23; 1), (12, 38; 1), (12, 40; 1),$
 $(12, 44; 1), (12, 56; 1), (12, 60; 2), (13, 17; 2), (13, 34; 1), (13, 40; 2), (13, 55; 2), (13, 60; 1),$
 $(13, 62; 2), (14, 19; 1), (14, 26; 1), (14, 27; 2), (14, 32; 1), (14, 36; 2), (14, 39; 1), (14, 40; 1),$
 $(14, 63; 1), (14, 64; 2), (16, 20; 2), (16, 22; 1), (16, 24; 2), (16, 25; 2), (16, 26; 1), (16, 36; 1),$
 $(16, 38; 1), (16, 51; 1), (16, 57; 2), (16, 62; 1), (16, 65; 2), (16, 71; 2), (16, 75; 1), (16, 78; 1),$
 $(16, 95; 2), (16, 96; 1), (17, 21; 2), (17, 24; 2), (17, 27; 2), (17, 37; 2), (17, 50; 2), (17, 59; 1),$
 $(17, 62; 2), (17, 65; 2), (17, 67; 1), (17, 68; 2), (17, 74; 1), (17, 76; 2), (17, 78; 1), (17, 79; 1),$
 $(17, 84; 2), (17, 93; 1), (17, 96; 2), (18, 21; 2), (18, 23; 1), (18, 24; 1), (18, 32; 1), (18, 34; 2),$
 $(18, 38; 2), (18, 39; 2), (18, 56; 2), (18, 60; 2), (18, 61; 2), (18, 62; 1), (18, 63; 1), (18, 64; 1),$
 $(18, 68; 1), (18, 70; 2), (18, 72; 1), (18, 78; 1), (18, 81; 1), (18, 89; 2), (19, 24; 2), (19, 29; 2),$
 $(19, 33; 1), (19, 40; 1), (19, 47; 1), (19, 56; 1), (19, 60; 1), (19, 61; 1), (19, 68; 1), (19, 75; 1),$
 $(19, 77; 2), (19, 78; 2), (19, 82; 2), (19, 84; 1), (19, 96; 2), (19, 97; 2), (20, 22; 1), (20, 23; 1),$
 $(20, 33; 1), (20, 39; 1), (20, 61; 1), (20, 62; 2), (21, 25; 2), (21, 60; 1), (22, 51; 2), (22, 55; 2),$
 $(22, 62; 1), (22, 64; 1), (23, 25; 2), (23, 46; 1), (23, 56; 2), (23, 63; 2), (24, 32; 1), (24, 37; 2),$
 $(24, 44; 2), (24, 56; 2), (24, 60; 1), (25, 60; 2), (26, 38; 2), (26, 40; 1), (26, 47; 1), (26, 57; 2),$
 $(26, 59; 2), (26, 60; 2), (26, 68; 1), (26, 72; 2), (26, 97; 2), (27, 39; 2), (27, 61; 1), (27, 64; 1),$
 $(27, 65; 2), (27, 67; 2), (27, 72; 1), (27, 84; 1), (27, 89; 2), (27, 93; 1), (27, 95; 1), (29, 37; 1),$
 $(29, 38; 2), (29, 56; 2), (29, 61; 1), (32, 34; 1), (32, 51; 2), (32, 56; 2), (32, 59; 1), (32, 60; 1),$
 $(32, 61; 2), (33, 37; 2), (33, 38; 1), (33, 42; 2), (33, 56; 2), (33, 64; 2), (33, 68; 2), (33, 71; 1),$
 $(33, 82; 1), (34, 36; 1), (34, 41; 1), (34, 56; 1), (34, 65; 2), (34, 74; 1), (34, 84; 2), (34, 97; 1),$
 $(36, 40; 2), (36, 60; 2), (36, 61; 2), (36, 63; 1), (36, 68; 2), (36, 78; 2), (36, 81; 1), (36, 96; 2),$
 $(37, 39; 1), (37, 60; 1), (37, 61; 1), (37, 62; 1), (37, 65; 2), (37, 67; 1), (37, 68; 2), (37, 72; 2),$
 $(38, 40; 2), (38, 61; 2), (39, 40; 1), (39, 47; 1), (39, 60; 2), (39, 65; 2), (39, 68; 1), (39, 70; 1),$
 $(39, 71; 2), (39, 72; 1), (39, 84; 2), (40, 61; 1), (40, 72; 1), (40, 74; 1), (40, 89; 2), (41, 57; 2),$
 $(41, 64; 1), (42, 93; 1), (42, 97; 2), (46, 50; 1), (47, 60; 1), (47, 61; 1), (47, 64; 2), (50, 51; 2),$
 $(50, 56; 1), (50, 59; 2), (50, 64; 1), (50, 76; 2), (50, 82; 2), (50, 96; 1), (51, 56; 2), (51, 62; 2),$
 $(51, 88; 1), (51, 95; 1), (51, 97; 2), (55, 57; 2), (55, 63; 1), (55, 65; 1), (55, 68; 2), (55, 71; 2),$
 $(55, 76; 2), (55, 93; 1), (56, 62; 2), (56, 70; 1), (56, 77; 2), (56, 78; 2), (56, 81; 2), (56, 82; 2),$
 $(56, 96; 1), (59, 61; 1), (60, 68; 2), (60, 69; 1), (60, 74; 1), (60, 81; 1), (60, 82; 2), (60, 84; 2),$
 $(60, 89; 1), (60, 95; 1), (60, 97; 2), (61, 63; 2), (61, 64; 1), (61, 65; 2), (61, 67; 1), (61, 68; 2),$
 $(61, 70; 2), (61, 71; 1), (61, 75; 1), (61, 78; 1), (61, 79; 2), (61, 82; 1), (61, 89; 1), (62, 64; 1),$
 $(63, 76; 2), (63, 82; 2), (65, 67; 1), (65, 71; 1), (65, 75; 1), (65, 82; 2), (65, 93; 1), (65, 96; 2),$
 $(67, 69; 2), (67, 75; 1), (67, 81; 2), (67, 82; 2), (67, 84; 2), (68, 78; 2), (68, 89; 2), (69, 71; 2),$
 $(69, 77; 2), (69, 97; 1), (70, 71; 1), (70, 74; 2), (70, 77; 2), (70, 84; 2), (70, 86; 2), (71, 96; 1),$
 $(72, 82; 2), (72, 97; 1), (74, 76; 2), (74, 77; 2), (74, 96; 1), (75, 79; 1), (76, 96; 2), (76, 97; 2),$

$(77, 79; 1)$, $(77, 88; 2)$, $(78, 82; 2)$, $(78, 86; 1)$, $(81, 84; 1)$, $(82, 95; 2)$, $(89, 97; 1)$, $(96, 97; 2)$. Let (ξ, η) denote (x, y) or (y, x) . We call a vertex i *linear* if $J_i = \varepsilon_i(\lambda_i \eta - s_i(\xi))$ for a unit ε_i of $\mathbb{Z}[\omega]$, an element $\lambda_i \in \mathbb{Z}[\omega]$ whose norm is a power of 3, and a polynomial s_i with coefficients in $\mathbb{Z}[\omega]$; the last column of Table 4 and its analogues gives $\lambda_i \eta = s_i(\xi)$ for the linear vertices, where $\lambda_i = 1$ if no factor is shown. For a linear vertex i , we put $f_i(t) = \lambda_i^d F$ and $g_{ij}(t) = \lambda_i^{d_j} J_j$ evaluated at $\xi = t$, $\eta = s_i(t)/\lambda_i$, where d and d_j are the degrees of F and J_j in η ; these are polynomials in t with coefficients in $\mathbb{Z}[\omega]$. For every linear vertex i , Table 7 gives an integer $D_i \geq 1$ and a polynomial $w_i(t)$ with coefficients in $\mathbb{Z}[\omega]$ such that

$$D_i^3 \lambda_i^{e_i} c_i \prod_j g_{ij}^{\rho_{ij}} - w_i^3 = f_i Q_i \quad (4)$$

for a polynomial $Q_i(t)$ with coefficients in $\mathbb{Z}[\omega]$, where $\rho_{ij} \in \{0, 1, 2\}$; the table also gives an integer $e_i \geq 0$ with $e_i + \sum_j \rho_{ij} d_j \equiv 0 \pmod{3}$, and $e_i = 0$ if $\lambda_i = 1$; Q_i is the quotient of the division of the left side by f_i , and the remainder is zero. For the edges (i, j) listed in Table 8, the table gives a linear endpoint $k \in \{i, j\}$ and the factorization of the norm of the resultant $R_{ij} = \text{Res}_t(f_k, g_{ki})$, where $\{k, l\} = \{i, j\}$; it is non-zero. These identities can be checked by expanding both sides and by polynomial division, using $\omega^2 = -1 - \omega$. They hold for each certificate, with its own data.

5 Proof of Theorem 1

Suppose that $(x, y) \in \mathbb{Z}^2$ is a solution of (1), so that $F(x, y) = 0$; from now on the polynomials denote their values at (x, y) .

Step 1: the values are non-zero. Checking the 2^2 residue pairs modulo 2, we find that F and H_4 have no common zero modulo 2, so $H_4 \neq 0$. Checking the 7^2 residue pairs modulo 7, we find that F and H_1 have no common zero modulo 7, so $H_1 \neq 0$. Checking the 17^2 residue pairs modulo 17, we find that F and H_3 have no common zero modulo 17, so $H_3 \neq 0$. The other non-constant H_i are non-zero by Step 2. For $i \in \{7, 8, 10, 11, 12, 13, 20, 21, 22, 23, 24, 25, 32, 33, 34, 35, 36, 37, 38, 39, 40, 43, 45, 46, 63, 66, 67, 69, 70, 71, 74, 75, 76, 77, 78, 79, 82, 83, 84, 85, 87, 88\}$ in the first certificate, there is no residue pair (a, b) modulo 2 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{2}$, as one checks for the 2^2 residue pairs; hence $J_i \neq 0$. For $i \in \{6, 9, 16, 17, 18, 19, 26, 27, 28, 29, 30, 31, 41, 42, 44, 47, 48, 49, 50, 51, 52, 53, 56, 57, 58, 59, 62, 64, 65, 68, 80, 81, 86, 89, 90, 91, 93, 94, 95, 96, 97\}$ in the first certificate, there is no residue pair (a, b) modulo 3 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{3}$, as one checks for the 3^2 residue pairs; hence $J_i \neq 0$. For $i \in \{14, 15, 60, 61, 72, 73\}$ in the first certificate, there is no residue pair (a, b) modulo 4 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{4}$, as one checks for the 4^2 residue pairs; hence $J_i \neq 0$. For $i \in \{2, 3, 5, 54, 92\}$ in the first certificate, there is no residue pair (a, b) modulo 5 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{5}$, as one checks for the 5^2 residue pairs; hence $J_i \neq 0$. For $i \in \{0\}$ in the first certificate, there is no residue pair (a, b) modulo 7 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{7}$, as one checks for the 7^2 residue pairs; hence $J_i \neq 0$. For $i \in \{4\}$ in the first certificate, there is no residue pair (a, b) modulo 9 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{9}$, as one checks for the 9^2 residue pairs; hence $J_i \neq 0$. For $i \in \{55\}$ in the first certificate, there is no residue pair (a, b) modulo 11 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{11}$, as one checks for the 11^2 residue pairs; hence $J_i \neq 0$. For $i \in \{1\}$ in the first certificate, there is no residue pair (a, b) modulo 17 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{17}$, as one checks for the 17^2 residue pairs; hence $J_i \neq 0$. For $i \in \{7, 8, 10, 11, 12, 13, 20, 21, 22, 23, 24, 25, 32, 33, 34, 35, 36, 37, 38, 39, 40, 43, 45, 46, 63, 66, 67, 69, 70, 71, 74, 75, 76, 77, 78, 79, 82, 83, 84, 85, 87, 88\}$ in the second certificate, there is no residue pair (a, b) modulo 2 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{2}$, as one checks for the 2^2 residue pairs; hence $J_i \neq 0$. For $i \in \{6, 9, 16, 17, 18, 19, 26, 27, 28, 29, 30, 31, 41, 42, 44, 47, 48, 49, 50, 51, 52, 53, 56, 57, 58, 59, 62, 64, 65, 68, 80, 81, 86, 89, 90, 91, 93, 94, 95, 96, 97\}$ in the second certificate,

there is no residue pair (a, b) modulo 3 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{3}$, as one checks for the 3^2 residue pairs; hence $J_i \neq 0$. For $i \in \{14, 15, 60, 61, 72, 73\}$ in the second certificate, there is no residue pair (a, b) modulo 4 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{4}$, as one checks for the 4^2 residue pairs; hence $J_i \neq 0$. For $i \in \{2, 3, 5, 54, 92\}$ in the second certificate, there is no residue pair (a, b) modulo 5 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{5}$, as one checks for the 5^2 residue pairs; hence $J_i \neq 0$. For $i \in \{0\}$ in the second certificate, there is no residue pair (a, b) modulo 7 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{7}$, as one checks for the 7^2 residue pairs; hence $J_i \neq 0$. For $i \in \{4\}$ in the second certificate, there is no residue pair (a, b) modulo 9 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{9}$, as one checks for the 9^2 residue pairs; hence $J_i \neq 0$. For $i \in \{55\}$ in the second certificate, there is no residue pair (a, b) modulo 11 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{11}$, as one checks for the 11^2 residue pairs; hence $J_i \neq 0$. For $i \in \{1\}$ in the second certificate, there is no residue pair (a, b) modulo 17 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{17}$, as one checks for the 17^2 residue pairs; hence $J_i \neq 0$. For $i \in \{7, 8, 10, 11, 12, 13, 20, 21, 22, 23, 24, 25, 32, 33, 34, 35, 36, 37, 38, 39, 40, 43, 45, 46, 63, 66, 67, 69, 70, 71, 74, 75, 76, 77, 78, 79, 82, 83, 84, 85, 87, 88\}$ in the third certificate, there is no residue pair (a, b) modulo 2 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{2}$, as one checks for the 2^2 residue pairs; hence $J_i \neq 0$. For $i \in \{6, 9, 16, 17, 18, 19, 26, 27, 28, 29, 30, 31, 41, 42, 44, 47, 48, 49, 50, 51, 52, 53, 56, 57, 58, 59, 62, 64, 65, 68, 80, 81, 86, 89, 90, 91, 93, 94, 95, 96, 97\}$ in the third certificate, there is no residue pair (a, b) modulo 3 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{3}$, as one checks for the 3^2 residue pairs; hence $J_i \neq 0$. For $i \in \{14, 15, 60, 61, 72, 73\}$ in the third certificate, there is no residue pair (a, b) modulo 4 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{4}$, as one checks for the 4^2 residue pairs; hence $J_i \neq 0$. For $i \in \{2, 3, 5, 54, 92\}$ in the third certificate, there is no residue pair (a, b) modulo 5 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{5}$, as one checks for the 5^2 residue pairs; hence $J_i \neq 0$. For $i \in \{0\}$ in the third certificate, there is no residue pair (a, b) modulo 7 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{7}$, as one checks for the 7^2 residue pairs; hence $J_i \neq 0$. For $i \in \{4\}$ in the third certificate, there is no residue pair (a, b) modulo 9 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{9}$, as one checks for the 9^2 residue pairs; hence $J_i \neq 0$. For $i \in \{55\}$ in the third certificate, there is no residue pair (a, b) modulo 11 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{11}$, as one checks for the 11^2 residue pairs; hence $J_i \neq 0$. For $i \in \{1\}$ in the third certificate, there is no residue pair (a, b) modulo 17 with $F(a, b) \equiv 0$ and $J_i(a, b) \equiv 0 \pmod{17}$, as one checks for the 17^2 residue pairs; hence $J_i \neq 0$. Also, all c_i are non-zero.

Step 2: signs. *The sign of H_{13} .* Suppose that $H_{13} \leq 0$, and put $Y = -H_{13}$, so that $Y \geq 0$. Since $H_{13} = x + y^2 - 1$, we have $x = -y^2 - Y + 1$. If $y \geq 0$, put $X = y$; if $y \leq -1$, put $X = -1 - y$. In both cases $X \geq 0$, and substituting into F gives $F(x, y) = G_{13,1}(X, Y)$ in the first case and $F(x, y) = G_{13,2}(X, Y)$ in the second, where

$$\begin{aligned} G_{13,1}(X, Y) &= X^8 + 4X^6Y - 4X^6 + 6X^4Y^2 - 12X^4Y + 6X^4 + 4X^2Y^3 \\ &\quad - 12X^2Y^2 + 12X^2Y - 4X^2 - XY + Y^4 - 4Y^3 + 6Y^2 - 4Y - 3, \\ G_{13,2}(X, Y) &= X^8 + 8X^7 + 4X^6Y + 24X^6 + 24X^5Y + 32X^5 + 6X^4Y^2 \\ &\quad + 48X^4Y + 16X^4 + 24X^3Y^2 + 32X^3Y + 4X^2Y^3 \\ &\quad + 24X^2Y^2 + 8XY^3 + XY + Y^4 + Y - 4. \end{aligned}$$

Expanding $G_{13,1}(3 + X, 3 + Y)$, and, for $0 \leq i < 3$, $G_{13,1}(i, 3 + Y)$ and $G_{13,1}(3 + X, i)$, gives in each case coefficients of one sign and a non-zero constant term; moreover $G_{13,1}(i, j) \neq 0$ for $0 \leq i, j < 3$. Hence $G_{13,1}(X, Y) \neq 0$ for all $X, Y \geq 0$. Expanding $G_{13,2}(2 + X, 2 + Y)$, and, for $0 \leq i < 2$, $G_{13,2}(i, 2 + Y)$ and $G_{13,2}(2 + X, i)$, gives in each case coefficients of one sign and a non-zero constant term; moreover $G_{13,2}(i, j) \neq 0$ for $0 \leq i, j < 2$. Hence $G_{13,2}(X, Y) \neq 0$ for all $X, Y \geq 0$. In both cases $F(x, y) \neq 0$, a contradiction. Hence $H_{13} > 0$.

The signs of H_i for $i \in I$. If $\Delta(x) < 0$, then by Lemma 7, applied to $F(x, z)$ as a polynomial in z , y is its only real root, and $F(x, z)$ has the sign of $(z - y)$ for $z \neq y$. For $i \in I$ and

$z = -r_i(x)/k_i$, we have $z - y = -H_i/k_i$, so if moreover $\varphi_i(x) = k_i^3 F(x, z) \neq 0$, then $H_i \neq 0$, and H_i has the sign of $-\varphi_i(x)$. Expanding $\Delta(7+t)$ and the $\varphi_i(7+t)$ as polynomials in t gives in each case coefficients of one sign and a non-zero constant term, and likewise for $\Delta(-62-t)$ and the $\varphi_i(-62-t)$. Hence, for $x \geq 7$ and for $x \leq -62$, we have $\Delta(x) < 0$, and $\varphi_i(x)$ is non-zero and has the sign of the constant term of the corresponding expansion; the resulting signs of the H_i are listed in Table 3. It remains to consider $-61 \leq x \leq 6$. Checking the residues of y , we find that for $x \in \{-60, -57, -54, -51, -48, -45, -42, -39, -36, -33, -30, -27, -24, -21, -18, -15, -12, -9, -6, -3, 0, 3, 6\}$, we have $F(x, y) \not\equiv 0 \pmod{3}$ for every y , for $x \in \{-61, -56, -46, -41, -31, -26, -16, -11, -1, 4\}$, we have $F(x, y) \not\equiv 0 \pmod{5}$ for every y , for $x \in \{-55, -34, -20, -13, 1\}$, we have $F(x, y) \not\equiv 0 \pmod{7}$ for every y , for $x \in \{-44, -35, -23, -22, -2\}$, we have $F(x, y) \not\equiv 0 \pmod{11}$ for every y , for $x \in \{-53, -52, -40, -38, -28, -25, -14, -5\}$, we have $F(x, y) \not\equiv 0 \pmod{13}$ for every y , for $x \in \{-49, -32, -4, 2\}$, we have $F(x, y) \not\equiv 0 \pmod{17}$ for every y , for $x \in \{-59, -58, -47, -43, -37, -19, -8, 5\}$, we have $F(x, y) \not\equiv 0 \pmod{19}$ for every y , for $x = -50$ and $x = -7$, we have $F(x, y) \not\equiv 0 \pmod{23}$ for every y , for $x = -17$, we have $F(x, y) \not\equiv 0 \pmod{29}$ for every y , for $x = -29$, we have $F(x, y) \not\equiv 0 \pmod{31}$ for every y , and for $x = -10$, we have $F(x, y) \not\equiv 0 \pmod{47}$ for every y . Hence (1) has no solution with $-61 \leq x \leq 6$.

For a place v , put $B_v^{(g)} = \prod_{\{i,j\}} (H_i, H_j)_v$, the product over the edges of the g -th graph. By Theorem 2(i), $(H_i, H_j)_\infty = -1$ exactly when $H_i < 0$ and $H_j < 0$. Here $H_i > 0$ for $i \in \{13, 14, 16, 17, 19\}$ and $H_i < 0$ for $i \in \{15, 18\}$. The other signs are given in Table 3 for $x \geq 7$ and for $x \leq -62$. In both cases and for all signs of H_1, H_3, H_4 , counting the edges of each graph whose two values are negative gives $(B_\infty^{(g)})_g = (1, 1)$.

Step 3: the quadratic graphs at the odd primes outside S . Let $S = \{2, 3, 5, 7, 11, 13, 19, 31, 37, 43, 61, 67, 73, 97, 109, 379\}$. It contains 2 and the primes dividing the $m_{g,i}$, the k_{ij} and the constant vertices. Let $p \notin S$ be an odd prime; we show that $B_p^{(g)} = 1$ for every g . For an edge $\{i, j\}$ of the g -th graph such that p divides neither H_i nor H_j , $(H_i, H_j)_p = 1$ by Theorem 2(ii). Now let i be a vertex with $p \mid H_i$. Then H_i is not constant, as $p \notin S$. For every edge $\{i, j\}$ of the g -th graph, $p \nmid H_j$: if H_j is constant, because $p \notin S$, and otherwise by (3), because $p \nmid k_{ij}$. In particular, no edge has both values divisible by p . By Theorem 2(v), the edges at i contribute $\prod_{j \sim_g i} (H_i, H_j)_p = (H_i, \Pi_i)_p$, where $\Pi_i = \prod_{j \sim_g i} H_j$ is not divisible by p . By (2), $m_{g,i}^2 \Pi_i \equiv W_{g,i}^2 \pmod{p}$, and $p \nmid m_{g,i}$, so Π_i is a non-zero square modulo p , and $(H_i, \Pi_i)_p = (\Pi_i, H_i)_p = 1$ by Lemma 3. Hence $B_p^{(g)} = 1$.

Step 4: cubic reciprocity. For every place v of K , put

$$\mathcal{B}_v^{(g)} = \sum_{(i,j)} w_{ij} \theta_v(J_i, J_j) + \sum_i \theta_v(J_i, c_i) \in \mathbb{Z}/3\mathbb{Z}, \quad (5)$$

the first sum over the edges of the g -th certificate, and, for a rational prime p , put $\mathcal{B}_p^{(g)} = \sum_{v|p} \mathcal{B}_v^{(g)}$. Applying (H3) to every pair (J_i, J_j) with an edge, multiplied by w_{ij} , and to every pair (J_i, c_i) , and adding, we find that $\mathcal{B}_v^{(g)} = 0$ for all but finitely many v and $\sum_p \mathcal{B}_p^{(g)} = 0$ for every g , as the complex place contributes 0. By (H1) and the definition of cl_p , $\mathcal{B}_p^{(g)} = \sum_{(i,j)} w_{ij} \theta_p(\text{cl}_p(J_i), \text{cl}_p(J_j)) + \sum_i \theta_p(\text{cl}_p(J_i), \text{cl}_p(c_i))$, where θ_p is the form of Lemma 8 if $p = 3$.

Step 5: the cubic certificates at the primes outside S . The set S also contains 3 and the primes dividing the norms of the constants c_i , of the D_i , of the R_{ij} . Let $p \notin S$, and let v be a place above p , with residue field k_v ; then all these numbers are units at v . First, no edge (i, j) has both values divisible by v . Indeed, suppose that v divides J_i and J_j . If R_{ij} is defined, let k be the linear endpoint used for it and $\{k, l\} = \{i, j\}$, and put $t = \xi$. As $J_k = \varepsilon_k(\lambda_k \eta - s_k(\xi))$ and λ_k is

a unit at v , we have $\eta \equiv s_k(t)/\lambda_k$ modulo v , hence $f_k(t) \equiv \lambda_k^d F(x, y) = 0$ and $g_{kl}(t) \equiv \lambda_k^{d_l} J_l \equiv 0$ modulo v . The resultant can be written as $R_{ij} = Af_k + Cg_{kl}$ with polynomials A and C with coefficients in $\mathbb{Z}[\omega]$, so v divides R_{ij} , a contradiction. By (H2), a term of (5) in which all arguments are units vanishes. Now let v divide J_i . Collecting the terms of (5) which contain J_i , and using (H1) and $\rho_{ji} = -w_{ij}$ for the edges (j, i) with $j < i$, we obtain $\theta_v(J_i, \Pi_i)$ with $\Pi_i = c_i \prod_j J_j^{\rho_{ij}}$, which is a unit. We claim that $D_i^3 \Pi_i \equiv W^3$ modulo v for some $W \in K$ integral at v . If i is linear, put $t = \xi$; then $g_{ij}(t) \equiv \lambda_i^{d_j} J_j$ and $f_i(t) \equiv 0$ modulo v as above, and (4) gives $D_i^3 \lambda_i^{E_i} \Pi_i \equiv w_i(t)^3$ modulo v , where $E_i = e_i + \sum_j \rho_{ij} d_j$ is divisible by 3; this gives the claim with $W = w_i(t)/\lambda_i^{E_i/3}$. Hence the reduction of Π_i is a non-zero cube in k_v , $\chi(\Pi_i) = 0$, and $\theta_v(J_i, \Pi_i) = v(J_i)\chi(\Pi_i) = 0$ by (H2). Hence $\mathcal{B}_v^{(g)} = 0$, and $\mathcal{B}_p^{(g)} = 0$.

Step 6: the primes in S . Let $p \in S$, and suppose that $x \equiv a$ and $y \equiv b \pmod{p^k}$. Lemma 5 gives, for every vertex H_i of the quadratic graphs, a vector b_i and a subspace D_i with $c_p(H_i) \in b_i + D_i$, and Lemma 9 gives the analogous data for the values J_i of the cubic certificates. If the hypotheses of Lemma 6 hold for every quadratic graph and those of Lemma 10 hold for every cubic certificate, then $(B_p^{(g)})_g$ and $(\mathcal{B}_p^{(g)})_g$ are determined by a, b and k . Starting from the solutions of (1) modulo p , we refine a residue class $x \equiv a, y \equiv b \pmod{p^k}$ to the classes modulo p^{k+1} that contain solutions of (1) modulo p^{k+1} , until all these hypotheses hold, or until no solution remains. Table 9 lists the resulting terminal classes together with the joint value $((B_p^{(g)})_g; (\mathcal{B}_p^{(g)})_g)$; every integer solution lies in one of them. The possible joint values are: for $p = 2$, $(2, 0, 0; -1, -1), (2, 0, 0; 1, -1), (2, 0, 0; 1, 1), (2, 1, 2; -1, 1), (2, 1, 2; 1, -1)$; for $p = 3$, $(0, 1, 1; -1, 1), (1, 0, 0; -1, 1), (2, 0, 0; -1, 1), (2, 1, 1; -1, 1), (2, 2, 2; -1, 1)$; for $p = 5$, $(0, 0, 0; 1, 1), (0, 1, 0; 1, 1)$; for $p = 7$, $(1, 2, 1; 1, 1)$; for $p = 11$, $(0, 0, 0; 1, 1)$; for $p = 13$, $(0, 0, 0; 1, 1)$; for $p = 19$, $(0, 1, 0; -1, 1), (0, 1, 0; 1, 1), (1, 1, 0; 1, 1), (2, 1, 0; -1, 1), (2, 1, 0; 1, 1)$; for $p = 31$, $(0, 0, 0; 1, 1)$; for $p = 37$, $(0, 0, 0; 1, 1)$; for $p = 43$, $(0, 0, 0; 1, 1)$; for $p = 61$, $(0, 0, 0; 1, 1)$; for $p = 67$, $(0, 0, 0; 1, 1)$; for $p = 73$, $(0, 0, 0; 1, 1)$; for $p = 97$, $(0, 0, 0; 1, 1)$; for $p = 109$, $(0, 0, 0; 1, 1)$; for $p = 379$, $(0, 0, 0; 1, 1)$.

Step 7: contradiction. By Steps 4 and 5, $(\sum_{p \in S} \mathcal{B}_p^{(g)})_g = (0, 0, 0)$. By Theorem 2(iv) and Step 3, $(\prod_{p \in S} B_p^{(g)})_g = (B_\infty^{(g)})_g = (1, 1)$ by Step 2. But by Step 6, the pair $((\sum_{p \in S} \mathcal{B}_p^{(g)})_g; (\prod_{p \in S} B_p^{(g)})_g)$ lies in the set consisting of the pairs $(0, 0, 0; -1, -1), (0, 0, 0; -1, 1), (0, 0, 0; 1, -1), (0, 0, 1; -1, -1), (0, 0, 1; -1, 1), (0, 0, 1; 1, -1), (0, 0, 1; 1, 1), (0, 0, 2; -1, -1), (0, 0, 2; 1, 1), (0, 1, 0; -1, -1), (0, 1, 0; -1, 1), (0, 1, 0; 1, -1), (0, 1, 0; 1, 1), (0, 1, 1; -1, -1), (0, 1, 1; -1, 1), (0, 1, 1; 1, -1), (0, 1, 1; 1, 1), (0, 1, 2; -1, -1), (0, 1, 2; -1, 1), (0, 1, 2; 1, -1), (0, 1, 2; 1, 1), (0, 2, 0; -1, -1), (0, 2, 0; -1, 1), (0, 2, 0; 1, -1), (0, 2, 0; 1, 1), (0, 2, 1; -1, -1), (0, 2, 1; -1, 1), (0, 2, 1; 1, -1), (0, 2, 1; 1, 1), (0, 2, 2; -1, -1), (0, 2, 2; -1, 1), (0, 2, 2; 1, -1), (0, 2, 2; 1, 1), (1, 0, 0; -1, -1), (1, 0, 0; -1, 1), (1, 0, 0; 1, -1), (1, 0, 0; 1, 1), (1, 0, 1; -1, -1), (1, 0, 1; -1, 1), (1, 0, 1; 1, -1), (1, 0, 1; 1, 1), (1, 0, 2; -1, -1), (1, 0, 2; -1, 1), (1, 0, 2; 1, -1), (1, 0, 2; 1, 1), (1, 1, 0; -1, -1), (1, 1, 0; -1, 1), (1, 1, 0; 1, -1), (1, 1, 0; 1, 1), (1, 1, 1; -1, -1), (1, 1, 1; -1, 1), (1, 1, 1; 1, -1), (1, 1, 1; 1, 1), (1, 1, 2; -1, -1), (1, 1, 2; -1, 1), (1, 1, 2; 1, -1), (1, 1, 2; 1, 1), (1, 2, 0; -1, -1), (1, 2, 0; -1, 1), (1, 2, 0; 1, -1), (1, 2, 0; 1, 1), (1, 2, 1; -1, -1), (1, 2, 1; -1, 1), (1, 2, 1; 1, -1), (1, 2, 1; 1, 1), (1, 2, 2; -1, -1), (1, 2, 2; -1, 1), (1, 2, 2; 1, -1), (1, 2, 2; 1, 1), (2, 0, 0; -1, -1), (2, 0, 0; -1, 1), (2, 0, 0; 1, -1), (2, 0, 0; 1, 1), (2, 0, 1; -1, -1), (2, 0, 1; -1, 1), (2, 0, 1; 1, -1), (2, 0, 1; 1, 1), (2, 0, 2; -1, -1), (2, 0, 2; -1, 1), (2, 0, 2; 1, -1), (2, 0, 2; 1, 1), (2, 1, 0; -1, -1), (2, 1, 0; -1, 1), (2, 1, 0; 1, -1), (2, 1, 0; 1, 1), (2, 1, 1; -1, -1), (2, 1, 1; -1, 1), (2, 1, 1; 1, -1), (2, 1, 1; 1, 1), (2, 1, 2; -1, -1), (2, 1, 2; -1, 1), (2, 1, 2; 1, -1), (2, 1, 2; 1, 1), (2, 2, 0; -1, -1), (2, 2, 0; -1, 1), (2, 2, 0; 1, -1), (2, 2, 0; 1, 1), (2, 2, 1; -1, -1), (2, 2, 1; -1, 1), (2, 2, 1; 1, -1), (2, 2, 1; 1, 1), (2, 2, 2; -1, -1), (2, 2, 2; -1, 1), (2, 2, 2; 1, -1), and $(2, 2, 2; 1, 1)$, which does not contain $(0, 0, 0; 1, 1)$. This contradiction proves the theorem. $\square$$

References

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Table 1: The data of the identities (2).

$g = 1, i = 0, m = 1, \text{neighbours } 2, 3, 7$	
W	$2x^3 - 4x^2$
A	$-4x^2 + 16x - 16$
B	$-4x^3 - 4x^2y^2 + 4x^2y + 14x^2 + 16xy^2 - 14xy - 30x - 16y^2 + 12y + 36$
$g = 1, i = 1, m = 1, \text{neighbours } 15, 5, 8, 9, 12, 13$	
W	$-6y^2 + 3y + 9$
A	$-12x^3 - 12x^2y^2 + 6x^2y + 48x^2 + 6xy^3 + 42xy^2 - 84xy + 18x - 63y^2 + 96y - 27$
B	$6x^3y^4 - 78x^3y^3 + 102x^3y^2 + 111x^3y - 171x^3 - 3x^2y^4$ $+ 78x^2y^3 - 123x^2y^2 + 33x^2y - 75x^2 + 27xy^5 - 99xy^4 + 45xy^3 + 81xy^2$ $- 135xy + 171x - 6y^6 - 30y^5 + 75y^4 + 15y^3 + 108y^2 - 351y + 201$
$g = 1, i = 2, m = 2, \text{neighbours } 0, 8, 11, 13$	
W	$-2x^3 + 6x^2 - 4x$
A	$-4x^2 + 24x + 4y^2 + 16y - 20$
B	$-2x^3y^2 - 12x^3y - 20x^3 - 2x^2y^3 + 4x^2y^2 + 20x^2y + 16x^2 + 2xy^3$ $- 8xy^2 - 18xy - 8x - 4y^4 - 8y^3 + 8y^2 + 16y + 36$
$g = 1, i = 3, m = 1, \text{neighbours } 0, 5, 9$	
W	$y^2 + y$
A	$-y - 1$
B	$x^3y + x^3 + 2xy^2 - 2x + 2y^2 + 3y + 1$
$g = 1, i = 4, m = 4, \text{neighbours } 15, 7, 8, 11$	
W	$16y^2 - 6y - 27$
A	$-256y + 240$
B	$128x^3y - 216x^3 - 64x^2y - 36x^2 + 24xy^2 - 280xy + 690x + 92y^2 - 4y - 441$
$g = 1, i = 5, m = 1, \text{neighbours } 1, 3, 10, 11$	
W	$8x + 4$
A	$x^2 - 3x + y - 7$
B	$x^3y + 8x^3 + x^2y^2 - x^2y - 15x^2 - 2xy^2 - 5xy - 27x + y^3 - 5y^2 - 13y - 23$
$g = 1, i = 6, m = 1, \text{neighbours } 16, 9, 11$	
W	$6x^2 + 2x - 6$
A	2
B	$-25x^2 + 8xy + 31x + 2y^2 - 13y - 3$
$g = 1, i = 7, m = 1, \text{neighbours } 17, 0, 4$	
W	$-2x^2 - 8x - 32$
A	-4
B	$-16x^2 - 8xy - 40x - 4y^2 - 28y - 154$
$g = 1, i = 8, m = 1, \text{neighbours } 18, 1, 2, 4, 11, 12$	
W	$7x^3 - 37x^2 + 42x - 6$
A	$-4x^3 - 4x^2y - 27x^2 + 6xy + 490x - 4y^2 + 224y - 447$
B	$-4x^3y^2 - 96x^3y - 761x^3 + 4x^2y^3 + 55x^2y^2 + 276x^2y + 841x^2$ $- 14xy^3 - 78xy^2 - 420xy - 315x + 4y^4 - 216y^3 + 15y^2 + 228y + 913$
$g = 1, i = 9, m = 1, \text{neighbours } 18, 1, 3, 6$	
W	$4x^3 - 6x^2 - 2x + 2$
A	$-16x^2 + 16x - 8y + 10$
B	$16x^3 + 4x^2y + x^2 + 4xy - 13x + 8y^3 - 2y^2 - 10y - 30$
$g = 1, i = 10, m = 13, \text{neighbours } 5, 13$	
W	$12x^5 + 21x^4 + 50x^3 + 79x^2 + 48x + 54$
A	$-144x^6 - 504x^5 - 1641x^4 + 360x^3y^2 - 192x^3y - 1817x^3 + 144x^2y^3$ $+ 360x^2y - 7546x^2 + 504xy^3 - 120xy^2 + 1249xy - 6601x + 144y^4 - 360y^3 + 969y^2$ $- 5249y - 5993$
B	$-360x^5y^2 + 336x^5y - 2179x^5 + 360x^4y^2 - 336x^4y + 2179x^4$ $+ 360x^3y^3 - 216x^3y^2 + 2067x^3y - 6627x^3 - 144x^2y^4 + 921x^2y^3$ $- 585x^2y^2 + 1515x^2y - 9432x^2 - 360xy^5 - 168xy^4 + 1112xy^3 - 378xy^2$ $+ 8006xy - 4193x - 144y^6 + 504y^5 - 1329y^4 + 6794y^3 - 1441y^2 + 406y - 27226$
$g = 1, i = 11, m = 1, \text{neighbours } 19, 2, 4, 5, 6, 8$	
W	$4x^4 - 6x^3 - 8x^2 - 2x$
A	$-16x^4 + 40x^3 + 28x^2 + 16xy - 68x + 16y^2 + 24y - 40$
B	$8x^5 + 8x^4 - 8x^3y + 68x^3 + 16x^2y^3 - 16x^2y^2 - 80x^2y + 44x^2$ $- 24xy^3 - 80xy^2 + 112xy + 72x - 16y^4 - 36y^3 - 12y^2 + 168y + 112$

$g = 1, i = 12, m = 40, \text{neighbours } 18, 1, 8$	
W	$-2x^5 + 22x^4 - 66x^3 + 62x^2 + 46x - 50$
A	$-4x^6 + 88x^5 - 748x^4 - 38x^3y^2 - 86x^3y + 2066x^3 + 4x^2y^3 - 92x^2y$ $-6916x^2 - 88xy^3 + 836xy + 15852x + 8y^4 - 264y^3 + 902y^2 - 13446y + 7650$
B	$38x^5y^2 + 90x^5y + 1086x^5 + 114x^4y^2 + 270x^4y + 3258x^4 - 76x^3y^3$ $+200x^3y^2 - 1272x^3y + 1320x^3 - 8x^2y^4 + 594x^2y^3 - 642x^2y^2$ $+570x^2y + 1004x^2 + 38xy^5 + 210xy^4 - 798xy^3 + 672xy^2 + 6436xy + 292x - 4y^6$ $+130y^5 - 382y^4 + 6416y^3 - 694y^2 - 5266y - 24900$
$g = 1, i = 13, m = 2, \text{neighbours } 1, 2, 10$	
W	$-y^6 - y^5 + 4y^4 - 5y^2 + 3y$
A	$-y^4 - 2y^3 + 3y^2$
B	$x^3y^4 + 2x^3y^3 - 3x^3y^2 + 8x^3 - x^2y^6 - 2x^2y^5 + 4x^2y^4$ $+2x^2y^3 - 11x^2y^2 + 8x^2y + xy^8 + 2xy^7 - 5xy^6 - 4xy^5 + 15xy^4$ $-6xy^3 - 11xy^2 + 16xy - y^{10} - 2y^9 + 6y^8 + 6y^7 - 20y^6 + 3y^5 + 28y^4$ $-25y^3 - 3y^2 - 8$
$g = 2, i = 20, m = 55, \text{neighbours } 19$	
W	$-16x^5 + 4x^4 - 16x^2 + 22x - 9$
A	$-256x^6 + 128x^5 - 16x^4 + 128x^3y^2 - 128x^3y - 784x^3 + 256x^2y^3$ $-384x^2y - 192x^2 - 128xy^3 - 192xy^2 + 720xy + 1272x - 256y^4 - 128y^3 + 1296y^2$ $+864y - 1772$
B	$-64x^5y^2 + 192x^5y + 136x^5 - 64x^3y^3 + 192x^3y^2 + 136x^3y - 408x^3$ $+128x^2y^4 + 8x^2y^3 - 584x^2y^2 - 48x^2y + 762x^2 - 64xy^5 - 128xy^4$ $+552xy^3 + 808xy^2 - 756xy - 1828x - 128y^6 + 128y^5 + 584y^4 + 132y^3 - 1476y^2$ $-810y + 373$

Table 2: The data of the identities (3).

$\{i, j\} = \{0, 2\}, k_{ij} = 2^2 \cdot 5$	
α	2
β	$-2x^3 - 4x^2 - 6x + 2y^2 - 2y - 14$
γ	$-x^3 - 2x^2 - 4x - 7$
$\{i, j\} = \{0, 3\}, k_{ij} = 2 \cdot 5$	
α	1
β	$x + y^2 - y$
γ	$-x^3 - 2x^2 - 4x - 7$
$\{i, j\} = \{0, 7\}, k_{ij} = 2^9$	
α	2
β	$x^3 - 4x^2 + 18x + 2y^2 - 2y - 65$
γ	$-x^3 + 4x^2 - 16x + 65$
$\{i, j\} = \{1, 5\}, k_{ij} = 2^3 \cdot 3$	
α	1
β	$-x^3 - x^2 - x - y^2 - 4y - 10$
γ	$y^2 + 3y + 9$
$\{i, j\} = \{1, 8\}, k_{ij} = 3$	
α	-1
β	$x^3 + x^2 + x - 2y^2 + y + 1$
γ	y^2
$\{i, j\} = \{1, 9\}, k_{ij} = 2$	
α	-1
β	$x^3 + x^2 - 2xy^2 - 2xy - x + y + 1$
γ	$y^2 + y + 1$
$\{i, j\} = \{1, 12\}, k_{ij} = 3$	
α	8
β	$-8x^3 - 8x^2 + 4xy^2 + 6xy + x - 8y^2 - 20y - 26$
γ	$-4y^2 - 6y - 9$
$\{i, j\} = \{1, 13\}, k_{ij} = 3$	
α	-1
β	$x^3 + x^2 + x + 1$

γ	y
$\{i, j\} = \{2, 8\}, k_{ij} = 2 \cdot 3$	
α	-2
β	$x^3 - x^2y + x^2 + xy^2 - 2xy + x + y^3 - 3y^2 + 8y - 7$
γ	$-2y^3 + 6y^2 - 10y + 8$
$\{i, j\} = \{2, 11\}, k_{ij} = 2^4 \cdot 5$	
α	$14x^4 + 98y - 118$
β	$-7x^7 + 7x^6y - 7x^6 - 7x^5y + 35x^5 - 28x^4y - 7x^4 + 38x^3 - 38x^2y + 38x^2$ $+ 49xy^2 - 32xy - 120x - 49y^3 + 21y^2 + 103y + 38$
γ	$-14x^4y^2 + 42x^4y - 84x^4 - 22y^2 - 88y + 316$
$\{i, j\} = \{2, 13\}, k_{ij} = 2^3 \cdot 3$	
α	$-2x - 6$
β	$x^4 - x^3y + 4x^3 + x^2y^2 - 5x^2y + 4x^2 - xy^3 + 5xy^2 - 6xy + x - 3y^3 + 6y^2$ $- y - 10$
γ	$2xy^2 - 4xy + 6x + 6y^2 - 12y + 20$
$\{i, j\} = \{3, 5\}, k_{ij} = 2^4 \cdot 5$	
α	1
β	$-x^3 - 2x^2 - 4x - y^2 - 5y - 25$
γ	$y^2 + 4y + 17$
$\{i, j\} = \{3, 9\}, k_{ij} = 2 \cdot 3^2$	
α	-1
β	$x^3 + 2x^2 - 2xy^2 + 6xy - 16x - 2y^2 + 7y - 12$
γ	$y^2 - 3y + 10$
$\{i, j\} = \{4, 7\}, k_{ij} = 3^2 \cdot 19^2$	
α	16
β	$-8x^3 + 4x^2 - 2x - 16y^2 - 104y - 551$
γ	$16y^2 + 96y + 552$
$\{i, j\} = \{4, 8\}, k_{ij} = 3^3 \cdot 11$	
α	16
β	$-8x^3 + 4x^2 - 2x + 16y^2 + 40y + 121$
γ	$-16y^2 - 48y - 120$
$\{i, j\} = \{4, 11\}, k_{ij} = 5^2 \cdot 11$	
α	-64
β	$32x^3 - 16x^2 - 32xy^2 - 8xy + 54x + 48y^2 + 44y - 73$
γ	$64y^2 + 16y - 92$
$\{i, j\} = \{5, 10\}, k_{ij} = 2^2 \cdot 3^3$	
α	$-x - 6$
β	$x^4 + x^3y + 4x^3 + 7x^2y - 20x^2 + 2xy - 48x - 24y + 4$
γ	$xy^2 - 5xy + 12x + 6y^2 - 30y + 76$
$\{i, j\} = \{5, 11\}, k_{ij} = 2^5 \cdot 5$	
α	$x^4 - 12y + 19$
β	$-x^7 - x^6y + 2x^6 + x^5y + 4x^5 + 4x^4y - 16x^4 + 9x^3 + 9x^2y - 18x^2$ $- 12xy^2 - 41xy - 40x - 12y^3 + 16y^2 + 28y + 156$
γ	$-x^4y^2 + 3x^4y - 8x^4 + 21y^2 + 5y + 76$
$\{i, j\} = \{6, 9\}, k_{ij} = 2^5 \cdot 7^2$	
α	$-20x^2 - 32$
β	$320x^5 + 80x^4y - 560x^4 + 412x^3 + 20x^2y^2 + 63x^2y - 451x^2 - 360x + 32y^2$ $- 204y + 892$
γ	$-630x^4 + 10x^3 + 40x^2y^2 - 85x^2y - 273x^2 + 16x + 64y^2 - 236y + 1236$
$\{i, j\} = \{6, 11\}, k_{ij} = 2 \cdot 7^2$	
α	$18528x^4 + 69468y + 50891$
β	$-79872x^7 + 96x^6y - 288x^6 - 5856x^5y + 31776x^5 + 18624x^4y^2 - 237384x^4y$ $+ 290712x^4 + 127552x^3 - 11166x^2y + 33498x^2 + 147456xy^2 - 31578xy - 478025x$ $+ 4260y^3 - 218803y^2 + 384220y + 71090$
γ	$300960x^6 - 80256x^5y + 541728x^5 + 96x^4y^2 - 358368x^4y + 716448x^4$ $+ 65208x^2y^2 - 122892x^2y - 642675x^2 + 65208xy^2 + 49324xy - 1159323x$ $- 65208y^3 - 347682y^2 + 762415y + 9608$
$\{i, j\} = \{8, 11\}, k_{ij} = 2^3 \cdot 3 \cdot 11$	
α	$4x^4 + 6y - 100$

β	$-2x^7 + x^6y - 2x^6 + 5x^5y + 2x^5 - 3x^4y^2 - x^4y - 2x^4 + 52x^3 - 26x^2y$ $+ 52x^2 + 12xy^2 - 152xy + 36x - 6y^3 + 86y^2 - 20y + 52$
γ	$-x^4y^2 - 6x^4y - 8x^4 + 2y^2 + 200y + 32$
$\{i, j\}$	$= \{8, 12\}, k_{ij} = 2^3 \cdot 3^5 \cdot 7^2 \cdot 13$
α	$268x^4 + 8978y - 388132$
β	$-134x^7 + 67x^6y - 134x^6 + 67x^5y + 1474x^5 - 134x^4y^2 - 603x^4y - 3350x^4$ $+ 189644x^3 - 94822x^2y + 189644x^2 + 17956xy^2 - 301048xy + 200492x - 8978y^3$ $+ 252356y^2 + 126984y + 167948$
γ	$-67x^4y^2 + 134x^4y - 3216x^4 + 58910y^2 + 222808y - 21696$
$\{i, j\}$	$= \{10, 13\}, k_{ij} = 2^3 \cdot 3$
α	$-22x^2y^2 + 33x^2 - 41y^2 - y + 34$
β	$22x^4y^2 + 11x^4 + 22x^3y^2 - 55x^3 - 22x^2y^3 - 3x^2y^2 - 10x^2y$ $+ 92x^2 + 41xy^2 - xy - 130x - 41y^3 - 105y^2 - 15y + 157$
γ	$-44x^5 - 82x^3 + 22x^2y^3 + 22x^2y^2 + 33x^2y + 11x^2 + 24x + 41y^3 + 42y^2$ $+ 112y - 3$

Table 3: The polynomials $\varphi_i(x) = k_i^3 F(x, -r_i(x)/k_i)$ of Step 2 and the signs of H_i for $x \geq 7$ and for $x \leq -62$.

i	k_i	$\varphi_i(x)$	$x \geq 7$	$x \leq -62$
0	-1	$-x^4 + x + 4$	+	+
2	2	$8x^4 - 8x^3 + 16x^2 - 8x - 32$	-	-
5	-1	$-x^4 - x^3 - 7x^2 - 13x - 2$	+	+
6	-1	$-x^4 - 64x^3 - 148x^2 - 107x - 20$	+	+
7	-1	$-x^4 - 8x^3 - 86x^2 - 299x - 332$	+	+
8	1	$x^4 - 8x^3 + 22x^2 - 20x + 2$	-	-
9	1	$-8x^6 + 24x^5 - 11x^4 - 18x^3 + 10x^2 + 5x - 4$	+	+
10	1	$-x^6 - 3x^5 - 5x^4 - 8x^3 - 6x^2 - 3x - 4$	+	+
11	1	$-x^6 + 3x^5 + x^4 - 6x^3 + 2x^2 + 3x - 4$	+	+
12	2	$-x^6 + 9x^5 - 16x^4 + 5x^3 + 40x^2 + x - 35$	+	+
20	-2	$-8x^6 + 28x^4 - 8x^3 - 46x^2 + 12x + 47$	+	+

Table 4: The polynomials J_i , the constants c_i and, for the linear vertices, $\eta = r_i(\xi)$ of the first certificate. Here $\langle A, B \rangle = A + B\omega$.

i	J_i	c_i	chart
0	$\langle x - 1, 0 \rangle$	$\langle 1, 0 \rangle$	$x = \langle 1, 0 \rangle$
1	$\langle x - 2, 0 \rangle$	$\langle 392, 147 \rangle$	$x = \langle 2, 0 \rangle$
2	$\langle x + 1, 0 \rangle$	$\langle 1, 0 \rangle$	$x = \langle -1, 0 \rangle$
3	$\langle -x - y + 2, 0 \rangle$	$\langle 57, -38 \rangle$	$y = \langle -x + 2, 0 \rangle$
4	$\langle x + 2, 0 \rangle$	$\langle 0, -1 \rangle$	$x = \langle -2, 0 \rangle$
5	$\langle -y, 0 \rangle$	$\langle 0, -1 \rangle$	$y = \langle 0, 0 \rangle$
6	$\langle x, -y + 1 \rangle$	$\langle 2, 3 \rangle$	$y = \langle -x + 1, -x \rangle$
7	$\langle -x - y - 1, -2x - 1 \rangle$	$\langle 2, -1 \rangle$	$y = \langle -x - 1, -2x - 1 \rangle$
8	$\langle -2x - y, -x + 1 \rangle$	$\langle 1, 5 \rangle$	$y = \langle -2x, -x + 1 \rangle$
9	$\langle -x - y, -x + 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x, -x + 1 \rangle$
10	$\langle 2x - y, x + 1 \rangle$	$\langle 28, 35 \rangle$	$y = \langle 2x, x + 1 \rangle$
11	$\langle x - y - 1, x - 1 \rangle$	$\langle 15, 24 \rangle$	$y = \langle x - 1, x - 1 \rangle$
12	$\langle -y, 1 \rangle$	$\langle 4, 0 \rangle$	$y = \langle 0, 1 \rangle$
13	$\langle x - 1, -1 \rangle$	$\langle 0, 3 \rangle$	$x = \langle 1, 1 \rangle$
14	$\langle -2x - y - 1, -x - 2 \rangle$	$\langle 3, 3 \rangle$	$y = \langle -2x - 1, -x - 2 \rangle$
15	$\langle x - y - 1, -x - 2 \rangle$	$\langle 2, 0 \rangle$	$y = \langle x - 1, -x - 2 \rangle$
16	$\langle -y - 1, -2 \rangle$	$\langle 4, 0 \rangle$	$y = \langle -1, -2 \rangle$
17	$\langle -y + 1, 2 \rangle$	$\langle 16, 0 \rangle$	$y = \langle 1, 2 \rangle$
18	$\langle x - 1, -2 \rangle$	$\langle 28, -14 \rangle$	$x = \langle 1, 2 \rangle$
19	$\langle x + 1, 2 \rangle$	$\langle 1568, 980 \rangle$	$x = \langle -1, -2 \rangle$
20	$\langle x + 2, 3 \rangle$	$\langle 0, 3 \rangle$	$x = \langle -2, -3 \rangle$
21	$\langle x + y - 2, -1 \rangle$	$\langle 12, 8 \rangle$	$y = \langle -x + 2, 1 \rangle$

i	J_i	c_i	chart
22	$\langle -x - y - 1, x + 1 \rangle$	$\langle 3, 9 \rangle$	$y = \langle -x - 1, x + 1 \rangle$
23	$\langle x - y - 1, 1 \rangle$	$\langle 3, 1 \rangle$	$y = \langle x - 1, 1 \rangle$
24	$\langle x - y - 1, x + 1 \rangle$	$\langle 16, 10 \rangle$	$y = \langle x - 1, x + 1 \rangle$
25	$\langle x - 2, -1 \rangle$	$\langle 31, 30 \rangle$	$x = \langle 2, 1 \rangle$
26	$\langle -y - 2, -x - 2 \rangle$	$\langle 6, -12 \rangle$	$y = \langle -2, -x - 2 \rangle$
27	$\langle x - y, x + 2 \rangle$	$\langle 180, 288 \rangle$	$y = \langle x, x + 2 \rangle$
28	$\langle 2x - y, 2x + 2 \rangle$	$\langle 10, 4 \rangle$	$y = \langle 2x, 2x + 2 \rangle$
29	$\langle -y - 2, -2 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -2, -2 \rangle$
30	$\langle -x - 3y - 6, x - 6 \rangle$	$\langle 15, 9 \rangle$	$\langle 1, -1 \rangle x = \langle -3y - 6, -6 \rangle$
31	$\langle -2x - 3y, -x + 6 \rangle$	$\langle -45, -189 \rangle$	$\langle 1, -1 \rangle x = \langle 6, 3y + 6 \rangle$
32	$\langle x + y + 1, 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x - 1, -1 \rangle$
33	$\langle x - y + 1, 1 \rangle$	$\langle 9, 6 \rangle$	$y = \langle x + 1, 1 \rangle$
34	$\langle x - y, -1 \rangle$	$\langle -27, -72 \rangle$	$y = \langle x, -1 \rangle$
35	$\langle -x - y + 1, -2x + 1 \rangle$	$\langle 4, 0 \rangle$	$y = \langle -x + 1, -2x + 1 \rangle$
36	$\langle x, -1 \rangle$	$\langle 9, -15 \rangle$	$x = \langle 0, 1 \rangle$
37	$\langle x + 1, 1 \rangle$	$\langle 351, 495 \rangle$	$x = \langle -1, -1 \rangle$
38	$\langle -x - y + 1, -x - 1 \rangle$	$\langle 4, 8 \rangle$	$y = \langle -x + 1, -x - 1 \rangle$
39	$\langle x - y + 2, 2x + 1 \rangle$	$\langle 22, 26 \rangle$	$y = \langle x + 2, 2x + 1 \rangle$
40	$\langle -x - y + 1, -2x - 1 \rangle$	$\langle -660, -468 \rangle$	$y = \langle -x + 1, -2x - 1 \rangle$
41	$\langle -y + 1, x - 2 \rangle$	$\langle 8, -4 \rangle$	$y = \langle 1, x - 2 \rangle$
42	$\langle -x - y + 3, -x + 2 \rangle$	$\langle 128, 80 \rangle$	$y = \langle -x + 3, -x + 2 \rangle$
43	$\langle x - y + 1, -y \rangle$	$\langle 1, 0 \rangle$	$y = \langle 0, -x - 1 \rangle$
44	$\langle -2x - y + 1, -x \rangle$	$\langle 6, 16 \rangle$	$y = \langle -2x + 1, -x \rangle$
45	$\langle 2x - y + 2, 2x - 1 \rangle$	$\langle 96, -30 \rangle$	$y = \langle 2x + 2, 2x - 1 \rangle$
46	$\langle x + 2, -1 \rangle$	$\langle 1, -2 \rangle$	$x = \langle -2, 1 \rangle$
47	$\langle x + y, y \rangle$	$\langle 16, 10 \rangle$	$y = \langle 0, x \rangle$
48	$\langle x^2 - y - 3, x^2 - 2x \rangle$	$\langle 2, -3 \rangle$	$y = \langle x^2 - 3, x^2 - 2x \rangle$
49	$\langle -x^2 - 3x - y - 6, -2 \rangle$	$\langle 21, 16 \rangle$	$y = \langle -x^2 - 3x - 6, -2 \rangle$
50	$\langle -x^2 - x - y + 1, -x^2 - 2x \rangle$	$\langle 20, -64 \rangle$	$y = \langle -x^2 - x + 1, -x^2 - 2x \rangle$
51	$\langle x - y + 1, x^2 + 2x \rangle$	$\langle 2960, 6656 \rangle$	$y = \langle x + 1, x^2 + 2x \rangle$
52	$\langle -x - y + 3, x^2 - 2 \rangle$	$\langle 64, -20 \rangle$	$y = \langle -x + 3, x^2 - 2 \rangle$
53	$\langle -x^2 - x - y + 5, -x^2 + 2 \rangle$	$\langle 6656, 2960 \rangle$	$y = \langle -x^2 - x + 5, -x^2 + 2 \rangle$
54	$\langle -2x^2 - 2x - y, -x^2 - 4x \rangle$	$\langle 48, -15 \rangle$	$y = \langle -2x^2 - 2x, -x^2 - 4x \rangle$
55	$\langle 2x + y + 1, 0 \rangle$	$\langle 14, -7 \rangle$	$y = \langle -2x - 1, 0 \rangle$
56	$\langle x^2 - y - 2, 0 \rangle$	$\langle -7, -21 \rangle$	$y = \langle x^2 - 2, 0 \rangle$
57	$\langle -y + 2, x \rangle$	$\langle 0, 3 \rangle$	$y = \langle 2, x \rangle$
58	$\langle -x - y + 1, -x^2 \rangle$	$\langle 1, 1 \rangle$	$y = \langle -x + 1, -x^2 \rangle$
59	$\langle x^2 - 2, y \rangle$	$\langle 2, 3 \rangle$	$y = \langle x^2 - 2, x^2 - 2 \rangle$
60	$\langle x^2 + y + 2, y \rangle$	$\langle 0, 12 \rangle$	$y = \langle 0, x^2 + 2 \rangle$
61	$\langle x^2 + 2, -y \rangle$	$\langle 0, 144 \rangle$	$y = \langle -x^2 - 2, -x^2 - 2 \rangle$
62	$\langle x - y, -y + 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle 1, -x + 1 \rangle$
63	$\langle -y, -x + 1 \rangle$	$\langle -495, -144 \rangle$	$y = \langle 0, -x + 1 \rangle$
64	$\langle x - y, -x \rangle$	$\langle 3, 1 \rangle$	$y = \langle x, -x \rangle$
65	$\langle x + y - 1, y - 1 \rangle$	$\langle -8, -3 \rangle$	$y = \langle 1, x \rangle$
66	$\langle x - y, 2x + 1 \rangle$	$\langle 8, 5 \rangle$	$y = \langle x, 2x + 1 \rangle$
67	$\langle -x - y - 1, x - 1 \rangle$	$\langle -9, 15 \rangle$	$y = \langle -x - 1, x - 1 \rangle$
68	$\langle -y - 1, x - 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -1, x - 1 \rangle$
69	$\langle x - y - 1, -x - 1 \rangle$	$\langle -1176, -735 \rangle$	$y = \langle x - 1, -x - 1 \rangle$
70	$\langle -y - 1, -1 \rangle$	$\langle 16, 0 \rangle$	$y = \langle -1, -1 \rangle$
71	$\langle x, 1 \rangle$	$\langle 0, 9 \rangle$	$x = \langle 0, -1 \rangle$
72	$\langle -x - y + 1, x + 2 \rangle$	$\langle -9, -9 \rangle$	$y = \langle -x + 1, x + 2 \rangle$
73	$\langle 2x - y + 1, x + 2 \rangle$	$\langle 4, 0 \rangle$	$y = \langle 2x + 1, x + 2 \rangle$
74	$\langle x - 1, -3 \rangle$	$\langle 0, 9 \rangle$	$x = \langle 1, 3 \rangle$
75	$\langle x + y - 1, 1 \rangle$	$\langle -48, -128 \rangle$	$y = \langle -x + 1, -1 \rangle$
76	$\langle -2x - y - 2, -x - 1 \rangle$	$\langle -45, 27 \rangle$	$y = \langle -2x - 2, -x - 1 \rangle$
77	$\langle x - y - 2, -1 \rangle$	$\langle 3, -5 \rangle$	$y = \langle x - 2, -1 \rangle$
78	$\langle -y - 2, -x - 1 \rangle$	$\langle -64, -220 \rangle$	$y = \langle -2, -x - 1 \rangle$

i	J_i	c_i	chart
79	$\langle x - 1, 1 \rangle$	$\langle -899, -960 \rangle$	$x = \langle 1, -1 \rangle$
80	$\langle -y - 2, -2x - 2 \rangle$	$\langle 20, -64 \rangle$	$y = \langle -2, -2x - 2 \rangle$
81	$\langle -y, 2 \rangle$	$\langle 1, 0 \rangle$	$y = \langle 0, 2 \rangle$
82	$\langle x + y, -1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x, 1 \rangle$
83	$\langle x - y, 2x - 1 \rangle$	$\langle 16, 0 \rangle$	$y = \langle x, 2x - 1 \rangle$
84	$\langle -y + 2, x + 1 \rangle$	$\langle -48, 0 \rangle$	$y = \langle 2, x + 1 \rangle$
85	$\langle x + 1, y \rangle$	$\langle 1, 0 \rangle$	$y = \langle x + 1, x + 1 \rangle$
86	$\langle -x - y + 1, x \rangle$	$\langle -156, 64 \rangle$	$y = \langle -x + 1, x \rangle$
87	$\langle -y + 3, -2x + 1 \rangle$	$\langle 14976, 6660 \rangle$	$y = \langle 3, -2x + 1 \rangle$
88	$\langle x + 3, 1 \rangle$	$\langle 5, 8 \rangle$	$x = \langle -3, -1 \rangle$
89	$\langle x, -y \rangle$	$\langle -64, -220 \rangle$	$y = \langle -x, -x \rangle$
90	$\langle 2x - y - 3, -x^2 + 2x \rangle$	$\langle 16, 21 \rangle$	$y = \langle 2x - 3, -x^2 + 2x \rangle$
91	$\langle -x^2 - 3x - y - 4, 2 \rangle$	$\langle -231, -416 \rangle$	$y = \langle -x^2 - 3x - 4, 2 \rangle$
92	$\langle -x^2 + 2x - y, x^2 + 4x \rangle$	$\langle 3744, 1665 \rangle$	$y = \langle -x^2 + 2x, x^2 + 4x \rangle$
93	$\langle -x - y + 2, -x \rangle$	$\langle 0, 9 \rangle$	$y = \langle -x + 2, -x \rangle$
94	$\langle x^2 - x - y + 1, x^2 \rangle$	$\langle -1, -1 \rangle$	$y = \langle x^2 - x + 1, x^2 \rangle$
95	$\langle x^2 - y - 2, -y \rangle$	$\langle -8, -3 \rangle$	$y = \langle 0, -x^2 + 2 \rangle$
96	$\langle x - 1, y - 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle x, x - 1 \rangle$
97	$\langle 2x - y, x \rangle$	$\langle 3, -5 \rangle$	$y = \langle 2x, x \rangle$

Table 5: The polynomials J_i , the constants c_i and, for the linear vertices, $\eta = r_i(\xi)$ of the second certificate. Here $\langle A, B \rangle = A + B\omega$.

i	J_i	c_i	chart
0	$\langle x - 1, 0 \rangle$	$\langle 1, 0 \rangle$	$x = \langle 1, 0 \rangle$
1	$\langle x - 2, 0 \rangle$	$\langle 21, 14 \rangle$	$x = \langle 2, 0 \rangle$
2	$\langle x + 1, 0 \rangle$	$\langle 1, 0 \rangle$	$x = \langle -1, 0 \rangle$
3	$\langle -x - y + 2, 0 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x + 2, 0 \rangle$
4	$\langle x + 2, 0 \rangle$	$\langle -7, -21 \rangle$	$x = \langle -2, 0 \rangle$
5	$\langle -y, 0 \rangle$	$\langle 1, 0 \rangle$	$y = \langle 0, 0 \rangle$
6	$\langle x, -y + 1 \rangle$	$\langle 2, 0 \rangle$	$y = \langle -x + 1, -x \rangle$
7	$\langle -x - y - 1, -2x - 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x - 1, -2x - 1 \rangle$
8	$\langle -2x - y, -x + 1 \rangle$	$\langle 3, 1 \rangle$	$y = \langle -2x, -x + 1 \rangle$
9	$\langle -x - y, -x + 1 \rangle$	$\langle 2, 13 \rangle$	$y = \langle -x, -x + 1 \rangle$
10	$\langle 2x - y, x + 1 \rangle$	$\langle 3, -6 \rangle$	$y = \langle 2x, x + 1 \rangle$
11	$\langle x - y - 1, x - 1 \rangle$	$\langle 2, 0 \rangle$	$y = \langle x - 1, x - 1 \rangle$
12	$\langle -y, 1 \rangle$	$\langle 1, 1 \rangle$	$y = \langle 0, 1 \rangle$
13	$\langle x - 1, -1 \rangle$	$\langle 3, 1 \rangle$	$x = \langle 1, 1 \rangle$
14	$\langle -2x - y - 1, -x - 2 \rangle$	$\langle 1, 1 \rangle$	$y = \langle -2x - 1, -x - 2 \rangle$
15	$\langle x - y - 1, -x - 2 \rangle$	$\langle 100, -100 \rangle$	$y = \langle x - 1, -x - 2 \rangle$
16	$\langle -y - 1, -2 \rangle$	$\langle 2, 0 \rangle$	$y = \langle -1, -2 \rangle$
17	$\langle -y + 1, 2 \rangle$	$\langle 4, 0 \rangle$	$y = \langle 1, 2 \rangle$
18	$\langle x - 1, -2 \rangle$	$\langle 20, -80 \rangle$	$x = \langle 1, 2 \rangle$
19	$\langle x + 1, 2 \rangle$	$\langle 3600, 9600 \rangle$	$x = \langle -1, -2 \rangle$
20	$\langle x + 2, 3 \rangle$	$\langle 3, 3 \rangle$	$x = \langle -2, -3 \rangle$
21	$\langle x + y - 2, -1 \rangle$	$\langle 0, 2 \rangle$	$y = \langle -x + 2, 1 \rangle$
22	$\langle -x - y - 1, x + 1 \rangle$	$\langle 50, 50 \rangle$	$y = \langle -x - 1, x + 1 \rangle$
23	$\langle x - y - 1, 1 \rangle$	$\langle 96, 36 \rangle$	$y = \langle x - 1, 1 \rangle$
24	$\langle x - y - 1, x + 1 \rangle$	$\langle 210, 140 \rangle$	$y = \langle x - 1, x + 1 \rangle$
25	$\langle x - 2, -1 \rangle$	$\langle 3, -5 \rangle$	$x = \langle 2, 1 \rangle$
26	$\langle -y - 2, -x - 2 \rangle$	$\langle 49, -49 \rangle$	$y = \langle -2, -x - 2 \rangle$
27	$\langle x - y, x + 2 \rangle$	$\langle 7203, 7203 \rangle$	$y = \langle x, x + 2 \rangle$
28	$\langle 2x - y, 2x + 2 \rangle$	$\langle 1, 0 \rangle$	$y = \langle 2x, 2x + 2 \rangle$
29	$\langle -y - 2, -2 \rangle$	$\langle 4, 0 \rangle$	$y = \langle -2, -2 \rangle$
30	$\langle -x - 3y - 6, x - 6 \rangle$	$\langle 940, -120 \rangle$	$\langle 1, -1 \rangle x = \langle -3y - 6, -6 \rangle$
31	$\langle -2x - 3y, -x + 6 \rangle$	$\langle 1109200, 240000 \rangle$	$\langle 1, -1 \rangle x = \langle 6, 3y + 6 \rangle$

i	J_i	c_i	chart
32	$\langle x + y + 1, 1 \rangle$	$\langle 2, -1 \rangle$	$y = \langle -x - 1, -1 \rangle$
33	$\langle x - y + 1, 1 \rangle$	$\langle 25, -25 \rangle$	$y = \langle x + 1, 1 \rangle$
34	$\langle x - y, -1 \rangle$	$\langle 1875, 1875 \rangle$	$y = \langle x, -1 \rangle$
35	$\langle -x - y + 1, -2x + 1 \rangle$	$\langle 1, 1 \rangle$	$y = \langle -x + 1, -2x + 1 \rangle$
36	$\langle x, -1 \rangle$	$\langle 2, -11 \rangle$	$x = \langle 0, 1 \rangle$
37	$\langle x + 1, 1 \rangle$	$\langle 48, 165 \rangle$	$x = \langle -1, -1 \rangle$
38	$\langle -x - y + 1, -x - 1 \rangle$	$\langle 5, -3 \rangle$	$y = \langle -x + 1, -x - 1 \rangle$
39	$\langle x - y + 2, 2x + 1 \rangle$	$\langle 50, 0 \rangle$	$y = \langle x + 2, 2x + 1 \rangle$
40	$\langle -x - y + 1, -2x - 1 \rangle$	$\langle 2500, 0 \rangle$	$y = \langle -x + 1, -2x - 1 \rangle$
41	$\langle -y + 1, x - 2 \rangle$	$\langle 700, -350 \rangle$	$y = \langle 1, x - 2 \rangle$
42	$\langle -x - y + 3, -x + 2 \rangle$	$\langle 980000, 612500 \rangle$	$y = \langle -x + 3, -x + 2 \rangle$
43	$\langle x - y + 1, -y \rangle$	$\langle 1, 0 \rangle$	$y = \langle 0, -x - 1 \rangle$
44	$\langle -2x - y + 1, -x \rangle$	$\langle 350, 525 \rangle$	$y = \langle -2x + 1, -x \rangle$
45	$\langle 2x - y + 2, 2x - 1 \rangle$	$\langle 925, 275 \rangle$	$y = \langle 2x + 2, 2x - 1 \rangle$
46	$\langle x + 2, -1 \rangle$	$\langle 1, 0 \rangle$	$x = \langle -2, 1 \rangle$
47	$\langle x + y, y \rangle$	$\langle 2, 2 \rangle$	$y = \langle 0, x \rangle$
48	$\langle x^2 - y - 3, x^2 - 2x \rangle$	$\langle 1, 0 \rangle$	$y = \langle x^2 - 3, x^2 - 2x \rangle$
49	$\langle -x^2 - 3x - y - 6, -2 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x^2 - 3x - 6, -2 \rangle$
50	$\langle -x^2 - x - y + 1, -x^2 - 2x \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x^2 - x + 1, -x^2 - 2x \rangle$
51	$\langle x - y + 1, x^2 + 2x \rangle$	$\langle 1, 0 \rangle$	$y = \langle x + 1, x^2 + 2x \rangle$
52	$\langle -x - y + 3, x^2 - 2 \rangle$	$\langle 1, 1 \rangle$	$y = \langle -x + 3, x^2 - 2 \rangle$
53	$\langle -x^2 - x - y + 5, -x^2 + 2 \rangle$	$\langle -1, -1 \rangle$	$y = \langle -x^2 - x + 5, -x^2 + 2 \rangle$
54	$\langle -2x^2 - 2x - y, -x^2 - 4x \rangle$	$\langle 1, 0 \rangle$	$y = \langle -2x^2 - 2x, -x^2 - 4x \rangle$
55	$\langle 2x + y + 1, 0 \rangle$	$\langle 392, 147 \rangle$	$y = \langle -2x - 1, 0 \rangle$
56	$\langle x^2 - y - 2, 0 \rangle$	$\langle 0, -1 \rangle$	$y = \langle x^2 - 2, 0 \rangle$
57	$\langle -y + 2, x \rangle$	$\langle 10, -6 \rangle$	$y = \langle 2, x \rangle$
58	$\langle -x - y + 1, -x^2 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x + 1, -x^2 \rangle$
59	$\langle x^2 - 2, y \rangle$	$\langle 4, 4 \rangle$	$y = \langle x^2 - 2, x^2 - 2 \rangle$
60	$\langle x^2 + y + 2, y \rangle$	$\langle 12, 0 \rangle$	$y = \langle 0, x^2 + 2 \rangle$
61	$\langle x^2 + 2, -y \rangle$	$\langle 144, 0 \rangle$	$y = \langle -x^2 - 2, -x^2 - 2 \rangle$
62	$\langle x - y, -y + 1 \rangle$	$\langle 2, -1 \rangle$	$y = \langle 1, -x + 1 \rangle$
63	$\langle -y, -x + 1 \rangle$	$\langle 4, 0 \rangle$	$y = \langle 0, -x + 1 \rangle$
64	$\langle x - y, -x \rangle$	$\langle 3, 0 \rangle$	$y = \langle x, -x \rangle$
65	$\langle x + y - 1, y - 1 \rangle$	$\langle 4, 0 \rangle$	$y = \langle 1, x \rangle$
66	$\langle x - y, 2x + 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle x, 2x + 1 \rangle$
67	$\langle -x - y - 1, x - 1 \rangle$	$\langle 3, -5 \rangle$	$y = \langle -x - 1, x - 1 \rangle$
68	$\langle -y - 1, x - 1 \rangle$	$\langle -48, 117 \rangle$	$y = \langle -1, x - 1 \rangle$
69	$\langle x - y - 1, -x - 1 \rangle$	$\langle 45, 72 \rangle$	$y = \langle x - 1, -x - 1 \rangle$
70	$\langle -y - 1, -1 \rangle$	$\langle -1, -1 \rangle$	$y = \langle -1, -1 \rangle$
71	$\langle x, 1 \rangle$	$\langle 3, -5 \rangle$	$x = \langle 0, -1 \rangle$
72	$\langle -x - y + 1, x + 2 \rangle$	$\langle -1, -1 \rangle$	$y = \langle -x + 1, x + 2 \rangle$
73	$\langle 2x - y + 1, x + 2 \rangle$	$\langle 30000, 30000 \rangle$	$y = \langle 2x + 1, x + 2 \rangle$
74	$\langle x - 1, -3 \rangle$	$\langle -9, -9 \rangle$	$x = \langle 1, 3 \rangle$
75	$\langle x + y - 1, 1 \rangle$	$\langle 0, 4 \rangle$	$y = \langle -x + 1, -1 \rangle$
76	$\langle -2x - y - 2, -x - 1 \rangle$	$\langle -2500, -2500 \rangle$	$y = \langle -2x - 2, -x - 1 \rangle$
77	$\langle x - y - 2, -1 \rangle$	$\langle 2304, -5616 \rangle$	$y = \langle x - 2, -1 \rangle$
78	$\langle -y - 2, -x - 1 \rangle$	$\langle -14700, -39200 \rangle$	$y = \langle -2, -x - 1 \rangle$
79	$\langle x - 1, 1 \rangle$	$\langle 39, 55 \rangle$	$x = \langle 1, -1 \rangle$
80	$\langle -y - 2, -2x - 2 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -2, -2x - 2 \rangle$
81	$\langle -y, 2 \rangle$	$\langle 16, 0 \rangle$	$y = \langle 0, 2 \rangle$
82	$\langle x + y, -1 \rangle$	$\langle 8, 5 \rangle$	$y = \langle -x, 1 \rangle$
83	$\langle x - y, 2x - 1 \rangle$	$\langle -1, -1 \rangle$	$y = \langle x, 2x - 1 \rangle$
84	$\langle -y + 2, x + 1 \rangle$	$\langle 55, 39 \rangle$	$y = \langle 2, x + 1 \rangle$
85	$\langle x + 1, y \rangle$	$\langle 1, 0 \rangle$	$y = \langle x + 1, x + 1 \rangle$
86	$\langle -x - y + 1, x \rangle$	$\langle -245000, -91875 \rangle$	$y = \langle -x + 1, x \rangle$
87	$\langle -y + 3, -2x + 1 \rangle$	$\langle 346875, -433125 \rangle$	$y = \langle 3, -2x + 1 \rangle$
88	$\langle x + 3, 1 \rangle$	$\langle 1, 0 \rangle$	$x = \langle -3, -1 \rangle$

i	J_i	c_i	chart
89	$\langle x, -y \rangle$	$\langle -4, -4 \rangle$	$y = \langle -x, -x \rangle$
90	$\langle 2x - y - 3, -x^2 + 2x \rangle$	$\langle 1, 0 \rangle$	$y = \langle 2x - 3, -x^2 + 2x \rangle$
91	$\langle -x^2 - 3x - y - 4, 2 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x^2 - 3x - 4, 2 \rangle$
92	$\langle -x^2 + 2x - y, x^2 + 4x \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x^2 + 2x, x^2 + 4x \rangle$
93	$\langle -x - y + 2, -x \rangle$	$\langle 220, 156 \rangle$	$y = \langle -x + 2, -x \rangle$
94	$\langle x^2 - x - y + 1, x^2 \rangle$	$\langle 1, 0 \rangle$	$y = \langle x^2 - x + 1, x^2 \rangle$
95	$\langle x^2 - y - 2, -y \rangle$	$\langle -16, -16 \rangle$	$y = \langle 0, -x^2 + 2 \rangle$
96	$\langle x - 1, y - 1 \rangle$	$\langle 8, 5 \rangle$	$y = \langle x, x - 1 \rangle$
97	$\langle 2x - y, x \rangle$	$\langle 9, 0 \rangle$	$y = \langle 2x, x \rangle$

Table 6: The polynomials J_i , the constants c_i and, for the linear vertices, $\eta = r_i(\xi)$ of the third certificate. Here $\langle A, B \rangle = A + B\omega$.

i	J_i	c_i	chart
0	$\langle x - 1, 0 \rangle$	$\langle 1, 0 \rangle$	$x = \langle 1, 0 \rangle$
1	$\langle x - 2, 0 \rangle$	$\langle 14, -7 \rangle$	$x = \langle 2, 0 \rangle$
2	$\langle x + 1, 0 \rangle$	$\langle 1, 0 \rangle$	$x = \langle -1, 0 \rangle$
3	$\langle -x - y + 2, 0 \rangle$	$\langle -1, -1 \rangle$	$y = \langle -x + 2, 0 \rangle$
4	$\langle x + 2, 0 \rangle$	$\langle 21, 14 \rangle$	$x = \langle -2, 0 \rangle$
5	$\langle -y, 0 \rangle$	$\langle 1, 0 \rangle$	$y = \langle 0, 0 \rangle$
6	$\langle x, -y + 1 \rangle$	$\langle 6, 16 \rangle$	$y = \langle -x + 1, -x \rangle$
7	$\langle -x - y - 1, -2x - 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x - 1, -2x - 1 \rangle$
8	$\langle -2x - y, -x + 1 \rangle$	$\langle 12, 0 \rangle$	$y = \langle -2x, -x + 1 \rangle$
9	$\langle -x - y, -x + 1 \rangle$	$\langle 2, -1 \rangle$	$y = \langle -x, -x + 1 \rangle$
10	$\langle 2x - y, x + 1 \rangle$	$\langle 4, 26 \rangle$	$y = \langle 2x, x + 1 \rangle$
11	$\langle x - y - 1, x - 1 \rangle$	$\langle 12, 36 \rangle$	$y = \langle x - 1, x - 1 \rangle$
12	$\langle -y, 1 \rangle$	$\langle 0, 3 \rangle$	$y = \langle 0, 1 \rangle$
13	$\langle x - 1, -1 \rangle$	$\langle 3, 1 \rangle$	$x = \langle 1, 1 \rangle$
14	$\langle -2x - y - 1, -x - 2 \rangle$	$\langle 2, 1 \rangle$	$y = \langle -2x - 1, -x - 2 \rangle$
15	$\langle x - y - 1, -x - 2 \rangle$	$\langle 1, 0 \rangle$	$y = \langle x - 1, -x - 2 \rangle$
16	$\langle -y - 1, -2 \rangle$	$\langle 3, 3 \rangle$	$y = \langle -1, -2 \rangle$
17	$\langle -y + 1, 2 \rangle$	$\langle -9, -9 \rangle$	$y = \langle 1, 2 \rangle$
18	$\langle x - 1, -2 \rangle$	$\langle 1, 3 \rangle$	$x = \langle 1, 2 \rangle$
19	$\langle x + 1, 2 \rangle$	$\langle -5, 3 \rangle$	$x = \langle -1, -2 \rangle$
20	$\langle x + 2, 3 \rangle$	$\langle 1, -1 \rangle$	$x = \langle -2, -3 \rangle$
21	$\langle x + y - 2, -1 \rangle$	$\langle 32, 12 \rangle$	$y = \langle -x + 2, 1 \rangle$
22	$\langle -x - y - 1, x + 1 \rangle$	$\langle 10, 2 \rangle$	$y = \langle -x - 1, x + 1 \rangle$
23	$\langle x - y - 1, 1 \rangle$	$\langle 5, 8 \rangle$	$y = \langle x - 1, 1 \rangle$
24	$\langle x - y - 1, x + 1 \rangle$	$\langle 7, 0 \rangle$	$y = \langle x - 1, x + 1 \rangle$
25	$\langle x - 2, -1 \rangle$	$\langle 8, 3 \rangle$	$x = \langle 2, 1 \rangle$
26	$\langle -y - 2, -x - 2 \rangle$	$\langle 2, -4 \rangle$	$y = \langle -2, -x - 2 \rangle$
27	$\langle x - y, x + 2 \rangle$	$\langle 20, 32 \rangle$	$y = \langle x, x + 2 \rangle$
28	$\langle 2x - y, 2x + 2 \rangle$	$\langle 1, 0 \rangle$	$y = \langle 2x, 2x + 2 \rangle$
29	$\langle -y - 2, -2 \rangle$	$\langle 4, 4 \rangle$	$y = \langle -2, -2 \rangle$
30	$\langle -x - 3y - 6, x - 6 \rangle$	$\langle 1, 0 \rangle$	$\langle 1, -1 \rangle x = \langle -3y - 6, -6 \rangle$
31	$\langle -2x - 3y, -x + 6 \rangle$	$\langle 1, 0 \rangle$	$\langle 1, -1 \rangle x = \langle 6, 3y + 6 \rangle$
32	$\langle x + y + 1, 1 \rangle$	$\langle 1, -1 \rangle$	$y = \langle -x - 1, -1 \rangle$
33	$\langle x - y + 1, 1 \rangle$	$\langle 1, 2 \rangle$	$y = \langle x + 1, 1 \rangle$
34	$\langle x - y, -1 \rangle$	$\langle -3, 0 \rangle$	$y = \langle x, -1 \rangle$
35	$\langle -x - y + 1, -2x + 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x + 1, -2x + 1 \rangle$
36	$\langle x, -1 \rangle$	$\langle 7, -28 \rangle$	$x = \langle 0, 1 \rangle$
37	$\langle x + 1, 1 \rangle$	$\langle 441, 1176 \rangle$	$x = \langle -1, -1 \rangle$
38	$\langle -x - y + 1, -x - 1 \rangle$	$\langle 2, 3 \rangle$	$y = \langle -x + 1, -x - 1 \rangle$
39	$\langle x - y + 2, 2x + 1 \rangle$	$\langle 4, 6 \rangle$	$y = \langle x + 2, 2x + 1 \rangle$
40	$\langle -x - y + 1, -2x - 1 \rangle$	$\langle -32, -12 \rangle$	$y = \langle -x + 1, -2x - 1 \rangle$
41	$\langle -y + 1, x - 2 \rangle$	$\langle 15, -9 \rangle$	$y = \langle 1, x - 2 \rangle$

i	J_i	c_i	chart
42	$\langle -x - y + 3, -x + 2 \rangle$	$\langle 495, 351 \rangle$	$y = \langle -x + 3, -x + 2 \rangle$
43	$\langle x - y + 1, -y \rangle$	$\langle 1, 0 \rangle$	$y = \langle 0, -x - 1 \rangle$
44	$\langle -2x - y + 1, -x \rangle$	$\langle 2, 13 \rangle$	$y = \langle -2x + 1, -x \rangle$
45	$\langle 2x - y + 2, 2x - 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle 2x + 2, 2x - 1 \rangle$
46	$\langle x + 2, -1 \rangle$	$\langle 1, 0 \rangle$	$x = \langle -2, 1 \rangle$
47	$\langle x + y, y \rangle$	$\langle 2, -4 \rangle$	$y = \langle 0, x \rangle$
48	$\langle x^2 - y - 3, x^2 - 2x \rangle$	$\langle 1, 0 \rangle$	$y = \langle x^2 - 3, x^2 - 2x \rangle$
49	$\langle -x^2 - 3x - y - 6, -2 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x^2 - 3x - 6, -2 \rangle$
50	$\langle -x^2 - x - y + 1, -x^2 - 2x \rangle$	$\langle 1, 1 \rangle$	$y = \langle -x^2 - x + 1, -x^2 - 2x \rangle$
51	$\langle x - y + 1, x^2 + 2x \rangle$	$\langle -1, -1 \rangle$	$y = \langle x + 1, x^2 + 2x \rangle$
52	$\langle -x - y + 3, x^2 - 2 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x + 3, x^2 - 2 \rangle$
53	$\langle -x^2 - x - y + 5, -x^2 + 2 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x^2 - x + 5, -x^2 + 2 \rangle$
54	$\langle -2x^2 - 2x - y, -x^2 - 4x \rangle$	$\langle 1, 0 \rangle$	$y = \langle -2x^2 - 2x, -x^2 - 4x \rangle$
55	$\langle 2x + y + 1, 0 \rangle$	$\langle -245, -392 \rangle$	$y = \langle -2x - 1, 0 \rangle$
56	$\langle x^2 - y - 2, 0 \rangle$	$\langle -7, -21 \rangle$	$y = \langle x^2 - 2, 0 \rangle$
57	$\langle -y + 2, x \rangle$	$\langle 1, -1 \rangle$	$y = \langle 2, x \rangle$
58	$\langle -x - y + 1, -x^2 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x + 1, -x^2 \rangle$
59	$\langle x^2 - 2, y \rangle$	$\langle 0, 1 \rangle$	$y = \langle x^2 - 2, x^2 - 2 \rangle$
60	$\langle x^2 + y + 2, y \rangle$	$\langle 0, 1 \rangle$	$y = \langle 0, x^2 + 2 \rangle$
61	$\langle x^2 + 2, -y \rangle$	$\langle 0, 1 \rangle$	$y = \langle -x^2 - 2, -x^2 - 2 \rangle$
62	$\langle x - y, -y + 1 \rangle$	$\langle 11, -2 \rangle$	$y = \langle 1, -x + 1 \rangle$
63	$\langle -y, -x + 1 \rangle$	$\langle -720, 432 \rangle$	$y = \langle 0, -x + 1 \rangle$
64	$\langle x - y, -x \rangle$	$\langle 0, 4 \rangle$	$y = \langle x, -x \rangle$
65	$\langle x + y - 1, y - 1 \rangle$	$\langle -156, 64 \rangle$	$y = \langle 1, x \rangle$
66	$\langle x - y, 2x + 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle x, 2x + 1 \rangle$
67	$\langle -x - y - 1, x - 1 \rangle$	$\langle 144, 0 \rangle$	$y = \langle -x - 1, x - 1 \rangle$
68	$\langle -y - 1, x - 1 \rangle$	$\langle 8, 5 \rangle$	$y = \langle -1, x - 1 \rangle$
69	$\langle x - y - 1, -x - 1 \rangle$	$\langle -192, 468 \rangle$	$y = \langle x - 1, -x - 1 \rangle$
70	$\langle -y - 1, -1 \rangle$	$\langle 0, 9 \rangle$	$y = \langle -1, -1 \rangle$
71	$\langle x, 1 \rangle$	$\langle 3, -5 \rangle$	$x = \langle 0, -1 \rangle$
72	$\langle -x - y + 1, x + 2 \rangle$	$\langle 0, -3 \rangle$	$y = \langle -x + 1, x + 2 \rangle$
73	$\langle 2x - y + 1, x + 2 \rangle$	$\langle 1, 0 \rangle$	$y = \langle 2x + 1, x + 2 \rangle$
74	$\langle x - 1, -3 \rangle$	$\langle 3, 3 \rangle$	$x = \langle 1, 3 \rangle$
75	$\langle x + y - 1, 1 \rangle$	$\langle 256, -624 \rangle$	$y = \langle -x + 1, -1 \rangle$
76	$\langle -2x - y - 2, -x - 1 \rangle$	$\langle 60, -36 \rangle$	$y = \langle -2x - 2, -x - 1 \rangle$
77	$\langle x - y - 2, -1 \rangle$	$\langle -55, -16 \rangle$	$y = \langle x - 2, -1 \rangle$
78	$\langle -y - 2, -x - 1 \rangle$	$\langle 49, 0 \rangle$	$y = \langle -2, -x - 1 \rangle$
79	$\langle x - 1, 1 \rangle$	$\langle 16, -39 \rangle$	$x = \langle 1, -1 \rangle$
80	$\langle -y - 2, -2x - 2 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -2, -2x - 2 \rangle$
81	$\langle -y, 2 \rangle$	$\langle -16, -16 \rangle$	$y = \langle 0, 2 \rangle$
82	$\langle x + y, -1 \rangle$	$\langle 3, 3 \rangle$	$y = \langle -x, 1 \rangle$
83	$\langle x - y, 2x - 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle x, 2x - 1 \rangle$
84	$\langle -y + 2, x + 1 \rangle$	$\langle -8, -3 \rangle$	$y = \langle 2, x + 1 \rangle$
85	$\langle x + 1, y \rangle$	$\langle 1, 0 \rangle$	$y = \langle x + 1, x + 1 \rangle$
86	$\langle -x - y + 1, x \rangle$	$\langle -48, 117 \rangle$	$y = \langle -x + 1, x \rangle$
87	$\langle -y + 3, -2x + 1 \rangle$	$\langle 1, 0 \rangle$	$y = \langle 3, -2x + 1 \rangle$
88	$\langle x + 3, 1 \rangle$	$\langle 1, 0 \rangle$	$x = \langle -3, -1 \rangle$
89	$\langle x, -y \rangle$	$\langle 20, 32 \rangle$	$y = \langle -x, -x \rangle$
90	$\langle 2x - y - 3, -x^2 + 2x \rangle$	$\langle 1, 0 \rangle$	$y = \langle 2x - 3, -x^2 + 2x \rangle$
91	$\langle -x^2 - 3x - y - 4, 2 \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x^2 - 3x - 4, 2 \rangle$
92	$\langle -x^2 + 2x - y, x^2 + 4x \rangle$	$\langle 1, 0 \rangle$	$y = \langle -x^2 + 2x, x^2 + 4x \rangle$
93	$\langle -x - y + 2, -x \rangle$	$\langle 3, 3 \rangle$	$y = \langle -x + 2, -x \rangle$
94	$\langle x^2 - x - y + 1, x^2 \rangle$	$\langle 1, 0 \rangle$	$y = \langle x^2 - x + 1, x^2 \rangle$
95	$\langle x^2 - y - 2, -y \rangle$	$\langle 0, 1 \rangle$	$y = \langle 0, -x^2 + 2 \rangle$
96	$\langle x - 1, y - 1 \rangle$	$\langle 165, 48 \rangle$	$y = \langle x, x - 1 \rangle$
97	$\langle 2x - y, x \rangle$	$\langle 0, 16 \rangle$	$y = \langle 2x, x \rangle$

Table 7: The data of the identities (4); the entry $j:e$ means $\rho_{ij} = e \neq 0$.

The first certificate.

$i = 0, D_i = 1, \rho_{ij}: 6:2, 9:2, 16:1, 17:2, 32:1, 33:1, 34:2, 44:1, 58:1, 64:1, 65:1, 68:1, 82:2, 86:2, 94:2, 97:2$
$w_i \quad \langle -48,$ $24t^2 - 24t - 24 \rangle$
$i = 1, D_i = 1, \rho_{ij}: 9:1, 11:2, 14:2, 23:2, 29:2, 30:2, 31:1, 39:2, 40:1, 43:1, 44:1, 48:2, 57:2, 59:2, 63:1, 64:1, 68:2, 72:1, 77:1, 81:1, 85:2, 86:2, 90:1, 93:1, 95:1, 97:2$
$w_i \quad \langle 479710t^2 + 3257030t - 10095960,$ $-406210t^2 - 31850t - 12589080 \rangle$
$i = 2, D_i = 1, \rho_{ij}: 23:2, 26:1, 27:2, 28:2, 33:2, 34:1, 38:1, 47:1, 54:1, 59:2, 60:1, 61:2, 62:1, 77:1, 80:1, 84:2, 89:2, 92:2, 95:1, 96:2$
$w_i \quad \langle 374t^2 + 654t + 666,$ $0 \rangle$
$i = 3, D_i = 1, \rho_{ij}: 6:1, 8:1, 18:1, 19:2, 20:2, 22:1, 29:1, 36:2, 37:1, 41:1, 42:2, 43:1, 52:2, 53:1, 54:2, 57:1, 60:2, 61:1, 62:1, 65:2, 67:2, 74:1, 76:2, 81:2, 85:2, 92:1, 93:2, 96:2$
$w_i \quad \langle -1387095t^3 + 983478t^2 + 1474875t - 423738,$ $-557802t^3 - 2401638t^2 + 5032530t - 1119252 \rangle$
$i = 4, D_i = 1, \rho_{ij}: 14:2, 15:1, 16:2, 17:1, 22:1, 30:2, 31:1, 41:1, 42:2, 49:1, 50:2, 51:1, 64:1, 72:1, 73:2, 76:2, 91:2, 97:2$
$w_i \quad \langle 1260t^2 - 4788t - 27216,$ $2520t^2 - 9576t - 54432 \rangle$
$i = 5, D_i = 2, \rho_{ij}: 9:2, 10:1, 11:1, 23:1, 26:1, 27:2, 43:1, 52:2, 53:1, 54:2, 63:2, 68:1, 69:2, 77:2, 85:2, 92:1$
$w_i \quad \langle -67t^3 - 46t^2 + 526t - 692,$ $-281t^3 + 202t^2 + 758t - 796 \rangle$
$i = 6, D_i = 1, \rho_{ij}: 0:1, 3:2, 7:2, 15:1, 17:1, 19:1, 20:1, 40:2, 44:1, 46:1, 48:2, 51:2, 53:1, 55:1, 57:1, 58:2, 59:1, 62:1$
$w_i \quad \langle -408t^3 + 12t^2 + 412t + 848,$ $-24t^3 - 388t^2 + 660t - 240 \rangle$
$i = 7, D_i = 1, \rho_{ij}: 6:1, 8:1, 16:2, 19:2, 30:2, 41:2, 43:1, 50:1, 54:2$
$w_i \quad \langle -2405t^3 - 3277t^2 - 1766t - 18,$ $-2224t^3 - 941t^2 + 71t - 978 \rangle$
$i = 8, D_i = 1, \rho_{ij}: 3:2, 7:2, 13:1, 16:1, 18:2, 19:1, 21:2, 26:1, 29:2, 32:1, 35:2, 42:2, 49:2, 53:2$
$w_i \quad \langle -371t^3 - 6217t^2 - 2143t - 365,$ $-2191t^3 - 4415t^2 + 4636t - 565 \rangle$
$i = 9, D_i = 1, \rho_{ij}: 0:1, 1:2, 5:1, 13:1, 14:1, 16:2, 18:1, 21:2, 24:1, 27:2, 34:1, 36:2, 37:2, 40:1, 44:2, 47:2, 51:1, 53:1, 55:1, 57:2, 59:2, 61:2, 63:1$
$w_i \quad \langle 30975t^3 - 735224t^2 + 840392t + 30009,$ $-479710t^3 + 362404t^2 + 229523t + 480823 \rangle$
$i = 10, D_i = 1, \rho_{ij}: 5:2, 13:2, 19:2, 20:1, 24:1, 25:2, 30:1, 31:1, 33:1, 34:2, 36:1, 61:2, 64:1$
$w_i \quad \langle -434t^3 - 2723t^2 - 385t - 1134,$ $1358t^3 - 1414t^2 - 2513t - 1113 \rangle$
$i = 11, D_i = 1, \rho_{ij}: 1:1, 5:2, 15:2, 18:1, 32:2, 39:2, 60:1, 68:1, 72:2, 75:1, 77:2, 81:2, 82:2, 84:2, 95:1$
$w_i \quad \langle 4263t^3 - 1062t^2 - 7494t + 936,$ $3108t^3 - 9003t^2 + 3306t + 5055 \rangle$
$i = 12, D_i = 1, \rho_{ij}: 13:1, 25:2, 32:1, 36:2, 38:1, 40:2, 42:1, 45:1, 47:2, 50:1, 56:1, 58:1, 60:2, 61:2, 62:2$
$w_i \quad \langle 12t^3 - 54t^2 + 36t - 90,$ $-24t^3 - 72t^2 + 4t - 100 \rangle$
$i = 13, D_i = 1, \rho_{ij}: 8:2, 9:2, 10:1, 12:2, 21:1, 23:2, 31:1, 35:2, 39:2, 40:1, 47:1, 59:1, 60:1, 61:1, 62:2, 64:2$
$w_i \quad \langle -23t^2 + 192t - 189,$ $53t^2 + 57t - 378 \rangle$
$i = 14, D_i = 1, \rho_{ij}: 1:1, 4:1, 9:2, 26:2, 36:2, 41:2, 50:2, 59:1, 60:1, 61:2, 63:2$
$w_i \quad \langle -79398t^3 - 314466t^2 - 108936t + 143088,$ $55512t^3 - 187464t^2 - 320364t + 37884 \rangle$
$i = 15, D_i = 1, \rho_{ij}: 4:2, 6:2, 11:1, 16:1, 17:2, 18:2, 22:1, 27:1, 33:1, 55:2, 59:2, 61:2$
$w_i \quad \langle -14496t^3 - 52482t^2 - 41034t - 5280,$ $-15624t^3 - 27066t^2 + 24390t + 24528 \rangle$
$i = 16, D_i = 1, \rho_{ij}: 0:2, 4:1, 7:1, 8:2, 9:1, 15:2, 18:2, 19:2, 22:2, 24:2, 33:1, 40:2, 41:2, 42:2, 47:2, 51:2, 53:2, 56:1, 58:2, 61:2, 64:2, 65:1, 73:2, 75:2, 77:1, 95:1, 97:1$
$w_i \quad \langle 696384t^3 - 800352t^2 - 1653696t + 3229056,$ $167184t^3 - 1536336t^2 + 1133136t + 2418048 \rangle$
$i = 17, D_i = 1, \rho_{ij}: 0:1, 4:2, 6:2, 15:1, 18:1, 19:1, 21:1, 23:2, 34:2, 39:1, 41:1, 42:1, 50:1, 52:1, 56:2, 59:2, 60:1, 64:2, 66:2, 67:1, 68:2, 73:1, 76:1, 78:1, 89:1, 94:1, 97:1$
$w_i \quad \langle 13968t^3 - 99504t^2 + 23760t + 240192,$ $44784t^3 - 28080t^2 - 110160t + 164736 \rangle$
$i = 18, D_i = 1, \rho_{ij}: 3:2, 8:1, 9:2, 11:2, 15:1, 16:1, 17:2, 21:1, 23:2, 24:1, 30:1, 31:2, 33:1, 35:1, 38:2, 43:2, 47:2, 51:1, 58:2, 62:1, 64:2, 65:1, 66:2, 67:1, 69:2, 77:2, 81:2, 83:2, 89:2, 91:2, 96:1, 97:2$
$w_i \quad \langle -6505632t^2 + 1323840t + 31924032,$ $7072464t^2 + 19718160t + 14301168 \rangle$
$i = 19, D_i = 1, \rho_{ij}: 3:1, 6:2, 7:1, 8:2, 10:1, 16:1, 17:2, 23:1, 29:1, 30:1, 31:2, 34:2, 35:1, 47:1, 49:1, 50:2, 62:2, 63:1, 64:1, 67:2, 68:1, 73:2, 75:2, 77:1, 78:2, 83:2, 84:1, 85:1, 89:1, 94:1, 96:2, 97:1$

w_i	$\langle 2436000t^2 - 14579040t + 9513504, \\ 1304352t^2 - 10284960t + 5184480 \rangle$
$i = 20, D_i = 1, \rho_{ij}: 3:1, 6:2, 10:2, 22:1, 30:1, 32:2, 34:2, 43:2, 60:1, 62:1$	
w_i	$\langle -8t^2 + 393t - 735, \\ 29t^2 + 285t + 870 \rangle$
$i = 21, D_i = 1, \rho_{ij}: 8:1, 9:1, 13:2, 17:2, 18:2, 25:2, 34:2, 36:2, 37:1, 43:2, 44:1, 53:2, 60:2, 63:1$	
w_i	$\langle -2484t^3 + 9676t^2 - 924t - 11100, \\ 1732t^3 + 8756t^2 - 14168t - 2808 \rangle$
$i = 22, D_i = 1, \rho_{ij}: 3:2, 4:2, 15:2, 16:1, 20:2, 27:2, 31:1, 32:2, 34:1, 40:1, 51:1, 55:2, 57:1, 60:1, 61:1, 62:2, 64:1$	
w_i	$\langle 6141135t^3 + 11337999t^2 + 5969724t + 5362626, \\ -4368189t^3 - 17704950t^2 - 18577635t - 2734338 \rangle$
$i = 23, D_i = 1, \rho_{ij}: 1:1, 2:1, 5:2, 13:1, 17:1, 18:1, 19:2, 25:1, 35:1, 37:1, 40:2, 46:1, 49:2, 51:2, 52:2, 56:2, 59:2, 60:2, 61:1, 63:2$	
w_i	$\langle 532t^3 + 252420t^2 - 574476t + 291396, \\ -233828t^3 + 599928t^2 - 689164t + 419160 \rangle$
$i = 24, D_i = 1, \rho_{ij}: 9:2, 10:2, 16:1, 18:2, 37:2, 38:2, 51:1, 56:1, 60:1$	
w_i	$\langle 28t^3 + 204t^2 + 24t - 138, \\ -70t^3 + 96t^2 + 190t - 102 \rangle$
$i = 25, D_i = 1, \rho_{ij}: 10:1, 12:1, 21:1, 23:2, 34:2, 39:1, 40:2, 41:1, 43:2, 47:2, 48:2, 55:1, 57:1, 58:1, 60:2, 61:2, 62:2$	
w_i	$\langle 6635t^2 - 22923t + 38480, \\ -2859t^2 - 3984t + 37509 \rangle$
$i = 26, D_i = 1, \rho_{ij}: 2:2, 5:2, 8:2, 14:1, 28:1, 32:1, 38:2, 40:2, 55:2, 57:1, 60:2, 68:2, 73:1, 76:2, 88:1, 91:2, 96:1, 97:2$	
w_i	$\langle -3711268t^3 - 9899884t^2 + 6914576t + 1775800, \\ -3361412t^3 - 2433128t^2 + 7924316t + 6309608 \rangle$
$i = 27, D_i = 1, \rho_{ij}: 2:1, 5:1, 9:1, 15:2, 22:1, 39:1, 46:2, 49:1, 55:1, 61:1, 62:2, 64:1, 67:1, 72:2, 80:2, 82:2, 84:1, 93:2$	
w_i	$\langle 676032t^3 + 2913444t^2 + 3673788t + 475776, \\ -513996t^3 + 824760t^2 + 3814836t + 1828344 \rangle$
$i = 28, D_i = 1, \rho_{ij}: 2:1, 26:2, 47:2, 54:2$	
w_i	$\langle -6t^3 + 2t^2 + 18t + 10, \\ -10t^3 - 22t^2 - 8t + 4 \rangle$
$i = 29, D_i = 1, \rho_{ij}: 1:1, 3:2, 8:1, 19:2, 34:1, 36:1, 37:2, 48:1, 49:2, 56:1, 59:2, 61:2, 63:2$	
w_i	$\langle 52t^3 - 64t^2 - 6t - 116, \\ 64t^3 + 8t^2 + 80t - 20 \rangle$
$i = 30, D_i = 14348907, e_i = 1, \rho_{ij}: 1:1, 4:1, 7:1, 10:2, 18:2, 19:2, 20:2, 39:1, 40:2, 41:1, 43:1, 53:1, 55:2, 56:1, 61:1, 69:1, 71:1, 76:1, 85:2, 87:1, 89:2$	
w_i	$\langle -1586098606623624t^3 - 4335235375888728t^2 + 9365524975281816t + 8733753916904520, \\ -587056696094592t^3 - 8883844612616520t^2 - 6315574244595696t + 5847617330112600 \rangle$
$i = 31, D_i = 1162261467, e_i = 2, \rho_{ij}: 1:2, 4:2, 10:2, 13:2, 18:1, 19:1, 22:2, 39:1, 40:2, 42:2, 43:1, 45:2, 47:1, 52:2, 55:1, 56:2, 60:2, 66:2, 69:1, 74:1, 85:2$	
w_i	$\langle 14202098550034261056t^3 - 25718513989392458496t^2 - 166028654350285772256t \\ -48512720679908758560, \\ -1031183642970985536t^3 - 90646136569060308000t^2 - 99383468116161417408t \\ +46492430982325473600 \rangle$
$i = 32, D_i = 1, \rho_{ij}: 0:2, 8:2, 11:1, 12:2, 20:1, 22:1, 26:2, 35:1, 46:2, 51:2, 60:1, 61:2, 62:1, 63:2$	
w_i	$\langle -876t^3 - 954t^2 + 788t - 1214, \\ -990t^3 - 1938t^2 - 758t - 394 \rangle$
$i = 33, D_i = 1, \rho_{ij}: 0:2, 2:1, 10:2, 15:2, 16:2, 18:2, 38:1, 39:1, 47:1, 50:2, 55:1, 56:1, 58:2, 59:1, 68:1, 69:2, 74:2, 75:2, 76:1, 79:2, 81:1$	
w_i	$\langle 5116t^3 - 15152t^2 - 19542t + 11446, \\ 8978t^3 + 10250t^2 - 21540t - 16384 \rangle$
$i = 34, D_i = 1, \rho_{ij}: 0:1, 2:2, 9:2, 10:1, 17:1, 19:1, 20:1, 21:1, 22:2, 25:1, 29:2, 40:2, 51:1, 55:2, 56:2, 69:1, 73:1, 84:2, 89:2, 94:1, 95:2$	
w_i	$\langle 172608t^3 + 165882t^2 - 395208t - 283878, \\ 150720t^3 - 254064t^2 - 404694t + 177198 \rangle$
$i = 35, D_i = 1, \rho_{ij}: 8:1, 13:1, 18:2, 19:2, 23:2, 32:2, 37:2, 46:1, 47:1, 48:1, 56:2$	
w_i	$\langle -888t^3 + 1044t^2 + 244t - 1104, \\ 1100t^3 - 2816t^2 + 2356t - 2048 \rangle$
$i = 36, D_i = 1, \rho_{ij}: 3:1, 9:1, 10:2, 12:1, 14:1, 21:1, 29:2, 48:2, 49:1, 57:2, 60:1, 64:2, 68:2, 75:1, 77:1, 78:2, 81:2, 83:2, 85:1, 95:2$	
w_i	$\langle 70t^2 + 1764t + 378, \\ 3794t^2 + 7959t + 7917 \rangle$
$i = 37, D_i = 1, \rho_{ij}: 3:2, 9:1, 21:2, 23:2, 24:1, 29:1, 35:1, 43:2, 59:1, 61:2, 68:2, 69:1, 70:2, 72:2, 75:2, 81:1, 90:1, 91:2, 93:1, 97:1$	
w_i	$\langle -1029t^2 - 1470t - 16317, \\ -1323t^2 + 4263t - 4410 \rangle$
$i = 38, D_i = 1, \rho_{ij}: 2:2, 12:2, 18:1, 24:1, 26:1, 33:2, 40:1, 47:2, 48:2, 63:2$	
w_i	$\langle 52t^3 + 122t^2 - 134t - 214, \\ -28t^3 + 178t^2 + 86t - 146 \rangle$
$i = 39, D_i = 1, \rho_{ij}: 1:1, 11:1, 13:1, 17:2, 25:2, 27:2, 30:2, 31:2, 33:2, 40:2, 46:2, 56:1, 57:1, 58:1, 61:1, 65:2, 68:1, 70:2, 71:1, 76:1, 77:2, 79:2, 84:1, 86:1, 96:2, 97:1$	
w_i	$\langle -451480176t^3 - 3231834066t^2 - 3565629144t - 1648435950, \\ \rangle$

$$\begin{aligned}
& -1035453384t^3 - 2969700372t^2 - 679372974t - 237311802) \\
i = 40, D_i = 1, \rho_{ij}: & 1:2, 6:1, 9:2, 12:1, 13:2, 16:1, 22:2, 23:1, 25:1, 26:1, 30:1, 31:1, 34:1, 38:2, 39:1, 44:2, 56:2, 60:2, 62:1, 63:2, \\
& 64:2, 71:2, 79:1, 88:1, 93:2, 94:2 \\
w_i & \langle -2876878980t^3 - 12739662720t^2 - 9461487654t - 4238434242, \\
& 554698548t^3 - 6323331654t^2 - 12992811876t - 6184754226 \rangle \\
i = 41, D_i = 1, \rho_{ij}: & 3:2, 4:2, 7:1, 14:1, 16:1, 17:2, 25:2, 30:2, 43:2, 52:2, 55:1, 64:1, 67:2, 70:1, 90:2 \\
w_i & \langle -68362t^3 + 273868t^2 - 703906t + 709744, \\
& 99610t^3 - 274708t^2 + 343882t - 113008 \rangle \\
i = 42, D_i = 1, \rho_{ij}: & 3:1, 4:1, 8:1, 12:2, 16:1, 17:2, 31:1, 48:1, 53:1, 55:2, 66:2, 72:2, 79:1, 85:1, 97:2 \\
w_i & \langle -24584t^3 + 75768t^2 - 41160t - 66976, \\
& 39956t^3 - 149184t^2 + 382788t - 367808 \rangle \\
i = 43, D_i = 1, \rho_{ij}: & 1:2, 3:2, 5:2, 7:2, 18:1, 20:1, 21:1, 25:1, 30:2, 31:2, 37:1, 41:1 \\
w_i & \langle -108t^3 - 138t^2 - 261t - 168, \\
& 6t^3 + 36t^2 - 24t + 153 \rangle \\
i = 44, D_i = 1, \rho_{ij}: & 0:2, 1:2, 6:2, 9:1, 21:2, 40:1, 55:2, 62:1, 64:2 \\
w_i & \langle -370t^3 + 382t^2 - 30t + 72, \\
& -156t^3 - 78t^2 + 116t + 248 \rangle \\
i = 45, D_i = 1, \rho_{ij}: & 12:2, 31:1 \\
w_i & \langle 2t + 30, \\
& 16t + 12 \rangle \\
i = 46, D_i = 1, \rho_{ij}: & 6:2, 23:2, 27:1, 32:1, 35:2, 39:1, 49:2, 53:1, 56:1 \\
w_i & \langle -12t^2 - 7t + 225, \\
& -26t^2 + 25t + 177 \rangle \\
i = 47, D_i = 1, \rho_{ij}: & 2:2, 9:1, 12:1, 13:2, 16:1, 18:1, 19:2, 25:1, 28:1, 31:2, 33:2, 35:2, 38:1, 56:1, 60:2, 61:2 \\
w_i & \langle 1012t^3 - 2240t^2 + 1444t - 792, \\
& 3464t^3 - 3808t^2 + 4448t - 6984 \rangle \\
i = 48, D_i = 1, \rho_{ij}: & 1:1, 6:1, 25:1, 29:2, 35:2, 36:1, 38:1, 42:2, 51:2, 56:2, 58:1 \\
w_i & \langle -98t^5 + 16t^4 + 876t^3 - 1522t^2 + 272t + 688, \\
& -62t^5 + 375t^4 - 117t^3 - 1403t^2 + 1359t + 108 \rangle \\
i = 49, D_i = 343, \rho_{ij}: & 4:2, 8:1, 19:2, 23:1, 27:2, 29:1, 36:2, 46:1, 54:1 \\
w_i & \langle -1715t^5 - 10633t^4 - 27097t^3 - 33271t^2 - 22638t - 13720, \\
& -686t^5 - 6860t^4 - 27783t^3 - 51107t^2 - 50764t - 31556 \rangle \\
i = 50, D_i = 16983563041, \rho_{ij}: & 4:1, 7:2, 12:2, 14:1, 17:2, 19:1, 33:1, 53:1, 59:1, 64:2, 65:2, 68:1, 76:1, 77:2, 78:1, 82:2, 90:2, \\
& 93:2, 96:2 \\
w_i & \langle 538412915525782t^5 - 53124585192248t^4 - 582841916441038t^3 + 558657322670654t^2 \\
& -439228907366342t + 153939015403624, \\
& 355703744330704t^5 + 384134228861338t^4 - 286716511258162t^3 + 82981689018326t^2 \\
& -63212821638602t + 218340686455096 \rangle \\
i = 51, D_i = 893871739, \rho_{ij}: & 4:2, 6:1, 9:2, 16:1, 18:2, 22:2, 23:1, 24:2, 32:1, 34:2, 48:1, 52:2, 57:1, 62:1, 66:1, 70:1, 72:2, 95:2, \\
& 97:1 \\
w_i & \langle 105344572184628t^5 + 229231619723072t^4 + 269570263560664t^3 + 28464451656716t^2 \\
& -209380516143360t - 11155519302720, \\
& 88618444204460t^5 + 227004091349484t^4 - 43420713593664t^3 - 307538359546428t^2 \\
& -69961553268052t - 166689201888720 \rangle \\
i = 52, D_i = 27, \rho_{ij}: & 3:1, 5:1, 17:2, 23:1, 31:1, 41:1, 51:1, 53:2, 62:2, 65:1, 67:2, 68:1, 75:2, 88:1, 93:2, 97:2 \\
w_i & \langle 14580t^5 - 90450t^4 + 2737692t^3 - 2543994t^2 - 12272364t + 15316128, \\
& 566028t^5 - 505062t^4 - 3070278t^3 + 4601394t^2 - 1785186t + 1662120 \rangle \\
i = 53, D_i = 27, \rho_{ij}: & 3:2, 5:2, 6:2, 8:1, 9:2, 16:1, 21:1, 30:2, 42:2, 46:2, 50:2, 52:1, 57:1, 64:1, 77:2, 96:1 \\
w_i & \langle -2118312t^5 + 1115748t^4 + 39658032t^3 - 43845516t^2 - 115582896t + 146512800, \\
& -5572800t^5 + 4507920t^4 + 47128932t^3 - 61277904t^2 - 55782324t + 75275568 \rangle \\
i = 54, D_i = 1, \rho_{ij}: & 2:2, 3:1, 5:1, 7:1, 28:1, 49:2 \\
w_i & \langle -7t^4 - 40t^3 - 23t^2 + 10t, \\
& t^4 - 35t^3 - 70t^2 - 34t \rangle \\
i = 55, D_i = 1, \rho_{ij}: & 6:2, 9:2, 15:1, 22:1, 25:2, 26:1, 27:2, 30:1, 31:2, 33:2, 34:1, 41:2, 42:1, 44:1, 57:2, 65:1, 68:1, 73:2, 76:2, \\
& 79:1, 86:2, 93:1 \\
w_i & \langle -133334109342t^3 - 203854927626t^2 - 75836734218t - 55970377848, \\
& -333302154192t^3 - 509586761376t^2 - 189573003648t - 139912069248 \rangle \\
i = 56, D_i = 1, \rho_{ij}: & 12:2, 16:2, 17:1, 23:1, 24:2, 29:2, 30:2, 31:1, 33:2, 34:1, 35:1, 39:2, 40:1, 46:2, 47:2, 48:1, 70:1, 77:2, 78:1, \\
& 81:1, 83:2, 88:1, 89:1, 90:2 \\
w_i & \langle 9751t^5 - 14119t^4 - 30940t^3 + 57806t^2 + 29176t - 60816, \\
& 11984t^5 - 20699t^4 - 27041t^3 + 76510t^2 + 21266t - 92484 \rangle \\
i = 57, D_i = 1, \rho_{ij}: & 1:1, 3:2, 6:2, 9:1, 22:2, 25:2, 26:2, 36:1, 39:2, 51:2, 53:2, 55:1, 60:1 \\
w_i & \langle -1900t^3 - 504t^2 + 9880t - 1956, \\
& -3620t^3 + 7896t^2 - 1336t - 2052 \rangle \\
i = 58, D_i = 1, \rho_{ij}: & 0:2, 6:1, 12:2, 16:1, 18:1, 25:2, 33:1, 39:2, 48:2 \\
w_i & \langle 5t^3 - 3t^2 - 6t, \\
& 3t^4 - t^3 - 6t + 8 \rangle \\
i = 59, D_i = 1, \rho_{ij}: & 1:1, 2:1, 6:2, 9:1, 13:2, 14:2, 15:1, 17:1, 23:1, 29:1, 33:2, 37:2, 50:2, 63:1
\end{aligned}$$

w_i	$\langle 21t^5 + 26t^4 - 77t^3 - 15t^2 + 91t - 32, \\ 60t^5 + 156t^4 - 160t^3 - 379t^2 + 101t + 134 \rangle$
$i = 60, D_i = 1, \rho_{ij}:$	2:2, 3:1, 11:2, 12:1, 13:2, 14:2, 17:2, 20:2, 21:1, 22:2, 23:1, 24:2, 25:1, 26:1, 31:1, 32:2, 36:2, 40:1, 47:1, 57:2, 62:1, 64:1, 68:2, 69:2, 70:2, 71:1, 72:2, 73:2, 76:1, 77:1, 79:2, 81:2, 82:2, 89:2, 97:1
w_i	$\langle -80077640t^5 - 134722572t^4 - 143498680t^3 - 369946700t^2 + 33266544t - 201061432, \\ 22506416t^5 - 208686396t^4 + 90255100t^3 - 509972128t^2 + 90519528t - 185321144 \rangle$
$i = 61, D_i = 1, \rho_{ij}:$	2:1, 3:2, 9:1, 10:1, 12:1, 13:2, 14:1, 15:1, 16:1, 22:2, 23:2, 25:1, 27:2, 29:1, 30:2, 32:1, 37:1, 39:2, 47:1, 63:1, 64:2, 70:2, 71:1, 72:1, 74:1, 75:2, 76:1, 77:2, 78:1, 79:2, 82:1, 89:2, 93:1, 96:2, 97:2
w_i	$\langle -14733576t^5 - 14168112t^4 - 20545176t^3 - 41528112t^2 + 17719536t - 26613168, \\ -34427040t^5 + 33250680t^4 - 84296616t^3 + 77567280t^2 - 30982272t + 22139616 \rangle$
$i = 62, D_i = 1, \rho_{ij}:$	2:2, 3:2, 6:2, 12:1, 13:1, 18:2, 19:1, 20:2, 22:1, 25:1, 27:1, 32:2, 40:2, 44:2, 51:2, 52:1, 60:2
w_i	$\langle -6005t^3 + 24324t^2 + 1732t - 30727, \\ -13180t^3 + 498t^2 + 30914t - 18806 \rangle$
$i = 63, D_i = 1, \rho_{ij}:$	1:2, 5:1, 9:2, 14:1, 19:2, 21:2, 23:1, 29:1, 32:1, 38:1, 40:1, 59:2, 61:2, 73:1, 82:1
w_i	$\langle 783t^3 + 978t^2 - 2445t - 522, \\ -264t^3 + 2706t^2 - 2061t - 1686 \rangle$
$i = 64, D_i = 1, \rho_{ij}:$	0:2, 1:2, 4:2, 10:2, 13:1, 16:1, 17:1, 18:1, 19:2, 22:2, 27:2, 36:1, 40:1, 41:2, 44:1, 50:1, 53:2, 60:2, 61:1
w_i	$\langle -71844852t^3 + 4479072t^2 + 10280484t - 27015960, \\ -67094100t^3 - 21096228t^2 + 31333728t + 31145616 \rangle$
$i = 65, D_i = 1, \rho_{ij}:$	0:2, 3:1, 16:2, 18:2, 39:1, 50:1, 52:2, 55:2, 66:1, 73:2, 74:2, 86:2, 88:2, 90:1, 93:2, 94:1, 95:2, 96:2
w_i	$\langle -1152t^3 + 3092t^2 - 3212t + 2256, \\ 744t^3 + 596t^2 - 1964t - 2192 \rangle$
$i = 66, D_i = 1, \rho_{ij}:$	17:1, 18:1, 31:1, 42:1, 51:2, 65:2, 67:2, 85:2, 92:1
w_i	$\langle -575t^3 + 21t^2 + 216t - 374, \\ -880t^3 - 966t^2 - 453t - 127 \rangle$
$i = 67, D_i = 1, \rho_{ij}:$	3:1, 17:2, 18:2, 19:1, 27:2, 41:1, 52:1, 66:1, 71:2, 75:1, 81:1, 82:2, 83:1, 91:1
w_i	$\langle 315t^3 + 1278t^2 - 759t - 59, \\ -189t^3 + 1419t^2 + 600t + 131 \rangle$
$i = 68, D_i = 1, \rho_{ij}:$	0:2, 1:1, 5:2, 11:2, 17:1, 19:2, 26:1, 33:2, 36:1, 37:1, 39:2, 50:2, 52:2, 55:2, 60:1, 71:2, 72:2, 75:1, 78:2, 86:1, 89:1, 93:1, 95:1
w_i	$\langle -399112t^3 + 652883t^2 - 231609t + 369278, \\ -228039t^3 - 46249t^2 + 369341t + 246127 \rangle$
$i = 69, D_i = 1, \rho_{ij}:$	5:1, 18:1, 30:2, 31:2, 33:1, 34:2, 37:2, 60:1, 71:1, 74:2, 78:2, 79:1, 97:2
w_i	$\langle 9702t^3 - 28812t^2 - 30135t - 14406, \\ -9555t^3 - 33075t^2 + 2940t - 11319 \rangle$
$i = 70, D_i = 1, \rho_{ij}:$	37:1, 39:1, 41:2, 51:2, 56:2, 60:1, 61:1, 71:2, 79:1, 82:2, 84:2, 87:2, 89:1, 94:2, 96:1
w_i	$\langle 20t^3 + 192t^2 + 204t + 280, \\ -28t^3 + 68t^2 + 112t - 112 \rangle$
$i = 71, D_i = 1, \rho_{ij}:$	30:2, 39:2, 40:1, 60:2, 61:2, 67:1, 68:1, 69:2, 70:1, 75:2, 77:1, 83:1, 89:2, 95:2, 96:1, 97:1
w_i	$\langle -243t^2 - 96t - 945, \\ -309t^2 + 69t \rangle$
$i = 72, D_i = 1, \rho_{ij}:$	1:2, 4:2, 11:1, 27:1, 37:1, 42:1, 51:1, 60:1, 61:2, 68:1, 95:2
w_i	$\langle -6921t^3 - 22344t^2 - 4599t + 10680, \\ -9630t^3 - 6078t^2 + 17820t + 6492 \rangle$
$i = 73, D_i = 1, \rho_{ij}:$	4:1, 16:1, 17:2, 19:1, 26:2, 34:2, 55:1, 60:1, 63:2, 65:1, 76:2, 95:1
w_i	$\langle 984t^3 - 5754t^2 - 18798t - 9120, \\ 4866t^3 + 9684t^2 - 4962t - 6528 \rangle$
$i = 74, D_i = 1, \rho_{ij}:$	3:2, 31:2, 33:1, 61:2, 65:1, 69:1, 76:2, 82:1, 85:1, 96:2
w_i	$\langle -3t^2 + 465t - 558, \\ 159t^2 - 303t + 1422 \rangle$
$i = 75, D_i = 1, \rho_{ij}:$	11:2, 16:1, 19:1, 33:1, 36:2, 37:1, 52:1, 61:1, 67:2, 68:2, 71:1, 79:1, 85:1, 86:2
w_i	$\langle 132t^3 + 2052t^2 - 2160t - 1392, \\ -656t^3 + 1580t^2 + 988t - 2312 \rangle$
$i = 76, D_i = 1, \rho_{ij}:$	3:1, 4:1, 17:2, 26:1, 30:2, 33:2, 39:2, 50:2, 55:1, 60:2, 61:2, 73:1, 74:1, 82:1, 93:2, 96:1, 97:2
w_i	$\langle 716511735t^3 + 1867989537t^2 + 1506927870t + 564664077, \\ 456117840t^3 + 1579222314t^2 + 1539256761t + 315844722 \rangle$
$i = 77, D_i = 1, \rho_{ij}:$	1:2, 2:2, 5:1, 11:1, 16:2, 18:1, 19:2, 36:2, 39:1, 50:1, 53:1, 56:1, 60:2, 61:1, 71:2, 79:2, 83:2, 88:2, 91:1, 95:1
w_i	$\langle 60480t^3 - 140252t^2 + 144452t - 91616, \\ 12908t^3 - 84924t^2 + 154616t - 82656 \rangle$
$i = 78, D_i = 1, \rho_{ij}:$	17:2, 19:1, 36:1, 50:2, 56:2, 61:2, 68:1, 69:1, 84:1
w_i	$\langle 266t^3 + 120t^2 - 522t + 30, \\ 112t^3 - 396t^2 - 404t + 342 \rangle$
$i = 79, D_i = 1, \rho_{ij}:$	33:1, 39:1, 40:2, 42:2, 55:2, 60:1, 61:1, 69:2, 70:2, 75:2, 77:1, 85:1, 89:1, 90:1, 93:2, 94:2, 96:1
w_i	$\langle -16187t^2 + 6314t + 29853, \\ -5027t^2 - 19978t + 95925 \rangle$
$i = 80, D_i = 1, \rho_{ij}:$	2:2, 27:1, 89:1, 92:1
w_i	$\langle -6t^2 - 2t + 4, \\ 4t^2 + 14t + 10 \rangle$

$i = 81, D_i = 1, \rho_{ij}: 1:2, 3:1, 11:1, 18:1, 33:2, 36:1, 37:2, 56:2, 60:1, 67:2, 90:2, 91:1, 95:1$
 $w_i \quad \langle -26t^3 + 64t^2 + 46t - 90, -42t^3 + 26t^2 + 54t - 144 \rangle$

$i = 82, D_i = 1, \rho_{ij}: 0:1, 11:1, 27:1, 50:1, 60:1, 61:2, 63:2, 67:1, 70:1, 74:2, 76:2, 83:2, 88:1, 96:2$
 $w_i \quad \langle -834t^3 - 764t^2 + 364t - 818, -54t^3 + 716t^2 + 1064t - 646 \rangle$

$i = 83, D_i = 1, \rho_{ij}: 18:1, 19:1, 36:1, 56:1, 67:2, 71:2, 77:1, 82:1, 88:2, 89:2, 90:2$
 $w_i \quad \langle 1288t^3 - 1620t^2 - 260t + 1648, 2828t^3 - 5540t^2 + 3168t - 1352 \rangle$

$i = 84, D_i = 1, \rho_{ij}: 2:1, 11:1, 19:2, 27:2, 34:1, 39:2, 70:1, 78:2, 89:1, 90:1$
 $w_i \quad \langle -52t^3 + 108t^2 + 64t - 76, 4t^3 + 60t^2 - 88t - 200 \rangle$

$i = 85, D_i = 1, \rho_{ij}: 1:1, 3:1, 5:1, 19:2, 30:1, 31:1, 36:2, 42:2, 66:1, 74:2, 75:2, 79:2$
 $w_i \quad \langle -15t^3 + 52t^2 + 50t + 115, -45t^3 - 127t^2 - 59t - 82 \rangle$

$i = 86, D_i = 1, \rho_{ij}: 0:1, 1:1, 39:2, 55:1, 65:1, 68:2, 75:1, 96:2, 97:1$
 $w_i \quad \langle 34t^3 - 136t^2 + 54t + 84, -108t^3 + 40t^2 + 50t + 84 \rangle$

$i = 87, D_i = 1, \rho_{ij}: 30:2, 70:1$
 $w_i \quad \langle -30t - 222, 96t - 156 \rangle$

$i = 88, D_i = 1, \rho_{ij}: 26:2, 40:2, 52:2, 56:2, 65:1, 77:1, 82:2, 83:1, 91:1$
 $w_i \quad \langle 44t^2 + 42t + 136, -29t^2 - 7t - 796 \rangle$

$i = 89, D_i = 1, \rho_{ij}: 2:1, 17:2, 18:1, 19:2, 30:1, 34:1, 56:2, 60:1, 61:1, 68:2, 70:2, 71:1, 79:2, 80:2, 83:1, 84:2$
 $w_i \quad \langle -876t^3 + 1908t^2 - 2096t + 1952, 1068t^3 - 240t^2 + 1076t - 2600 \rangle$

$i = 90, D_i = 1, \rho_{ij}: 1:2, 37:2, 41:1, 50:1, 56:1, 65:2, 79:2, 81:1, 83:1, 84:2, 94:2$
 $w_i \quad \langle -62t^5 + 375t^4 - 117t^3 - 1403t^2 + 1359t + 108, -98t^5 + 16t^4 + 876t^3 - 1522t^2 + 272t + 688 \rangle$

$i = 91, D_i = 343, \rho_{ij}: 4:1, 18:1, 26:1, 37:1, 67:2, 77:2, 81:2, 88:2, 92:2$
 $w_i \quad \langle 5488t^5 + 41846t^4 + 137543t^3 + 219863t^2 + 197568t + 122108, 7203t^5 + 39445t^4 + 79919t^3 + 64141t^2 + 11662t + 5488 \rangle$

$i = 92, D_i = 1, \rho_{ij}: 2:1, 3:2, 5:2, 66:2, 80:2, 91:1$
 $w_i \quad \langle -63t^5 - 75t^4 + 306t^3 + 318t^2, -15t^5 + 240t^4 + 537t^3 + 282t^2 \rangle$

$i = 93, D_i = 1, \rho_{ij}: 1:2, 3:1, 27:1, 37:2, 40:1, 50:1, 52:1, 55:2, 61:2, 65:1, 68:2, 76:1, 79:1$
 $w_i \quad \langle -1080t^3 + 822t^2 + 2862t - 876, 246t^3 - 2700t^2 + 4638t - 276 \rangle$

$i = 94, D_i = 1, \rho_{ij}: 0:1, 17:2, 19:2, 34:2, 40:1, 65:2, 70:1, 79:1, 90:1$
 $w_i \quad \langle 3t^4 - t^3 - 6t + 8, 5t^3 - 3t^2 - 6t \rangle$

$i = 95, D_i = 1, \rho_{ij}: 1:2, 2:2, 11:2, 16:2, 34:1, 36:1, 51:1, 65:1, 68:2, 71:1, 72:1, 73:2, 77:2, 81:2$
 $w_i \quad \langle 24t^5 - 214t^4 - 132t^3 + 602t^2 + 126t - 306, 12t^5 - 220t^4 - 12t^3 + 598t^2 - 66t - 232 \rangle$

$i = 96, D_i = 1, \rho_{ij}: 2:1, 3:1, 18:2, 19:1, 26:2, 39:1, 50:1, 53:2, 61:1, 65:1, 70:2, 71:2, 74:1, 76:2, 79:2, 82:1, 86:1$
 $w_i \quad \langle -291t^3 - 5379t^2 + 3602t + 4358, -2307t^3 - 2232t^2 + 6799t - 1286 \rangle$

$i = 97, D_i = 1, \rho_{ij}: 0:1, 1:1, 4:1, 16:2, 17:2, 18:1, 19:2, 26:1, 37:2, 39:2, 42:1, 51:2, 52:1, 60:2, 61:1, 69:1, 71:2, 76:1, 86:2$
 $w_i \quad \langle 20919708t^3 + 9733560t^2 - 12482784t - 16748916, 30816108t^3 + 1026408t^2 - 6943248t + 5014500 \rangle$

The second certificate.

$i = 0, D_i = 1, \rho_{ij}: 6:1, 12:1, 14:2, 15:1, 16:1, 17:2, 24:1, 39:2, 40:1, 41:1, 42:2, 44:1, 60:2, 61:1, 62:1, 65:2, 70:2, 72:1, 73:2, 78:2, 86:2, 96:2$
 $w_i \quad \langle -36t^2 - 972t - 324, -324t^2 - 108t - 36 \rangle$

$i = 1, D_i = 1, \rho_{ij}: 6:1, 8:1, 9:2, 14:2, 23:2, 26:1, 27:2, 30:2, 31:1, 33:1, 34:2, 39:1, 40:2, 44:2, 47:2, 60:1, 61:2, 65:2, 67:2, 68:1, 72:1, 77:1, 86:1, 89:1$
 $w_i \quad \langle -2496410t^2 - 7587230t - 3753960, -2804830t^2 - 7549150t - 2641800 \rangle$

$i = 2, D_i = 1, \rho_{ij}: 6:2, 10:1, 21:1, 23:2, 26:2, 27:1, 38:1, 39:1, 40:2, 47:2, 57:1, 59:1, 60:2, 61:1, 65:1, 69:2, 75:2, 77:1, 84:2, 89:1, 93:2, 95:2$
 $w_i \quad \langle -43t^2 + 793t - 1245, 429t^2 - 815t - 21 \rangle$

$i = 3, D_i = 1, \rho_{ij}: 8:2, 18:1, 19:2, 20:2, 23:2, 29:2, 36:1, 37:2, 57:1, 67:1, 74:1, 77:1, 81:1, 93:2$
 $w_i \quad \langle 22t^3 + 39t^2 - 17t + 10, 17t^3 + 66t^2 - 3t + 38 \rangle$

$i = 4, D_i = 1, \rho_{ij}: 26:1, 27:2, 30:2, 31:1, 39:1, 40:2, 43:1, 45:2, 47:1, 59:2, 85:2, 87:1, 89:2, 95:1$

w_i	$\langle 490t^2 - 2527t + 4956,$ $1043t^2 - 3479t + 7980 \rangle$
$i = 5, D_i = 2, \rho_{ij}:$	6:2, 9:1, 10:2, 22:1, 23:1, 25:1, 26:1, 27:2, 36:2, 37:1, 41:1, 42:2, 43:1, 45:2, 57:2, 64:2, 65:1, 68:2, 69:1, 76:2, 77:2, 79:2, 85:2, 87:1, 93:1, 97:1
w_i	$\langle 486t^3 - 4968t^2 + 792t - 4176,$ $237t^3 - 4056t^2 + 114t + 3408 \rangle$
$i = 6, D_i = 1, \rho_{ij}:$	0:2, 1:2, 2:1, 5:1, 8:1, 18:2, 19:2, 20:1, 32:2, 43:1, 44:1, 47:2, 55:1, 60:2, 63:2
w_i	$\langle -52t^3 - 312t^2 + 240t + 244,$ $452t^3 - 824t^2 + 476t - 560 \rangle$
$i = 7, D_i = 1, \rho_{ij}:$	--
w_i	$\langle 1,$ $0 \rangle$
$i = 8, D_i = 1, \rho_{ij}:$	1:2, 3:1, 6:2, 9:1, 13:1, 16:2, 18:2, 25:2, 26:1, 29:1, 36:2, 38:2, 42:1, 60:2
w_i	$\langle -13t^3 - 300t^2 + 33t - 136,$ $-104t^3 - 173t^2 + 210t - 141 \rangle$
$i = 9, D_i = 1, \rho_{ij}:$	1:1, 5:2, 8:2, 10:2, 13:2, 14:1, 16:1, 17:2, 19:1, 24:2, 27:1, 32:1, 36:1, 37:1, 40:2, 45:2, 55:2, 59:2, 60:2, 61:1
w_i	$\langle 40257t^3 - 121030t^2 + 96082t - 31213,$ $-23730t^3 - 30968t^2 + 77252t + 31066 \rangle$
$i = 10, D_i = 1, \rho_{ij}:$	2:2, 5:1, 9:1, 13:1, 16:1, 20:2, 25:1, 30:1, 33:2, 34:2, 37:1, 41:2, 45:2, 59:2, 61:1, 64:2
w_i	$\langle 6717t^3 + 22153t^2 - 3762t + 7255,$ $-7764t^3 + 18764t^2 + 20103t + 10592 \rangle$
$i = 11, D_i = 1, \rho_{ij}:$	13:1, 15:1, 18:2, 19:1, 21:2, 22:2, 25:1, 26:1, 32:2, 39:2, 45:1, 59:1, 60:1, 61:2, 62:2, 63:1, 65:2, 72:1, 74:2, 95:1, 97:2
w_i	$\langle 6924t^3 + 50722t^2 - 67208t - 26762,$ $26694t^3 - 6454t^2 - 47596t + 4946 \rangle$
$i = 12, D_i = 1, \rho_{ij}:$	0:2, 13:2, 20:1, 24:2, 33:2, 36:2, 37:2, 40:1, 42:1
w_i	$\langle -t^2 - 4t + 5,$ $t^2 - 6t - 3 \rangle$
$i = 13, D_i = 1, \rho_{ij}:$	8:2, 9:1, 10:2, 11:2, 12:1, 16:1, 22:1, 34:1, 40:2, 47:2, 53:2, 55:1, 56:2, 59:2, 60:1, 64:1
w_i	$\langle -480t^2 - 1006t - 1330,$ $-125t^2 - 522t - 1169 \rangle$
$i = 14, D_i = 1, \rho_{ij}:$	0:1, 1:1, 9:2, 16:2, 19:1, 21:1, 27:1, 32:1, 55:2, 59:1, 60:1, 61:1, 63:1
w_i	$\langle 1493t^3 - 2248t^2 - 5909t + 88,$ $2428t^3 + 3460t^2 - 2728t - 2272 \rangle$
$i = 15, D_i = 1, \rho_{ij}:$	0:2, 11:2, 17:2, 22:2, 27:1, 33:2, 34:1, 37:2, 41:1, 42:1, 45:2, 55:2, 59:1, 61:1
w_i	$\langle 40320t^3 + 772320t^2 + 1540140t + 636840,$ $372150t^3 + 1213380t^2 + 735510t - 4800 \rangle$
$i = 16, D_i = 1, \rho_{ij}:$	0:2, 8:1, 9:2, 10:2, 13:2, 14:1, 19:2, 24:1, 25:2, 26:2, 30:1, 33:2, 36:2, 37:2, 38:2, 40:2, 41:1, 43:1, 44:2, 57:1, 59:1, 60:2, 68:2, 73:2, 76:2, 78:2
w_i	$\langle 367200t^3 - 654600t^2 + 976680t - 1313448,$ $197856t^3 - 171816t^2 - 75816t + 270456 \rangle$
$i = 17, D_i = 1, \rho_{ij}:$	0:1, 9:1, 15:1, 18:1, 22:1, 24:1, 27:1, 31:2, 34:1, 36:1, 37:1, 39:1, 42:2, 61:1, 67:2, 68:1, 69:1, 71:1, 72:2, 78:2, 79:1, 84:1, 85:2, 86:1, 93:2, 95:2
w_i	$\langle -7980t^3 + 11868t^2 - 3612t + 17760,$ $-12144t^3 + 22872t^2 - 41568t + 60432 \rangle$
$i = 18, D_i = 1, \rho_{ij}:$	3:2, 6:1, 8:1, 11:1, 17:2, 21:2, 23:1, 30:2, 31:2, 38:1, 39:2, 41:2, 43:1, 53:2, 56:2, 60:2, 61:1, 62:1, 63:1, 65:2, 68:1, 72:1, 81:1, 86:1, 87:1, 89:1, 96:1
w_i	$\langle 189840t^2 - 1128960t + 1268400,$ $263760t^2 - 1058400t + 853440 \rangle$
$i = 19, D_i = 1, \rho_{ij}:$	3:1, 6:1, 9:2, 11:2, 14:2, 16:1, 29:2, 30:1, 31:1, 40:1, 42:1, 44:2, 45:2, 47:2, 52:1, 56:1, 60:2, 61:1, 62:2, 63:2, 65:2, 67:2, 75:1, 77:2, 84:2, 85:2, 96:2
w_i	$\langle -3312960t^2 + 2251200t - 11309760,$ $-4734240t^2 + 6904800t - 11131680 \rangle$
$i = 20, D_i = 1, \rho_{ij}:$	3:1, 6:2, 10:1, 12:2, 22:1, 30:1, 32:2, 55:2, 60:2, 63:2
w_i	$\langle -165t^2 + 231t + 2268,$ $-39t^2 + 21t + 1494 \rangle$
$i = 21, D_i = 1, \rho_{ij}:$	2:2, 11:1, 14:2, 18:1, 36:2, 39:1, 41:2, 42:1, 43:1, 45:2, 59:1, 60:1, 61:2
w_i	$\langle 224t^3 - 794t^2 - 34t + 1220,$ $-98t^3 - 814t^2 + 1368t + 384 \rangle$
$i = 22, D_i = 1, \rho_{ij}:$	5:2, 11:1, 13:2, 15:1, 17:2, 20:2, 24:1, 31:2, 32:2, 39:1, 41:1, 57:2, 59:1, 61:2, 62:2
w_i	$\langle -7011540t^3 - 15380790t^2 - 10436670t - 5848680,$ $1543680t^3 + 9686580t^2 + 11663370t + 584400 \rangle$
$i = 23, D_i = 1, \rho_{ij}:$	1:1, 2:1, 3:1, 5:2, 18:2, 25:2, 27:2, 31:2, 35:2, 40:2, 43:2, 47:2, 60:1
w_i	$\langle 25152t^3 - 44052t^2 + 27624t - 21252,$ $6240t^3 + 11316t^2 - 42876t + 19824 \rangle$
$i = 24, D_i = 1, \rho_{ij}:$	0:2, 9:1, 12:1, 16:2, 17:2, 22:2, 37:1, 44:1, 56:2, 62:2
w_i	$\langle -210t^3 - 770t^2 - 630t + 700,$ $280t^3 - 350t^2 - 910t + 70 \rangle$

$i = 25, D_i = 1, \rho_{ij}: 5:2, 8:1, 10:2, 11:2, 16:1, 23:1, 39:1, 40:1, 41:2, 55:1, 57:2, 60:2, 61:2, 62:2$
 $w_i \quad \langle -277t^2 + 436t - 631, 723t^2 + 482t - 941 \rangle$

$i = 26, D_i = 1, \rho_{ij}: 1:2, 2:1, 4:2, 5:2, 8:2, 11:2, 16:1, 29:1, 30:1, 32:1, 33:2, 38:1, 40:1, 45:2, 55:2, 64:2, 68:1, 72:1, 73:1, 77:2, 85:1, 87:2$
 $w_i \quad \langle -2718940t^3 + 3866422t^2 + 19047070t + 9282392, 3302614t^3 + 14642474t^2 + 18793544t + 2601844 \rangle$

$i = 27, D_i = 1, \rho_{ij}: 1:1, 2:2, 4:1, 5:1, 9:2, 14:2, 15:2, 17:2, 23:1, 31:2, 34:1, 39:2, 43:2, 45:1, 55:1, 63:1, 67:1, 81:2, 82:2, 84:2, 87:1, 97:1$
 $w_i \quad \langle -487746t^3 + 40714296t^2 + 90214194t + 30444876, -21923874t^3 - 21192402t^2 + 39885804t + 37151604 \rangle$

$i = 28, D_i = 1, \rho_{ij}: --$
 $w_i \quad \langle 1, 0 \rangle$

$i = 29, D_i = 1, \rho_{ij}: 3:1, 8:2, 19:1, 26:2, 37:1, 38:2, 56:2$
 $w_i \quad \langle 6t^3 - 2t^2 - 6t + 16, 4t^3 - 4t^2 - 10t + 24 \rangle$

$i = 30, D_i = 1594323, e_i = 2, \rho_{ij}: 1:1, 4:1, 10:2, 16:2, 18:1, 19:2, 20:2, 26:2, 32:1, 39:1, 43:2, 61:1, 76:2, 77:2, 85:1, 87:1, 89:2$
 $w_i \quad \langle 5903736166080t^3 + 45354335274720t^2 - 4337435634480t - 42550647289200, -4304916694080t^3 + 37634200605360t^2 + 67700006200080t - 9818158586400 \rangle$

$i = 31, D_i = 531441, e_i = 1, \rho_{ij}: 1:2, 4:2, 17:1, 18:1, 19:2, 22:1, 23:1, 27:1, 40:2, 43:2, 45:2, 47:1, 60:2, 69:1, 74:1, 82:2, 85:1$
 $w_i \quad \langle 13715568362880t^3 + 63494035947000t^2 - 44564145784920t - 80715513938400, 13535500084920t^3 - 16520799378960t^2 - 143892297658440t - 49360009716000 \rangle$

$i = 32, D_i = 1, \rho_{ij}: 6:1, 9:2, 11:1, 14:2, 20:1, 22:1, 26:2, 30:2, 35:1, 55:2, 60:1, 61:2$
 $w_i \quad \langle 138t^3 + 146t^2 - 74t + 90, 36t^3 + 256t^2 + 2t + 18 \rangle$

$i = 33, D_i = 1, \rho_{ij}: 1:2, 10:1, 12:1, 15:1, 16:1, 26:1, 38:2, 41:2, 55:2, 59:2, 61:1, 69:2, 71:1, 73:1, 93:1$
 $w_i \quad \langle 1050t^3 - 900t^2 - 2850t + 180, 1020t^3 + 1500t^2 - 1650t - 2040 \rangle$

$i = 34, D_i = 1, \rho_{ij}: 1:1, 10:1, 13:2, 15:2, 17:2, 27:2, 42:1, 55:1, 57:2, 60:2, 69:2, 70:2, 73:2, 84:1, 95:1$
 $w_i \quad \langle 5130t^3 + 6270t^2 - 10530t - 10410, 4650t^3 - 7110t^2 - 14400t + 2280 \rangle$

$i = 35, D_i = 1, \rho_{ij}: 23:1, 32:2$
 $w_i \quad \langle 2t, 2 \rangle$

$i = 36, D_i = 1, \rho_{ij}: 3:2, 5:1, 8:1, 9:2, 12:1, 16:1, 17:2, 21:1, 47:2, 68:1, 69:1, 70:2, 73:2, 78:1, 81:1, 85:1, 86:1$
 $w_i \quad \langle -36t^2 + 39t + 95, 30t^2 + 153t + 223 \rangle$

$i = 37, D_i = 1, \rho_{ij}: 3:1, 5:2, 9:2, 10:2, 12:1, 15:1, 16:1, 17:2, 24:2, 29:2, 43:2, 44:2, 67:2, 68:1, 70:2, 75:2, 89:1$
 $w_i \quad \langle -155t^2 - 773t - 906, 95t^2 + 194t - 9 \rangle$

$i = 38, D_i = 1, \rho_{ij}: 2:2, 8:1, 16:1, 18:2, 26:2, 29:1, 33:1, 40:2, 61:1$
 $w_i \quad \langle 2t^3 + t^2 - 26t - 19, -4t^3 + 5t^2 - 4t - 18 \rangle$

$i = 39, D_i = 1, \rho_{ij}: 0:1, 1:2, 2:2, 4:2, 11:1, 17:2, 18:1, 21:2, 22:2, 25:2, 27:1, 30:2, 40:2, 41:1, 45:2, 47:2, 57:2, 61:1, 68:2, 70:1, 71:2, 77:2, 79:1, 84:2, 95:1$
 $w_i \quad \langle -262265760t^3 - 3221349360t^2 - 4339412820t - 2029388340, -1143199440t^3 - 3864658620t^2 - 1738792620t - 729224160 \rangle$

$i = 40, D_i = 1, \rho_{ij}: 0:2, 1:1, 2:1, 4:1, 9:1, 12:2, 13:1, 16:1, 19:2, 23:1, 25:2, 26:2, 31:1, 38:1, 39:1, 42:2, 59:2, 60:2, 63:2, 75:1, 76:1, 79:1, 87:1, 89:1, 93:1$
 $w_i \quad \langle -38950500t^3 - 10199400t^2 + 145900500t + 72093000, -57929700t^3 - 181808700t^2 - 64389600t - 25792200 \rangle$

$i = 41, D_i = 1, \rho_{ij}: 0:2, 5:2, 10:1, 15:2, 16:2, 18:1, 21:1, 22:2, 25:1, 33:1, 39:2, 42:1, 45:1, 55:1, 64:2, 67:1, 70:1, 73:1, 75:1$
 $w_i \quad \langle -547120t^3 + 1643040t^2 - 2431380t + 1412320, -534170t^3 + 1929060t^2 - 3829770t + 3082520 \rangle$

$i = 42, D_i = 1, \rho_{ij}: 0:1, 5:1, 8:2, 12:2, 15:2, 17:1, 19:2, 21:2, 34:2, 40:1, 41:2, 55:2, 69:2, 73:1, 75:2, 76:1, 79:2, 87:2, 97:1$
 $w_i \quad \langle -151145400t^3 + 550091640t^2 - 1101324000t + 896311920, -164281320t^3 + 494693220t^2 - 749191380t + 454712160 \rangle$

$i = 43, D_i = 1, \rho_{ij}: 4:2, 5:2, 6:2, 16:2, 18:2, 21:2, 23:1, 27:1, 30:1, 31:1, 37:1, 45:2$
 $w_i \quad \langle -283t^3 - 554t^2 - 479t - 430, 142t^3 + 152t^2 + 401t + 496 \rangle$

$i = 44, D_i = 1, \rho_{ij}: 0:2, 1:1, 6:2, 16:1, 19:1, 24:2, 37:1$
 $w_i \quad \langle -35t^3 + 35t^2, -35t^2 + 35t \rangle$

$i = 45, D_i = 1, \rho_{ij}: 4:1, 5:1, 9:1, 10:1, 11:2, 15:1, 19:1, 21:1, 26:1, 27:2, 31:1, 39:1, 41:2, 43:1, 47:2, 52:2$
 $w_i \quad \langle 700420t^3 + 54034475t^2 + 37694260t - 21083295, 14208935t^3 + 43526110t^2 - 31644835t - 21270630 \rangle$

$i = 46, D_i = 1, \rho_{ij}: --$

w_i	$\langle 1, 0 \rangle$
$i = 47, D_i = 1, \rho_{ij}: 1:1, 2:1, 4:2, 6:1, 13:1, 19:1, 23:1, 31:2, 36:1, 39:1, 45:1, 61:1, 62:1, 64:1$	
w_i	$\langle 6t^3 - 46t^2 - 16t - 16, -6t^3 - 44t^2 + 22t + 16 \rangle$
$i = 48, D_i = 1, \rho_{ij}: --$	
w_i	$\langle 1, 0 \rangle$
$i = 49, D_i = 1, \rho_{ij}: --$	
w_i	$\langle 1, 0 \rangle$
$i = 50, D_i = 1, \rho_{ij}: --$	
w_i	$\langle 1, 0 \rangle$
$i = 51, D_i = 1, \rho_{ij}: --$	
w_i	$\langle 1, 0 \rangle$
$i = 52, D_i = 1, \rho_{ij}: 19:2, 45:1, 71:2$	
w_i	$\langle -t^2 + 2t + 1, -t^2 - t + 2 \rangle$
$i = 53, D_i = 1, \rho_{ij}: 13:1, 18:1, 87:2$	
w_i	$\langle -t^2 - t + 2, -t^2 + 2t + 1 \rangle$
$i = 54, D_i = 1, \rho_{ij}: --$	
w_i	$\langle 1, 0 \rangle$
$i = 55, D_i = 1, \rho_{ij}: 6:2, 9:1, 13:2, 14:1, 15:1, 20:1, 25:2, 26:1, 27:2, 32:1, 33:1, 34:2, 41:2, 42:1, 57:1, 65:1, 68:2, 71:1, 72:2, 73:2, 74:2, 79:1, 82:2, 93:2$	
w_i	$\langle 1981002090270t^3 + 3028760704410t^2 + 1126738950330t + 831576099480, 396200418054t^3 + 605752140882t^2 + 225347790066t + 166315219896 \rangle$
$i = 56, D_i = 1, \rho_{ij}: 13:1, 18:1, 19:2, 24:1, 29:1, 62:1, 71:2, 78:2, 81:2, 96:2$	
w_i	$\langle -6t^5 + 13t^4 + 13t^3 - 30t^2 - 12t + 38, 0 \rangle$
$i = 57, D_i = 1, \rho_{ij}: 2:2, 3:2, 5:1, 16:2, 22:1, 25:1, 34:1, 39:1, 55:2, 59:1$	
w_i	$\langle 10t^3 - 52t^2 + 26t - 20, -34t^3 - 8t^2 + 4t - 16 \rangle$
$i = 58, D_i = 1, \rho_{ij}: --$	
w_i	$\langle 1, 0 \rangle$
$i = 59, D_i = 31, \rho_{ij}: 2:2, 4:1, 9:1, 10:1, 11:2, 13:1, 14:2, 15:2, 16:2, 21:2, 22:2, 33:1, 40:1, 57:2, 61:2, 62:1, 63:1$	
w_i	$\langle -14100t^5 - 55140t^4 + 38136t^3 + 134396t^2 - 22476t - 43024, -52980t^5 - 103052t^4 + 118208t^3 + 274044t^2 - 21892t - 125464 \rangle$
$i = 60, D_i = 43, \rho_{ij}: 0:1, 1:2, 2:1, 6:1, 8:1, 9:1, 11:2, 13:2, 14:2, 16:1, 18:1, 19:1, 20:1, 21:2, 23:2, 25:1, 31:1, 32:2, 34:1, 40:1, 62:1, 63:2, 68:1, 69:1, 72:1, 73:1, 75:2, 76:2, 79:2, 82:2, 84:1, 89:1, 95:2, 96:1, 97:2$	
w_i	$\langle 83223160t^5 + 39281928t^4 + 200320472t^3 + 99254320t^2 + 68166264t + 40712192, 2424944t^5 + 36813192t^4 - 6058088t^3 + 74320856t^2 - 23654976t + 1472536 \rangle$
$i = 61, D_i = 43, \rho_{ij}: 0:2, 1:1, 2:2, 9:2, 10:2, 11:1, 14:2, 15:2, 17:2, 18:2, 19:2, 21:1, 22:1, 25:1, 30:2, 32:1, 33:2, 38:2, 39:2, 47:2, 59:1, 62:2, 63:1, 64:1, 65:2, 67:2, 68:2, 71:1, 72:1, 74:2, 75:1, 77:1, 79:2, 82:1, 96:2$	
w_i	$\langle 1226168304t^5 - 830648232t^4 + 3069019296t^3 - 1808549040t^2 + 1228851408t - 307294728, 348055488t^5 - 1404739704t^4 + 1291718640t^3 - 3427636296t^2 + 1196230848t - 1244338608 \rangle$
$i = 62, D_i = 1, \rho_{ij}: 0:2, 11:1, 18:2, 19:1, 22:1, 24:1, 25:1, 47:2, 56:2, 59:2, 60:2, 61:1$	
w_i	$\langle 295t^3 - 353t^2 - 564t + 824, 290t^3 + 411t^2 - 1013t + 148 \rangle$
$i = 63, D_i = 1, \rho_{ij}: 6:1, 11:2, 14:2, 18:2, 19:1, 20:1, 27:2, 40:1, 59:2, 60:1, 61:2, 64:1, 71:2, 73:2, 75:1, 76:1, 79:2, 82:1, 87:2, 95:2, 96:1$	
w_i	$\langle 9354t^3 - 59344t^2 + 41860t + 34098, 26064t^3 - 24854t^2 - 26614t + 15762 \rangle$
$i = 64, D_i = 1, \rho_{ij}: 5:1, 10:1, 13:2, 26:1, 41:1, 47:2, 61:2, 63:2$	
w_i	$\langle 47t^3 + 18t^2 - 23t - 24, -17t^3 + 27t^2 - 10t - 36 \rangle$
$i = 65, D_i = 1, \rho_{ij}: 0:1, 1:1, 2:2, 5:2, 11:1, 18:1, 19:1, 55:2, 61:1, 67:2, 74:2, 82:1, 85:2, 86:2, 89:1$	
w_i	$\langle -22t^3 - 136t^2 + 54t + 8, -118t^3 + 84t^2 - 14t + 160 \rangle$
$i = 66, D_i = 1, \rho_{ij}: --$	
w_i	$\langle 1, 0 \rangle$
$i = 67, D_i = 1, \rho_{ij}: 1:1, 3:2, 17:1, 19:1, 27:2, 37:1, 41:2, 61:1, 65:1, 68:2, 71:2, 79:1, 81:2, 84:1$	
w_i	$\langle -61t^3 + 203t^2 + 133t + 89, 0 \rangle$

	$-113t^3 - 42t^2 + 147t + 15)$
$i = 68, D_i = 1, \rho_{ij}: 1:2, 5:1, 16:1, 17:2, 18:2, 26:2, 36:2, 37:2, 39:1, 55:1, 60:2, 61:1, 67:1, 69:1, 71:1, 72:2, 78:1, 82:2, 87:1, 95:1$	
w_i	$\langle -85253t^3 + 114807t^2 - 18816t + 80206,$ $-32746t^3 - 48510t^2 + 100401t + 36323 \rangle$
$i = 69, D_i = 1, \rho_{ij}: 2:1, 5:2, 17:2, 31:2, 33:1, 34:1, 36:2, 42:1, 60:2, 68:2, 71:2, 74:1, 79:2, 87:1, 95:1, 97:1$	
w_i	$\langle -98763t^3 - 58968t^2 + 146013t + 1113,$ $-71526t^3 + 109158t^2 + 161091t + 67431 \rangle$
$i = 70, D_i = 1, \rho_{ij}: 0:1, 34:1, 36:1, 37:1, 39:2, 41:2, 71:1, 74:2, 78:1$	
w_i	$\langle 4t^3 - 5t^2 - t + 4,$ $-2t^3 - 8t^2 + t + 5 \rangle$
$i = 71, D_i = 1, \rho_{ij}: 17:2, 33:2, 39:1, 52:1, 55:2, 56:1, 61:2, 63:1, 67:1, 68:2, 69:1, 70:2, 76:2, 89:1, 95:1, 97:2$	
w_i	$\langle 244t^2 + 286t + 994,$ $242t^2 + 361t + 917 \rangle$
$i = 72, D_i = 1, \rho_{ij}: 0:2, 1:2, 11:2, 17:1, 18:2, 26:2, 55:1, 60:2, 61:2, 68:1, 75:2, 82:2, 95:2$	
w_i	$\langle -1861722t^3 + 2087872t^2 + 6653610t + 149216,$ $870222t^3 + 6388292t^2 + 4047978t - 2564156 \rangle$
$i = 73, D_i = 1, \rho_{ij}: 0:1, 16:1, 26:2, 33:2, 34:1, 36:1, 41:2, 42:2, 55:1, 60:2, 63:1, 76:1, 87:1, 95:2$	
w_i	$\langle -440040t^3 - 18780t^2 + 2354700t + 1414320,$ $-795780t^3 - 2068980t^2 - 418320t + 505680 \rangle$
$i = 74, D_i = 1, \rho_{ij}: 3:2, 11:1, 31:2, 55:1, 61:1, 65:1, 69:2, 70:1, 76:2, 82:1$	
w_i	$\langle -75t^2 + 516t - 1899,$ $-30t^2 + 177t - 855 \rangle$
$i = 75, D_i = 1, \rho_{ij}: 2:1, 19:2, 37:1, 40:2, 41:2, 42:1, 60:1, 61:2, 63:2, 72:1, 85:2, 87:1, 95:2$	
w_i	$\langle 1192t^3 - 2364t^2 - 1656t + 3468,$ $1144t^3 + 1060t^2 - 5004t + 1408 \rangle$
$i = 76, D_i = 1, \rho_{ij}: 5:1, 16:1, 30:1, 40:2, 42:2, 60:1, 63:2, 71:1, 73:2, 74:1, 78:2, 82:1, 93:1, 95:2, 96:1$	
w_i	$\langle 7183500t^3 + 20619300t^2 + 17918400t + 5451900,$ $1905600t^3 + 9565800t^2 + 10841100t + 985800 \rangle$
$i = 77, D_i = 1, \rho_{ij}: 1:2, 2:2, 3:2, 5:1, 19:1, 26:1, 30:1, 39:1, 61:2, 79:1, 83:1, 85:1, 89:1$	
w_i	$\langle 4116t^3 - 13464t^2 + 18936t - 11100,$ $-2940t^3 - 624t^2 + 9336t - 3924 \rangle$
$i = 78, D_i = 1, \rho_{ij}: 0:1, 16:1, 17:1, 36:2, 56:1, 68:2, 70:2, 76:1, 86:2, 96:1$	
w_i	$\langle -1190t^3 - 1610t^2 - 70t + 1960,$ $-1050t^3 + 280t^2 + 2100t + 490 \rangle$
$i = 79, D_i = 1, \rho_{ij}: 5:1, 17:2, 39:2, 40:2, 42:1, 55:2, 60:1, 61:1, 63:1, 67:2, 69:1, 77:2, 93:1, 96:1$	
w_i	$\langle 703t^2 + 2154t + 3309,$ $323t^2 + 1481t - 584 \rangle$
$i = 80, D_i = 1, \rho_{ij}: --$	
w_i	$\langle 1,$ $0 \rangle$
$i = 81, D_i = 1, \rho_{ij}: 3:2, 18:2, 27:1, 36:2, 56:1, 67:1, 84:1$	
w_i	$\langle 12t^2 - 4t + 4,$ $6t^3 - 4t + 12 \rangle$
$i = 82, D_i = 1, \rho_{ij}: 27:1, 31:1, 55:1, 60:1, 61:2, 63:2, 65:2, 68:1, 72:1, 74:2, 76:2, 83:2$	
w_i	$\langle 310t^3 + 4t^2 - 338t + 348,$ $-88t^3 - 610t^2 - 346t + 144 \rangle$
$i = 83, D_i = 1, \rho_{ij}: 77:2, 82:1$	
w_i	$\langle -2t + 2,$ $2 \rangle$
$i = 84, D_i = 1, \rho_{ij}: 2:1, 17:2, 19:1, 27:1, 34:2, 39:1, 60:2, 67:2, 81:2$	
w_i	$\langle -11t^3 + 27t^2 + 132t - 32,$ $2t^3 + 53t^2 + 25t - 110 \rangle$
$i = 85, D_i = 1, \rho_{ij}: 4:1, 5:1, 17:1, 19:1, 26:2, 30:2, 31:2, 36:2, 65:1, 75:1, 77:2, 87:1$	
w_i	$\langle -579t^3 - 975t^2 - 1152t - 1305,$ $-255t^3 - 351t^2 - 645t - 747 \rangle$
$i = 86, D_i = 1, \rho_{ij}: 0:1, 1:2, 17:2, 18:2, 36:2, 65:1, 78:1$	
w_i	$\langle -350t^3 + 875t^2 - 525t,$ $175t^3 + 175t^2 - 350t \rangle$
$i = 87, D_i = 1, \rho_{ij}: 4:2, 5:2, 18:2, 26:1, 27:2, 30:2, 40:2, 42:1, 53:1, 63:1, 68:2, 69:2, 73:2, 75:2, 85:2, 89:1$	
w_i	$\langle -3111998400t^3 - 212159184675t^2 - 146343249600t + 82917441675,$ $52724348325t^3 - 42476876550t^2 - 271788565875t - 163241700 \rangle$
$i = 88, D_i = 1, \rho_{ij}: --$	
w_i	$\langle 1,$ $0 \rangle$
$i = 89, D_i = 1, \rho_{ij}: 1:2, 2:2, 4:1, 18:2, 30:1, 37:2, 40:2, 60:2, 65:2, 71:2, 77:2, 87:2, 96:2, 97:2$	
w_i	$\langle -308t^3 + 852t^2 - 536t + 656,$ $56t^3 + 120t^2 + 296t - 608 \rangle$
$i = 90, D_i = 1, \rho_{ij}: --$	

w_i	$\langle 1,$ $0 \rangle$
$i = 91, D_i = 1, \rho_{ij}:$	---
w_i	$\langle 1,$ $0 \rangle$
$i = 92, D_i = 1, \rho_{ij}:$	---
w_i	$\langle 1,$ $0 \rangle$
$i = 93, D_i = 1, \rho_{ij}:$	2:1, 3:1, 5:2, 17:1, 33:2, 40:2, 55:1, 76:2, 79:2, 95:2
w_i	$\langle 122t^3 - 796t^2 + 866t + 60,$ $352t^3 - 764t^2 + 190t + 96 \rangle$
$i = 94, D_i = 1, \rho_{ij}:$	---
w_i	$\langle 1,$ $0 \rangle$
$i = 95, D_i = 1, \rho_{ij}:$	2:1, 4:2, 11:2, 17:1, 34:2, 39:2, 60:1, 63:1, 68:2, 69:2, 71:2, 72:1, 73:1, 75:1, 76:1, 93:1, 96:2
w_i	$\langle 376t^5 + 1400t^4 - 168t^3 - 3844t^2 - 916t + 2088,$ $1228t^5 + 2664t^4 - 2876t^3 - 6892t^2 + 924t + 3128 \rangle$
$i = 96, D_i = 1, \rho_{ij}:$	0:1, 18:2, 19:1, 56:1, 60:2, 61:1, 63:2, 76:2, 78:2, 79:2, 89:1, 95:1
w_i	$\langle 319t^3 - 503t^2 - 323t + 649,$ $-22t^3 - 716t^2 + 555t + 571 \rangle$
$i = 97, D_i = 1, \rho_{ij}:$	5:2, 11:1, 27:2, 42:2, 60:1, 69:2, 71:1, 89:1
w_i	$\langle -30t^3 - 45t^2 + 33t + 60,$ $-111t^3 - 9t^2 + 36t + 12 \rangle$
<i>The third certificate.</i>	
$i = 0, D_i = 1, \rho_{ij}:$	6:1, 11:2, 16:1, 17:2, 23:1, 32:1, 59:1, 60:2, 61:1, 63:1, 65:2, 77:2, 82:2, 95:2
w_i	$\langle 36t^2 + 60t + 84,$ $6t^2 + 18t + 6 \rangle$
$i = 1, D_i = 1, \rho_{ij}:$	8:2, 11:1, 22:1, 39:1, 40:2, 47:2, 57:1, 59:1, 63:2, 64:2, 67:1, 76:2, 89:1, 93:2, 95:2, 97:1
w_i	$\langle 1890t^2 - 5600t - 1680,$ $4690t^2 - 3815t + 9660 \rangle$
$i = 2, D_i = 1, \rho_{ij}:$	12:1, 57:2, 59:2, 60:2, 61:1, 70:2, 93:1, 95:1
w_i	$\langle -2t^2 + 6t + 6,$ $3t^2 + 7t + 9 \rangle$
$i = 3, D_i = 1, \rho_{ij}:$	8:2, 18:1, 19:2, 20:2, 23:2, 29:2, 36:1, 37:2, 62:1, 67:1, 74:1, 77:1, 81:1, 96:2
w_i	$\langle -18t^3 + 6t^2 + 88t - 94,$ $-63t^3 - 42t^2 + 5t + 148 \rangle$
$i = 4, D_i = 1, \rho_{ij}:$	16:1, 17:2, 26:1, 27:2, 32:2, 39:1, 40:2, 47:2, 50:1, 51:2, 59:1, 64:2, 82:1, 89:1, 95:2, 97:1
w_i	$\langle 294t^2 - 294t + 1176,$ $294t^2 - 1176t + 2352 \rangle$
$i = 5, D_i = 1, \rho_{ij}:$	8:1, 10:1, 47:1, 64:2, 67:2, 69:2, 89:2, 97:1
w_i	$\langle -t^2 - 12,$ $-2t^2 - 24 \rangle$
$i = 6, D_i = 1, \rho_{ij}:$	0:2, 8:2, 13:2, 16:2, 17:2, 21:2, 26:2, 32:1, 33:2, 36:2, 40:2, 55:1, 57:2, 60:2, 62:1, 63:2
w_i	$\langle -2872t^3 + 2724t^2 - 1348t + 5904,$ $788t^3 - 2900t^2 + 2052t + 596 \rangle$
$i = 7, D_i = 1, \rho_{ij}:$	---
w_i	$\langle 1,$ $0 \rangle$
$i = 8, D_i = 1, \rho_{ij}:$	1:1, 3:1, 5:2, 6:1, 10:1, 16:1, 21:2, 26:2, 29:1, 32:1, 36:1, 38:1, 60:1
w_i	$\langle 66t^3 - 10t^2 - 108t - 30,$ $12t^3 - 164t^2 - 66t - 108 \rangle$
$i = 9, D_i = 1, \rho_{ij}:$	16:2, 18:1, 19:1, 24:1, 27:1, 34:2, 36:2, 37:2, 40:1, 47:1, 55:2, 60:2, 61:2
w_i	$\langle 547t^3 - 1448t^2 + 386t - 665,$ $10t^3 - 1145t^2 + 983t - 168 \rangle$
$i = 10, D_i = 1, \rho_{ij}:$	5:2, 8:2, 13:1, 23:1, 61:1, 64:2
w_i	$\langle 6t^3 + 30t^2 + 8t,$ $-24t^3 + 6t^2 + 10t \rangle$
$i = 11, D_i = 1, \rho_{ij}:$	0:1, 1:2, 19:1, 22:1, 32:1, 37:1, 55:1, 60:2, 65:2, 72:1, 77:2
w_i	$\langle 22t^3 - 46t^2 + 30t,$ $-4t^3 + 16t^2 - 36t \rangle$
$i = 12, D_i = 1, \rho_{ij}:$	2:2, 13:2, 19:2, 20:1, 23:1, 38:1, 40:1, 44:1, 56:1, 60:2
w_i	$\langle 8t^3 - 3t^2 + 9,$ $10t^3 + 6t^2 - 3t + 3 \rangle$
$i = 13, D_i = 1, \rho_{ij}:$	6:1, 10:2, 12:1, 17:2, 34:1, 40:2, 55:2, 60:1, 62:2
w_i	$\langle -7t^2 - 12t + 91,$ $-14t^2 + 10t + 56 \rangle$
$i = 14, D_i = 1, \rho_{ij}:$	19:1, 26:1, 27:2, 32:1, 36:2, 39:1, 40:1, 63:1, 64:2
w_i	$\langle 60t^3 - 111t^2 - 252t + 15,$

	$114t^3 + 135t^2 - 120t - 93\rangle$
$i = 15, D_i = 1, \rho_{ij}: --$	
w_i	$\langle 1,$ $0\rangle$
$i = 16, D_i = 1, \rho_{ij}: 0:2, 4:2, 6:1, 8:2, 9:1, 20:2, 22:1, 24:2, 25:2, 26:1, 36:1, 38:1, 51:1, 57:2, 62:1, 65:2, 71:2, 75:1, 78:1, 95:2,$ $96:1$	
w_i	$\langle 2124t^3 + 2580t^2 - 8196t + 2232,$ $2856t^3 - 3060t^2 - 5316t + 10668\rangle$
$i = 17, D_i = 1, \rho_{ij}: 0:1, 4:1, 6:1, 13:1, 21:2, 24:2, 27:2, 37:2, 50:2, 59:1, 62:2, 65:2, 67:1, 68:2, 74:1, 76:2, 78:1, 79:1, 84:2, 93:1,$ $96:2$	
w_i	$\langle -2400t^3 - 14664t^2 + 16152t + 24240,$ $6900t^3 - 13200t^2 - 14784t + 45708\rangle$
$i = 18, D_i = 1, \rho_{ij}: 3:2, 9:2, 21:2, 23:1, 24:1, 32:1, 34:2, 38:2, 39:2, 56:2, 60:2, 61:2, 62:1, 63:1, 64:1, 68:1, 70:2, 72:1, 78:1,$ $81:1, 89:2$	
w_i	$\langle -4340t^2 + 6860t - 10640,$ $-980t^2 - 280t - 2100\rangle$
$i = 19, D_i = 1, \rho_{ij}: 3:1, 9:2, 11:2, 12:1, 14:2, 24:2, 29:2, 33:1, 40:1, 47:1, 56:1, 60:1, 61:1, 68:1, 75:1, 77:2, 78:2, 82:2, 84:1,$ $96:2, 97:2$	
w_i	$\langle 2240t^2 + 11200t - 7560,$ $1400t^2 + 5040t + 11200\rangle$
$i = 20, D_i = 1, \rho_{ij}: 3:1, 12:2, 16:1, 22:1, 23:1, 33:1, 39:1, 61:1, 62:2$	
w_i	$\langle -6t^2 + 34t + 6,$ $5t^2 - t - 42\rangle$
$i = 21, D_i = 1, \rho_{ij}: 6:1, 8:1, 17:1, 18:1, 25:2, 60:1$	
w_i	$\langle 6t^2 - 4t - 10,$ $4t^2 - 12t - 2\rangle$
$i = 22, D_i = 1, \rho_{ij}: 1:2, 11:2, 16:2, 20:2, 51:2, 55:2, 62:1, 64:1$	
w_i	$\langle 624t^3 + 1860t^2 + 1686t + 456,$ $654t^3 + 1548t^2 + 1194t + 528\rangle$
$i = 23, D_i = 1, \rho_{ij}: 0:2, 3:1, 10:2, 12:2, 18:2, 20:2, 25:2, 46:1, 56:2, 63:2$	
w_i	$\langle -157t^3 + 233t^2 - 75t + 75,$ $-30t^3 - 92t^2 + 321t - 181\rangle$
$i = 24, D_i = 1, \rho_{ij}: 9:2, 16:1, 17:1, 18:2, 19:1, 32:1, 37:2, 44:2, 56:2, 60:1$	
w_i	$\langle -70t^3 + 175t^2 + 280t - 175,$ $-98t^3 - 133t^2 + 336t - 14\rangle$
$i = 25, D_i = 1, \rho_{ij}: 16:1, 21:1, 23:1, 60:2$	
w_i	$\langle -2t - 4,$ $t + 9\rangle$
$i = 26, D_i = 1, \rho_{ij}: 4:2, 6:1, 8:1, 14:2, 16:2, 38:2, 40:1, 47:1, 57:2, 59:2, 60:2, 68:1, 72:2, 97:2$	
w_i	$\langle 16040t^3 + 90552t^2 + 138176t + 29504,$ $40048t^3 + 83384t^2 + 26664t - 35264\rangle$
$i = 27, D_i = 1, \rho_{ij}: 4:1, 9:2, 14:1, 17:1, 39:2, 61:1, 64:1, 65:2, 67:2, 72:1, 84:1, 89:2, 93:1, 95:1$	
w_i	$\langle 1812t^3 - 696t^2 - 4716t - 2472,$ $2412t^3 + 4436t^2 + 1336t - 976\rangle$
$i = 28, D_i = 1, \rho_{ij}: --$	
w_i	$\langle 1,$ $0\rangle$
$i = 29, D_i = 1, \rho_{ij}: 3:1, 8:2, 19:1, 37:1, 38:2, 56:2, 61:1$	
w_i	$\langle -6t^3 + 2t^2 + 6t - 16,$ $-4t^3 + 4t^2 + 10t - 24\rangle$
$i = 30, D_i = 1, e_i = 0, \rho_{ij}: --$	
w_i	$\langle 1,$ $0\rangle$
$i = 31, D_i = 1, e_i = 0, \rho_{ij}: --$	
w_i	$\langle 1,$ $0\rangle$
$i = 32, D_i = 1, \rho_{ij}: 0:2, 4:1, 6:2, 8:2, 11:2, 14:2, 18:2, 24:2, 34:1, 51:2, 56:2, 59:1, 60:1, 61:2$	
w_i	$\langle 3644t^3 + 6834t^2 + 2182t + 1476,$ $292t^3 + 4164t^2 + 4838t - 2586\rangle$
$i = 33, D_i = 1, \rho_{ij}: 6:1, 19:2, 20:2, 37:2, 38:1, 42:2, 56:2, 64:2, 68:2, 71:1, 82:1$	
w_i	$\langle -282t^3 + 158t^2 + 733t + 51,$ $-204t^3 - 473t^2 + 386t + 621\rangle$
$i = 34, D_i = 1, \rho_{ij}: 9:1, 13:2, 18:1, 32:2, 36:1, 41:1, 56:1, 65:2, 74:1, 84:2, 97:1$	
w_i	$\langle 85t^3 + 83t^2 - 187t - 161,$ $80t^3 - 119t^2 - 206t + 65\rangle$
$i = 35, D_i = 1, \rho_{ij}: --$	
w_i	$\langle 1,$ $0\rangle$

$i = 36, D_i = 1, \rho_{ij}: 3:2, 6:1, 8:2, 9:1, 14:1, 16:2, 34:2, 40:2, 60:2, 61:2, 63:1, 68:2, 78:2, 81:1, 96:2$
 $w_i \quad \langle 203t^2 + 126t - 861, 7t^2 + 630t - 273 \rangle$

$i = 37, D_i = 1, \rho_{ij}: 3:1, 9:1, 11:2, 17:1, 24:1, 29:2, 33:1, 39:1, 60:1, 61:1, 62:1, 65:2, 67:1, 68:2, 72:2$
 $w_i \quad \langle -175t^2 + 147t + 798, -140t^2 + 147t + 903 \rangle$

$i = 38, D_i = 1, \rho_{ij}: 8:2, 12:2, 16:2, 18:1, 26:1, 29:1, 33:2, 40:2, 61:2$
 $w_i \quad \langle -17t^3 + 16t^2 + 70t - 31, -15t^3 - 4t^2 + 56t + 97 \rangle$

$i = 39, D_i = 1, \rho_{ij}: 1:2, 4:2, 14:2, 18:1, 20:2, 27:1, 37:2, 40:1, 47:1, 60:2, 65:2, 68:1, 70:1, 71:2, 72:1, 84:2$
 $w_i \quad \langle 62364t^3 + 154716t^2 + 2790t - 3906, 26136t^3 - 51594t^2 - 194034t - 93744 \rangle$

$i = 40, D_i = 1, \rho_{ij}: 1:1, 4:1, 6:1, 9:2, 12:2, 13:1, 14:2, 19:2, 26:2, 36:1, 38:1, 39:2, 61:1, 72:1, 74:1, 89:2$
 $w_i \quad \langle -15564t^3 - 12870t^2 + 44256t + 22050, -19728t^3 - 67926t^2 - 31914t - 13524 \rangle$

$i = 41, D_i = 1, \rho_{ij}: 34:2, 57:2, 64:1$
 $w_i \quad \langle -3t + 3, 6t - 6 \rangle$

$i = 42, D_i = 1, \rho_{ij}: 33:1, 93:1, 97:2$
 $w_i \quad \langle -9t + 9, 15t - 15 \rangle$

$i = 43, D_i = 1, \rho_{ij}: --$
 $w_i \quad \langle 1, 0 \rangle$

$i = 44, D_i = 1, \rho_{ij}: 12:2, 24:1$
 $w_i \quad \langle 4t - 5, 5t - 1 \rangle$

$i = 45, D_i = 1, \rho_{ij}: --$
 $w_i \quad \langle 1, 0 \rangle$

$i = 46, D_i = 1, \rho_{ij}: 23:2, 50:1$
 $w_i \quad \langle -t - 3, 2 \rangle$

$i = 47, D_i = 1, \rho_{ij}: 1:1, 4:1, 5:2, 9:2, 19:2, 26:2, 39:2, 60:1, 61:1, 64:2$
 $w_i \quad \langle -20t^3 - 26t^2 + 2t + 8, -2t^3 - 46t^2 - 4t - 72 \rangle$

$i = 48, D_i = 1, \rho_{ij}: --$
 $w_i \quad \langle 1, 0 \rangle$

$i = 49, D_i = 1, \rho_{ij}: --$
 $w_i \quad \langle 1, 0 \rangle$

$i = 50, D_i = 1, \rho_{ij}: 4:2, 17:1, 46:2, 51:2, 56:1, 59:2, 64:1, 76:2, 82:2, 96:1$
 $w_i \quad \langle -71t^5 - 204t^4 - 48t^3 + 226t^2 + 149t + 100, -31t^5 - 111t^4 - 231t^3 - 97t^2 + 130t + 32 \rangle$

$i = 51, D_i = 1, \rho_{ij}: 4:1, 16:2, 22:1, 32:1, 50:1, 56:2, 62:2, 88:1, 95:1, 97:2$
 $w_i \quad \langle -7t^5 - 3t^4 + 116t^3 + 100t^2 - 18t + 64, 25t^5 + 81t^4 + 79t^3 + 2t^2 - 9t - 16 \rangle$

$i = 52, D_i = 1, \rho_{ij}: --$
 $w_i \quad \langle 1, 0 \rangle$

$i = 53, D_i = 1, \rho_{ij}: --$
 $w_i \quad \langle 1, 0 \rangle$

$i = 54, D_i = 1, \rho_{ij}: --$
 $w_i \quad \langle 1, 0 \rangle$

$i = 55, D_i = 1, \rho_{ij}: 6:2, 9:1, 11:2, 13:1, 22:1, 57:2, 63:1, 65:1, 68:2, 71:2, 76:2, 93:1$
 $w_i \quad \langle 94444t^3 + 144298t^2 + 53718t + 39634, 2744t^3 + 4298t^2 + 1638t + 1190 \rangle$

$i = 56, D_i = 1, \rho_{ij}: 12:2, 18:1, 19:2, 23:1, 24:1, 29:1, 32:1, 33:1, 34:2, 50:2, 51:1, 62:2, 70:1, 77:2, 78:2, 81:2, 82:2, 96:1$
 $w_i \quad \langle -14t^5 - 259t^4 + 217t^3 + 1050t^2 - 308t - 854, -133t^5 - 329t^4 + 287t^3 + 1323t^2 + 98t - 910 \rangle$

$i = 57, D_i = 1, \rho_{ij}: 1:2, 2:1, 6:1, 16:1, 26:1, 41:1, 55:1$
 $w_i \quad \langle -t^3 + t^2 + 2t - 8, t^3 + 2t^2 - 5t - 4 \rangle$

$i = 58, D_i = 1, \rho_{ij}: --$
 $w_i \quad \langle 1, 0 \rangle$

$0\rangle$	
$i = 59, D_i = 1, \rho_{ij}: 0:2, 1:2, 2:1, 4:2, 17:2, 26:1, 32:2, 50:1, 61:1$	
$w_i \langle -2t^5 + 2t^4 + 8t^3 - 10t^2 - 6t + 8,$ $-2t^5 + 2t^4 + 8t^3 - 14t^2 - 10t + 12 \rangle$	
$i = 60, D_i = 1, \rho_{ij}: 0:1, 2:1, 6:1, 8:2, 9:1, 11:1, 12:1, 13:2, 18:1, 19:2, 21:2, 24:2, 25:1, 26:1, 32:2, 36:1, 37:2, 39:1, 47:2, 68:2,$ $69:1, 74:1, 81:1, 82:2, 84:2, 89:1, 95:1, 97:2$	
$w_i \langle 14684t^5 + 1416t^4 + 47732t^3 - 3236t^2 + 36800t - 12532,$ $12280t^5 - 31548t^4 + 38492t^3 - 86352t^2 + 27756t - 46836 \rangle$	
$i = 61, D_i = 43, \rho_{ij}: 0:2, 2:2, 9:1, 10:2, 18:1, 19:2, 20:2, 27:2, 29:2, 32:1, 36:1, 37:2, 38:1, 40:2, 47:2, 59:2, 63:2, 64:1, 65:2,$ $67:1, 68:2, 70:2, 71:1, 75:1, 78:1, 79:2, 82:1, 89:1$	
$w_i \langle -4789048t^5 - 22032484t^4 - 13728612t^3 - 56454872t^2 - 8536328t - 24618816,$ $6972860t^5 - 18294412t^4 + 22230384t^3 - 51402544t^2 + 16349512t - 29834872 \rangle$	
$i = 62, D_i = 1, \rho_{ij}: 3:2, 6:2, 13:1, 16:2, 17:1, 18:2, 20:1, 22:2, 37:2, 51:1, 56:1, 64:1$	
$w_i \langle 134t^3 - 113t^2 - 251t + 224,$ $115t^3 + 200t^2 - 340t - 119 \rangle$	
$i = 63, D_i = 1, \rho_{ij}: 0:2, 1:1, 6:1, 14:2, 18:2, 23:1, 36:2, 55:2, 61:1, 76:2, 82:2$	
$w_i \langle -72t^3 + 276t^2 + 288t - 96,$ $228t^3 - 216t^2 + 12t - 60 \rangle$	
$i = 64, D_i = 1, \rho_{ij}: 1:1, 4:1, 5:1, 10:1, 14:1, 18:2, 22:2, 27:2, 33:1, 41:2, 47:1, 50:2, 61:2, 62:2$	
$w_i \langle -154212t^3 - 26808t^2 + 53484t + 23280,$ $-46896t^3 - 53736t^2 + 55320t + 108672 \rangle$	
$i = 65, D_i = 1, \rho_{ij}: 0:1, 11:1, 16:1, 17:1, 27:1, 34:1, 37:1, 39:1, 55:2, 61:1, 67:1, 71:1, 75:1, 82:2, 93:1, 96:2$	
$w_i \langle 162t^3 - 364t^2 + 274t - 224,$ $-74t^3 + 84t^2 + 66t + 308 \rangle$	
$i = 66, D_i = 1, \rho_{ij}: --$	
$w_i \langle 1,$ $0 \rangle$	
$i = 67, D_i = 1, \rho_{ij}: 1:2, 3:2, 5:1, 17:2, 27:1, 37:2, 61:2, 65:2, 69:2, 75:1, 81:2, 82:2, 84:2$	
$w_i \langle -1464t^3 + 1248t^2 + 3600t - 672,$ $-1512t^3 - 2988t^2 + 2892t - 1596 \rangle$	
$i = 68, D_i = 1, \rho_{ij}: 17:1, 18:2, 19:2, 26:2, 33:1, 36:1, 37:1, 39:2, 55:1, 60:1, 61:1, 78:2, 89:2$	
$w_i \langle 129t^3 - 158t^2 + 164t + 49,$ $253t^3 - 34t^2 + 274t - 21 \rangle$	
$i = 69, D_i = 1, \rho_{ij}: 5:1, 60:2, 67:1, 71:2, 77:2, 97:1$	
$w_i \langle 12t^3 - 66t^2 - 26t,$ $-66t^3 - 78t^2 - 4t \rangle$	
$i = 70, D_i = 1, \rho_{ij}: 2:1, 18:1, 39:2, 56:2, 61:1, 71:1, 74:2, 77:2, 84:2, 86:2$	
$w_i \langle 12t^3 - 81t^2 + 9t + 126,$ $-42t^3 - 42t^2 + 129t + 63 \rangle$	
$i = 71, D_i = 1, \rho_{ij}: 16:1, 33:2, 39:1, 55:1, 61:2, 65:2, 69:1, 70:2, 96:1$	
$w_i \langle -20t^2 + 7t + 21,$ $-11t^2 - 7t + 42 \rangle$	
$i = 72, D_i = 1, \rho_{ij}: 11:2, 18:2, 26:1, 27:2, 37:1, 39:2, 40:2, 82:2, 97:1$	
$w_i \langle 456t^3 + 3627t^2 + 2424t - 1467,$ $1551t^3 + 2547t^2 - 1389t - 1593 \rangle$	
$i = 73, D_i = 1, \rho_{ij}: --$	
$w_i \langle 1,$ $0 \rangle$	
$i = 74, D_i = 1, \rho_{ij}: 3:2, 17:2, 34:2, 40:2, 60:2, 70:1, 76:2, 77:2, 96:1$	
$w_i \langle 108t^2 - 537t + 2043,$ $123t^2 - 1020t + 3159 \rangle$	
$i = 75, D_i = 1, \rho_{ij}: 16:2, 19:2, 61:2, 65:2, 67:2, 79:1$	
$w_i \langle 32t^3 - 28t^2 - 108t + 8,$ $20t^3 + 56t^2 - 92t - 44 \rangle$	
$i = 76, D_i = 1, \rho_{ij}: 1:1, 17:1, 50:1, 55:1, 63:1, 74:1, 96:2, 97:2$	
$w_i \langle 84t^3 + 234t^2 + 192t + 48,$ $36t^2 - 6t - 12 \rangle$	
$i = 77, D_i = 1, \rho_{ij}: 0:1, 3:2, 11:1, 19:1, 56:1, 69:1, 70:1, 74:1, 79:1, 88:2$	
$w_i \langle t^3 - 26t^2 + 35t - 12,$ $-9t^3 - 39t^2 + 91t - 39 \rangle$	
$i = 78, D_i = 1, \rho_{ij}: 16:2, 17:2, 18:2, 19:1, 36:1, 56:1, 61:2, 68:1, 82:2, 86:1$	
$w_i \langle -224t^3 + 273t^2 + 553t - 91,$ $84t^3 + 560t^2 + 252t - 420 \rangle$	
$i = 79, D_i = 1, \rho_{ij}: 17:2, 61:1, 75:2, 77:2$	
$w_i \langle 5t + 31,$ $-3t + 1 \rangle$	
$i = 80, D_i = 1, \rho_{ij}: --$	
$w_i \langle 1,$	

$i = 81, D_i = 1, \rho_{ij}: 3:2, 18:2, 36:2, 56:1, 60:2, 67:1, 84:1$
$w_i \quad \langle -12t^3 + 4t^2 - 16t + 12, 4t^2 - 24t + 36 \rangle$
$i = 82, D_i = 1, \rho_{ij}: 0:1, 4:2, 19:1, 33:2, 50:1, 56:1, 60:1, 61:2, 63:1, 65:1, 67:1, 72:1, 78:1, 95:2$
$w_i \quad \langle -570t^3 - 660t^2 + 218t - 524, -162t^3 + 354t^2 + 670t - 478 \rangle$
$i = 83, D_i = 1, \rho_{ij}: --$
$w_i \quad \langle 1, 0 \rangle$
$i = 84, D_i = 1, \rho_{ij}: 17:1, 19:2, 27:2, 34:1, 39:1, 60:1, 67:1, 70:1, 81:2$
$w_i \quad \langle -17t^3 + 10t^2 + 20t + 58, -9t^3 + 9t^2 + 32t - 15 \rangle$
$i = 85, D_i = 1, \rho_{ij}: --$
$w_i \quad \langle 1, 0 \rangle$
$i = 86, D_i = 1, \rho_{ij}: 70:1, 78:2$
$w_i \quad \langle -2t + 13, 11t + 2 \rangle$
$i = 87, D_i = 1, \rho_{ij}: --$
$w_i \quad \langle 1, 0 \rangle$
$i = 88, D_i = 1, \rho_{ij}: 51:2, 77:1$
$w_i \quad \langle 2, -t - 3 \rangle$
$i = 89, D_i = 1, \rho_{ij}: 1:2, 4:2, 5:1, 18:1, 27:1, 40:1, 60:2, 61:2, 68:1, 97:1$
$w_i \quad \langle -18t^3 + 20t^2 + 6t + 80, 2t^3 + 46t^2 + 4t + 72 \rangle$
$i = 90, D_i = 1, \rho_{ij}: --$
$w_i \quad \langle 1, 0 \rangle$
$i = 91, D_i = 1, \rho_{ij}: --$
$w_i \quad \langle 1, 0 \rangle$
$i = 92, D_i = 1, \rho_{ij}: --$
$w_i \quad \langle 1, 0 \rangle$
$i = 93, D_i = 1, \rho_{ij}: 1:1, 2:2, 17:2, 27:2, 42:2, 55:2, 65:2$
$w_i \quad \langle -12t^3 - 36t + 12, -18t^3 + 12t^2 - 48t + 18 \rangle$
$i = 94, D_i = 1, \rho_{ij}: --$
$w_i \quad \langle 1, 0 \rangle$
$i = 95, D_i = 1, \rho_{ij}: 0:1, 1:1, 2:2, 4:1, 16:1, 27:2, 51:2, 60:2, 82:1$
$w_i \quad \langle -4t^5 - 8t^4 + 2t^3 + 6t^2, -4t^5 - 2t^4 + 8t^3 + 2t^2 \rangle$
$i = 96, D_i = 1, \rho_{ij}: 3:1, 16:2, 17:1, 19:1, 36:1, 50:2, 56:2, 65:1, 71:2, 74:2, 76:1, 97:2$
$w_i \quad \langle 124t^3 - 874t^2 + 132t + 1070, -241t^3 - 779t^2 + 891t + 289 \rangle$
$i = 97, D_i = 1, \rho_{ij}: 1:2, 4:2, 5:2, 19:1, 26:1, 34:2, 42:1, 51:1, 60:1, 69:2, 72:2, 76:1, 89:2, 96:1$
$w_i \quad \langle 122988t^3 + 26256t^2 - 46500t - 29088, 98400t^3 - 17580t^2 - 3924t + 62928 \rangle$

Table 8: The edges (i, j) with a linear endpoint, the linear endpoint k used, and the norm of $R_{ij} = \text{Res}_t(f_k, g_{kl})$.

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
<i>The first certificate.</i>					
$(0, 6)$	0	2^4	$(23, 63)$	23	$2^2 \cdot 7$
$(0, 9)$	0	2^4	$(24, 37)$	24	2^4
$(0, 16)$	0	$2^2 \cdot 3^2$	$(24, 38)$	24	$2^8 \cdot 5^2$
$(0, 17)$	0	$2^2 \cdot 3^2$	$(24, 51)$	24	$2^2 \cdot 5^2 \cdot 7^3$
$(0, 32)$	0	$2^2 \cdot 3^2$	$(24, 56)$	24	$5^2 \cdot 7^3$
$(0, 33)$	0	$2^2 \cdot 3^2$	$(24, 60)$	24	7^2

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(0, 34)	0	$2^2 \cdot 3^2$	(25, 34)	25	$7 \cdot 13$
(0, 44)	0	2^2	(25, 39)	25	$19 \cdot 61$
(0, 58)	0	2^4	(25, 40)	25	$2^2 \cdot 5^2 \cdot 7 \cdot 13$
(0, 64)	0	$2^2 \cdot 3^2$	(25, 41)	25	$7 \cdot 19$
(0, 65)	0	2^4	(25, 43)	25	$2^4 \cdot 19$
(0, 68)	0	2^4	(25, 47)	25	$13 \cdot 19$
(0, 82)	0	$2^2 \cdot 3^2$	(25, 48)	25	$13 \cdot 19$
(0, 86)	0	2^2	(25, 55)	25	$2^2 \cdot 7 \cdot 13 \cdot 19$
(0, 94)	0	2^4	(25, 57)	25	$5^2 \cdot 7$
(0, 97)	0	$2^2 \cdot 3^2$	(25, 58)	25	7^2
(1, 9)	1	$3 \cdot 7^2$	(25, 60)	25	$2^6 \cdot 7 \cdot 19$
(1, 11)	1	$7 \cdot 19$	(25, 61)	25	$7 \cdot 19 \cdot 61$
(1, 14)	1	$2^2 \cdot 3 \cdot 7^2 \cdot 19$	(25, 62)	25	$7 \cdot 13$
(1, 23)	1	$7 \cdot 19$	(26, 28)	26	$2^2 \cdot 7^3 \cdot 19$
(1, 29)	1	$2^2 \cdot 7 \cdot 13$	(26, 32)	26	$2^4 \cdot 7^2$
(1, 30)	1	$2^4 \cdot 3^3 \cdot 5^2 \cdot 19$	(26, 38)	26	$2^2 \cdot 7^2 \cdot 31$
(1, 31)	1	$2^4 \cdot 3^3 \cdot 5^2 \cdot 19$	(26, 40)	26	$2^2 \cdot 3^2 \cdot 7^2$
(1, 39)	1	$3 \cdot 5^2 \cdot 7 \cdot 13$	(26, 55)	26	$2^2 \cdot 5^2 \cdot 97$
(1, 40)	1	$3 \cdot 5^2 \cdot 7 \cdot 13$	(26, 57)	26	$2^{10} \cdot 7$
(1, 43)	1	$3^3 \cdot 7$	(26, 60)	26	$2^4 \cdot 3^4 \cdot 7^2$
(1, 44)	1	$2^2 \cdot 5^2 \cdot 7$	(26, 68)	26	$3 \cdot 7 \cdot 31$
(1, 48)	1	$2^2 \cdot 7^2$	(26, 73)	26	$3^2 \cdot 7 \cdot 97$
(1, 57)	1	$2^2 \cdot 7$	(26, 76)	26	$3 \cdot 7 \cdot 31$
(1, 59)	1	$2^2 \cdot 7$	(26, 88)	26	$2^2 \cdot 3 \cdot 7 \cdot 31$
(1, 63)	1	$7 \cdot 19$	(26, 91)	26	$2^6 \cdot 3^2 \cdot 7^2 \cdot 31$
(1, 64)	1	$2^2 \cdot 3 \cdot 5^2 \cdot 7$	(26, 96)	26	$3^4 \cdot 7$
(1, 68)	1	$3 \cdot 7^2$	(26, 97)	26	$2^{10} \cdot 3$
(1, 72)	1	$2^2 \cdot 3 \cdot 7^2 \cdot 19$	(27, 39)	27	$2^2 \cdot 3^2 \cdot 7^2$
(1, 77)	1	$7 \cdot 19$	(27, 46)	27	$2^2 \cdot 3 \cdot 7 \cdot 31$
(1, 81)	1	$2^2 \cdot 7 \cdot 13$	(27, 49)	27	$2^6 \cdot 3^2 \cdot 7^2 \cdot 31$
(1, 85)	1	$3^3 \cdot 7$	(27, 55)	27	$2^2 \cdot 5^2 \cdot 97$
(1, 86)	1	$2^2 \cdot 5^2 \cdot 7$	(27, 61)	27	$2^4 \cdot 3^4 \cdot 7^2$
(1, 90)	1	$2^2 \cdot 7^2$	(27, 62)	27	$3^4 \cdot 7$
(1, 93)	1	$2^2 \cdot 7$	(27, 64)	27	$2^{10} \cdot 3$
(1, 95)	1	$2^2 \cdot 7$	(27, 67)	27	$7^2 \cdot 19$
(1, 97)	1	$2^2 \cdot 3 \cdot 5^2 \cdot 7$	(27, 72)	27	$3 \cdot 5^2 \cdot 7^3$
(2, 23)	2	2^8	(27, 80)	27	$2^2 \cdot 7^3 \cdot 19$
(2, 26)	2	$2^2 \cdot 3$	(27, 82)	27	$2^4 \cdot 7^2$
(2, 27)	2	$2^2 \cdot 3$	(27, 84)	27	$2^2 \cdot 7^2 \cdot 31$
(2, 28)	2	7^2	(27, 93)	27	$2^{10} \cdot 7$
(2, 33)	2	2^2	(28, 47)	28	$2^2 \cdot 7 \cdot 19$
(2, 34)	2	2^2	(28, 54)	28	$2^4 \cdot 3^3 \cdot 7 \cdot 19^2$
(2, 38)	2	1	(29, 34)	29	13^2
(2, 47)	2	$2^2 \cdot 7$	(29, 36)	29	7^2
(2, 54)	2	$2^2 \cdot 3^3 \cdot 7$	(29, 37)	29	19
(2, 59)	2	2^2	(29, 48)	29	$7^2 \cdot 19^2$
(2, 60)	2	$2^2 \cdot 3^3 \cdot 7$	(29, 49)	29	$2^2 \cdot 7 \cdot 19^2$
(2, 61)	2	$2^2 \cdot 3^3 \cdot 7$	(29, 56)	29	$2^4 \cdot 19$
(2, 62)	2	$2^2 \cdot 3 \cdot 7$	(29, 59)	29	$2^4 \cdot 7^2$
(2, 77)	2	2^8	(29, 61)	29	$2^4 \cdot 7^2$
(2, 80)	2	7^2	(29, 63)	29	7^4
(2, 84)	2	1	(30, 39)	30	$2^4 \cdot 3^8 \cdot 7 \cdot 13$
(2, 89)	2	$2^2 \cdot 7$	(30, 40)	30	$2^4 \cdot 3^8 \cdot 7^3 \cdot 31$
(2, 92)	2	$2^2 \cdot 3^3 \cdot 7$	(30, 41)	30	$2^2 \cdot 3^8 \cdot 5^2 \cdot 13 \cdot 19 \cdot 31$
(2, 95)	2	2^2	(30, 43)	30	$2^4 \cdot 3^8 \cdot 19 \cdot 31$
(2, 96)	2	$2^2 \cdot 3 \cdot 7$	(30, 53)	30	$2^6 \cdot 3^{18} \cdot 7^2 \cdot 13 \cdot 19^2 \cdot 31$
(3, 6)	3	$3 \cdot 13$	(30, 55)	30	$2^4 \cdot 3^8 \cdot 5^4 \cdot 7 \cdot 13$

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(3, 8)	3	13^2	(30, 56)	30	$2^2 \cdot 3^{18} \cdot 7^4 \cdot 19 \cdot 31$
(3, 18)	3	$2^2 \cdot 31$	(30, 61)	30	$2^4 \cdot 3^{18} \cdot 5^2 \cdot 7 \cdot 31$
(3, 19)	3	$2^2 \cdot 31$	(30, 69)	30	$3^8 \cdot 7^4 \cdot 13^2$
(3, 20)	3	13^2	(30, 71)	30	$2^4 \cdot 3^9 \cdot 7$
(3, 22)	3	$3^2 \cdot 7 \cdot 19 \cdot 37$	(30, 76)	30	$2^4 \cdot 3^8 \cdot 5^4 \cdot 31$
(3, 29)	3	$2^2 \cdot 13 \cdot 19^2$	(30, 85)	30	$3^8 \cdot 7 \cdot 19^4$
(3, 36)	3	$7 \cdot 19$	(30, 87)	30	$2^2 \cdot 3^8 \cdot 7 \cdot 19^4$
(3, 37)	3	$7 \cdot 19$	(30, 89)	30	$2^8 \cdot 3^8 \cdot 13 \cdot 19$
(3, 41)	3	$7 \cdot 19$	(31, 39)	31	$2^4 \cdot 3^8 \cdot 7^3 \cdot 31$
(3, 42)	3	$7 \cdot 19$	(31, 40)	31	$2^4 \cdot 3^8 \cdot 7 \cdot 13$
(3, 43)	3	$3^5 \cdot 13$	(31, 42)	31	$2^2 \cdot 3^8 \cdot 5^2 \cdot 13 \cdot 19 \cdot 31$
(3, 52)	3	$3 \cdot 19^2 \cdot 31$	(31, 43)	31	$3^8 \cdot 7 \cdot 19^4$
(3, 53)	3	$3 \cdot 19^2 \cdot 31$	(31, 45)	31	$2^2 \cdot 3^8 \cdot 7 \cdot 19^4$
(3, 54)	3	$2^4 \cdot 3^3 \cdot 19^2$	(31, 47)	31	$2^8 \cdot 3^8 \cdot 13 \cdot 19$
(3, 57)	3	2^2	(31, 52)	31	$2^6 \cdot 3^{18} \cdot 7^2 \cdot 13 \cdot 19^2 \cdot 31$
(3, 60)	3	$2^2 \cdot 3^3 \cdot 37$	(31, 55)	31	$2^4 \cdot 3^8 \cdot 5^4 \cdot 7 \cdot 13$
(3, 61)	3	$2^2 \cdot 3^3 \cdot 37$	(31, 56)	31	$2^2 \cdot 3^{18} \cdot 7^4 \cdot 19 \cdot 31$
(3, 62)	3	$2^2 \cdot 3^4$	(31, 60)	31	$2^4 \cdot 3^{18} \cdot 5^2 \cdot 7 \cdot 31$
(3, 65)	3	$3 \cdot 13$	(31, 66)	31	$2^4 \cdot 3^8 \cdot 19^3$
(3, 67)	3	13^2	(31, 69)	31	$2^4 \cdot 3^8 \cdot 5^4 \cdot 7^2$
(3, 74)	3	13^2	(31, 74)	31	$2^4 \cdot 3^9 \cdot 5^2 \cdot 7 \cdot 19$
(3, 76)	3	$3^2 \cdot 7 \cdot 19 \cdot 37$	(31, 85)	31	$2^4 \cdot 3^8 \cdot 19 \cdot 31$
(3, 81)	3	$2^2 \cdot 13 \cdot 19^2$	(32, 35)	32	$2^{10} \cdot 7$
(3, 85)	3	$3^5 \cdot 13$	(32, 46)	32	$2^6 \cdot 7^2$
(3, 92)	3	$2^4 \cdot 3^3 \cdot 19^2$	(32, 51)	32	$2^4 \cdot 5^2 \cdot 7^2$
(3, 93)	3	2^2	(32, 60)	32	$2^6 \cdot 3 \cdot 7$
(3, 96)	3	$2^2 \cdot 3^4$	(32, 61)	32	$2^4 \cdot 3 \cdot 7$
(4, 14)	4	$2^2 \cdot 3^2 \cdot 5^2$	(32, 62)	32	$5^2 \cdot 7$
(4, 15)	4	$2^2 \cdot 3^2$	(32, 63)	32	$2^2 \cdot 3^4 \cdot 7$
(4, 16)	4	$2^2 \cdot 3^2 \cdot 7$	(33, 38)	33	$5^2 \cdot 31$
(4, 17)	4	$2^2 \cdot 3^2 \cdot 7$	(33, 39)	33	$3 \cdot 13^2 \cdot 37$
(4, 22)	4	$3^2 \cdot 7$	(33, 47)	33	$2^4 \cdot 7 \cdot 13$
(4, 30)	4	$2^6 \cdot 3^3 \cdot 7 \cdot 31$	(33, 50)	33	$2^2 \cdot 5^2 \cdot 7^2$
(4, 31)	4	$2^6 \cdot 3^3 \cdot 7 \cdot 31$	(33, 55)	33	$2^2 \cdot 3^4 \cdot 5^2 \cdot 13$
(4, 41)	4	$2^2 \cdot 3^2 \cdot 7 \cdot 31$	(33, 56)	33	$7^2 \cdot 61$
(4, 42)	4	$2^2 \cdot 3^2 \cdot 7 \cdot 31$	(33, 58)	33	$2^2 \cdot 7 \cdot 37 \cdot 61$
(4, 49)	4	$2^2 \cdot 3^2 \cdot 7^2$	(33, 59)	33	$2^4 \cdot 7 \cdot 97$
(4, 50)	4	$2^2 \cdot 7^2$	(33, 68)	33	$5^2 \cdot 31$
(4, 51)	4	$2^2 \cdot 7^2$	(33, 69)	33	$7 \cdot 13$
(4, 64)	4	$2^2 \cdot 3^2 \cdot 7^2$	(33, 74)	33	$3^3 \cdot 5^2$
(4, 72)	4	$2^2 \cdot 3^2 \cdot 5^2$	(33, 75)	33	2^{12}
(4, 73)	4	$2^2 \cdot 3^2$	(33, 76)	33	$2^6 \cdot 3^2 \cdot 31$
(4, 76)	4	$3^2 \cdot 7$	(33, 79)	33	$7 \cdot 13$
(4, 91)	4	$2^2 \cdot 3^2 \cdot 7^2$	(33, 81)	33	13^2
(4, 97)	4	$2^2 \cdot 3^2 \cdot 7^2$	(34, 40)	34	$3 \cdot 13^2 \cdot 37$
(5, 9)	5	$3 \cdot 7$	(34, 51)	34	$2^2 \cdot 5^2 \cdot 7^2$
(5, 10)	5	$5^2 \cdot 7^2$	(34, 55)	34	$2^2 \cdot 3^4 \cdot 5^2 \cdot 13$
(5, 11)	5	3^2	(34, 56)	34	$7^2 \cdot 61$
(5, 23)	5	$7 \cdot 19$	(34, 69)	34	2^4
(5, 26)	5	$2^4 \cdot 3 \cdot 7$	(34, 73)	34	$2^2 \cdot 3 \cdot 97$
(5, 27)	5	$2^4 \cdot 3 \cdot 7$	(34, 84)	34	$5^2 \cdot 31$
(5, 43)	5	3^2	(34, 89)	34	$2^4 \cdot 7 \cdot 13$
(5, 52)	5	$3 \cdot 5^2 \cdot 7^2 \cdot 19$	(34, 94)	34	$2^2 \cdot 7 \cdot 37 \cdot 61$
(5, 53)	5	$3 \cdot 5^2 \cdot 7^2 \cdot 19$	(34, 95)	34	$2^4 \cdot 7 \cdot 97$
(5, 54)	5	$2^8 \cdot 3^5 \cdot 7$	(35, 37)	35	7^2
(5, 63)	5	3^2	(35, 46)	35	$2^4 \cdot 7 \cdot 13 \cdot 31$
(5, 68)	5	$3 \cdot 7$	(35, 47)	35	$2^2 \cdot 7 \cdot 19 \cdot 73$

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(5, 69)	5	$5^2 \cdot 7^2$	(35, 48)	35	$2^2 \cdot 7 \cdot 19^2$
(5, 77)	5	$7 \cdot 19$	(35, 56)	35	$7 \cdot 31 \cdot 73$
(5, 85)	5	3^2	(36, 48)	36	$3 \cdot 7$
(5, 92)	5	$2^8 \cdot 3^5 \cdot 7$	(36, 49)	36	$2^6 \cdot 3 \cdot 7^2$
(6, 7)	6	$2^4 \cdot 19$	(36, 57)	36	$3 \cdot 13$
(6, 15)	6	$2^8 \cdot 3 \cdot 13$	(36, 60)	36	$2^2 \cdot 3 \cdot 7$
(6, 17)	6	$2^2 \cdot 3 \cdot 7^2$	(36, 64)	36	$2^2 \cdot 3 \cdot 7$
(6, 19)	6	2^6	(36, 68)	36	7^2
(6, 20)	6	$13^2 \cdot 19$	(36, 75)	36	$2^2 \cdot 7 \cdot 13$
(6, 40)	6	$2^4 \cdot 3$	(36, 77)	36	$7 \cdot 13$
(6, 44)	6	2^4	(36, 78)	36	2^4
(6, 46)	6	$3 \cdot 37$	(36, 81)	36	19
(6, 48)	6	$2^8 \cdot 37$	(36, 83)	36	7^2
(6, 51)	6	$2^{10} \cdot 19$	(36, 85)	36	13
(6, 53)	6	$2^2 \cdot 3 \cdot 7 \cdot 13^2$	(36, 95)	36	$2^2 \cdot 7^2$
(6, 55)	6	$2^2 \cdot 7 \cdot 13 \cdot 19$	(37, 43)	37	13
(6, 57)	6	$2^6 \cdot 7$	(37, 59)	37	$2^2 \cdot 7^2$
(6, 58)	6	2^8	(37, 61)	37	$2^2 \cdot 3 \cdot 7$
(6, 59)	6	$7^2 \cdot 13$	(37, 68)	37	7^2
(6, 62)	6	$3 \cdot 7$	(37, 69)	37	$2^2 \cdot 7^2$
(7, 8)	7	$2^2 \cdot 3^4 \cdot 7$	(37, 70)	37	13
(7, 16)	7	2^{12}	(37, 72)	37	$3 \cdot 7^3$
(7, 19)	7	$2^2 \cdot 7^3$	(37, 75)	37	7
(7, 30)	7	$2^4 \cdot 3^4 \cdot 19^3$	(37, 81)	37	7^2
(7, 41)	7	$7^5 \cdot 19$	(37, 90)	37	$3 \cdot 7$
(7, 43)	7	$2^2 \cdot 7$	(37, 91)	37	$2^6 \cdot 3 \cdot 7^2$
(7, 50)	7	$2^2 \cdot 7^5$	(37, 93)	37	$3 \cdot 13$
(7, 54)	7	$2^4 \cdot 19^3$	(37, 97)	37	$2^2 \cdot 3 \cdot 7$
(8, 13)	8	7	(38, 40)	38	7^2
(8, 16)	8	$2^2 \cdot 13$	(38, 47)	38	$2^2 \cdot 3^4 \cdot 7$
(8, 18)	8	13	(38, 48)	38	$5^2 \cdot 7^3$
(8, 19)	8	$7^2 \cdot 11^2$	(38, 63)	38	7^4
(8, 21)	8	7^2	(39, 40)	39	$3^8 \cdot 5^2$
(8, 26)	8	$7^2 \cdot 19$	(39, 46)	39	$3 \cdot 7^2$
(8, 29)	8	$7^2 \cdot 13$	(39, 56)	39	$3^2 \cdot 7^3 \cdot 31$
(8, 32)	8	$2^4 \cdot 7$	(39, 57)	39	$2^4 \cdot 3 \cdot 7$
(8, 35)	8	$2^6 \cdot 7 \cdot 13$	(39, 58)	39	$2^2 \cdot 37$
(8, 42)	8	$5^2 \cdot 7^2 \cdot 13$	(39, 61)	39	$2^4 \cdot 3^5 \cdot 61$
(8, 49)	8	$2^2 \cdot 7^3 \cdot 11^2 \cdot 19$	(39, 65)	39	$2^4 \cdot 3$
(8, 53)	8	$2^2 \cdot 5^4 \cdot 7^2 \cdot 13$	(39, 68)	39	$3 \cdot 7^2$
(9, 13)	9	$3 \cdot 7$	(39, 70)	39	$2^2 \cdot 7^2 \cdot 19$
(9, 14)	9	$3 \cdot 5^2 \cdot 109$	(39, 71)	39	$3^4 \cdot 19$
(9, 16)	9	$2^2 \cdot 3 \cdot 7 \cdot 31$	(39, 76)	39	$3^4 \cdot 5^2 \cdot 7 \cdot 19$
(9, 18)	9	$2^2 \cdot 7$	(39, 77)	39	$2^{10} \cdot 7$
(9, 21)	9	$7 \cdot 43$	(39, 79)	39	$2^2 \cdot 5^2 \cdot 7 \cdot 13$
(9, 24)	9	$7^2 \cdot 31$	(39, 84)	39	7^2
(9, 27)	9	$3 \cdot 7 \cdot 31$	(39, 86)	39	$7 \cdot 13$
(9, 34)	9	$5^2 \cdot 31$	(39, 96)	39	$3^2 \cdot 13$
(9, 36)	9	7^2	(39, 97)	39	$2^2 \cdot 3^5 \cdot 7$
(9, 37)	9	7^2	(40, 44)	40	$7 \cdot 13$
(9, 40)	9	$3 \cdot 7^2$	(40, 56)	40	$3^2 \cdot 7^3 \cdot 31$
(9, 44)	9	7^3	(40, 60)	40	$2^4 \cdot 3^5 \cdot 61$
(9, 47)	9	$5^2 \cdot 7^2$	(40, 62)	40	$3^2 \cdot 13$
(9, 51)	9	$5^2 \cdot 7^2 \cdot 61$	(40, 63)	40	$2^2 \cdot 3^3 \cdot 7$
(9, 53)	9	$3 \cdot 5^2 \cdot 7 \cdot 37 \cdot 43$	(40, 64)	40	$2^2 \cdot 3^5 \cdot 7$
(9, 55)	9	$2^2 \cdot 7^2$	(40, 71)	40	$2^2 \cdot 3 \cdot 7$
(9, 57)	9	$37 \cdot 61$	(40, 79)	40	$19 \cdot 61$

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(9, 59)	9	$7^2 \cdot 109$	(40, 88)	40	$3 \cdot 7^2$
(9, 61)	9	$3^2 \cdot 7^3$	(40, 93)	40	$2^4 \cdot 3 \cdot 7$
(9, 63)	9	7	(40, 94)	40	$2^2 \cdot 37$
(10, 13)	10	7^3	(41, 43)	41	$7^2 \cdot 13 \cdot 31$
(10, 19)	10	7^2	(41, 52)	41	$2^2 \cdot 5^2 \cdot 7$
(10, 20)	10	$2^2 \cdot 7^2$	(41, 55)	41	$2^2 \cdot 3^4 \cdot 7 \cdot 13^2$
(10, 24)	10	7^2	(41, 64)	41	$2^2 \cdot 3^2 \cdot 7^4$
(10, 25)	10	$2^2 \cdot 7^2 \cdot 13$	(41, 67)	41	$5^2 \cdot 7^2 \cdot 13$
(10, 30)	10	$2^4 \cdot 3^4 \cdot 5^4 \cdot 7^2$	(41, 70)	41	$2^2 \cdot 7^2 \cdot 13$
(10, 31)	10	$3^4 \cdot 7^4 \cdot 13^2$	(41, 90)	41	$2^4 \cdot 3 \cdot 7^3 \cdot 13 \cdot 19$
(10, 33)	10	2^4	(42, 48)	42	$2^4 \cdot 3 \cdot 7^3 \cdot 13 \cdot 19$
(10, 34)	10	$7 \cdot 13$	(42, 53)	42	$2^2 \cdot 5^2 \cdot 7$
(10, 36)	10	$2^2 \cdot 7^2$	(42, 55)	42	$2^2 \cdot 3^4 \cdot 7 \cdot 13^2$
(10, 61)	10	$5^2 \cdot 7^3 \cdot 37$	(42, 66)	42	$7^5 \cdot 19$
(10, 64)	10	$2^2 \cdot 7 \cdot 37$	(42, 72)	42	$2^8 \cdot 3 \cdot 7^3$
(11, 15)	11	$3^4 \cdot 13$	(42, 79)	42	$7 \cdot 19$
(11, 18)	11	13	(42, 85)	42	$7^2 \cdot 13 \cdot 31$
(11, 32)	11	$3^4 \cdot 7$	(42, 97)	42	$2^2 \cdot 3^2 \cdot 7^4$
(11, 39)	11	$2^2 \cdot 3^3 \cdot 7$	(44, 55)	44	$2^2 \cdot 7 \cdot 13^2$
(11, 60)	11	$3^2 \cdot 7^2$	(44, 62)	44	$7^3 \cdot 13$
(11, 68)	11	7	(44, 64)	44	$2^2 \cdot 5^4 \cdot 7^2$
(11, 72)	11	$3^2 \cdot 7^2 \cdot 19$	(46, 49)	46	$2^2 \cdot 3 \cdot 7 \cdot 31$
(11, 75)	11	7^4	(46, 53)	46	$2^2 \cdot 3^3 \cdot 13$
(11, 77)	11	$2^2 \cdot 7$	(46, 56)	46	$31^2 \cdot 37$
(11, 81)	11	7^4	(47, 56)	47	$2^2 \cdot 73$
(11, 82)	11	$2^2 \cdot 3^4 \cdot 7$	(47, 60)	47	$2^4 \cdot 5^2 \cdot 7$
(11, 84)	11	7^4	(47, 61)	47	$2^4 \cdot 7$
(11, 95)	11	7^4	(48, 51)	48	$2^{12} \cdot 19^2$
(12, 13)	12	19	(48, 56)	48	$5^2 \cdot 37 \cdot 61$
(12, 25)	12	$7 \cdot 19$	(48, 58)	48	$2^8 \cdot 61$
(12, 32)	12	2^6	(49, 54)	49	$2^6 \cdot 3^4 \cdot 7 \cdot 19$
(12, 36)	12	13	(50, 53)	50	$2^{16} \cdot 5^2 \cdot 7 \cdot 19$
(12, 38)	12	7^4	(50, 59)	50	$2^4 \cdot 3^2 \cdot 7^2 \cdot 31$
(12, 40)	12	$2^2 \cdot 7^2 \cdot 19$	(50, 64)	50	$2^{10} \cdot 3^2 \cdot 7$
(12, 42)	12	$2^2 \cdot 7^2 \cdot 13$	(50, 65)	50	$2^{10} \cdot 19$
(12, 45)	12	$2^{10} \cdot 7$	(50, 68)	50	$5^2 \cdot 7^2 \cdot 61$
(12, 47)	12	2^2	(50, 76)	50	$2^2 \cdot 3^2 \cdot 7 \cdot 379$
(12, 50)	12	19	(50, 77)	50	$2^2 \cdot 7^4$
(12, 56)	12	7	(50, 78)	50	$2^2 \cdot 5^2 \cdot 7^3$
(12, 58)	12	$2^2 \cdot 7^2$	(50, 82)	50	$2^4 \cdot 5^2 \cdot 7^2$
(12, 60)	12	2^2	(50, 90)	50	$2^{12} \cdot 19^2$
(12, 61)	12	7^3	(50, 93)	50	$2^8 \cdot 61$
(12, 62)	12	7	(50, 96)	50	$5^2 \cdot 7^2 \cdot 379$
(13, 21)	13	2^4	(51, 52)	51	$2^{16} \cdot 5^2 \cdot 7 \cdot 19$
(13, 23)	13	13	(51, 57)	51	$2^8 \cdot 61$
(13, 31)	13	$2^4 \cdot 3^3 \cdot 7$	(51, 62)	51	$5^2 \cdot 7^2 \cdot 379$
(13, 35)	13	7	(51, 66)	51	$2^2 \cdot 7^5$
(13, 39)	13	$2^2 \cdot 3 \cdot 7$	(51, 70)	51	19
(13, 40)	13	$3^4 \cdot 19$	(51, 72)	51	$2^6 \cdot 3^6$
(13, 47)	13	$2^2 \cdot 7$	(51, 95)	51	$2^4 \cdot 3^2 \cdot 7^2 \cdot 31$
(13, 59)	13	$2^2 \cdot 7 \cdot 13$	(51, 97)	51	$2^{10} \cdot 3^2 \cdot 7$
(13, 60)	13	3^3	(52, 53)	52	$2^2 \cdot 3^6 \cdot 5^4 \cdot 7^2$
(13, 61)	13	$2^2 \cdot 3^2$	(52, 62)	52	$3 \cdot 7^2 \cdot 13 \cdot 31$
(13, 62)	13	$3^2 \cdot 7$	(52, 65)	52	$2^2 \cdot 3 \cdot 7 \cdot 13^2$
(13, 64)	13	$3^2 \cdot 7$	(52, 67)	52	$2^2 \cdot 5^4 \cdot 7^2 \cdot 13$
(14, 26)	14	$3 \cdot 5^2 \cdot 7^3$	(52, 68)	52	$3 \cdot 5^2 \cdot 7 \cdot 37 \cdot 43$
(14, 36)	14	$3 \cdot 7^3$	(52, 75)	52	$2^2 \cdot 7 \cdot 19 \cdot 43$

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(14, 41)	14	$2^8 \cdot 3 \cdot 7^3$	(52, 88)	52	$2^2 \cdot 3^3 \cdot 13$
(14, 50)	14	$2^6 \cdot 3^6$	(52, 93)	52	$7 \cdot 13 \cdot 31 \cdot 37$
(14, 59)	14	$2^4 \cdot 3^2 \cdot 7 \cdot 109$	(52, 97)	52	$2^8 \cdot 3^2 \cdot 31^2$
(14, 60)	14	$3^4 \cdot 5^2 \cdot 7^2$	(53, 57)	53	$7 \cdot 13 \cdot 31 \cdot 37$
(14, 61)	14	$2^6 \cdot 3^5$	(53, 64)	53	$2^8 \cdot 3^2 \cdot 31^2$
(14, 63)	14	$3^2 \cdot 7^2 \cdot 19$	(53, 77)	53	$2^2 \cdot 13 \cdot 19 \cdot 31^2$
(15, 16)	15	$2^6 \cdot 3^4$	(53, 96)	53	$3 \cdot 7^2 \cdot 13 \cdot 31$
(15, 17)	15	$2^2 \cdot 3^6 \cdot 7$	(55, 57)	55	$2^4 \cdot 3^4 \cdot 7^2$
(15, 18)	15	$2^2 \cdot 13$	(55, 65)	55	$2^2 \cdot 7 \cdot 13 \cdot 19$
(15, 22)	15	$3^4 \cdot 7^2 \cdot 43$	(55, 68)	55	$2^2 \cdot 7^2$
(15, 27)	15	$3^2 \cdot 7 \cdot 97$	(55, 73)	55	$2^2 \cdot 3^2 \cdot 7 \cdot 13 \cdot 97$
(15, 33)	15	$2^2 \cdot 3 \cdot 97$	(55, 76)	55	$2^2 \cdot 3 \cdot 7$
(15, 55)	15	$2^2 \cdot 3^2 \cdot 7 \cdot 13 \cdot 97$	(55, 79)	55	$2^2 \cdot 7 \cdot 13 \cdot 19$
(15, 59)	15	$3^2 \cdot 7 \cdot 13 \cdot 97$	(55, 86)	55	$2^2 \cdot 7 \cdot 13^2$
(15, 61)	15	$3^5 \cdot 43$	(55, 93)	55	$2^4 \cdot 3^4 \cdot 7^2$
(16, 18)	16	$2^4 \cdot 7$	(56, 70)	56	7
(16, 19)	16	$2^2 \cdot 13$	(56, 77)	56	$7^2 \cdot 31$
(16, 22)	16	$2^2 \cdot 3^4 \cdot 13$	(56, 78)	56	$5^2 \cdot 7^3$
(16, 24)	16	$2^2 \cdot 7^2 \cdot 31$	(56, 81)	56	$2^4 \cdot 19$
(16, 33)	16	$2^2 \cdot 3^3 \cdot 13$	(56, 83)	56	$7 \cdot 31 \cdot 73$
(16, 40)	16	$2^2 \cdot 3^5 \cdot 7$	(56, 88)	56	$31^2 \cdot 37$
(16, 41)	16	$2^2 \cdot 3 \cdot 13$	(56, 89)	56	$2^2 \cdot 73$
(16, 42)	16	$2^4 \cdot 3^4 \cdot 7^2$	(56, 90)	56	$5^2 \cdot 37 \cdot 61$
(16, 47)	16	$2^2 \cdot 13$	(57, 60)	57	$2^4 \cdot 3 \cdot 7^2 \cdot 31$
(16, 51)	16	$2^6 \cdot 3^2 \cdot 31$	(59, 63)	59	7^4
(16, 53)	16	$2^4 \cdot 3^4 \cdot 13^2$	(60, 62)	60	$3^3 \cdot 31$
(16, 56)	16	$2^2 \cdot 3^2 \cdot 7$	(60, 64)	60	$2^4 \cdot 3^5 \cdot 7^2$
(16, 58)	16	$2^4 \cdot 7^2$	(60, 68)	60	$3^2 \cdot 7^3$
(16, 61)	16	$2^4 \cdot 3^3 \cdot 7$	(60, 69)	60	$5^2 \cdot 7^3 \cdot 37$
(16, 64)	16	$2^2 \cdot 3^5 \cdot 7$	(60, 70)	60	7^3
(16, 65)	16	$2^2 \cdot 3 \cdot 7^2$	(60, 71)	60	$2^2 \cdot 3^2$
(16, 73)	16	$2^2 \cdot 3^6 \cdot 7$	(60, 72)	60	$2^6 \cdot 3^5$
(16, 75)	16	$2^2 \cdot 7^2$	(60, 73)	60	$3^5 \cdot 43$
(16, 77)	16	$2^4 \cdot 7$	(60, 76)	60	$2^2 \cdot 3^5 \cdot 7 \cdot 43$
(16, 95)	16	$2^4 \cdot 3^2 \cdot 31$	(60, 77)	60	$2^4 \cdot 7 \cdot 19 \cdot 31$
(16, 97)	16	$2^2 \cdot 3^5 \cdot 7$	(60, 79)	60	$7 \cdot 19 \cdot 61$
(17, 18)	17	$2^2 \cdot 13$	(60, 81)	60	$2^4 \cdot 7^2$
(17, 19)	17	$2^4 \cdot 7$	(60, 82)	60	$2^4 \cdot 3 \cdot 7$
(17, 21)	17	$2^2 \cdot 7^2$	(60, 89)	60	$2^4 \cdot 7$
(17, 23)	17	$2^4 \cdot 7$	(60, 97)	60	$2^4 \cdot 3^3 \cdot 37$
(17, 34)	17	$2^2 \cdot 3^3 \cdot 13$	(61, 63)	61	$3^2 \cdot 7^2$
(17, 39)	17	$2^2 \cdot 3^5 \cdot 7$	(61, 64)	61	$2^4 \cdot 3^3 \cdot 37$
(17, 41)	17	$2^4 \cdot 3^4 \cdot 7^2$	(61, 70)	61	2^2
(17, 42)	17	$2^2 \cdot 3 \cdot 13$	(61, 71)	61	3^3
(17, 50)	17	$2^6 \cdot 3^2 \cdot 31$	(61, 72)	61	$3^4 \cdot 5^2 \cdot 7^2$
(17, 52)	17	$2^4 \cdot 3^4 \cdot 13^2$	(61, 74)	61	$2^4 \cdot 3^3 \cdot 31$
(17, 56)	17	$2^2 \cdot 3^2 \cdot 7$	(61, 75)	61	$2^4 \cdot 7^2 \cdot 19$
(17, 59)	17	$2^4 \cdot 3^2 \cdot 31$	(61, 76)	61	$3^5 \cdot 31 \cdot 37$
(17, 60)	17	$2^4 \cdot 3^3 \cdot 7$	(61, 77)	61	$2^6 \cdot 19$
(17, 64)	17	$2^2 \cdot 3^5 \cdot 7$	(61, 78)	61	7^2
(17, 66)	17	2^{12}	(61, 79)	61	$2^6 \cdot 7 \cdot 19$
(17, 67)	17	$2^2 \cdot 13$	(61, 82)	61	$2^6 \cdot 3 \cdot 7$
(17, 68)	17	$2^2 \cdot 3 \cdot 7 \cdot 31$	(61, 89)	61	$2^4 \cdot 5^2 \cdot 7$
(17, 73)	17	$2^6 \cdot 3^4$	(61, 93)	61	$2^4 \cdot 3 \cdot 7^2 \cdot 31$
(17, 76)	17	$2^2 \cdot 3^4 \cdot 13$	(61, 96)	61	$3^3 \cdot 31$
(17, 78)	17	$2^2 \cdot 7^2 \cdot 31$	(61, 97)	61	$2^4 \cdot 3^5 \cdot 7^2$
(17, 89)	17	$2^2 \cdot 13$	(63, 73)	63	$3^4 \cdot 13$

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(17, 94)	17	$2^4 \cdot 7^2$	(63, 82)	63	$3^4 \cdot 7$
(17, 97)	17	$2^2 \cdot 3^5 \cdot 7$	(65, 66)	65	$2^4 \cdot 19$
(18, 21)	18	$2^2 \cdot 7$	(65, 73)	65	$2^8 \cdot 3 \cdot 13$
(18, 23)	18	$2^2 \cdot 7 \cdot 31$	(65, 74)	65	$13^2 \cdot 19$
(18, 24)	18	$5^2 \cdot 7^2$	(65, 86)	65	2^4
(18, 30)	18	$2^2 \cdot 3^6 \cdot 19^2$	(65, 88)	65	$3 \cdot 37$
(18, 31)	18	$2^2 \cdot 3^6 \cdot 7$	(65, 90)	65	$2^8 \cdot 37$
(18, 33)	18	$2^4 \cdot 5^2$	(65, 93)	65	$2^6 \cdot 7$
(18, 35)	18	$2^2 \cdot 7^2 \cdot 13$	(65, 94)	65	2^8
(18, 38)	18	$5^2 \cdot 7^2$	(65, 95)	65	$7^2 \cdot 13$
(18, 43)	18	$7 \cdot 19$	(65, 96)	65	$3 \cdot 7$
(18, 47)	18	$2^4 \cdot 7$	(66, 67)	66	$2^2 \cdot 3^4 \cdot 7$
(18, 51)	18	$2^2 \cdot 5^2 \cdot 7$	(66, 85)	66	$2^2 \cdot 7$
(18, 58)	18	$2^2 \cdot 7$	(66, 92)	66	$2^4 \cdot 19^3$
(18, 62)	18	$2^4 \cdot 5^2$	(67, 71)	67	7
(18, 64)	18	$2^4 \cdot 7^2$	(67, 75)	67	7^2
(18, 65)	18	2^6	(67, 81)	67	$7^2 \cdot 13$
(18, 66)	18	$2^2 \cdot 7^3$	(67, 82)	67	$2^4 \cdot 7$
(18, 67)	18	$7^2 \cdot 11^2$	(67, 83)	67	$2^6 \cdot 7 \cdot 13$
(18, 69)	18	7^2	(67, 91)	67	$2^2 \cdot 7^3 \cdot 11^2 \cdot 19$
(18, 77)	18	$2^2 \cdot 7$	(68, 71)	68	$3 \cdot 7$
(18, 81)	18	$7 \cdot 19$	(68, 72)	68	$3 \cdot 5^2 \cdot 109$
(18, 83)	18	$2^2 \cdot 7 \cdot 13$	(68, 75)	68	$7 \cdot 43$
(18, 89)	18	$2^2 \cdot 7$	(68, 78)	68	$7^2 \cdot 31$
(18, 91)	18	$7^2 \cdot 11^2$	(68, 86)	68	7^3
(18, 96)	18	$2^2 \cdot 7$	(68, 89)	68	$5^2 \cdot 7^2$
(18, 97)	18	$2^2 \cdot 7 \cdot 31$	(68, 93)	68	$37 \cdot 61$
(19, 23)	19	$2^2 \cdot 7$	(68, 95)	68	$7^2 \cdot 109$
(19, 29)	19	$7 \cdot 19$	(69, 71)	69	7^3
(19, 30)	19	$2^2 \cdot 3^6 \cdot 7$	(69, 74)	69	$2^2 \cdot 7^2$
(19, 31)	19	$2^2 \cdot 3^6 \cdot 19^2$	(69, 78)	69	7^2
(19, 34)	19	$2^4 \cdot 5^2$	(69, 79)	69	$2^2 \cdot 7^2 \cdot 13$
(19, 35)	19	$2^2 \cdot 7 \cdot 13$	(69, 97)	69	$2^2 \cdot 7 \cdot 37$
(19, 47)	19	$2^2 \cdot 7$	(70, 71)	70	19
(19, 49)	19	$7^2 \cdot 11^2$	(70, 79)	70	$7 \cdot 19$
(19, 50)	19	$2^2 \cdot 5^2 \cdot 7$	(70, 82)	70	2^6
(19, 62)	19	$2^2 \cdot 7$	(70, 84)	70	7^4
(19, 63)	19	13	(70, 87)	70	$2^{10} \cdot 7$
(19, 64)	19	$2^2 \cdot 7 \cdot 31$	(70, 89)	70	2^2
(19, 67)	19	13	(70, 94)	70	$2^2 \cdot 7^2$
(19, 68)	19	$2^2 \cdot 7$	(70, 96)	70	7
(19, 73)	19	$2^2 \cdot 13$	(71, 75)	71	2^4
(19, 75)	19	$2^2 \cdot 7$	(71, 77)	71	13
(19, 77)	19	$2^2 \cdot 7 \cdot 31$	(71, 83)	71	7
(19, 78)	19	$5^2 \cdot 7^2$	(71, 89)	71	$2^2 \cdot 7$
(19, 83)	19	$2^2 \cdot 7^2 \cdot 13$	(71, 95)	71	$2^2 \cdot 7 \cdot 13$
(19, 84)	19	$5^2 \cdot 7^2$	(71, 96)	71	$3^2 \cdot 7$
(19, 85)	19	$7 \cdot 19$	(71, 97)	71	$3^2 \cdot 7$
(19, 89)	19	$2^4 \cdot 7$	(72, 95)	72	$2^4 \cdot 3^2 \cdot 7 \cdot 109$
(19, 94)	19	$2^2 \cdot 7$	(73, 76)	73	$3^4 \cdot 7^2 \cdot 43$
(19, 96)	19	$2^4 \cdot 5^2$	(73, 95)	73	$3^2 \cdot 7 \cdot 13 \cdot 97$
(19, 97)	19	$2^4 \cdot 7^2$	(74, 76)	74	$3^3 \cdot 13 \cdot 31$
(20, 22)	20	$3^3 \cdot 13 \cdot 31$	(74, 82)	74	$2^4 \cdot 3^3 \cdot 7$
(20, 30)	20	$2^4 \cdot 3^3 \cdot 5^2 \cdot 7 \cdot 19$	(74, 85)	74	$2^8 \cdot 13$
(20, 32)	20	$2^4 \cdot 3^3 \cdot 7$	(74, 96)	74	$7^2 \cdot 31$
(20, 34)	20	$3^3 \cdot 5^2$	(75, 79)	75	$7 \cdot 19$
(20, 43)	20	$2^8 \cdot 13$	(75, 85)	75	13

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(20, 60)	20	$2^4 \cdot 3^3 \cdot 31$	(75, 86)	75	$2^2 \cdot 7^2$
(20, 62)	20	$7^2 \cdot 31$	(76, 82)	76	$3^2 \cdot 7$
(21, 25)	21	$7 \cdot 19$	(76, 93)	76	$2^4 \cdot 3 \cdot 5^4$
(21, 34)	21	2^{12}	(76, 96)	76	$3 \cdot 31 \cdot 379$
(21, 36)	21	7	(76, 97)	76	$3^8 \cdot 5^2$
(21, 37)	21	$2^2 \cdot 7 \cdot 13$	(77, 79)	77	$2^4 \cdot 7$
(21, 43)	21	13	(77, 83)	77	$2^{10} \cdot 7$
(21, 44)	21	$2^2 \cdot 7^2$	(77, 88)	77	$2^4 \cdot 31$
(21, 53)	21	$2^2 \cdot 7 \cdot 19 \cdot 43$	(77, 91)	77	$2^2 \cdot 7^2 \cdot 19^2$
(21, 60)	21	$2^4 \cdot 7^2 \cdot 19$	(77, 95)	77	$2^4 \cdot 7^2 \cdot 13$
(21, 63)	21	7^4	(78, 84)	78	$2^8 \cdot 5^2$
(22, 27)	22	$3 \cdot 7 \cdot 31$	(79, 85)	79	$2^4 \cdot 19$
(22, 31)	22	$2^4 \cdot 3^4 \cdot 5^4 \cdot 31$	(79, 89)	79	$13 \cdot 19$
(22, 32)	22	$3^2 \cdot 7$	(79, 90)	79	$13 \cdot 19$
(22, 34)	22	$2^6 \cdot 3^2 \cdot 31$	(79, 93)	79	$5^2 \cdot 7$
(22, 40)	22	$3^4 \cdot 5^2 \cdot 7 \cdot 19$	(79, 94)	79	7^2
(22, 51)	22	$2^2 \cdot 3^2 \cdot 7 \cdot 379$	(79, 96)	79	$7 \cdot 13$
(22, 55)	22	$2^2 \cdot 3 \cdot 7$	(80, 89)	80	$2^2 \cdot 7 \cdot 19$
(22, 57)	22	$2^4 \cdot 3 \cdot 5^4$	(80, 92)	80	$2^4 \cdot 3^3 \cdot 7 \cdot 19^2$
(22, 60)	22	$3^5 \cdot 31 \cdot 37$	(81, 90)	81	$7^2 \cdot 19^2$
(22, 61)	22	$2^2 \cdot 3^5 \cdot 7 \cdot 43$	(81, 91)	81	$2^2 \cdot 7 \cdot 19^2$
(22, 62)	22	$3 \cdot 31 \cdot 379$	(81, 95)	81	$2^4 \cdot 7^2$
(22, 64)	22	$3^8 \cdot 5^2$	(82, 83)	82	$2^{10} \cdot 7$
(23, 25)	23	$2^4 \cdot 7$	(82, 88)	82	$2^6 \cdot 7^2$
(23, 35)	23	$2^{10} \cdot 7$	(82, 96)	82	$5^2 \cdot 7$
(23, 37)	23	$7 \cdot 13$	(83, 88)	83	$2^4 \cdot 7 \cdot 13 \cdot 31$
(23, 40)	23	$2^{10} \cdot 7$	(83, 89)	83	$2^2 \cdot 7 \cdot 19 \cdot 73$
(23, 46)	23	$2^4 \cdot 31$	(83, 90)	83	$2^2 \cdot 7 \cdot 19^2$
(23, 49)	23	$2^2 \cdot 7^2 \cdot 19^2$	(84, 89)	84	$2^2 \cdot 3^4 \cdot 7$
(23, 51)	23	$2^2 \cdot 7^4$	(84, 90)	84	$5^2 \cdot 7^3$
(23, 52)	23	$2^2 \cdot 13 \cdot 19 \cdot 31^2$	(86, 96)	86	$7^3 \cdot 13$
(23, 56)	23	$7^2 \cdot 31$	(86, 97)	86	$2^2 \cdot 5^4 \cdot 7^2$
(23, 59)	23	$2^4 \cdot 7^2 \cdot 13$	(88, 91)	88	$2^2 \cdot 3 \cdot 7 \cdot 31$
(23, 60)	23	$2^6 \cdot 19$	(90, 94)	90	$2^8 \cdot 61$
(23, 61)	23	$2^4 \cdot 7 \cdot 19 \cdot 31$	(91, 92)	91	$2^6 \cdot 3^4 \cdot 7 \cdot 19$
<i>The second certificate.</i>					
(0, 6)	0	2^4	(19, 61)	19	$2^2 \cdot 7$
(0, 12)	0	2^2	(19, 62)	19	$2^2 \cdot 7$
(0, 14)	0	$2^6 \cdot 3^2$	(19, 63)	19	13
(0, 15)	0	$2^2 \cdot 3^2 \cdot 5^2$	(19, 65)	19	$2^4 \cdot 19$
(0, 16)	0	$2^2 \cdot 3^2$	(19, 67)	19	13
(0, 17)	0	$2^2 \cdot 3^2$	(19, 75)	19	$2^2 \cdot 7$
(0, 24)	0	5^2	(19, 77)	19	$2^2 \cdot 7 \cdot 31$
(0, 39)	0	$2^2 \cdot 3^2 \cdot 5^2$	(19, 84)	19	$5^2 \cdot 7^2$
(0, 40)	0	$2^2 \cdot 3^2 \cdot 5^2$	(19, 85)	19	$7 \cdot 19$
(0, 41)	0	$2^2 \cdot 3^2$	(19, 96)	19	$2^4 \cdot 5^2$
(0, 42)	0	$2^2 \cdot 3^2$	(20, 22)	20	$3^3 \cdot 13 \cdot 31$
(0, 44)	0	2^2	(20, 30)	20	$2^4 \cdot 3^3 \cdot 5^2 \cdot 7 \cdot 19$
(0, 60)	0	$2^6 \cdot 3^2$	(20, 32)	20	$2^4 \cdot 3^3 \cdot 7$
(0, 61)	0	$2^6 \cdot 3^2$	(20, 55)	20	$2^2 \cdot 3^3 \cdot 5^2 \cdot 13$
(0, 62)	0	2^2	(20, 60)	20	$2^4 \cdot 3^3 \cdot 31$
(0, 65)	0	2^4	(20, 63)	20	$2^2 \cdot 3^3 \cdot 7$
(0, 70)	0	2^2	(21, 36)	21	7
(0, 72)	0	$2^6 \cdot 3^2$	(21, 39)	21	$2^8 \cdot 7^2$
(0, 73)	0	$2^2 \cdot 3^2 \cdot 5^2$	(21, 41)	21	$2^2 \cdot 7^2 \cdot 13$
(0, 78)	0	5^2	(21, 42)	21	$7 \cdot 19$
(0, 86)	0	2^2	(21, 43)	21	13

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(0, 96)	0	2^2	(21, 45)	21	$5^2 \cdot 19 \cdot 43$
(1, 6)	1	$2^2 \cdot 3 \cdot 13$	(21, 59)	21	$2^4 \cdot 5^2 \cdot 7^2$
(1, 8)	1	$7 \cdot 13 \cdot 19$	(21, 60)	21	$2^4 \cdot 7^2 \cdot 19$
(1, 9)	1	$3 \cdot 7^2$	(21, 61)	21	$2^6 \cdot 7 \cdot 19$
(1, 14)	1	$2^2 \cdot 3 \cdot 7^2 \cdot 19$	(22, 24)	22	$2^8 \cdot 5^2$
(1, 23)	1	$7 \cdot 19$	(22, 31)	22	$2^4 \cdot 3^4 \cdot 5^4 \cdot 31$
(1, 26)	1	$2^2 \cdot 3 \cdot 7 \cdot 19$	(22, 32)	22	$3^2 \cdot 7$
(1, 27)	1	$2^2 \cdot 3 \cdot 7 \cdot 19$	(22, 39)	22	$2^2 \cdot 3^4 \cdot 13$
(1, 30)	1	$2^4 \cdot 3^3 \cdot 5^2 \cdot 19$	(22, 41)	22	$3^2 \cdot 5^2 \cdot 7^2 \cdot 19$
(1, 31)	1	$2^4 \cdot 3^3 \cdot 5^2 \cdot 19$	(22, 57)	22	$2^4 \cdot 3 \cdot 5^4$
(1, 33)	1	$13 \cdot 67$	(22, 59)	22	$2^2 \cdot 3^2 \cdot 7 \cdot 379$
(1, 34)	1	$13 \cdot 67$	(22, 61)	22	$2^2 \cdot 3^5 \cdot 7 \cdot 43$
(1, 39)	1	$3 \cdot 5^2 \cdot 7 \cdot 13$	(22, 62)	22	$3 \cdot 31 \cdot 379$
(1, 40)	1	$3 \cdot 5^2 \cdot 7 \cdot 13$	(23, 25)	23	$2^4 \cdot 7$
(1, 44)	1	$2^2 \cdot 5^2 \cdot 7$	(23, 27)	23	$2^2 \cdot 7^2$
(1, 47)	1	$2^2 \cdot 7 \cdot 13$	(23, 31)	23	$2^4 \cdot 3^4 \cdot 7 \cdot 19$
(1, 60)	1	$2^2 \cdot 3^3 \cdot 7 \cdot 67$	(23, 35)	23	$2^{10} \cdot 7$
(1, 61)	1	$2^2 \cdot 3^3 \cdot 7 \cdot 67$	(23, 40)	23	$2^{10} \cdot 7$
(1, 65)	1	$2^2 \cdot 3 \cdot 13$	(23, 43)	23	$2^8 \cdot 13$
(1, 67)	1	$7 \cdot 13 \cdot 19$	(23, 47)	23	$2^2 \cdot 3^4$
(1, 68)	1	$3 \cdot 7^2$	(23, 60)	23	$2^6 \cdot 19$
(1, 72)	1	$2^2 \cdot 3 \cdot 7^2 \cdot 19$	(24, 37)	24	2^4
(1, 77)	1	$7 \cdot 19$	(24, 44)	24	7^5
(1, 86)	1	$2^2 \cdot 5^2 \cdot 7$	(24, 56)	24	$5^2 \cdot 7^3$
(1, 89)	1	$2^2 \cdot 7 \cdot 13$	(24, 62)	24	$5^2 \cdot 7$
(2, 6)	2	$2^4 \cdot 3$	(25, 39)	25	$19 \cdot 61$
(2, 10)	2	7^2	(25, 40)	25	$2^2 \cdot 5^2 \cdot 7 \cdot 13$
(2, 21)	2	$2^2 \cdot 7^2$	(25, 41)	25	$7 \cdot 19$
(2, 23)	2	2^8	(25, 55)	25	$2^2 \cdot 7 \cdot 13 \cdot 19$
(2, 26)	2	$2^2 \cdot 3$	(25, 57)	25	$5^2 \cdot 7$
(2, 27)	2	$2^2 \cdot 3$	(25, 60)	25	$2^6 \cdot 7 \cdot 19$
(2, 38)	2	1	(25, 61)	25	$7 \cdot 19 \cdot 61$
(2, 39)	2	$2^4 \cdot 3$	(25, 62)	25	$7 \cdot 13$
(2, 40)	2	$2^4 \cdot 3$	(26, 29)	26	$2^2 \cdot 7$
(2, 47)	2	$2^2 \cdot 7$	(26, 30)	26	$2^2 \cdot 3^4 \cdot 7^5$
(2, 57)	2	$2^2 \cdot 7^2$	(26, 32)	26	$2^4 \cdot 7^2$
(2, 59)	2	2^2	(26, 33)	26	$2^2 \cdot 7^2 \cdot 31$
(2, 60)	2	$2^2 \cdot 3^3 \cdot 7$	(26, 38)	26	$2^2 \cdot 7^2 \cdot 31$
(2, 61)	2	$2^2 \cdot 3^3 \cdot 7$	(26, 40)	26	$2^2 \cdot 3^2 \cdot 7^2$
(2, 65)	2	$2^4 \cdot 3$	(26, 45)	26	$2^4 \cdot 3^2 \cdot 5^2 \cdot 19$
(2, 69)	2	7^2	(26, 55)	26	$2^2 \cdot 5^2 \cdot 97$
(2, 75)	2	$2^2 \cdot 7^2$	(26, 64)	26	$2^2 \cdot 3^4 \cdot 7$
(2, 77)	2	2^8	(26, 68)	26	$3 \cdot 7 \cdot 31$
(2, 84)	2	1	(26, 72)	26	$2^4 \cdot 3^2 \cdot 5^2 \cdot 19$
(2, 89)	2	$2^2 \cdot 7$	(26, 73)	26	$3^2 \cdot 7 \cdot 97$
(2, 93)	2	$2^2 \cdot 7^2$	(26, 77)	26	$2^2 \cdot 7^2$
(2, 95)	2	2^2	(26, 85)	26	$2^2 \cdot 3^4 \cdot 31$
(3, 8)	3	13^2	(26, 87)	26	$3^2 \cdot 5^2 \cdot 31^2$
(3, 18)	3	$2^2 \cdot 31$	(27, 31)	27	$2^2 \cdot 3^4 \cdot 7^5$
(3, 19)	3	$2^2 \cdot 31$	(27, 34)	27	$2^2 \cdot 7^2 \cdot 31$
(3, 20)	3	13^2	(27, 39)	27	$2^2 \cdot 3^2 \cdot 7^2$
(3, 23)	3	$7^2 \cdot 13 \cdot 31$	(27, 43)	27	$2^2 \cdot 3^4 \cdot 31$
(3, 29)	3	$2^2 \cdot 13 \cdot 19^2$	(27, 45)	27	$3^2 \cdot 5^2 \cdot 31^2$
(3, 36)	3	$7 \cdot 19$	(27, 55)	27	$2^2 \cdot 5^2 \cdot 97$
(3, 37)	3	$7 \cdot 19$	(27, 63)	27	$2^2 \cdot 19$
(3, 57)	3	2^2	(27, 67)	27	$7^2 \cdot 19$
(3, 67)	3	13^2	(27, 81)	27	$2^2 \cdot 7$

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(3, 74)	3	13^2	(27, 82)	27	$2^4 \cdot 7^2$
(3, 77)	3	$7^2 \cdot 13 \cdot 31$	(27, 84)	27	$2^2 \cdot 7^2 \cdot 31$
(3, 81)	3	$2^2 \cdot 13 \cdot 19^2$	(27, 87)	27	$2^4 \cdot 3^2 \cdot 5^2 \cdot 19$
(3, 93)	3	2^2	(27, 97)	27	$2^2 \cdot 3^4 \cdot 7$
(4, 26)	4	$2^2 \cdot 5^2$	(29, 37)	29	19
(4, 27)	4	$2^2 \cdot 5^2$	(29, 38)	29	7^4
(4, 30)	4	$2^6 \cdot 3^3 \cdot 7 \cdot 31$	(29, 56)	29	$2^4 \cdot 19$
(4, 31)	4	$2^6 \cdot 3^3 \cdot 7 \cdot 31$	(30, 32)	30	$2^4 \cdot 3^8 \cdot 7$
(4, 39)	4	$3^2 \cdot 7^2$	(30, 39)	30	$2^4 \cdot 3^8 \cdot 7 \cdot 13$
(4, 40)	4	$3^2 \cdot 7^2$	(30, 43)	30	$2^4 \cdot 3^8 \cdot 19 \cdot 31$
(4, 43)	4	$7 \cdot 31$	(30, 61)	30	$2^4 \cdot 3^{18} \cdot 5^2 \cdot 7 \cdot 31$
(4, 45)	4	$5^2 \cdot 7^3$	(30, 76)	30	$2^4 \cdot 3^8 \cdot 5^4 \cdot 31$
(4, 47)	4	$2^2 \cdot 7$	(30, 77)	30	$2^4 \cdot 3^8 \cdot 7 \cdot 19$
(4, 59)	4	$2^2 \cdot 7$	(30, 85)	30	$3^8 \cdot 7 \cdot 19^4$
(4, 85)	4	$7 \cdot 31$	(30, 87)	30	$2^2 \cdot 3^8 \cdot 7 \cdot 19^4$
(4, 87)	4	$5^2 \cdot 7^3$	(30, 89)	30	$2^8 \cdot 3^8 \cdot 13 \cdot 19$
(4, 89)	4	$2^2 \cdot 7$	(31, 40)	31	$2^4 \cdot 3^8 \cdot 7 \cdot 13$
(4, 95)	4	$2^2 \cdot 7$	(31, 43)	31	$3^8 \cdot 7 \cdot 19^4$
(5, 6)	5	$3 \cdot 7$	(31, 45)	31	$2^2 \cdot 3^8 \cdot 7 \cdot 19^4$
(5, 9)	5	$3 \cdot 7$	(31, 47)	31	$2^8 \cdot 3^8 \cdot 13 \cdot 19$
(5, 10)	5	$5^2 \cdot 7^2$	(31, 60)	31	$2^4 \cdot 3^{18} \cdot 5^2 \cdot 7 \cdot 31$
(5, 22)	5	3^6	(31, 69)	31	$2^4 \cdot 3^8 \cdot 5^4 \cdot 7^2$
(5, 23)	5	$7 \cdot 19$	(31, 74)	31	$2^4 \cdot 3^9 \cdot 5^2 \cdot 7 \cdot 19$
(5, 25)	5	$7 \cdot 19$	(31, 82)	31	$2^4 \cdot 3^8 \cdot 7$
(5, 26)	5	$2^4 \cdot 3 \cdot 7$	(31, 85)	31	$2^4 \cdot 3^8 \cdot 19 \cdot 31$
(5, 27)	5	$2^4 \cdot 3 \cdot 7$	(32, 35)	32	$2^{10} \cdot 7$
(5, 36)	5	$3 \cdot 7$	(32, 55)	32	$2^2 \cdot 3 \cdot 7$
(5, 37)	5	$3 \cdot 7$	(32, 60)	32	$2^6 \cdot 3 \cdot 7$
(5, 41)	5	$3 \cdot 5^2 \cdot 31$	(32, 61)	32	$2^4 \cdot 3 \cdot 7$
(5, 42)	5	$3 \cdot 5^2 \cdot 31$	(33, 38)	33	$5^2 \cdot 31$
(5, 43)	5	3^2	(33, 41)	33	$3^4 \cdot 7 \cdot 13$
(5, 45)	5	$3^2 \cdot 7 \cdot 31$	(33, 55)	33	$2^2 \cdot 3^4 \cdot 5^2 \cdot 13$
(5, 57)	5	$2^4 \cdot 3 \cdot 7$	(33, 59)	33	$2^4 \cdot 7 \cdot 97$
(5, 64)	5	$2^4 \cdot 3^4$	(33, 61)	33	$2^4 \cdot 3 \cdot 67$
(5, 65)	5	$3 \cdot 7$	(33, 69)	33	$7 \cdot 13$
(5, 68)	5	$3 \cdot 7$	(33, 71)	33	2^4
(5, 69)	5	$5^2 \cdot 7^2$	(33, 73)	33	$3 \cdot 7$
(5, 76)	5	3^6	(33, 93)	33	$2^4 \cdot 3^4$
(5, 77)	5	$7 \cdot 19$	(34, 42)	34	$3^4 \cdot 7 \cdot 13$
(5, 79)	5	$7 \cdot 19$	(34, 55)	34	$2^2 \cdot 3^4 \cdot 5^2 \cdot 13$
(5, 85)	5	3^2	(34, 57)	34	$2^4 \cdot 3^4$
(5, 87)	5	$3^2 \cdot 7 \cdot 31$	(34, 60)	34	$2^4 \cdot 3 \cdot 67$
(5, 93)	5	$2^4 \cdot 3 \cdot 7$	(34, 69)	34	2^4
(5, 97)	5	$2^4 \cdot 3^4$	(34, 70)	34	2^2
(6, 8)	6	$7 \cdot 13$	(34, 73)	34	$2^2 \cdot 3 \cdot 97$
(6, 18)	6	$2^4 \cdot 19$	(34, 84)	34	$5^2 \cdot 31$
(6, 19)	6	2^6	(34, 95)	34	$2^4 \cdot 7 \cdot 97$
(6, 20)	6	$13^2 \cdot 19$	(36, 47)	36	7
(6, 32)	6	2^6	(36, 68)	36	7^2
(6, 43)	6	$3 \cdot 13$	(36, 69)	36	13
(6, 44)	6	2^4	(36, 70)	36	7
(6, 47)	6	$2^4 \cdot 7 \cdot 13$	(36, 73)	36	$3 \cdot 5^2 \cdot 7$
(6, 55)	6	$2^2 \cdot 7 \cdot 13 \cdot 19$	(36, 78)	36	2^4
(6, 60)	6	$2^4 \cdot 3^2 \cdot 7$	(36, 81)	36	19
(6, 63)	6	$7 \cdot 19$	(36, 85)	36	13
(8, 9)	8	7	(36, 86)	36	$5^2 \cdot 7$
(8, 13)	8	7	(37, 43)	37	13

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(8, 16)	8	$2^2 \cdot 13$	(37, 44)	37	$5^2 \cdot 7$
(8, 18)	8	13	(37, 67)	37	$2^2 \cdot 7$
(8, 25)	8	$2^4 \cdot 7$	(37, 68)	37	7^2
(8, 26)	8	$7^2 \cdot 19$	(37, 70)	37	13
(8, 29)	8	$7^2 \cdot 13$	(37, 75)	37	7
(8, 36)	8	$2^2 \cdot 7$	(37, 89)	37	7
(8, 38)	8	7^2	(38, 40)	38	7^2
(8, 42)	8	$5^2 \cdot 7^2 \cdot 13$	(38, 61)	38	$2^4 \cdot 7^3$
(8, 60)	8	$2^4 \cdot 5^2$	(39, 40)	39	$3^8 \cdot 5^2$
(9, 10)	9	7^5	(39, 41)	39	$2^2 \cdot 3 \cdot 7 \cdot 13 \cdot 19$
(9, 13)	9	$3 \cdot 7$	(39, 45)	39	$3^2 \cdot 13$
(9, 14)	9	$3 \cdot 5^2 \cdot 109$	(39, 47)	39	$2^2 \cdot 13$
(9, 16)	9	$2^2 \cdot 3 \cdot 7 \cdot 31$	(39, 57)	39	$2^4 \cdot 3 \cdot 7$
(9, 17)	9	$2^2 \cdot 3 \cdot 7$	(39, 61)	39	$2^4 \cdot 3^5 \cdot 61$
(9, 19)	9	$2^2 \cdot 5^2 \cdot 7$	(39, 68)	39	$3 \cdot 7^2$
(9, 24)	9	$7^2 \cdot 31$	(39, 70)	39	$2^2 \cdot 7^2 \cdot 19$
(9, 27)	9	$3 \cdot 7 \cdot 31$	(39, 71)	39	$3^4 \cdot 19$
(9, 32)	9	7^3	(39, 77)	39	$2^{10} \cdot 7$
(9, 36)	9	7^2	(39, 79)	39	$2^2 \cdot 5^2 \cdot 7 \cdot 13$
(9, 37)	9	7^2	(39, 84)	39	7^2
(9, 40)	9	$3 \cdot 7^2$	(39, 95)	39	$2^4 \cdot 3^2 \cdot 5^2 \cdot 7^2$
(9, 45)	9	$3^4 \cdot 5^2 \cdot 31$	(40, 42)	40	$2^2 \cdot 3 \cdot 7 \cdot 13 \cdot 19$
(9, 55)	9	$2^2 \cdot 7^2$	(40, 59)	40	$2^4 \cdot 3^2 \cdot 5^2 \cdot 7^2$
(9, 59)	9	$7^2 \cdot 109$	(40, 60)	40	$2^4 \cdot 3^5 \cdot 61$
(9, 60)	9	$3^2 \cdot 5^2$	(40, 63)	40	$2^2 \cdot 3^3 \cdot 7$
(9, 61)	9	$3^2 \cdot 7^3$	(40, 75)	40	$2^8 \cdot 7^2$
(10, 13)	10	7^3	(40, 76)	40	$2^2 \cdot 3^4 \cdot 13$
(10, 16)	10	$2^2 \cdot 7 \cdot 97$	(40, 79)	40	$19 \cdot 61$
(10, 20)	10	$2^2 \cdot 7^2$	(40, 87)	40	$3^2 \cdot 13$
(10, 25)	10	$2^2 \cdot 7^2 \cdot 13$	(40, 89)	40	$2^2 \cdot 13$
(10, 30)	10	$2^4 \cdot 3^4 \cdot 5^4 \cdot 7^2$	(40, 93)	40	$2^4 \cdot 3 \cdot 7$
(10, 33)	10	2^4	(41, 42)	41	$2^2 \cdot 3^4 \cdot 7^2$
(10, 34)	10	$7 \cdot 13$	(41, 45)	41	$7 \cdot 13 \cdot 31 \cdot 37$
(10, 37)	10	13	(41, 55)	41	$2^2 \cdot 3^4 \cdot 7 \cdot 13^2$
(10, 41)	10	$5^2 \cdot 7 \cdot 13$	(41, 64)	41	$2^2 \cdot 3^2 \cdot 7^4$
(10, 45)	10	7^6	(41, 67)	41	$5^2 \cdot 7^2 \cdot 13$
(10, 59)	10	$7^3 \cdot 97$	(41, 70)	41	$2^2 \cdot 7^2 \cdot 13$
(10, 61)	10	$5^2 \cdot 7^3 \cdot 37$	(41, 73)	41	$2^{10} \cdot 3 \cdot 5^2$
(10, 64)	10	$2^2 \cdot 7 \cdot 37$	(41, 75)	41	$7 \cdot 19$
(11, 13)	11	$2^2 \cdot 7$	(42, 55)	42	$2^2 \cdot 3^4 \cdot 7 \cdot 13^2$
(11, 15)	11	$3^4 \cdot 13$	(42, 69)	42	$5^2 \cdot 7 \cdot 13$
(11, 18)	11	13	(42, 73)	42	$2^2 \cdot 3 \cdot 7^2 \cdot 13 \cdot 37$
(11, 19)	11	7^2	(42, 75)	42	$2^2 \cdot 7^2 \cdot 13$
(11, 21)	11	$2^2 \cdot 7 \cdot 43$	(42, 76)	42	$3^2 \cdot 5^2 \cdot 7^2 \cdot 19$
(11, 22)	11	$2^8 \cdot 3 \cdot 7$	(42, 79)	42	$7 \cdot 19$
(11, 25)	11	$7 \cdot 19$	(42, 87)	42	$7 \cdot 13 \cdot 31 \cdot 37$
(11, 26)	11	$2^2 \cdot 19$	(42, 97)	42	$2^2 \cdot 3^2 \cdot 7^4$
(11, 32)	11	$3^4 \cdot 7$	(43, 45)	43	$3^2 \cdot 7 \cdot 19 \cdot 31$
(11, 39)	11	$2^2 \cdot 3^3 \cdot 7$	(45, 47)	45	$2^2 \cdot 7^2 \cdot 19$
(11, 45)	11	$7 \cdot 19 \cdot 43$	(45, 52)	45	$2^4 \cdot 3^2 \cdot 5^2 \cdot 7^3 \cdot 19^2$
(11, 59)	11	2^6	(47, 61)	47	$2^4 \cdot 7$
(11, 60)	11	$3^2 \cdot 7^2$	(47, 62)	47	$7^3 \cdot 13$
(11, 61)	11	$2^2 \cdot 3^2 \cdot 7^2$	(47, 64)	47	$2^4 \cdot 7^4$
(11, 62)	11	7^4	(52, 71)	52	$3^2 \cdot 7^2 \cdot 19$
(11, 63)	11	3^6	(53, 87)	53	$2^4 \cdot 3^2 \cdot 5^2 \cdot 7^3 \cdot 19^2$
(11, 65)	11	$7 \cdot 19$	(55, 57)	55	$2^4 \cdot 3^4 \cdot 7^2$
(11, 72)	11	$3^2 \cdot 7^2 \cdot 19$	(55, 65)	55	$2^2 \cdot 7 \cdot 13 \cdot 19$

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(11, 74)	11	$2^2 \cdot 3^3 \cdot 7$	(55, 68)	55	$2^2 \cdot 7^2$
(11, 95)	11	7^4	(55, 71)	55	$2^2 \cdot 7 \cdot 13$
(11, 97)	11	$2^2 \cdot 3 \cdot 7$	(55, 72)	55	$2^2 \cdot 3^2 \cdot 5^2$
(12, 13)	12	19	(55, 73)	55	$2^2 \cdot 3^2 \cdot 7 \cdot 13 \cdot 97$
(12, 20)	12	7^4	(55, 74)	55	$2^2 \cdot 3^3 \cdot 5^2 \cdot 13$
(12, 24)	12	7	(55, 79)	55	$2^2 \cdot 7 \cdot 13 \cdot 19$
(12, 33)	12	2^2	(55, 82)	55	$2^2 \cdot 3 \cdot 7$
(12, 36)	12	13	(55, 93)	55	$2^4 \cdot 3^4 \cdot 7^2$
(12, 37)	12	7	(56, 62)	56	$2^2 \cdot 7^3$
(12, 40)	12	$2^2 \cdot 7^2 \cdot 19$	(56, 71)	56	7^3
(12, 42)	12	$2^2 \cdot 7^2 \cdot 13$	(56, 78)	56	$5^2 \cdot 7^3$
(13, 16)	13	$2^2 \cdot 3^2$	(56, 81)	56	$2^4 \cdot 19$
(13, 22)	13	$2^4 \cdot 3 \cdot 19$	(56, 96)	56	$2^2 \cdot 7^3$
(13, 34)	13	2^4	(57, 59)	57	$2^6 \cdot 7^3$
(13, 40)	13	$3^4 \cdot 19$	(59, 61)	59	$2^{16} \cdot 7^2$
(13, 47)	13	$2^2 \cdot 7$	(59, 62)	59	$7^2 \cdot 379$
(13, 53)	13	$3^2 \cdot 7^2 \cdot 19$	(59, 63)	59	7^4
(13, 55)	13	$2^2 \cdot 7 \cdot 13$	(60, 62)	60	$3^3 \cdot 31$
(13, 56)	13	7^3	(60, 63)	60	$2^2 \cdot 3^2 \cdot 7^2$
(13, 59)	13	$2^2 \cdot 7 \cdot 13$	(60, 68)	60	$3^2 \cdot 7^3$
(13, 60)	13	3^3	(60, 69)	60	$5^2 \cdot 7^3 \cdot 37$
(13, 64)	13	$3^2 \cdot 7$	(60, 72)	60	$2^6 \cdot 3^5$
(14, 16)	14	$2^6 \cdot 3^4$	(60, 73)	60	$3^5 \cdot 43$
(14, 19)	14	$2^4 \cdot 7$	(60, 75)	60	$2^6 \cdot 7 \cdot 19$
(14, 21)	14	$2^6 \cdot 7^3$	(60, 76)	60	$2^2 \cdot 3^5 \cdot 7 \cdot 43$
(14, 27)	14	$2^4 \cdot 3^2 \cdot 5^2 \cdot 19$	(60, 79)	60	$7 \cdot 19 \cdot 61$
(14, 32)	14	$3 \cdot 7$	(60, 82)	60	$2^4 \cdot 3 \cdot 7$
(14, 55)	14	$2^2 \cdot 3^2 \cdot 5^2$	(60, 84)	60	$2^4 \cdot 7^3$
(14, 59)	14	$2^4 \cdot 3^2 \cdot 7 \cdot 109$	(60, 89)	60	$2^4 \cdot 7$
(14, 60)	14	$3^4 \cdot 5^2 \cdot 7^2$	(60, 95)	60	$2^{16} \cdot 7^2$
(14, 61)	14	$2^6 \cdot 3^5$	(60, 96)	60	$3^2 \cdot 7^3$
(14, 63)	14	$3^2 \cdot 7^2 \cdot 19$	(60, 97)	60	$2^4 \cdot 3^3 \cdot 37$
(15, 17)	15	$2^2 \cdot 3^6 \cdot 7$	(61, 62)	61	$3^2 \cdot 7^3$
(15, 22)	15	$3^4 \cdot 7^2 \cdot 43$	(61, 63)	61	$3^2 \cdot 7^2$
(15, 27)	15	$3^2 \cdot 7 \cdot 97$	(61, 64)	61	$2^4 \cdot 3^3 \cdot 37$
(15, 33)	15	$2^2 \cdot 3 \cdot 97$	(61, 65)	61	$2^4 \cdot 3^2 \cdot 7$
(15, 34)	15	$3 \cdot 7$	(61, 67)	61	$2^4 \cdot 5^2$
(15, 37)	15	$3 \cdot 5^2 \cdot 7$	(61, 68)	61	$3^2 \cdot 5^2$
(15, 41)	15	$2^2 \cdot 3 \cdot 7^2 \cdot 13 \cdot 37$	(61, 71)	61	3^3
(15, 42)	15	$2^{10} \cdot 3 \cdot 5^2$	(61, 72)	61	$3^4 \cdot 5^2 \cdot 7^2$
(15, 45)	15	$2^2 \cdot 3^2 \cdot 5^2 \cdot 37$	(61, 74)	61	$2^4 \cdot 3^3 \cdot 31$
(15, 55)	15	$2^2 \cdot 3^2 \cdot 7 \cdot 13 \cdot 97$	(61, 75)	61	$2^4 \cdot 7^2 \cdot 19$
(15, 59)	15	$3^2 \cdot 7 \cdot 13 \cdot 97$	(61, 77)	61	$2^6 \cdot 19$
(15, 61)	15	$3^5 \cdot 43$	(61, 79)	61	$2^6 \cdot 7 \cdot 19$
(16, 19)	16	$2^2 \cdot 13$	(61, 82)	61	$2^6 \cdot 3 \cdot 7$
(16, 24)	16	$2^2 \cdot 7^2 \cdot 31$	(61, 96)	61	$3^3 \cdot 31$
(16, 25)	16	$2^2 \cdot 7^2$	(63, 64)	63	$2^2 \cdot 3 \cdot 7$
(16, 26)	16	$2^2 \cdot 3^2$	(63, 71)	63	$2^2 \cdot 7$
(16, 30)	16	$2^4 \cdot 3^4 \cdot 5^2$	(63, 73)	63	$3^4 \cdot 13$
(16, 33)	16	$2^2 \cdot 3^3 \cdot 13$	(63, 75)	63	$2^2 \cdot 7 \cdot 43$
(16, 36)	16	$2^2 \cdot 3 \cdot 7$	(63, 76)	63	$2^8 \cdot 3 \cdot 7$
(16, 37)	16	$2^2 \cdot 3 \cdot 7$	(63, 79)	63	$7 \cdot 19$
(16, 38)	16	$2^4 \cdot 7$	(63, 82)	63	$3^4 \cdot 7$
(16, 40)	16	$2^2 \cdot 3^5 \cdot 7$	(63, 87)	63	$7 \cdot 19 \cdot 43$
(16, 41)	16	$2^2 \cdot 3 \cdot 13$	(63, 95)	63	2^6
(16, 43)	16	$2^2 \cdot 3 \cdot 7$	(63, 96)	63	7^4
(16, 44)	16	$2^2 \cdot 5^2 \cdot 7^2$	(65, 67)	65	$7 \cdot 13$

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(16, 57)	16	$2^4 \cdot 3^3 \cdot 7$	(65, 74)	65	$13^2 \cdot 19$
(16, 59)	16	$2^2 \cdot 3^2 \cdot 97$	(65, 82)	65	2^6
(16, 60)	16	$2^2 \cdot 3^4 \cdot 7$	(65, 85)	65	$3 \cdot 13$
(16, 68)	16	$2^2 \cdot 3 \cdot 7$	(65, 86)	65	2^4
(16, 73)	16	$2^2 \cdot 3^6 \cdot 7$	(65, 89)	65	$2^4 \cdot 7 \cdot 13$
(16, 76)	16	$2^2 \cdot 3^5 \cdot 7$	(67, 68)	67	7
(16, 78)	16	$2^2 \cdot 7^2$	(67, 71)	67	7
(17, 18)	17	$2^2 \cdot 13$	(67, 79)	67	$2^4 \cdot 7$
(17, 22)	17	$2^2 \cdot 3^5 \cdot 7$	(67, 81)	67	$7^2 \cdot 13$
(17, 24)	17	$2^2 \cdot 7^2$	(67, 84)	67	7^2
(17, 27)	17	$2^2 \cdot 3^2$	(68, 69)	68	7^5
(17, 31)	17	$2^4 \cdot 3^4 \cdot 5^2$	(68, 71)	68	$3 \cdot 7$
(17, 34)	17	$2^2 \cdot 3^3 \cdot 13$	(68, 72)	68	$3 \cdot 5^2 \cdot 109$
(17, 36)	17	$2^2 \cdot 3 \cdot 7$	(68, 78)	68	$7^2 \cdot 31$
(17, 37)	17	$2^2 \cdot 3 \cdot 7$	(68, 82)	68	7^3
(17, 39)	17	$2^2 \cdot 3^5 \cdot 7$	(68, 87)	68	$3^4 \cdot 5^2 \cdot 31$
(17, 42)	17	$2^2 \cdot 3 \cdot 13$	(68, 95)	68	$7^2 \cdot 109$
(17, 61)	17	$2^2 \cdot 3^4 \cdot 7$	(69, 71)	69	7^3
(17, 67)	17	$2^2 \cdot 13$	(69, 74)	69	$2^2 \cdot 7^2$
(17, 68)	17	$2^2 \cdot 3 \cdot 7 \cdot 31$	(69, 79)	69	$2^2 \cdot 7^2 \cdot 13$
(17, 69)	17	$2^2 \cdot 7 \cdot 97$	(69, 87)	69	7^6
(17, 71)	17	$2^2 \cdot 3^2$	(69, 95)	69	$7^3 \cdot 97$
(17, 72)	17	$2^6 \cdot 3^4$	(69, 97)	69	$2^2 \cdot 7 \cdot 37$
(17, 78)	17	$2^2 \cdot 7^2 \cdot 31$	(70, 71)	70	19
(17, 79)	17	$2^2 \cdot 7^2$	(70, 74)	70	7^4
(17, 84)	17	$2^4 \cdot 7$	(70, 78)	70	7
(17, 85)	17	$2^2 \cdot 3 \cdot 7$	(71, 76)	71	$2^4 \cdot 3 \cdot 19$
(17, 86)	17	$2^2 \cdot 5^2 \cdot 7^2$	(71, 89)	71	$2^2 \cdot 7$
(17, 93)	17	$2^4 \cdot 3^3 \cdot 7$	(71, 95)	71	$2^2 \cdot 7 \cdot 13$
(17, 95)	17	$2^2 \cdot 3^2 \cdot 97$	(71, 97)	71	$3^2 \cdot 7$
(18, 21)	18	$2^2 \cdot 7$	(72, 75)	72	$2^6 \cdot 7^3$
(18, 23)	18	$2^2 \cdot 7 \cdot 31$	(72, 82)	72	$3 \cdot 7$
(18, 30)	18	$2^2 \cdot 3^6 \cdot 19^2$	(72, 95)	72	$2^4 \cdot 3^2 \cdot 7 \cdot 109$
(18, 31)	18	$2^2 \cdot 3^6 \cdot 7$	(73, 76)	73	$3^4 \cdot 7^2 \cdot 43$
(18, 38)	18	$5^2 \cdot 7^2$	(73, 87)	73	$2^2 \cdot 3^2 \cdot 5^2 \cdot 37$
(18, 39)	18	$2^2 \cdot 7$	(73, 95)	73	$3^2 \cdot 7 \cdot 13 \cdot 97$
(18, 41)	18	$2^2 \cdot 7 \cdot 13$	(74, 76)	74	$3^3 \cdot 13 \cdot 31$
(18, 43)	18	$7 \cdot 19$	(74, 82)	74	$2^4 \cdot 3^3 \cdot 7$
(18, 53)	18	$2^4 \cdot 5^2 \cdot 7 \cdot 19$	(75, 85)	75	13
(18, 56)	18	$2^2 \cdot 5^2 \cdot 7 \cdot 19$	(75, 87)	75	$5^2 \cdot 19 \cdot 43$
(18, 60)	18	$2^2 \cdot 7$	(75, 95)	75	$2^4 \cdot 5^2 \cdot 7^2$
(18, 61)	18	2^2	(76, 78)	76	$2^8 \cdot 5^2$
(18, 62)	18	$2^4 \cdot 5^2$	(76, 82)	76	$3^2 \cdot 7$
(18, 63)	18	7^2	(76, 93)	76	$2^4 \cdot 3 \cdot 5^4$
(18, 65)	18	2^6	(76, 95)	76	$2^2 \cdot 3^2 \cdot 7 \cdot 379$
(18, 68)	18	$2^2 \cdot 5^2 \cdot 7$	(76, 96)	76	$3 \cdot 31 \cdot 379$
(18, 72)	18	$2^4 \cdot 7$	(77, 79)	77	$2^4 \cdot 7$
(18, 81)	18	$7 \cdot 19$	(77, 83)	77	$2^{10} \cdot 7$
(18, 86)	18	$2^2 \cdot 5^2 \cdot 7$	(77, 85)	77	$2^8 \cdot 13$
(18, 87)	18	$2^4 \cdot 5^2 \cdot 7 \cdot 19$	(77, 89)	77	$2^2 \cdot 3^4$
(18, 89)	18	$2^2 \cdot 7$	(78, 86)	78	7^5
(18, 96)	18	$2^2 \cdot 7$	(78, 96)	78	$5^2 \cdot 7$
(19, 29)	19	$7 \cdot 19$	(79, 93)	79	$5^2 \cdot 7$
(19, 30)	19	$2^2 \cdot 3^6 \cdot 7$	(79, 96)	79	$7 \cdot 13$
(19, 31)	19	$2^2 \cdot 3^6 \cdot 19^2$	(81, 84)	81	7^4
(19, 40)	19	$2^2 \cdot 7$	(82, 83)	82	$2^{10} \cdot 7$
(19, 42)	19	$2^2 \cdot 7 \cdot 13$	(85, 87)	85	$3^2 \cdot 7 \cdot 19 \cdot 31$

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(19, 44)	19	$2^2 \cdot 5^2 \cdot 7$	(87, 89)	87	$2^2 \cdot 7^2 \cdot 19$
(19, 45)	19	$2^4 \cdot 5^2 \cdot 7 \cdot 19$	(89, 96)	89	$7^3 \cdot 13$
(19, 47)	19	$2^2 \cdot 7$	(89, 97)	89	$2^4 \cdot 7^4$
(19, 52)	19	$2^4 \cdot 5^2 \cdot 7 \cdot 19$	(93, 95)	93	$2^6 \cdot 7^3$
(19, 56)	19	$2^2 \cdot 5^2 \cdot 7 \cdot 19$	(95, 96)	95	$7^2 \cdot 379$
(19, 60)	19	2^2			
<i>The third certificate.</i>					
(0, 6)	0	2^4	(19, 47)	19	$2^2 \cdot 7$
(0, 11)	0	3^2	(19, 56)	19	$2^2 \cdot 5^2 \cdot 7 \cdot 19$
(0, 16)	0	$2^2 \cdot 3^2$	(19, 60)	19	2^2
(0, 17)	0	$2^2 \cdot 3^2$	(19, 61)	19	$2^2 \cdot 7$
(0, 23)	0	2^2	(19, 68)	19	$2^2 \cdot 7$
(0, 32)	0	$2^2 \cdot 3^2$	(19, 75)	19	$2^2 \cdot 7$
(0, 59)	0	2^2	(19, 77)	19	$2^2 \cdot 7 \cdot 31$
(0, 60)	0	$2^6 \cdot 3^2$	(19, 78)	19	$5^2 \cdot 7^2$
(0, 61)	0	$2^6 \cdot 3^2$	(19, 82)	19	$2^2 \cdot 5^2 \cdot 7$
(0, 63)	0	3^2	(19, 84)	19	$5^2 \cdot 7^2$
(0, 65)	0	2^4	(19, 96)	19	$2^4 \cdot 5^2$
(0, 77)	0	2^2	(19, 97)	19	$2^4 \cdot 7^2$
(0, 82)	0	$2^2 \cdot 3^2$	(20, 22)	20	$3^3 \cdot 13 \cdot 31$
(0, 95)	0	2^2	(20, 23)	20	$2^8 \cdot 13$
(1, 8)	1	$7 \cdot 13 \cdot 19$	(20, 33)	20	$2^2 \cdot 3^3 \cdot 13$
(1, 11)	1	$7 \cdot 19$	(20, 39)	20	$3^3 \cdot 13^3$
(1, 22)	1	$3^2 \cdot 5^2 \cdot 7 \cdot 13$	(20, 61)	20	$3^3 \cdot 7^3$
(1, 39)	1	$3 \cdot 5^2 \cdot 7 \cdot 13$	(20, 62)	20	$7^2 \cdot 31$
(1, 40)	1	$3 \cdot 5^2 \cdot 7 \cdot 13$	(21, 25)	21	$7 \cdot 19$
(1, 47)	1	$2^2 \cdot 7 \cdot 13$	(21, 60)	21	$2^4 \cdot 7^2 \cdot 19$
(1, 57)	1	$2^2 \cdot 7$	(22, 51)	22	$2^2 \cdot 3^2 \cdot 7 \cdot 379$
(1, 59)	1	$2^2 \cdot 7$	(22, 55)	22	$2^2 \cdot 3 \cdot 7$
(1, 63)	1	$7 \cdot 19$	(22, 62)	22	$3 \cdot 31 \cdot 379$
(1, 64)	1	$2^2 \cdot 3 \cdot 5^2 \cdot 7$	(22, 64)	22	$3^8 \cdot 5^2$
(1, 67)	1	$7 \cdot 13 \cdot 19$	(23, 25)	23	$2^4 \cdot 7$
(1, 76)	1	$3^2 \cdot 5^2 \cdot 7 \cdot 13$	(23, 46)	23	$2^4 \cdot 31$
(1, 89)	1	$2^2 \cdot 7 \cdot 13$	(23, 56)	23	$7^2 \cdot 31$
(1, 93)	1	$2^2 \cdot 7$	(23, 63)	23	$2^2 \cdot 7$
(1, 95)	1	$2^2 \cdot 7$	(24, 32)	24	$5^2 \cdot 7$
(1, 97)	1	$2^2 \cdot 3 \cdot 5^2 \cdot 7$	(24, 37)	24	2^4
(2, 12)	2	2^2	(24, 44)	24	7^5
(2, 57)	2	$2^2 \cdot 7^2$	(24, 56)	24	$5^2 \cdot 7^3$
(2, 59)	2	2^2	(24, 60)	24	7^2
(2, 60)	2	$2^2 \cdot 3^3 \cdot 7$	(25, 60)	25	$2^6 \cdot 7 \cdot 19$
(2, 61)	2	$2^2 \cdot 3^3 \cdot 7$	(26, 38)	26	$2^2 \cdot 7^2 \cdot 31$
(2, 70)	2	2^2	(26, 40)	26	$2^2 \cdot 3^2 \cdot 7^2$
(2, 93)	2	$2^2 \cdot 7^2$	(26, 47)	26	$2^8 \cdot 7$
(2, 95)	2	2^2	(26, 57)	26	$2^{10} \cdot 7$
(3, 8)	3	13^2	(26, 59)	26	$2^4 \cdot 7^3$
(3, 18)	3	$2^2 \cdot 31$	(26, 60)	26	$2^4 \cdot 3^4 \cdot 7^2$
(3, 19)	3	$2^2 \cdot 31$	(26, 68)	26	$3 \cdot 7 \cdot 31$
(3, 20)	3	13^2	(26, 72)	26	$2^4 \cdot 3^2 \cdot 5^2 \cdot 19$
(3, 23)	3	$7^2 \cdot 13 \cdot 31$	(26, 97)	26	$2^{10} \cdot 3$
(3, 29)	3	$2^2 \cdot 13 \cdot 19^2$	(27, 39)	27	$2^2 \cdot 3^2 \cdot 7^2$
(3, 36)	3	$7 \cdot 19$	(27, 61)	27	$2^4 \cdot 3^4 \cdot 7^2$
(3, 37)	3	$7 \cdot 19$	(27, 64)	27	$2^{10} \cdot 3$
(3, 62)	3	$2^2 \cdot 3^4$	(27, 65)	27	$2^8 \cdot 3 \cdot 7$
(3, 67)	3	13^2	(27, 67)	27	$7^2 \cdot 19$
(3, 74)	3	13^2	(27, 72)	27	$3 \cdot 5^2 \cdot 7^3$
(3, 77)	3	$7^2 \cdot 13 \cdot 31$	(27, 84)	27	$2^2 \cdot 7^2 \cdot 31$

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(3, 81)	3	$2^2 \cdot 13 \cdot 19^2$	(27, 89)	27	$2^8 \cdot 7$
(3, 96)	3	$2^2 \cdot 3^4$	(27, 93)	27	$2^{10} \cdot 7$
(4, 16)	4	$2^2 \cdot 3^2 \cdot 7$	(27, 95)	27	$2^4 \cdot 7^3$
(4, 17)	4	$2^2 \cdot 3^2 \cdot 7$	(29, 37)	29	19
(4, 26)	4	$2^2 \cdot 5^2$	(29, 38)	29	7^4
(4, 27)	4	$2^2 \cdot 5^2$	(29, 56)	29	$2^4 \cdot 19$
(4, 32)	4	$3^2 \cdot 7$	(29, 61)	29	$2^4 \cdot 7^2$
(4, 39)	4	$3^2 \cdot 7^2$	(32, 34)	32	$3^4 \cdot 61$
(4, 40)	4	$3^2 \cdot 7^2$	(32, 51)	32	$2^4 \cdot 5^2 \cdot 7^2$
(4, 47)	4	$2^2 \cdot 7$	(32, 56)	32	$7^2 \cdot 61$
(4, 50)	4	$2^2 \cdot 7^2$	(32, 59)	32	$2^4 \cdot 7^2$
(4, 51)	4	$2^2 \cdot 7^2$	(32, 60)	32	$2^6 \cdot 3 \cdot 7$
(4, 59)	4	$2^2 \cdot 7$	(32, 61)	32	$2^4 \cdot 3 \cdot 7$
(4, 64)	4	$2^2 \cdot 3^2 \cdot 7^2$	(33, 37)	33	$3 \cdot 7$
(4, 82)	4	$3^2 \cdot 7$	(33, 38)	33	$5^2 \cdot 31$
(4, 89)	4	$2^2 \cdot 7$	(33, 42)	33	$2^2 \cdot 3^6$
(4, 95)	4	$2^2 \cdot 7$	(33, 56)	33	$7^2 \cdot 61$
(4, 97)	4	$2^2 \cdot 3^2 \cdot 7^2$	(33, 64)	33	$2^2 \cdot 3 \cdot 7$
(5, 8)	5	$5^2 \cdot 7^2$	(33, 68)	33	$5^2 \cdot 31$
(5, 10)	5	$5^2 \cdot 7^2$	(33, 71)	33	2^4
(5, 47)	5	2^4	(33, 82)	33	$3^4 \cdot 61$
(5, 64)	5	$2^4 \cdot 3^4$	(34, 36)	34	$3 \cdot 7$
(5, 67)	5	$5^2 \cdot 7^2$	(34, 41)	34	$2^2 \cdot 3^6$
(5, 69)	5	$5^2 \cdot 7^2$	(34, 56)	34	$7^2 \cdot 61$
(5, 89)	5	2^4	(34, 65)	34	$2^4 \cdot 7 \cdot 13$
(5, 97)	5	$2^4 \cdot 3^4$	(34, 74)	34	$2^2 \cdot 3^3 \cdot 13$
(6, 8)	6	$7 \cdot 13$	(34, 84)	34	$5^2 \cdot 31$
(6, 13)	6	$3 \cdot 13$	(34, 97)	34	$2^2 \cdot 3 \cdot 7$
(6, 16)	6	$2^2 \cdot 3 \cdot 13$	(36, 40)	36	$2^4 \cdot 3 \cdot 7$
(6, 17)	6	$2^2 \cdot 3 \cdot 7^2$	(36, 60)	36	$2^2 \cdot 3 \cdot 7$
(6, 21)	6	7	(36, 61)	36	$3 \cdot 7$
(6, 26)	6	$2^8 \cdot 3 \cdot 7$	(36, 63)	36	$2^2 \cdot 3 \cdot 7$
(6, 32)	6	2^6	(36, 68)	36	7^2
(6, 33)	6	$2^4 \cdot 7 \cdot 13$	(36, 78)	36	2^4
(6, 36)	6	7	(36, 81)	36	19
(6, 40)	6	$2^4 \cdot 3$	(36, 96)	36	7
(6, 55)	6	$2^2 \cdot 7 \cdot 13 \cdot 19$	(37, 39)	37	$2^4 \cdot 3 \cdot 7$
(6, 57)	6	$2^6 \cdot 7$	(37, 60)	37	$3 \cdot 7$
(6, 60)	6	$2^4 \cdot 3^2 \cdot 7$	(37, 61)	37	$2^2 \cdot 3 \cdot 7$
(6, 62)	6	$3 \cdot 7$	(37, 62)	37	7
(6, 63)	6	$7 \cdot 19$	(37, 65)	37	7
(8, 10)	8	$2^8 \cdot 3^4 \cdot 7$	(37, 67)	37	$2^2 \cdot 7$
(8, 16)	8	$2^2 \cdot 13$	(37, 68)	37	7^2
(8, 21)	8	7^2	(37, 72)	37	$3 \cdot 7^3$
(8, 26)	8	$7^2 \cdot 19$	(38, 40)	38	7^2
(8, 29)	8	$7^2 \cdot 13$	(38, 61)	38	$2^4 \cdot 7^3$
(8, 32)	8	$2^4 \cdot 7$	(39, 40)	39	$3^8 \cdot 5^2$
(8, 36)	8	$2^2 \cdot 7$	(39, 47)	39	$2^2 \cdot 13$
(8, 38)	8	7^2	(39, 60)	39	$2^4 \cdot 3^4 \cdot 7^3$
(8, 60)	8	$2^4 \cdot 5^2$	(39, 65)	39	$2^4 \cdot 3$
(9, 16)	9	$2^2 \cdot 3 \cdot 7 \cdot 31$	(39, 68)	39	$3 \cdot 7^2$
(9, 18)	9	$2^2 \cdot 7$	(39, 70)	39	$2^2 \cdot 7^2 \cdot 19$
(9, 19)	9	$2^2 \cdot 5^2 \cdot 7$	(39, 71)	39	$3^4 \cdot 19$
(9, 24)	9	$7^2 \cdot 31$	(39, 72)	39	$3^5 \cdot 7$
(9, 27)	9	$3 \cdot 7 \cdot 31$	(39, 84)	39	7^2
(9, 34)	9	$5^2 \cdot 31$	(40, 61)	40	$2^4 \cdot 3^4 \cdot 7^3$
(9, 36)	9	7^2	(40, 72)	40	$2^4 \cdot 3^9$

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(9, 37)	9	7^2	(40, 74)	40	$3^3 \cdot 13^3$
(9, 40)	9	$3 \cdot 7^2$	(40, 89)	40	$2^2 \cdot 13$
(9, 47)	9	$5^2 \cdot 7^2$	(41, 57)	41	3^4
(9, 55)	9	$2^2 \cdot 7^2$	(41, 64)	41	$2^2 \cdot 3^2 \cdot 7^4$
(9, 60)	9	$3^2 \cdot 5^2$	(42, 93)	42	3^4
(9, 61)	9	$3^2 \cdot 7^3$	(42, 97)	42	$2^2 \cdot 3^2 \cdot 7^4$
(10, 13)	10	7^3	(46, 50)	46	$2^4 \cdot 31$
(10, 23)	10	$2^2 \cdot 7^2$	(47, 60)	47	$2^4 \cdot 5^2 \cdot 7$
(10, 61)	10	$5^2 \cdot 7^3 \cdot 37$	(47, 61)	47	$2^4 \cdot 7$
(10, 64)	10	$2^2 \cdot 7 \cdot 37$	(47, 64)	47	$2^4 \cdot 7^4$
(11, 19)	11	7^2	(50, 51)	50	$2^6 \cdot 3^6 \cdot 7^2$
(11, 22)	11	$2^8 \cdot 3 \cdot 7$	(50, 56)	50	$3^6 \cdot 31$
(11, 32)	11	$3^4 \cdot 7$	(50, 59)	50	$2^4 \cdot 3^2 \cdot 7^2 \cdot 31$
(11, 37)	11	$2^2 \cdot 3 \cdot 7$	(50, 64)	50	$2^{10} \cdot 3^2 \cdot 7$
(11, 55)	11	$2^6 \cdot 3^2 \cdot 19$	(50, 76)	50	$2^2 \cdot 3^2 \cdot 7 \cdot 379$
(11, 60)	11	$3^2 \cdot 7^2$	(50, 82)	50	$2^4 \cdot 5^2 \cdot 7^2$
(11, 65)	11	$7 \cdot 19$	(50, 96)	50	$5^2 \cdot 7^2 \cdot 379$
(11, 72)	11	$3^2 \cdot 7^2 \cdot 19$	(51, 56)	51	$3^6 \cdot 31$
(11, 77)	11	$2^2 \cdot 7$	(51, 62)	51	$5^2 \cdot 7^2 \cdot 379$
(12, 13)	12	19	(51, 88)	51	$2^4 \cdot 31$
(12, 19)	12	2^6	(51, 95)	51	$2^4 \cdot 3^2 \cdot 7^2 \cdot 31$
(12, 20)	12	7^4	(51, 97)	51	$2^{10} \cdot 3^2 \cdot 7$
(12, 23)	12	2^2	(55, 57)	55	$2^4 \cdot 3^4 \cdot 7^2$
(12, 38)	12	7^4	(55, 63)	55	$2^6 \cdot 3^2 \cdot 19$
(12, 40)	12	$2^2 \cdot 7^2 \cdot 19$	(55, 65)	55	$2^2 \cdot 7 \cdot 13 \cdot 19$
(12, 44)	12	$3^4 \cdot 7$	(55, 68)	55	$2^2 \cdot 7^2$
(12, 56)	12	7	(55, 71)	55	$2^2 \cdot 7 \cdot 13$
(12, 60)	12	2^2	(55, 76)	55	$2^2 \cdot 3 \cdot 7$
(13, 17)	13	$2^2 \cdot 3 \cdot 7$	(55, 93)	55	$2^4 \cdot 3^4 \cdot 7^2$
(13, 34)	13	2^4	(56, 62)	56	$2^2 \cdot 7^3$
(13, 40)	13	$3^4 \cdot 19$	(56, 70)	56	7
(13, 55)	13	$2^2 \cdot 7 \cdot 13$	(56, 77)	56	$7^2 \cdot 31$
(13, 60)	13	3^3	(56, 78)	56	$5^2 \cdot 7^3$
(13, 62)	13	$3^2 \cdot 7$	(56, 81)	56	$2^4 \cdot 19$
(14, 19)	14	$2^4 \cdot 7$	(56, 82)	56	$7^2 \cdot 61$
(14, 26)	14	$3 \cdot 5^2 \cdot 7^3$	(56, 96)	56	$2^2 \cdot 7^3$
(14, 27)	14	$2^4 \cdot 3^2 \cdot 5^2 \cdot 19$	(59, 61)	59	$2^{16} \cdot 7^2$
(14, 32)	14	$3 \cdot 7$	(60, 68)	60	$3^2 \cdot 7^3$
(14, 36)	14	$3 \cdot 7^3$	(60, 69)	60	$5^2 \cdot 7^3 \cdot 37$
(14, 39)	14	$2^4 \cdot 3^9$	(60, 74)	60	$3^3 \cdot 7^3$
(14, 40)	14	$3^5 \cdot 7$	(60, 81)	60	$2^4 \cdot 7^2$
(14, 63)	14	$3^2 \cdot 7^2 \cdot 19$	(60, 82)	60	$2^4 \cdot 3 \cdot 7$
(14, 64)	14	$2^4 \cdot 3^4 \cdot 7^2$	(60, 84)	60	$2^4 \cdot 7^3$
(16, 20)	16	$2^2 \cdot 3^3 \cdot 13$	(60, 89)	60	$2^4 \cdot 7$
(16, 22)	16	$2^2 \cdot 3^4 \cdot 13$	(60, 95)	60	$2^{16} \cdot 7^2$
(16, 24)	16	$2^2 \cdot 7^2 \cdot 31$	(60, 97)	60	$2^4 \cdot 3^3 \cdot 37$
(16, 25)	16	$2^2 \cdot 7^2$	(61, 63)	61	$3^2 \cdot 7^2$
(16, 26)	16	$2^2 \cdot 3^2$	(61, 64)	61	$2^4 \cdot 3^3 \cdot 37$
(16, 36)	16	$2^2 \cdot 3 \cdot 7$	(61, 65)	61	$2^4 \cdot 3^2 \cdot 7$
(16, 38)	16	$2^4 \cdot 7$	(61, 67)	61	$2^4 \cdot 5^2$
(16, 51)	16	$2^6 \cdot 3^2 \cdot 31$	(61, 68)	61	$3^2 \cdot 5^2$
(16, 57)	16	$2^4 \cdot 3^3 \cdot 7$	(61, 70)	61	2^2
(16, 62)	16	$2^4 \cdot 3^3 \cdot 7$	(61, 71)	61	3^3
(16, 65)	16	$2^2 \cdot 3 \cdot 7^2$	(61, 75)	61	$2^4 \cdot 7^2 \cdot 19$
(16, 71)	16	$2^2 \cdot 3 \cdot 7$	(61, 78)	61	7^2
(16, 75)	16	$2^2 \cdot 7^2$	(61, 79)	61	$2^6 \cdot 7 \cdot 19$
(16, 78)	16	$2^2 \cdot 7^2$	(61, 82)	61	$2^6 \cdot 3 \cdot 7$

(i, j)	k	$N(R_{ij})$	(i, j)	k	$N(R_{ij})$
(16, 95)	16	$2^4 \cdot 3^2 \cdot 31$	(61, 89)	61	$2^4 \cdot 5^2 \cdot 7$
(16, 96)	16	$2^2 \cdot 3 \cdot 7$	(62, 64)	62	$3^2 \cdot 7$
(17, 21)	17	$2^2 \cdot 7^2$	(63, 76)	63	$2^8 \cdot 3 \cdot 7$
(17, 24)	17	$2^2 \cdot 7^2$	(63, 82)	63	$3^4 \cdot 7$
(17, 27)	17	$2^2 \cdot 3^2$	(65, 67)	65	$7 \cdot 13$
(17, 37)	17	$2^2 \cdot 3 \cdot 7$	(65, 71)	65	$3 \cdot 13$
(17, 50)	17	$2^6 \cdot 3^2 \cdot 31$	(65, 75)	65	7
(17, 59)	17	$2^4 \cdot 3^2 \cdot 31$	(65, 82)	65	2^6
(17, 62)	17	$2^2 \cdot 3 \cdot 7$	(65, 93)	65	$2^6 \cdot 7$
(17, 65)	17	$2^2 \cdot 3 \cdot 13$	(65, 96)	65	$3 \cdot 7$
(17, 67)	17	$2^2 \cdot 13$	(67, 69)	67	$2^8 \cdot 3^4 \cdot 7$
(17, 68)	17	$2^2 \cdot 3 \cdot 7 \cdot 31$	(67, 75)	67	7^2
(17, 74)	17	$2^2 \cdot 3^3 \cdot 13$	(67, 81)	67	$7^2 \cdot 13$
(17, 76)	17	$2^2 \cdot 3^4 \cdot 13$	(67, 82)	67	$2^4 \cdot 7$
(17, 78)	17	$2^2 \cdot 7^2 \cdot 31$	(67, 84)	67	7^2
(17, 79)	17	$2^2 \cdot 7^2$	(68, 78)	68	$7^2 \cdot 31$
(17, 84)	17	$2^4 \cdot 7$	(68, 89)	68	$5^2 \cdot 7^2$
(17, 93)	17	$2^4 \cdot 3^3 \cdot 7$	(69, 71)	69	7^3
(17, 96)	17	$2^4 \cdot 3^3 \cdot 7$	(69, 77)	69	$2^2 \cdot 7^2$
(18, 21)	18	$2^2 \cdot 7$	(69, 97)	69	$2^2 \cdot 7 \cdot 37$
(18, 23)	18	$2^2 \cdot 7 \cdot 31$	(70, 71)	70	19
(18, 24)	18	$5^2 \cdot 7^2$	(70, 74)	70	7^4
(18, 32)	18	$2^2 \cdot 5^2 \cdot 7$	(70, 77)	70	2^2
(18, 34)	18	2^2	(70, 84)	70	7^4
(18, 38)	18	$5^2 \cdot 7^2$	(70, 86)	70	$3^4 \cdot 7$
(18, 39)	18	$2^2 \cdot 7$	(71, 96)	71	$3^2 \cdot 7$
(18, 56)	18	$2^2 \cdot 5^2 \cdot 7 \cdot 19$	(72, 82)	72	$3 \cdot 7$
(18, 60)	18	$2^2 \cdot 7$	(72, 97)	72	$2^4 \cdot 3^4 \cdot 7^2$
(18, 61)	18	2^2	(74, 76)	74	$3^3 \cdot 13 \cdot 31$
(18, 62)	18	$2^4 \cdot 5^2$	(74, 77)	74	$2^8 \cdot 13$
(18, 63)	18	7^2	(74, 96)	74	$7^2 \cdot 31$
(18, 64)	18	$2^4 \cdot 7^2$	(75, 79)	75	$7 \cdot 19$
(18, 68)	18	$2^2 \cdot 5^2 \cdot 7$	(76, 96)	76	$3 \cdot 31 \cdot 379$
(18, 70)	18	2^6	(76, 97)	76	$3^8 \cdot 5^2$
(18, 72)	18	$2^4 \cdot 7$	(77, 79)	77	$2^4 \cdot 7$
(18, 78)	18	5^2	(77, 88)	77	$2^4 \cdot 31$
(18, 81)	18	$7 \cdot 19$	(78, 82)	78	$5^2 \cdot 7$
(18, 89)	18	$2^2 \cdot 7$	(78, 86)	78	7^5
(19, 24)	19	5^2	(81, 84)	81	7^4
(19, 29)	19	$7 \cdot 19$	(82, 95)	82	$2^4 \cdot 7^2$
(19, 33)	19	2^2	(89, 97)	89	$2^4 \cdot 7^4$
(19, 40)	19	$2^2 \cdot 7$	(96, 97)	96	$3^2 \cdot 7$

Table 9: The terminal residue classes $x \equiv a, y \equiv b \pmod{p^k}$ of Step 6, grouped by p, k and the joint value $((\mathcal{B}_p^{(g)})_g; (B_p^{(g)})_g)$.

p	k	value	classes (a, b)
2	4	$(2, 0, 0; 1, -1)$	$(0, 12), (4, 12), (6, 4), (8, 12), (12, 12), (14, 4)$
2	5	$(2, 0, 0; 1, 1)$	$(4, 3), (4, 19), (12, 7), (12, 23), (20, 11), (20, 27)$
2	5	$(2, 1, 2; -1, 1)$	$(17, 11), (21, 23)$
2	5	$(2, 1, 2; 1, -1)$	$(3, 17), (7, 13), (11, 9), (15, 5), (19, 1), (23, 29), (27, 25), (31, 21)$
2	6	$(2, 0, 0; -1, -1)$	$(4, 1), (4, 33), (12, 29), (20, 9), (20, 41), (28, 37), (36, 17),$ $(36, 49), (52, 25), (52, 57)$
2	6	$(2, 0, 0; 1, -1)$	$(10, 20), (26, 20), (42, 20), (58, 20)$
2	6	$(2, 1, 2; -1, 1)$	$(1, 59), (5, 39), (9, 19), (13, 63), (25, 3), (29, 47), (33, 27),$ $(37, 7), (41, 51), (45, 31), (57, 35), (61, 15)$

p	k	value	classes (a, b)
2	7	$(2, 0, 0; -1, -1)$	$(28, 5), (28, 69), (44, 45), (44, 109), (60, 21), (60, 85), (76, 61), (76, 125), (108, 13), (108, 77), (124, 53), (124, 117)$
2	7	$(2, 0, 0; 1, -1)$	$(2, 52), (18, 116), (34, 52), (50, 116), (66, 52), (82, 116), (98, 52), (114, 116)$
2	7	$(2, 0, 0; 1, 1)$	$(28, 47), (28, 111), (60, 31), (60, 95), (92, 15), (92, 79)$
2	9	$(2, 0, 0; 1, 1)$	$(124, 63), (124, 319), (380, 191), (380, 447), (508, 127), (508, 383)$
2	11	$(2, 0, 0; 1, 1)$	$(764, 767), (764, 1791), (1788, 255), (1788, 1279)$
2	12	$(2, 0, 0; 1, 1)$	$(252, 2047), (252, 4095), (1276, 1535), (1276, 3583), (2300, 1023), (2300, 3071), (3324, 511), (3324, 2559)$
3	4	$(0, 1, 1; -1, 1)$	$(2, 15), (11, 78), (20, 60), (29, 42), (38, 24), (47, 6), (56, 69), (65, 51), (74, 33)$
3	4	$(1, 0, 0; -1, 1)$	$(8, 48), (17, 3), (26, 39), (35, 75), (44, 30), (53, 66), (62, 21), (71, 57), (80, 12)$
3	4	$(2, 0, 0; -1, 1)$	$(13, 21), (13, 48), (13, 75), (40, 12), (40, 39), (40, 66), (67, 3), (67, 30), (67, 57)$
3	4	$(2, 1, 1; -1, 1)$	$(4, 6), (4, 33), (4, 60), (31, 24), (31, 51), (31, 78), (58, 15), (58, 42), (58, 69)$
3	4	$(2, 2, 2; -1, 1)$	$(5, 27), (14, 36), (22, 0), (22, 27), (22, 54), (23, 45), (32, 54), (41, 63), (49, 18), (49, 45), (49, 72), (50, 72), (59, 0), (68, 9), (76, 9), (76, 36), (76, 63), (77, 18)$
5	1	$(0, 0, 0; 1, 1)$	$(0, 3)$
5	1	$(0, 1, 0; 1, 1)$	$(3, 1)$
5	2	$(0, 0, 0; 1, 1)$	$(2, 9), (7, 14), (17, 24), (22, 4)$
5	2	$(0, 1, 0; 1, 1)$	$(3, 13), (8, 18), (13, 23), (18, 3), (23, 8)$
5	3	$(0, 0, 0; 1, 1)$	$(12, 69), (37, 94), (62, 119), (87, 19), (112, 44)$
5	3	$(0, 1, 0; 1, 1)$	$(1, 87), (6, 122), (11, 107), (16, 42), (21, 52), (26, 12), (31, 47), (36, 32), (41, 92), (46, 102), (51, 62), (56, 97), (61, 82), (66, 17), (71, 27), (76, 112), (81, 22), (86, 7), (91, 67), (96, 77), (101, 37), (106, 72), (111, 57), (116, 117), (121, 2)$
7	2	$(1, 2, 1; 1, 1)$	$(3, 35), (4, 5), (5, 2), (9, 1), (10, 0), (13, 12), (18, 28), (23, 8), (25, 12), (26, 23), (27, 40), (32, 9), (39, 0), (39, 16), (40, 6), (40, 13), (40, 27), (40, 37), (40, 41), (41, 19), (44, 43), (46, 19)$

p	k	value	classes (a, b)
7	3	(1, 2, 1; 1, 1)	(0, 277), (2, 267), (4, 128), (4, 259), (6, 145), (7, 144), (11, 217), (11, 233), (11, 236), (12, 205), (16, 78), (17, 161), (18, 173), (18, 191), (19, 65), (20, 124), (21, 25), (24, 175), (25, 2), (25, 231), (28, 39), (30, 330), (31, 189), (32, 47), (32, 287), (33, 30), (34, 250), (35, 102), (37, 64), (38, 203), (39, 327), (40, 34), (40, 83), (40, 181), (40, 230), (40, 279), (40, 328), (45, 217), (46, 154), (46, 268), (47, 93), (48, 180), (49, 32), (51, 120), (53, 177), (53, 308), (55, 243), (56, 242), (60, 138), (60, 266), (60, 282), (61, 254), (65, 274), (66, 259), (67, 75), (67, 240), (68, 114), (69, 222), (70, 123), (73, 273), (74, 51), (74, 280), (77, 137), (79, 183), (80, 287), (81, 292), (81, 336), (82, 79), (83, 5), (84, 200), (86, 260), (87, 301), (88, 229), (94, 315), (95, 317), (96, 142), (97, 278), (98, 130), (100, 316), (102, 14), (102, 226), (104, 341), (105, 340), (109, 40), (109, 315), (109, 331), (110, 303), (114, 127), (115, 14), (116, 289), (116, 320), (117, 163), (118, 320), (119, 221), (122, 28), (123, 100), (123, 329), (126, 235), (128, 36), (129, 42), (130, 42), (130, 194), (131, 128), (132, 103), (133, 298), (135, 113), (136, 56), (137, 131), (138, 48), (138, 97), (138, 146), (138, 195), (138, 244), (138, 293), (138, 342), (143, 70), (144, 23), (144, 252), (145, 191), (146, 33), (147, 228), (149, 169), (151, 63), (151, 275), (153, 96), (154, 95), (158, 21), (158, 37), (158, 285), (159, 9), (163, 323), (164, 112), (165, 222), (165, 338), (166, 212), (167, 75), (168, 319), (171, 126), (172, 35), (172, 149), (175, 333), (177, 232), (178, 140), (179, 91), (179, 96), (180, 177), (181, 201), (182, 53), (184, 309), (185, 154), (186, 33), (187, 20), (187, 69), (187, 118), (187, 167), (187, 216), (187, 314), (192, 168), (193, 72), (193, 301), (194, 240), (195, 131), (196, 326), (198, 22), (200, 112), (200, 324), (202, 194), (203, 193), (207, 70), (207, 86), (207, 187), (208, 58), (212, 176), (213, 210), (214, 44), (214, 124), (215, 261), (216, 173), (217, 74), (220, 224), (221, 84), (221, 198), (224, 88), (226, 85), (227, 238), (228, 140), (228, 341), (229, 226), (231, 151), (233, 162), (234, 252), (235, 278), (241, 266), (242, 7), (243, 289), (244, 229), (245, 81), (247, 218), (249, 30), (249, 161), (251, 292), (252, 291), (256, 89), (256, 119), (256, 135), (257, 107), (261, 29), (262, 308), (263, 26), (263, 93), (264, 310), (265, 271), (266, 172), (269, 322), (270, 133), (270, 247), (273, 186), (275, 281), (276, 336), (277, 189), (277, 243), (278, 275), (279, 54), (280, 249), (282, 15), (283, 7), (284, 180), (290, 21), (291, 56), (291, 170), (292, 338), (293, 327), (294, 179), (296, 71), (298, 79), (300, 47), (301, 46), (305, 168), (305, 184), (305, 334), (306, 156), (310, 225), (311, 63), (312, 142), (312, 271), (313, 16), (314, 26), (315, 270), (318, 77), (319, 182), (319, 296), (322, 284), (324, 134), (325, 91), (326, 145), (326, 238), (327, 324), (328, 152), (329, 4), (331, 211), (332, 105), (333, 82), (339, 119), (340, 105), (340, 219), (341, 44), (342, 82)

p	k	value	classes (a, b)
7	4	$(1, 2, 1; 1, 1)$	$(14, 403), (42, 1929), (63, 1187), (91, 1684), (95, 1232),$ $(112, 1971), (140, 1439), (161, 354), (187, 265), (187, 608),$ $(187, 1294), (187, 1637), (187, 1980), (187, 2323), (189, 1194),$ $(210, 1138), (230, 1671), (238, 949), (242, 121), (259, 1922),$ $(298, 553), (336, 459), (357, 1089), (385, 214), (406, 1873),$ $(434, 2370), (438, 1575), (455, 256), (483, 2125), (504, 1040),$ $(532, 1880), (553, 1824), (573, 2357), (581, 1635), (585, 464),$ $(602, 207), (641, 896), (679, 1145), (700, 1775), (728, 900),$ $(749, 158), (777, 655), (781, 1918), (798, 942), (826, 410),$ $(847, 1726), (875, 165), (896, 109), (916, 642), (924, 2321),$ $(928, 807), (945, 893), (973, 2076), (984, 1239), (1022, 1831),$ $(1043, 60), (1071, 1586), (1092, 844), (1120, 1341), (1124, 2261),$ $(1141, 1628), (1169, 1096), (1190, 11), (1218, 851), (1239, 795),$ $(1259, 1328), (1267, 606), (1271, 1150), (1288, 1579),$ $(1327, 1582), (1337, 2363), (1365, 116), (1386, 746),$ $(1414, 2272), (1435, 1530), (1463, 2027), (1467, 203),$ $(1484, 2314), (1512, 1782), (1533, 697), (1561, 1537),$ $(1582, 1481), (1602, 2014), (1610, 1292), (1614, 1493),$ $(1631, 2265), (1670, 1925), (1680, 648), (1708, 802),$ $(1729, 1432), (1757, 557), (1778, 2216), (1806, 312), (1810, 546),$ $(1827, 599), (1855, 67), (1876, 1383), (1904, 2223), (1925, 2167),$ $(1945, 299), (1953, 1978), (1957, 1836), (1974, 550),$ $(2013, 2268), (2051, 1488), (2072, 2118), (2098, 475),$ $(2098, 818), (2098, 1161), (2098, 1504), (2098, 1847),$ $(2098, 2190), (2100, 1243), (2121, 501), (2149, 998), (2153, 889),$ $(2170, 1285), (2198, 753), (2219, 2069), (2247, 508), (2268, 452),$ $(2288, 985), (2296, 263), (2300, 2179), (2317, 1236), (2345, 18),$ $(2356, 210), (2394, 2174)$
7	5	$(1, 2, 1; 1, 1)$	$(187, 951), (187, 5753), (187, 8154), (187, 10555), (187, 12956),$ $(187, 15357), (287, 10308), (308, 305), (630, 8593), (651, 5793),$ $(994, 11281), (1316, 5163), (1659, 3448), (2002, 1733),$ $(2023, 10938), (2366, 16426), (2688, 15110), (2709, 5107),$ $(3031, 13395), (3052, 10595), (3395, 16083), (3717, 9965),$ $(4060, 8250), (4403, 6535), (4424, 15740), (4767, 4421),$ $(5089, 3105), (5110, 9909), (5432, 1390), (5453, 15397),$ $(5796, 4078), (6118, 14767), (6461, 13052), (6804, 11337),$ $(6825, 3735), (7168, 9223), (7490, 7907), (7511, 14711),$ $(7833, 6192), (7854, 3392), (8197, 8880), (8519, 2762),$ $(8862, 1047), (9205, 16139), (9226, 8537), (9569, 14025),$ $(9891, 12709), (9912, 2706), (10234, 10994), (10255, 8194),$ $(10598, 13682), (10920, 7564), (11263, 5849), (11606, 4134),$ $(11627, 13339), (11970, 2020), (12292, 704), (12313, 7508),$ $(12635, 15796), (12656, 12996), (12999, 1677), (13321, 12366),$ $(13664, 10651), (14007, 8936), (14028, 1334), (14103, 132),$ $(14103, 2533), (14103, 4934), (14103, 9736), (14103, 12137),$ $(14103, 14538), (14371, 6822), (14693, 5506), (14714, 12310),$ $(15036, 3791), (15057, 991), (15400, 6479), (15722, 361),$ $(16065, 15453), (16408, 13738), (16429, 6136), (16772, 11624)$
7	6	$(1, 2, 1; 1, 1)$	$(47717, 7335), (47717, 24142), (47717, 40949), (47717, 57756),$ $(47717, 74563), (47717, 91370), (101029, 3352),$ $(101029, 20159), (101029, 36966), (101029, 53773),$ $(101029, 70580), (101029, 87387), (101029, 104194)$
7	7	$(1, 2, 1; 1, 1)$	$(283015, 108177), (283015, 225826), (283015, 343475),$ $(283015, 461124), (283015, 578773), (283015, 696422),$ $(283015, 814071)$
11	1	$(0, 0, 0; 1, 1)$	$(1, 9), (2, 2), (3, 0), (3, 3), (3, 8), (4, 7), (5, 9), (5, 10), (6, 1),$ $(7, 3), (7, 5), (8, 0), (8, 9)$

p	k	value	classes (a, b)
11	2	$(0, 0, 0; 1, 1)$	$(5, 91), (16, 14), (27, 58), (38, 102), (49, 25), (60, 69), (71, 113), (82, 36), (93, 80), (104, 3), (115, 47)$
11	3	$(0, 0, 0; 1, 1)$	$(8, 486), (19, 420), (30, 354), (41, 288), (52, 222), (63, 156), (74, 90), (85, 24), (96, 1289), (107, 1223), (118, 1157), (129, 1091), (140, 1025), (151, 959), (162, 893), (173, 827), (184, 761), (195, 695), (206, 629), (217, 563), (228, 497), (239, 431), (250, 365), (261, 299), (272, 233), (283, 167), (294, 101), (305, 35), (316, 1300), (327, 1234), (338, 1168), (349, 1102), (360, 1036), (371, 970), (382, 904), (393, 838), (404, 772), (415, 706), (426, 640), (437, 574), (448, 508), (459, 442), (470, 376), (481, 310), (492, 244), (503, 178), (514, 112), (525, 46), (536, 1311), (547, 1245), (558, 1179), (569, 1113), (580, 1047), (591, 981), (602, 915), (613, 849), (624, 783), (635, 717), (646, 651), (657, 585), (668, 519), (679, 453), (690, 387), (701, 321), (712, 255), (723, 189), (734, 123), (745, 57), (756, 1322), (767, 1256), (778, 1190), (789, 1124), (800, 1058), (811, 992), (822, 926), (833, 860), (844, 794), (855, 728), (866, 662), (877, 596), (888, 530), (899, 464), (910, 398), (921, 332), (932, 266), (943, 200), (954, 134), (965, 68), (976, 2), (987, 1267), (998, 1201), (1009, 1135), (1020, 1069), (1031, 1003), (1042, 937), (1053, 871), (1064, 805), (1075, 739), (1086, 673), (1097, 607), (1108, 541), (1119, 475), (1130, 409), (1141, 343), (1152, 277), (1163, 211), (1174, 145), (1185, 79), (1196, 13), (1207, 1278), (1218, 1212), (1229, 1146), (1240, 1080), (1251, 1014), (1262, 948), (1273, 882), (1284, 816), (1295, 750), (1306, 684), (1317, 618), (1328, 552)$
13	2	$(0, 0, 0; 1, 1)$	$(2, 162), (3, 102), (5, 15), (6, 102), (7, 33), (9, 113), (10, 57), (15, 97), (16, 76), (18, 93), (19, 89), (20, 150), (22, 113), (23, 83), (28, 32), (29, 50), (32, 76), (33, 98), (35, 113), (36, 109), (41, 136), (42, 24), (45, 63), (46, 46), (48, 113), (49, 135), (54, 71), (55, 167), (57, 158), (58, 50), (59, 163), (61, 113), (62, 161), (68, 141), (70, 67), (71, 37), (72, 111), (74, 113), (75, 18), (80, 110), (81, 115), (83, 145), (84, 24), (85, 59), (87, 113), (88, 44), (93, 45), (94, 89), (96, 54), (97, 11), (98, 7), (100, 113), (106, 149), (107, 63), (109, 132), (110, 167), (111, 124), (113, 113), (114, 96), (120, 37), (122, 41), (123, 154), (124, 72), (126, 113), (127, 122), (132, 19), (133, 11), (135, 119), (136, 141), (137, 20), (139, 113), (140, 148), (146, 154), (148, 28), (149, 128), (150, 137), (152, 113), (153, 5), (158, 58), (159, 128), (161, 106), (162, 115), (163, 85), (165, 113), (166, 31)$

p	k	value	classes (a, b)
13	3	$(0, 0, 0; 1, 1)$	(31, 1185), (44, 1094), (67, 2034), (101, 408), (119, 1605), (145, 1644), (200, 2), (213, 2108), (236, 1189), (270, 746), (288, 760), (314, 799), (369, 1016), (382, 925), (405, 344), (439, 1084), (457, 2112), (483, 2151), (538, 2030), (551, 1939), (574, 1696), (608, 1422), (626, 1267), (652, 1306), (707, 847), (720, 756), (743, 851), (777, 1760), (795, 422), (821, 461), (876, 1861), (889, 1770), (946, 2098), (964, 1774), (990, 1813), (1045, 678), (1058, 587), (1081, 1358), (1115, 239), (1133, 929), (1159, 968), (1214, 1692), (1227, 1601), (1250, 513), (1284, 577), (1302, 84), (1328, 123), (1383, 509), (1396, 418), (1419, 1865), (1453, 915), (1471, 1436), (1497, 1475), (1552, 1523), (1565, 1432), (1588, 1020), (1622, 1253), (1640, 591), (1666, 630), (1721, 340), (1734, 249), (1757, 175), (1791, 1591), (1809, 1943), (1835, 1982), (1890, 1354), (1903, 1263), (1926, 1527), (1960, 1929), (1978, 1098), (2004, 1137), (2059, 171), (2072, 80), (2095, 682), (2129, 70), (2147, 253), (2173, 292)
13	4	$(0, 0, 0; 1, 1)$	(912, 6), (3109, 17582), (5306, 6597), (7503, 24173), (9700, 13188), (11897, 2203), (14094, 19779), (16291, 8794), (18488, 26370), (20685, 15385), (22882, 4400), (25079, 21976), (27276, 10991)
19	1	$(2, 1, 0; 1, 1)$	(12, 6)
19	2	$(0, 1, 0; -1, 1)$	(13, 57), (32, 342), (51, 266), (70, 190), (89, 114), (108, 38), (127, 323), (146, 247), (165, 171), (184, 95), (203, 19), (222, 304), (241, 228), (260, 152), (279, 76), (298, 0), (317, 285), (336, 209), (355, 133)
19	2	$(0, 1, 0; 1, 1)$	(8, 292), (27, 83), (46, 235), (65, 26), (84, 178), (103, 330), (122, 121), (141, 273), (160, 64), (179, 216), (198, 7), (217, 159), (236, 311), (255, 102), (274, 254), (293, 45), (312, 197), (331, 349), (350, 140)
19	2	$(1, 1, 0; 1, 1)$	(2, 312), (9, 25), (13, 65), (15, 337), (21, 179), (32, 350), (34, 52), (40, 46), (47, 101), (51, 274), (53, 128), (59, 274), (66, 139), (70, 198), (72, 204), (78, 141), (85, 177), (89, 122), (91, 280), (97, 8), (104, 215), (108, 46), (110, 356), (116, 236), (127, 331), (129, 71), (135, 103), (146, 255), (148, 147), (154, 331), (165, 179), (167, 223), (173, 198), (184, 103), (186, 299), (192, 65), (203, 27), (205, 14), (211, 293), (218, 82), (222, 312), (224, 90), (230, 160), (237, 120), (241, 236), (243, 166), (249, 27), (260, 160), (262, 242), (268, 255), (279, 84), (287, 122), (298, 8), (300, 33), (306, 350), (317, 293), (319, 109), (325, 217), (332, 310), (336, 217), (338, 185), (344, 84), (355, 141), (357, 261)
19	2	$(2, 1, 0; -1, 1)$	(4, 249), (6, 247), (13, 239), (23, 211), (25, 285), (32, 30), (42, 173), (44, 323), (51, 182), (61, 135), (63, 0), (70, 334), (80, 97), (82, 38), (89, 125), (99, 59), (101, 76), (108, 277), (118, 21), (120, 114), (127, 68), (137, 344), (139, 152), (146, 220), (156, 306), (158, 190), (165, 11), (175, 268), (177, 228), (184, 163), (194, 230), (196, 266), (203, 315), (213, 192), (215, 304), (222, 106), (232, 154), (234, 342), (241, 258), (251, 116), (253, 19), (260, 49), (270, 78), (272, 57), (279, 201), (289, 40), (291, 95), (298, 353), (308, 2), (310, 133), (317, 144), (327, 325), (329, 171), (335, 16), (335, 35), (335, 54), (335, 73), (335, 92), (335, 111), (335, 130), (335, 149), (335, 168), (335, 187), (335, 206), (335, 225), (335, 263), (335, 282), (335, 301), (335, 320), (335, 339), (335, 358), (336, 296), (346, 287), (348, 209), (355, 87)

p	k	value	classes (a, b)
19	2	$(2, 1, 0; 1, 1)$	$(7, 52), (26, 128), (45, 204), (64, 280), (83, 356), (102, 71),$ $(121, 147), (159, 299), (178, 14), (197, 90), (216, 166),$ $(235, 242), (254, 318), (273, 33), (292, 109), (311, 185),$ $(330, 261), (349, 337)$

p	k	value	classes (a, b)
19	3	(1, 1, 0; 1, 1)	(28, 5839), (123, 975), (142, 5345), (161, 6466), (180, 4338), (199, 5820), (256, 4490), (275, 6694), (281, 2123), (294, 5649), (313, 1355), (351, 3597), (389, 6561), (484, 1697), (503, 6067), (522, 329), (541, 5060), (560, 6542), (617, 5212), (636, 557), (642, 3567), (655, 6371), (674, 2077), (712, 4319), (750, 424), (845, 2419), (864, 6789), (883, 1051), (902, 5782), (921, 405), (978, 5934), (997, 1279), (1003, 5011), (1016, 234), (1035, 2799), (1073, 5041), (1111, 1146), (1206, 3141), (1225, 652), (1244, 1773), (1263, 6504), (1282, 1127), (1339, 6656), (1358, 2001), (1364, 6455), (1377, 956), (1396, 3521), (1434, 5763), (1472, 1868), (1567, 3863), (1586, 1374), (1605, 2495), (1624, 367), (1643, 1849), (1700, 519), (1719, 2723), (1725, 1040), (1738, 1678), (1757, 4243), (1795, 6485), (1833, 2590), (1928, 4585), (1947, 2096), (1966, 3217), (1985, 1089), (2004, 2571), (2061, 1241), (2080, 3445), (2086, 2484), (2099, 2400), (2118, 4965), (2156, 348), (2194, 3312), (2289, 5307), (2308, 2818), (2327, 3939), (2346, 1811), (2365, 3293), (2422, 1963), (2441, 4167), (2447, 3928), (2460, 3122), (2479, 5687), (2517, 1070), (2555, 4034), (2650, 6029), (2669, 3540), (2688, 4661), (2707, 2533), (2726, 4015), (2783, 2685), (2802, 4889), (2808, 5372), (2821, 3844), (2840, 6409), (2878, 1792), (2916, 4756), (3011, 6751), (3030, 4262), (3049, 5383), (3068, 3255), (3087, 4737), (3144, 3407), (3163, 5611), (3169, 6816), (3182, 4566), (3201, 272), (3239, 2514), (3277, 5478), (3372, 614), (3391, 4984), (3410, 6105), (3429, 3977), (3448, 5459), (3505, 4129), (3524, 6333), (3530, 1401), (3543, 5288), (3562, 994), (3600, 3236), (3638, 6200), (3733, 1336), (3752, 5706), (3771, 6827), (3790, 4699), (3809, 6181), (3866, 4851), (3885, 196), (3891, 2845), (3904, 6010), (3923, 1716), (3961, 3958), (3999, 63), (4094, 2058), (4113, 6428), (4132, 690), (4151, 5421), (4170, 44), (4227, 5573), (4246, 918), (4252, 4289), (4265, 6732), (4284, 2438), (4322, 4680), (4360, 785), (4455, 2780), (4474, 291), (4493, 1412), (4512, 6143), (4531, 766), (4588, 6295), (4607, 1640), (4613, 5733), (4626, 595), (4645, 3160), (4683, 5402), (4721, 1507), (4816, 3502), (4835, 1013), (4854, 2134), (4873, 6), (4892, 1488), (4949, 158), (4968, 2362), (4974, 318), (4987, 1317), (5006, 3882), (5044, 6124), (5082, 2229), (5177, 4224), (5196, 1735), (5215, 2856), (5234, 728), (5253, 2210), (5310, 880), (5329, 3084), (5335, 1762), (5348, 2039), (5367, 4604), (5405, 6846), (5443, 2951), (5538, 4946), (5557, 2457), (5576, 3578), (5595, 1450), (5614, 2932), (5671, 1602), (5690, 3806), (5696, 3206), (5709, 2761), (5728, 5326), (5766, 709), (5804, 3673), (5899, 5668), (5918, 3179), (5937, 4300), (5956, 2172), (5975, 3654), (6032, 2324), (6051, 4528), (6070, 3483), (6089, 6048), (6127, 1431), (6165, 4395), (6260, 6390), (6279, 3901), (6298, 5022), (6317, 2894), (6336, 4376), (6393, 3046), (6412, 5250), (6418, 6094), (6431, 4205), (6450, 6770), (6488, 2153), (6526, 5117), (6621, 253), (6640, 4623), (6659, 5744), (6678, 3616), (6697, 5098), (6754, 3768), (6773, 5972), (6779, 679), (6792, 4927), (6811, 633), (6849, 2875)

p	k	value	classes (a, b)
19	3	$(2, 1, 0; -1, 1)$	(2501, 244), (2501, 605), (2501, 966), (2501, 1327), (2501, 1688), (2501, 2049), (2501, 2410), (2501, 2771), (2501, 3132), (2501, 3493), (2501, 3854), (2501, 4215), (2501, 4576), (2501, 4937), (2501, 5298), (2501, 5659), (2501, 6020), (2501, 6381), (2501, 6742)
19	3	$(2, 1, 0; 1, 1)$	(140, 945), (501, 2389), (862, 3833), (1223, 5277), (1584, 6721), (1945, 1306), (2306, 2750), (2667, 4194), (3028, 5638), (3389, 223), (3750, 1667), (4111, 3111), (4472, 4555), (4833, 5999), (5194, 584), (5555, 2028), (5916, 3472), (6277, 4916), (6638, 6360)
19	4	$(1, 1, 0; 1, 1)$	(6057, 11509), (12916, 38945), (19775, 66381), (26634, 93817), (33493, 121253), (40352, 18368), (47211, 45804), (54070, 73240), (60929, 100676), (67788, 128112), (74647, 25227), (81506, 52663), (88365, 80099), (95224, 107535), (102083, 4650), (108942, 32086), (115801, 59522), (122660, 86958), (129519, 114394)
31	1	$(0, 0, 0; 1, 1)$	(0, 26), (3, 12), (4, 18), (5, 7), (6, 19), (9, 25), (11, 3), (18, 17), (21, 2), (21, 15), (23, 3), (24, 10), (26, 8), (26, 15), (27, 1), (28, 28), (29, 13), (29, 24), (30, 16)

p	k	value	classes (a, b)
31	2	(0, 0, 0; 1, 1)	(3, 720), (4, 278), (4, 417), (8, 651), (13, 888), (14, 308), (20, 44), (21, 696), (23, 152), (23, 341), (28, 476), (28, 550), (29, 428), (34, 193), (35, 76), (35, 402), (39, 93), (44, 206), (45, 308), (51, 695), (52, 169), (54, 214), (54, 868), (59, 426), (59, 910), (60, 366), (65, 627), (66, 526), (66, 696), (70, 496), (75, 485), (76, 308), (82, 385), (83, 603), (85, 276), (85, 434), (90, 302), (90, 383), (91, 304), (96, 100), (97, 355), (97, 650), (101, 899), (106, 764), (107, 308), (113, 75), (114, 76), (116, 0), (116, 338), (121, 178), (121, 817), (122, 242), (127, 534), (128, 14), (128, 774), (132, 341), (137, 82), (138, 308), (144, 726), (145, 510), (147, 400), (147, 527), (152, 54), (152, 290), (153, 180), (158, 7), (159, 634), (159, 898), (163, 744), (168, 361), (169, 308), (175, 416), (176, 944), (178, 93), (178, 462), (183, 724), (183, 891), (184, 118), (189, 441), (190, 61), (190, 293), (194, 186), (199, 640), (200, 308), (207, 417), (209, 524), (209, 620), (214, 197), (214, 767), (215, 56), (220, 875), (221, 185), (221, 913), (225, 589), (230, 919), (231, 308), (237, 757), (238, 851), (240, 186), (240, 586), (245, 631), (245, 643), (246, 955), (251, 348), (252, 309), (252, 572), (256, 31), (261, 237), (262, 308), (268, 447), (269, 324), (271, 648), (271, 713), (276, 104), (276, 519), (277, 893), (282, 782), (283, 231), (283, 433), (287, 434), (292, 516), (293, 308), (299, 137), (300, 758), (302, 279), (302, 710), (307, 395), (307, 538), (308, 831), (313, 255), (314, 557), (314, 851), (318, 837), (323, 795), (324, 308), (330, 788), (331, 231), (333, 772), (333, 806), (338, 11), (338, 271), (339, 769), (344, 689), (345, 510), (345, 681), (349, 279), (354, 113), (355, 308), (361, 478), (362, 665), (364, 372), (364, 834), (369, 147), (369, 445), (370, 707), (375, 162), (376, 169), (376, 805), (380, 682), (385, 392), (386, 308), (392, 168), (393, 138), (395, 896), (395, 899), (400, 23), (400, 879), (401, 645), (406, 596), (407, 789), (407, 929), (411, 124), (416, 671), (417, 308), (423, 819), (424, 572), (426, 465), (426, 958), (431, 352), (431, 860), (432, 583), (437, 69), (438, 92), (438, 448), (442, 527), (447, 950), (448, 308), (454, 509), (455, 45), (457, 31), (457, 59), (462, 736), (462, 786), (463, 521), (468, 503), (469, 107), (469, 216), (473, 930), (478, 268), (479, 308), (485, 199), (486, 479), (488, 121), (488, 558), (493, 259), (493, 612), (494, 459), (499, 937), (500, 340), (500, 727), (504, 372), (509, 547), (510, 308), (516, 850), (517, 913), (519, 124), (519, 183), (524, 488), (524, 693), (525, 397), (530, 410), (531, 386), (531, 464), (535, 775), (540, 826), (541, 308), (547, 540), (548, 386), (550, 245), (550, 651), (555, 166), (555, 364), (556, 335), (561, 844), (562, 45), (562, 588), (566, 217), (571, 144), (572, 308), (578, 230), (579, 820), (581, 217), (581, 307), (586, 240), (586, 600), (587, 273), (592, 317), (593, 665), (593, 712), (597, 620), (602, 423), (603, 308), (609, 881), (610, 293), (612, 369), (612, 744), (617, 73), (617, 116), (618, 211), (623, 751), (624, 324), (624, 836), (628, 62), (633, 702), (634, 308), (640, 571), (641, 727), (643, 310), (643, 431), (648, 507), (648, 953), (649, 149), (654, 224), (655, 944), (655, 960), (659, 465), (664, 20), (665, 308), (671, 261), (672, 200), (674, 493), (674, 837), (679, 829), (679, 941), (680, 87), (685, 658), (686, 123), (686, 603), (690, 868), (695, 299), (696, 308), (702, 912), (703, 634), (705, 403), (705, 555), (710, 414), (710, 705), (711, 25), (716, 131), (717, 247), (717, 262), (721, 310), (726, 578), (727, 308), (733, 602), (734, 607), (736, 617), (736, 930), (741, 581), (741, 848), (742, 924), (747, 565), (748, 371), (748, 882), (752, 713), (757, 857), (758, 308), (764, 292), (765, 541), (767, 496), (767, 679), (772, 321), (772, 457), (773, 862), (778, 28), (779, 495), (779, 541), (783, 155), (788, 175),

p	k	value	classes (a, b)
31	3	$(0, 0, 0; 1, 1)$	(206, 28936), (1167, 19326), (2128, 9716), (3089, 106), (4050, 20287), (5011, 10677), (5972, 1067), (6933, 21248), (7894, 11638), (8855, 2028), (9816, 22209), (10777, 12599), (11738, 2989), (12699, 23170), (13660, 13560), (14621, 3950), (15582, 24131), (16543, 14521), (17504, 4911), (18465, 25092), (19426, 15482), (20387, 5872), (21348, 26053), (22309, 16443), (23270, 6833), (24231, 27014), (25192, 17404), (26153, 7794), (27114, 27975), (28075, 18365), (29036, 8755)
37	1	$(0, 0, 0; 1, 1)$	(3, 14), (3, 24), (5, 24), (6, 17), (6, 28), (6, 29), (7, 3), (8, 9), (8, 30), (8, 35), (9, 7), (12, 15), (12, 23), (12, 36), (14, 11), (15, 3), (15, 8), (15, 26), (17, 23), (19, 28), (23, 12), (24, 26), (24, 29), (25, 1), (25, 3), (25, 33), (26, 11), (27, 3), (28, 31), (29, 27), (30, 35), (31, 24), (33, 4), (33, 18), (34, 1), (35, 24)
37	2	$(0, 0, 0; 1, 1)$	(3, 258), (20, 781), (24, 1277), (32, 836), (33, 1014), (40, 1109), (57, 189), (61, 1277), (69, 799), (70, 163), (77, 591), (94, 966), (98, 1277), (106, 762), (107, 681), (114, 73), (131, 374), (135, 1277), (143, 725), (144, 1199), (151, 924), (168, 1151), (172, 1277), (180, 688), (181, 348), (188, 406), (205, 559), (209, 1277), (217, 651), (218, 866), (225, 1257), (242, 1336), (246, 1277), (254, 614), (255, 15), (262, 739), (279, 744), (283, 1277), (291, 577), (292, 533), (299, 221), (316, 152), (320, 1277), (328, 540), (329, 1051), (336, 1072), (353, 929), (357, 1277), (365, 503), (366, 200), (373, 554), (390, 337), (394, 1277), (402, 466), (403, 718), (410, 36), (427, 1114), (431, 1277), (439, 429), (440, 1236), (447, 887), (464, 522), (468, 1277), (476, 392), (477, 385), (484, 369), (501, 1299), (505, 1277), (513, 355), (514, 903), (521, 1220), (538, 707), (542, 1277), (550, 318), (551, 52), (558, 702), (575, 115), (579, 1277), (587, 281), (588, 570), (595, 184), (612, 892), (616, 1277), (624, 244), (625, 1088), (632, 1035), (649, 300), (653, 1277), (661, 207), (662, 237), (669, 517), (686, 1077), (690, 1277), (698, 170), (699, 755), (706, 1368), (723, 485), (727, 1277), (735, 133), (736, 1273), (743, 850), (760, 1262), (764, 1277), (772, 96), (773, 422), (780, 332), (797, 670), (801, 1277), (809, 59), (810, 940), (817, 1183), (834, 78), (838, 1277), (846, 22), (847, 89), (854, 665), (871, 855), (875, 1277), (883, 1354), (884, 607), (891, 147), (908, 263), (912, 1277), (920, 1317), (921, 1125), (928, 998), (945, 1040), (949, 1277), (957, 1280), (958, 274), (965, 480), (982, 448), (986, 1277), (994, 1243), (995, 792), (1002, 1331), (1019, 1225), (1023, 1277), (1031, 1206), (1032, 1310), (1039, 813), (1056, 633), (1060, 1277), (1068, 1169), (1069, 459), (1076, 295), (1093, 41), (1097, 1277), (1105, 1132), (1106, 977), (1113, 1146), (1130, 818), (1134, 1277), (1142, 1095), (1143, 126), (1150, 628), (1167, 226), (1171, 1277), (1179, 1058), (1180, 644), (1187, 110), (1204, 1003), (1208, 1277), (1216, 1021), (1217, 1162), (1224, 961), (1241, 411), (1245, 1277), (1253, 984), (1254, 311), (1261, 443), (1278, 1188), (1282, 1277), (1290, 947), (1291, 829), (1298, 1294), (1315, 596), (1319, 1277), (1327, 910), (1328, 1347), (1335, 776), (1352, 4), (1356, 1277), (1364, 873), (1365, 496)
43	1	$(0, 0, 0; 1, 1)$	(0, 31), (6, 4), (9, 13), (11, 7), (14, 4), (16, 0), (17, 3), (17, 37), (19, 17), (20, 30), (23, 3), (23, 15), (23, 25), (25, 23), (27, 0), (27, 19), (27, 24), (32, 18), (33, 39), (34, 22), (35, 33), (36, 39), (37, 21), (37, 28), (37, 37), (38, 22)

p	k	value	classes (a, b)
43	2	$(0, 0, 0; 1, 1)$	$(18, 205), (24, 39), (31, 93), (61, 635), (67, 1501), (74, 308),$ $(104, 1065), (110, 1114), (117, 523), (147, 1495), (153, 727),$ $(160, 738), (190, 76), (196, 340), (203, 953), (233, 506),$ $(239, 1802), (246, 1168), (276, 936), (282, 1415), (289, 1383),$ $(319, 1366), (325, 1028), (332, 1598), (362, 1796), (368, 641),$ $(375, 1813), (405, 377), (411, 254), (418, 179), (448, 807),$ $(454, 1716), (461, 394), (491, 1237), (497, 1329), (504, 609),$ $(534, 1667), (540, 942), (547, 824), (577, 248), (583, 555),$ $(590, 1039), (620, 678), (626, 168), (633, 1254), (663, 1108),$ $(669, 1630), (676, 1469), (706, 1538), (712, 1243), (719, 1684),$ $(749, 119), (755, 856), (762, 50), (792, 549), (798, 469),$ $(805, 265), (835, 979), (841, 82), (848, 480), (878, 1409),$ $(884, 1544), (891, 695), (921, 1839), (927, 1157), (934, 910),$ $(964, 420), (970, 770), (977, 1125), (1007, 850), (1013, 383),$ $(1020, 1340), (1050, 1280), (1056, 1845), (1063, 1555),$ $(1093, 1710), (1099, 1458), (1106, 1770), (1136, 291),$ $(1142, 1071), (1149, 136), (1179, 721), (1185, 684), (1192, 351),$ $(1222, 1151), (1228, 297), (1235, 566), (1265, 1581),$ $(1271, 1759), (1278, 781), (1308, 162), (1314, 1372), (1321, 996),$ $(1351, 592), (1357, 985), (1364, 1211), (1394, 1022), (1400, 598),$ $(1407, 1426), (1437, 1452), (1443, 211), (1450, 1641), (1480, 33),$ $(1486, 1673), (1493, 7), (1523, 463), (1529, 1286), (1536, 222),$ $(1566, 893), (1572, 899), (1579, 437), (1609, 1323), (1615, 512),$ $(1622, 652), (1652, 1753), (1658, 125), (1665, 867), (1695, 334),$ $(1701, 1587), (1708, 1082), (1738, 764), (1744, 1200),$ $(1751, 1297), (1781, 1194), (1787, 813), (1794, 1512),$ $(1824, 1624), (1830, 426), (1837, 1727)$
61	1	$(0, 0, 0; 1, 1)$	$(1, 4), (1, 5), (1, 52), (3, 6), (4, 41), (7, 48), (8, 33), (8, 39),$ $(8, 50), (10, 51), (12, 42), (13, 43), (14, 21), (16, 36), (19, 51),$ $(20, 53), (22, 14), (22, 18), (22, 29), (23, 26), (23, 39), (24, 59),$ $(25, 7), (25, 57), (25, 58), (26, 33), (26, 41), (26, 48), (28, 33),$ $(28, 56), (29, 6), (31, 9), (31, 22), (31, 30), (33, 36), (35, 31),$ $(36, 41), (37, 8), (38, 50), (39, 12), (40, 11), (42, 39), (43, 9),$ $(43, 24), (43, 28), (44, 4), (47, 58), (49, 15), (49, 39), (53, 10),$ $(53, 14), (53, 37), (54, 7), (55, 7), (55, 23), (55, 31), (56, 41),$ $(57, 43), (58, 10), (59, 5), (60, 33)$

p	k	value	classes (a, b)
61	2	$(0, 0, 0; 1, 1)$	(23, 1582), (30, 1203), (49, 800), (84, 850), (91, 3399), (110, 861), (145, 118), (152, 1874), (171, 922), (206, 3107), (213, 349), (232, 983), (267, 2375), (274, 2545), (293, 1044), (328, 1643), (335, 1020), (354, 1105), (389, 911), (396, 3216), (415, 1166), (450, 179), (457, 1691), (476, 1227), (511, 3168), (518, 166), (537, 1288), (572, 2436), (579, 2362), (598, 1349), (633, 1704), (640, 837), (659, 1410), (694, 972), (701, 3033), (720, 1471), (755, 240), (762, 1508), (781, 1532), (816, 3229), (823, 3704), (842, 1593), (877, 2497), (884, 2179), (903, 1654), (938, 1765), (945, 654), (964, 1715), (999, 1033), (1006, 2850), (1025, 1776), (1060, 301), (1067, 1325), (1086, 1837), (1121, 3290), (1128, 3521), (1147, 1898), (1182, 2558), (1189, 1996), (1208, 1959), (1243, 1826), (1250, 471), (1269, 2020), (1304, 1094), (1311, 2667), (1330, 2081), (1365, 362), (1372, 1142), (1391, 2142), (1426, 3351), (1433, 3338), (1452, 2203), (1487, 2619), (1494, 1813), (1513, 2264), (1548, 1887), (1555, 288), (1574, 2325), (1609, 1155), (1616, 2484), (1635, 2386), (1670, 423), (1677, 959), (1696, 2447), (1731, 3412), (1738, 3155), (1757, 2508), (1792, 2680), (1799, 1630), (1818, 2569), (1853, 1948), (1860, 105), (1879, 2630), (1914, 1216), (1921, 2301), (1940, 2691), (1975, 484), (1982, 776), (2001, 2752), (2036, 3473), (2043, 2972), (2062, 2813), (2097, 2741), (2104, 1447), (2123, 2874), (2158, 2009), (2165, 3643), (2184, 2935), (2219, 1277), (2226, 2118), (2245, 2996), (2280, 545), (2287, 593), (2306, 3057), (2341, 3534), (2348, 2789), (2367, 3118), (2402, 2802), (2409, 1264), (2428, 3179), (2463, 2070), (2470, 3460), (2489, 3240), (2524, 1338), (2531, 1935), (2550, 3301), (2585, 606), (2592, 410), (2611, 3362), (2646, 3595), (2653, 2606), (2672, 3423), (2707, 2863), (2714, 1081), (2733, 3484), (2768, 2131), (2775, 3277), (2794, 3545), (2829, 1399), (2836, 1752), (2855, 3606), (2890, 667), (2897, 227), (2916, 3667), (2951, 3656), (2958, 2423), (2977, 7), (3012, 2924), (3019, 898), (3038, 68), (3073, 2192), (3080, 3094), (3099, 129), (3134, 1460), (3141, 1569), (3160, 190), (3195, 728), (3202, 44), (3221, 251), (3256, 3717), (3263, 2240), (3282, 312), (3317, 2985), (3324, 715), (3343, 373), (3378, 2253), (3385, 2911), (3404, 434), (3439, 1521), (3446, 1386), (3465, 495), (3500, 789), (3507, 3582), (3526, 556), (3561, 57), (3568, 2057), (3587, 617), (3622, 3046), (3629, 532), (3648, 678), (3683, 2314), (3690, 2728), (3709, 739)
67	1	$(0, 0, 0; 1, 1)$	(0, 41), (1, 18), (1, 53), (1, 63), (3, 21), (3, 55), (3, 58), (5, 52), (8, 18), (9, 1), (10, 3), (10, 31), (10, 33), (11, 37), (12, 10), (13, 19), (14, 54), (15, 65), (16, 28), (16, 53), (18, 34), (18, 39), (18, 61), (19, 31), (19, 40), (19, 63), (20, 0), (21, 5), (22, 9), (25, 2), (25, 23), (25, 42), (26, 3), (27, 65), (29, 4), (31, 12), (33, 26), (33, 46), (33, 62), (34, 47), (35, 7), (36, 2), (36, 29), (36, 36), (37, 21), (38, 5), (39, 7), (39, 27), (39, 33), (40, 14), (40, 15), (40, 38), (41, 50), (44, 3), (45, 23), (46, 14), (46, 57), (46, 63), (47, 0), (47, 17), (47, 50), (48, 37), (48, 60), (49, 56), (50, 32), (50, 40), (50, 62), (52, 57), (53, 10), (54, 3), (54, 17), (54, 47), (56, 34), (56, 46), (56, 54), (58, 15), (58, 53), (58, 66), (59, 53), (60, 16), (60, 25), (60, 26), (61, 11), (61, 28), (63, 40), (66, 6)

p	k	value	classes (a, b)
67	2	$(0, 0, 0; 1, 1)$	$(2, 1849), (69, 107), (136, 2854), (203, 1112), (270, 3859),$ $(337, 2117), (404, 375), (471, 3122), (538, 1380), (605, 4127),$ $(672, 2385), (739, 643), (806, 3390), (873, 1648), (940, 4395),$ $(1007, 2653), (1074, 911), (1141, 3658), (1208, 1916),$ $(1275, 174), (1342, 2921), (1409, 1179), (1476, 3926),$ $(1543, 2184), (1610, 442), (1677, 3189), (1744, 1447),$ $(1811, 4194), (1878, 2452), (1945, 710), (2012, 3457),$ $(2079, 1715), (2146, 4462), (2213, 2720), (2280, 978),$ $(2347, 3725), (2414, 1983), (2481, 241), (2548, 2988),$ $(2615, 1246), (2682, 3993), (2749, 2251), (2816, 509),$ $(2883, 3256), (2950, 1514), (3017, 4261), (3084, 2519),$ $(3151, 777), (3218, 3524), (3285, 1782), (3352, 40), (3419, 2787),$ $(3486, 1045), (3553, 3792), (3620, 2050), (3687, 308),$ $(3754, 3055), (3821, 1313), (3888, 4060), (3955, 2318),$ $(4022, 576), (4089, 3323), (4156, 1581), (4223, 4328),$ $(4290, 2586), (4357, 844), (4424, 3591)$
73	1	$(0, 0, 0; 1, 1)$	$(0, 39), (1, 25), (1, 54), (1, 67), (2, 10), (2, 12), (2, 51), (3, 31),$ $(5, 43), (6, 21), (7, 13), (8, 32), (10, 63), (11, 59), (12, 0),$ $(13, 42), (14, 12), (14, 62), (14, 72), (15, 17), (16, 24), (16, 53),$ $(16, 69), (18, 10), (18, 23), (18, 40), (19, 63), (20, 1), (20, 16),$ $(20, 56), (21, 53), (22, 18), (23, 20), (24, 64), (27, 33), (28, 12),$ $(29, 12), (30, 45), (31, 37), (32, 0), (35, 24), (36, 37), (37, 42),$ $(38, 9), (39, 6), (39, 61), (40, 66), (41, 0), (42, 25), (43, 9),$ $(43, 27), (43, 37), (44, 40), (44, 41), (44, 65), (45, 25), (47, 7),$ $(48, 22), (50, 11), (50, 65), (50, 70), (51, 6), (51, 16), (51, 51),$ $(52, 26), (53, 5), (53, 69), (53, 72), (54, 22), (55, 31), (57, 55),$ $(58, 25), (59, 1), (59, 17), (59, 55), (60, 6), (61, 0), (62, 26),$ $(62, 56), (62, 64), (64, 15), (66, 5), (69, 6), (70, 49), (72, 20),$ $(72, 23), (72, 30)$
73	2	$(0, 0, 0; 1, 1)$	$(34, 3563), (107, 4074), (180, 4585), (253, 5096), (326, 278),$ $(399, 789), (472, 1300), (545, 1811), (618, 2322), (691, 2833),$ $(764, 3344), (837, 3855), (910, 4366), (983, 4877), (1056, 59),$ $(1129, 570), (1202, 1081), (1275, 1592), (1348, 2103),$ $(1421, 2614), (1494, 3125), (1567, 3636), (1640, 4147),$ $(1713, 4658), (1786, 5169), (1859, 351), (1932, 862),$ $(2005, 1373), (2078, 1884), (2151, 2395), (2224, 2906),$ $(2297, 3417), (2370, 3928), (2443, 4439), (2516, 4950),$ $(2589, 132), (2662, 643), (2735, 1154), (2808, 1665),$ $(2881, 2176), (2954, 2687), (3027, 3198), (3100, 3709),$ $(3173, 4220), (3246, 4731), (3319, 5242), (3392, 424),$ $(3465, 935), (3538, 1446), (3611, 1957), (3684, 2468),$ $(3757, 2979), (3830, 3490), (3903, 4001), (3976, 4512),$ $(4049, 5023), (4122, 205), (4195, 716), (4268, 1227),$ $(4341, 1738), (4414, 2249), (4487, 2760), (4560, 3271),$ $(4633, 3782), (4706, 4293), (4779, 4804), (4852, 5315),$ $(4925, 497), (4998, 1008), (5071, 1519), (5144, 2030),$ $(5217, 2541), (5290, 3052)$

p	k	value	classes (a, b)
97	1	$(0, 0, 0; 1, 1)$	$(0, 30), (5, 34), (5, 66), (5, 94), (6, 84), (7, 77), (11, 8), (12, 22),$ $(13, 41), (13, 63), (13, 90), (14, 0), (15, 54), (17, 0), (17, 9),$ $(17, 88), (18, 96), (20, 41), (20, 60), (20, 93), (21, 28), (22, 54),$ $(25, 94), (26, 94), (27, 93), (28, 17), (30, 49), (30, 62), (30, 83),$ $(31, 3), (31, 41), (31, 53), (32, 60), (33, 41), (34, 59), (34, 76),$ $(36, 28), (37, 38), (38, 30), (38, 74), (38, 90), (39, 95), (41, 94),$ $(42, 40), (43, 32), (45, 5), (46, 69), (47, 10), (47, 43), (47, 44),$ $(48, 38), (50, 23), (51, 76), (53, 51), (54, 27), (56, 19), (57, 32),$ $(57, 74), (57, 88), (58, 85), (59, 16), (60, 7), (61, 47), (62, 21),$ $(63, 15), (63, 35), (63, 47), (64, 19), (65, 11), (68, 52), (69, 3),$ $(69, 30), (69, 64), (70, 48), (71, 89), (73, 73), (74, 15), (74, 36),$ $(74, 46), (75, 26), (75, 83), (75, 85), (79, 1), (79, 24), (79, 72),$ $(80, 0), (80, 42), (80, 55), (81, 5), (82, 55), (83, 0), (84, 22),$ $(85, 72), (86, 79), (87, 30), (88, 35), (92, 17), (93, 16), (94, 63),$ $(95, 36), (95, 78), (95, 80), (96, 57)$

p	k	value	classes (a, b)
97	2	(0, 0, 0; 1, 1)	(16, 4138), (24, 5130), (91, 2981), (113, 355), (121, 183), (188, 3466), (210, 5981), (218, 4645), (285, 3951), (307, 2198), (315, 9107), (382, 4436), (404, 7824), (412, 4160), (479, 4921), (501, 4041), (509, 8622), (576, 5406), (598, 258), (606, 3675), (673, 5891), (695, 5884), (703, 8137), (770, 6376), (792, 2101), (800, 3190), (867, 6861), (889, 7727), (897, 7652), (964, 7346), (986, 3944), (994, 2705), (1061, 7831), (1083, 161), (1091, 7167), (1158, 8316), (1180, 5787), (1188, 2220), (1255, 8801), (1277, 2004), (1285, 6682), (1352, 9286), (1374, 7630), (1382, 1735), (1449, 362), (1471, 3847), (1479, 6197), (1546, 847), (1568, 64), (1576, 1250), (1643, 1332), (1665, 5690), (1673, 5712), (1740, 1817), (1762, 1907), (1770, 765), (1837, 2302), (1859, 7533), (1867, 5227), (1934, 2787), (1956, 3750), (1964, 280), (2031, 3272), (2053, 9376), (2061, 4742), (2128, 3757), (2150, 5593), (2158, 9204), (2225, 4242), (2247, 1810), (2255, 4257), (2322, 4727), (2344, 7436), (2352, 8719), (2419, 5212), (2441, 3653), (2449, 3772), (2516, 5697), (2538, 9279), (2546, 8234), (2613, 6182), (2635, 5496), (2643, 3287), (2710, 6667), (2732, 1713), (2740, 7749), (2807, 7152), (2829, 7339), (2837, 2802), (2904, 7637), (2926, 3556), (2934, 7264), (3001, 8122), (3023, 9182), (3031, 2317), (3098, 8607), (3120, 5399), (3128, 6779), (3195, 9092), (3217, 1616), (3225, 1832), (3292, 168), (3314, 7242), (3322, 6294), (3389, 653), (3411, 3459), (3419, 1347), (3486, 1138), (3508, 9085), (3516, 5809), (3583, 1623), (3605, 5302), (3613, 862), (3680, 2108), (3702, 1519), (3710, 5324), (3777, 2593), (3799, 7145), (3807, 377), (3874, 3078), (3896, 3362), (3904, 4839), (3971, 3563), (3993, 8988), (4001, 9301), (4068, 4048), (4090, 5205), (4098, 4354), (4165, 4533), (4187, 1422), (4195, 8816), (4262, 5018), (4284, 7048), (4292, 3869), (4359, 5503), (4381, 3265), (4389, 8331), (4456, 5988), (4478, 8891), (4486, 3384), (4553, 6473), (4575, 5108), (4583, 7846), (4650, 6958), (4672, 1325), (4680, 2899), (4747, 7443), (4769, 6951), (4777, 7361), (4844, 7928), (4866, 3168), (4874, 2414), (4941, 8413), (4963, 8794), (4971, 6876), (5038, 8898), (5060, 5011), (5068, 1929), (5135, 9383), (5157, 1228), (5165, 6391), (5232, 459), (5254, 6854), (5262, 1444), (5329, 944), (5351, 3071), (5359, 5906), (5426, 1429), (5448, 8697), (5456, 959), (5523, 1914), (5545, 4914), (5553, 5421), (5620, 2399), (5642, 1131), (5650, 474), (5717, 2884), (5739, 6757), (5747, 4936), (5814, 3369), (5836, 2974), (5844, 9398), (5911, 3854), (5933, 8600), (5941, 4451), (6008, 4339), (6030, 4817), (6038, 8913), (6105, 4824), (6127, 1034), (6135, 3966), (6202, 5309), (6224, 6660), (6232, 8428), (6299, 5794), (6321, 2877), (6329, 3481), (6396, 6279), (6418, 8503), (6426, 7943), (6493, 6764), (6515, 4720), (6523, 2996), (6590, 7249), (6612, 937), (6620, 7458), (6687, 7734), (6709, 6563), (6717, 2511), (6784, 8219), (6806, 2780), (6814, 6973), (6881, 8704), (6903, 8406), (6911, 2026), (6978, 9189), (7000, 4623), (7008, 6488), (7075, 265), (7097, 840), (7105, 1541), (7172, 750), (7194, 6466), (7202, 6003), (7269, 1235), (7291, 2683), (7299, 1056), (7366, 1720), (7388, 8309), (7396, 5518), (7463, 2205), (7485, 4526), (7493, 571), (7560, 2690), (7582, 743), (7590, 5033), (7657, 3175), (7679, 6369), (7687, 86), (7754, 3660), (7776, 2586), (7784, 4548), (7851, 4145), (7873, 8212), (7881, 9010), (7948, 4630), (7970, 4429), (7978, 4063), (8045, 5115), (8067, 646), (8075, 8525), (8142, 5600), (8164, 6272), (8172, 3578), (8239, 6085), (8261, 2489), (8269, 8040), (8336, 6570), (8358, 8115), (8366, 3093), (8423, 7055), (8455, 4222), (8463, 7555), (8520, 7540),

p	k	value	classes (a, b)
109	1	$(0, 0, 0; 1, 1)$	$(0, 13), (2, 22), (3, 59), (4, 74), (5, 92), (6, 34), (6, 92), (7, 17),$ $(9, 83), (10, 81), (13, 30), (15, 26), (16, 23), (16, 25), (16, 61),$ $(18, 55), (19, 21), (19, 31), (19, 57), (20, 64), (21, 18), (21, 92),$ $(21, 108), (22, 99), (23, 101), (24, 1), (26, 44), (26, 80), (26, 94),$ $(27, 76), (28, 3), (28, 31), (28, 75), (30, 28), (31, 103), (32, 46),$ $(33, 96), (34, 59), (35, 4), (35, 43), (35, 62), (36, 2), (36, 16),$ $(36, 91), (37, 21), (38, 1), (39, 39), (39, 85), (39, 94), (41, 108),$ $(42, 12), (43, 42), (44, 105), (46, 6), (47, 37), (48, 51), (48, 70),$ $(48, 97), (49, 2), (50, 25), (50, 42), (51, 42), (52, 107), (57, 22),$ $(58, 49), (61, 61), (62, 71), (64, 71), (65, 28), (67, 17), (68, 1),$ $(69, 24), (69, 34), (69, 51), (70, 54), (71, 108), (73, 49), (74, 7),$ $(74, 42), (74, 60), (75, 30), (76, 64), (77, 48), (77, 78), (77, 92),$ $(78, 3), (79, 102), (80, 7), (80, 24), (80, 78), (82, 47), (82, 83),$ $(83, 4), (83, 46), (83, 59), (85, 37), (85, 73), (85, 108), (87, 37),$ $(88, 1), (90, 29), (90, 36), (90, 44), (91, 77), (92, 19), (94, 104),$ $(95, 8), (98, 59), (101, 35), (102, 72), (103, 23), (104, 93),$ $(106, 26), (108, 20), (108, 37), (108, 52)$
109	2	$(0, 0, 0; 1, 1)$	$(82, 9571), (191, 7282), (300, 4993), (409, 2704), (518, 415),$ $(627, 10007), (736, 7718), (845, 5429), (954, 3140), (1063, 851),$ $(1172, 10443), (1281, 8154), (1390, 5865), (1499, 3576),$ $(1608, 1287), (1717, 10879), (1826, 8590), (1935, 6301),$ $(2044, 4012), (2153, 1723), (2262, 11315), (2371, 9026),$ $(2480, 6737), (2589, 4448), (2698, 2159), (2807, 11751),$ $(2916, 9462), (3025, 7173), (3134, 4884), (3243, 2595),$ $(3352, 306), (3461, 9898), (3570, 7609), (3679, 5320),$ $(3788, 3031), (3897, 742), (4006, 10334), (4115, 8045),$ $(4224, 5756), (4333, 3467), (4442, 1178), (4551, 10770),$ $(4660, 8481), (4769, 6192), (4878, 3903), (4987, 1614),$ $(5096, 11206), (5205, 8917), (5314, 6628), (5423, 4339),$ $(5532, 2050), (5641, 11642), (5750, 9353), (5859, 7064),$ $(5968, 4775), (6077, 2486), (6186, 197), (6295, 9789),$ $(6404, 7500), (6513, 5211), (6622, 2922), (6731, 633),$ $(6840, 10225), (6949, 7936), (7058, 5647), (7167, 3358),$ $(7276, 1069), (7385, 10661), (7494, 8372), (7603, 6083),$ $(7712, 3794), (7821, 1505), (7930, 11097), (8039, 8808),$ $(8148, 6519), (8257, 4230), (8366, 1941), (8475, 11533),$ $(8584, 9244), (8693, 6955), (8802, 4666), (8911, 2377),$ $(9020, 88), (9129, 9680), (9238, 7391), (9347, 5102),$ $(9456, 2813), (9565, 524), (9674, 10116), (9783, 7827),$ $(9892, 5538), (10001, 3249), (10110, 960), (10219, 10552),$ $(10328, 8263), (10437, 5974), (10546, 3685), (10655, 1396),$ $(10764, 10988), (10873, 8699), (10982, 6410), (11091, 4121),$ $(11200, 1832), (11309, 11424), (11418, 9135), (11527, 6846),$ $(11636, 4557), (11745, 2268), (11854, 11860)$

p	k	value	classes (a, b)
379	1	$(0, 0, 0; 1, 1)$	$(0, 238), (3, 307), (4, 355), (6, 114), (8, 113), (9, 226), (12, 337),$ $(15, 56), (16, 121), (17, 81), (18, 254), (19, 182), (20, 246),$ $(21, 58), (21, 85), (21, 236), (23, 78), (25, 184), (26, 3),$ $(27, 323), (28, 26), (28, 117), (28, 236), (29, 80), (30, 324),$ $(33, 317), (34, 171), (35, 51), (35, 125), (35, 203), (37, 243),$ $(37, 253), (37, 262), (38, 101), (38, 303), (38, 354), (39, 80),$ $(41, 41), (41, 108), (41, 230), (43, 323), (44, 133), (45, 3),$ $(47, 156), (49, 39), (50, 330), (51, 217), (52, 149), (53, 91),$ $(54, 100), (56, 166), (58, 86), (58, 327), (58, 345), (59, 330),$ $(60, 172), (60, 222), (60, 364), (61, 318), (62, 126), (63, 81),$ $(63, 312), (63, 365), (65, 85), (65, 121), (65, 173), (66, 97),$ $(69, 359), (70, 275), (72, 52), (74, 19), (74, 135), (74, 225),$ $(76, 192), (77, 89), (77, 115), (77, 175), (78, 361), (80, 179),$ $(82, 66), (83, 112), (83, 171), (86, 324), (87, 48), (87, 346),$ $(87, 364), (88, 233), (89, 174), (89, 274), (89, 310), (90, 300),$ $(91, 35), (92, 209), (92, 340), (95, 99), (96, 95), (97, 208),$ $(98, 40), (98, 129), (98, 210), (99, 39), (99, 115), (99, 225),$ $(101, 28), (101, 45), (101, 306), (104, 167), (106, 162), (108, 12),$ $(108, 82), (108, 285), (110, 326), (111, 81), (112, 192), (115, 94),$ $(115, 323), (115, 341), (117, 353), (120, 0), (121, 85), (122, 78),$ $(123, 311), (125, 331), (126, 59), (127, 235), (129, 212),$ $(130, 305), (131, 63), (131, 80), (131, 236), (132, 162),$ $(133, 106), (134, 185), (136, 51), (138, 174), (139, 26),$ $(140, 111), (142, 366), (143, 99), (144, 11), (145, 335),$ $(146, 144), (146, 261), (146, 353), (147, 121), (148, 181),$ $(150, 338), (151, 121), (154, 67), (155, 234), (159, 55),$ $(160, 242), (161, 45), (161, 74), (161, 260), (162, 48), (162, 123),$ $(162, 208), (163, 162), (165, 293), (166, 43), (167, 296),$ $(168, 91), (169, 173), (171, 172), (172, 85), (173, 129),$ $(174, 231), (175, 11), (179, 264), (180, 11), (180, 80), (180, 288),$ $(181, 249), (183, 376), (186, 255), (187, 42), (188, 81),$ $(191, 152), (192, 311), (193, 221), (194, 323), (195, 143),$ $(195, 240), (195, 375), (196, 104), (198, 29), (198, 170),$ $(198, 180), (199, 236), (201, 138), (202, 209), (202, 216),$ $(202, 333), (203, 177), (205, 377), (206, 317), (207, 127),$ $(208, 71), (210, 321), (211, 155), (211, 283), (211, 320),$ $(212, 266), (215, 151), (216, 272), (218, 272), (219, 8), (221, 97),$ $(221, 107), (221, 175), (222, 314), (223, 201), (223, 229),$ $(223, 328), (224, 208), (225, 319), (226, 134), (226, 279),$ $(226, 345), (227, 203), (228, 213), (229, 211), (229, 244),$ $(229, 303), (231, 229), (231, 233), (231, 296), (234, 180),$ $(235, 25), (235, 154), (235, 200), (236, 106), (237, 227),$ $(238, 211), (238, 221), (238, 326), (240, 349), (241, 68),$ $(243, 45), (244, 41), (244, 355), (244, 362), (245, 66), (246, 333),$ $(248, 14), (248, 145), (248, 220), (250, 279), (252, 7), (253, 18),$ $(253, 45), (253, 316), (254, 24), (255, 235), (256, 53), (258, 158),$ $(259, 0), (259, 11), (259, 368), (260, 74), (261, 160), (261, 257),$ $(261, 341), (262, 280), (265, 297), (268, 182), (269, 57),$ $(270, 183), (271, 192), (272, 188), (272, 240), (272, 330),$ $(275, 73), (275, 98), (275, 208), (277, 185), (277, 245),$ $(277, 328), (278, 170), (278, 246), (278, 342), (279, 357),$ $(280, 337), (282, 349), (285, 344), (286, 150), (287, 349),$ $(288, 205), (288, 271), (288, 282), (289, 120), (290, 26),$ $(290, 76), (290, 277), (291, 87), (292, 152), (294, 320),$ $(295, 319), (296, 244), (297, 30), (298, 6), (299, 192), (299, 273),$ $(299, 293), (300, 301), (301, 26), (302, 35), (303, 217),$ $(304, 215), (305, 199), (306, 336), (307, 279), (308, 367),$ $(309, 229), (309, 252), (309, 277), (310, 79), (311, 65),$ $(314, 357), (315, 63), (317, 112), (317, 278), (317, 368),$ $(318, 63), (319, 215), (321, 62), (322, 254), (323, 138), (324, 61),$ $(326, 269), (327, 224), (328, 349), (329, 24), (329, 108),$ $(329, 247), (332, 136), (333, 285), (334, 44), (334, 92),$ $(334, 243), (335, 299), (336, 222), (337, 3), (339, 331),$

p	k	value	classes (a, b)
379	2	(0, 0, 0; 1, 1)	(83, 94467), (462, 46713), (841, 142600), (1220, 94846), (1599, 47092), (1978, 142979), (2357, 95225), (2736, 47471), (3115, 143358), (3494, 95604), (3873, 47850), (4252, 96), (4631, 95983), (5010, 48229), (5389, 475), (5768, 96362), (6147, 48608), (6526, 854), (6905, 96741), (7284, 48987), (7663, 1233), (8042, 97120), (8421, 49366), (8800, 1612), (9179, 97499), (9558, 49745), (9937, 1991), (10316, 97878), (10695, 50124), (11074, 2370), (11453, 98257), (11832, 50503), (12211, 2749), (12590, 98636), (12969, 50882), (13348, 3128), (13727, 99015), (14106, 51261), (14485, 3507), (14864, 99394), (15243, 51640), (15622, 3886), (16001, 99773), (16380, 52019), (16759, 4265), (17138, 100152), (17517, 52398), (17896, 4644), (18275, 100531), (18654, 52777), (19033, 5023), (19412, 100910), (19791, 53156), (20170, 5402), (20549, 101289), (20928, 53535), (21307, 5781), (21686, 101668), (22065, 53914), (22444, 6160), (22823, 102047), (23202, 54293), (23581, 6539), (23960, 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